Seminar on Algebraic Geometry at Bois Marie 1962

Local Cohomology of Coherent Sheaves and Local and Global Lefschetz Theorems (SGA 2)

Alexander Grothendieck
(written up by a group of participants)
With an additional exposé by Michèle Raynaud

July 22, 2026

Abstract

This volume is a newly updated edition of the book *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)*, Advanced Studies in Pure Mathematics 2, North-Holland Publishing Company, Amsterdam, 1968, by A. Grothendieck *et al.*

The monograph gives necessary and sufficient conditions for the finiteness of the local cohomology sheaves of coherent sheaves. These results yield algebraization theorems which, together with purity results also proved in the text, lead to Lefschetz-type theorems for both the fundamental group and the Picard group.

Keywords. Algebraization, Lefschetz theorem, local duality, local cohomology, depth, homotopical depth, fundamental group, Picard group.

2020 Mathematics Subject Classification. 14-02, 14B10, 14B15, 14B20, 14C22, 14J70, 32S20.

Editor's Preface

The present text is a new updated edition of the book "Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)", Advanced Studies in Pure Mathematics 2, North-Holland Publishing Company, Amsterdam, 1968, by A. Grothendieck *et al.* It is the second part of the SGA project initiated by B. Edixhoven, who prepared a new edition of SGA 1. This version is intended to reproduce the original text as closely as possible, while incorporating minor typographical corrections and numbered editorial footnotes explaining the current status of questions raised in the first edition. Further details have also been supplied for some proofs. To avoid confusion, original notes retain star-based numbering, whereas new editorial notes use Arabic numbers. The page numbers of the original edition are shown in the margins.

I thank the mathematicians who carried out most of the initial typesetting in LaTeX 2_ε: L. Bayle, N. Borne, O. Brinon, J. Buresi, M. Chardin, F. Ducrot, P. Graftiaux, F. Han, P. Karwasz, L. Koelblen, D. Madore, S. Morel, D. Naie, B. Osserman, J. Riou, and V. Sécherre, as well as C. Sabbah for adapting the text to the SMF format. I also thank J.-B. Bost, P. Colmez, O. Gabber, W. Fulton, S. Kleiman, F. Orgogozo, M. Raynaud, and J.-P. Serre for their comments and advice.

Yves Laszlo, editor

Contents

Introduction

We present here, in revised and expanded form, a photo-offset reissue of the second Algebraic Geometry Seminar of the Institut des Hautes Études Scientifiques, held in 1962 and originally issued in mimeographed form. 1

The reader is referred to the Introduction to the first of these seminars (cited below as SGA 1) for the aims pursued by these seminars and their relation to the Éléments de Géométrie Algébrique.

The text of Exposés I through XI was written up as the seminar proceeded, from my oral lectures and handwritten notes, by a group of participants comprising I. Giorgiutti, J. Giraud, Mlle M. Jaffe (now Mme M. Hakim), and A. Laudal. These notes were originally regarded as provisional and intended for very limited circulation, pending their absorption into the EGA—an absorption that has now become problematic, to say the least, as it has for the other parts of the SGA. As the warning to the original edition said, the "confidential"character of the notes was meant to excuse certain "stylistic weaknesses", doubtless more apparent in the present seminar, SGA 2, than in the others. I have tried, as far as possible, to remedy them in the present reissue by a fairly thorough revision of the original text. In particular, I have harmonized the systems used to number statements in the different Exposés, introducing everywhere the same decimal system, already used in most of the original Exposés in the present SGA 2 as well as in all the other parts of the SGA. This led me, in particular, to revise completely the numbering[1] of the statements in Exposés III through VIII (and consequently 2
the references to those Exposés)[2]. I have also tried to remove from the original text the principal typing and syntactical errors, which were numerous and troublesome. Moreover, Mme M. Hakim kindly undertook to rewrite Exposé IV in a less telegraphic style than the original. As in the other SGA reissues, I have also added a number of footnotes, either to give additional references or to indicate the status of questions on which progress has been made since the original text was written. Finally, the seminar has been enlarged by a new Exposé, namely Exposé XIV, written by Mme Michèle Raynaud in 1967, which takes up and completes suggestions contained in the "Comments on Exposé XIII"(XIII XIII.6), written in March 1963. This Exposé treats Lefschetz-type theorems from the viewpoint of étale cohomology, using the results on étale cohomology presented in SGA 4 and SGA 5 (to appear in the same Series in Pure Mathematics)[3]; it is therefore less "elementary"in nature than the other Exposés of the present volume, which use little more than the substance of Chapters I through III of the EGA. Here is an outline of the contents of the present volume. Exposé I contains the sorites of "cohomology with support in Y", $H^*_Y(X, F)$, where Y is a closed subset of a space X, cohomology that may be interpreted as the cohomology of X *modulo* the open subset $X - Y$, and that is the abutment of a very useful "*local-to-global spectral sequence*"I 2.6, involving

[1] *Editor's note.* The original numbering has been preserved as far as possible, with the adverb *bis* added when ambiguous duplicates occurred here and there.

[2] It goes without saying that every reference to SGA 2 appearing in those parts of the SGA published in the Series in Pure Mathematics refers to the present volume, not to the original edition of SGA 2!

[3] *Editor's note.* In fact, these seminars were published by Springer (numbers 269, 279, 305, and 589), but unfortunately are out of print.

the *sheaves* of cohomology "with support in Y", $\underline{H}^*_Y(F)$[4]. In many questions this formalism can play a "localization"role analogous to that played by tubular neighborhoods of Y in 3
differential geometry. Exposé II studies the preceding notions for quasi-coherent sheaves on preschemes, and Exposé III relates them to the classical notion of *depth* (III 3.3).

Exposés IV and V give forms of *local duality*, which may be compared with Serre's projective duality theorem (XII XII.1.1). We note that both kinds of duality theorem are substantially generalized in Hartshorne's seminar (cited in the footnote at the end of Exp. IV) [5]

Exposés VI and VII give elementary technical notions used in Exposé VIII to prove the *finiteness theorem* (VIII VIII.2.3). This theorem gives necessary and sufficient conditions, for a coherent sheaf F on a noetherian scheme X, for the local cohomology sheaves $\underline{H}^i_Y(F)$ to be coherent for $i \leq n$ (or, equivalently, for the sheaves $\mathrm{R}^i f_*(F|X-Y)$ to be coherent for $i \leq n-1$, where $f\colon X-Y \to X$ is the inclusion). It is one of the central technical results of the seminar, and in Exposé IX we show how a theorem of this kind can be used to establish a "comparison theorem"and an "existence theorem"in formal geometry, following and generalizing the use made in (EGA III §§4 and 5) of the finiteness theorem for a proper morphism.

These last results are applied in X and XI, devoted respectively to Lefschetz-type theorems for the fundamental group and for the Picard group. These theorems compare, under certain 4
conditions, the invariants (π_1 or Pic) associated respectively with a scheme X and a subscheme Y, which plays the role of a hyperplane section, and in particular give conditions under which they are isomorphic. Roughly speaking, the hypotheses allow one to pass from Y to the formal completion of X along Y, and then to apply the results of IX in order to pass from there to an open *neighborhood* U of Y in X. To pass from U to X, one also needs information of a "purity"or "parafactoriality"type for the local rings of X at the points of $Z = X - U$, which is a finite discrete set in the cases under consideration. This explains the interaction, in the proofs of Exposés X, XI, and XII, between local and global results, particularly in certain inductions. The main results obtained in X and XI are the theorems of *local nature* X 3.4 (the *purity theorem*) and XI XI.3.15 (the *parafactoriality theorem*). Notice that these theorems are proved by cohomological techniques of an essentially global nature. In XII, using the preceding local results, one obtains global variants for projective schemes over a field, or more generally over a more or less arbitrary base scheme; typical statements include XII XII.3.5 and XII XII.3.7.

In XIII, we review some of the many problems and conjectures suggested by the results and methods of the seminar. Perhaps the most interesting concern cohomological and homotopical Lefschetz-type theorems for complex analytic spaces; cf. XIII, pages 26 and following [6]. In the context of the étale cohomology of schemes , the corresponding conjectures are proved 5
in XIV by a duality technique that should apply equally in the complex analytic case (cf. the comments in XIII, p. 25, and XIV XIV.6.4). But the corresponding homotopical statements for analytic spaces, and more particularly those involving the fundamental group, seem to require entirely new techniques (cf. XIV XIV.6.4).

I am happy to thank all those who, in various capacities, helped bring the present volume

[4] *Editor's note.* The underlined notation has been retained for sheafified versions of functors, since the calligraphic analogue of Γ is not clear.

[5] *Editor's note.* Hartshorne's book contains sign errors and, more importantly, does not actually prove the compatibility of the trace with base change. Conrad completely redid this work and proved this crucial and highly nontrivial compatibility (*Grothendieck duality and base change*, Lect. Notes in Math., vol. 1750, Springer-Verlag, Berlin, 2000). Unfortunately, errors remain (cf. two preprints: Conrad B., "Clarifications and corrections to "*Grothendieck duality and base change*"" and "An addendum to Chapter 5 of "*Grothendieck duality and base change*""). For a more concrete aspect, with particular attention to the notion of residue, see Lipman's work, especially Lipman J., *Dualizing sheaves, differentials and residues on algebraic varieties*, Astérisque, vol. 117, Société mathématique de France, 1984. A categorical proof of the duality theorem, based on Brown's representability theorem, was obtained by Neeman (Neeman A., "The Grothendieck duality theorem via Bousfield's techniques and Brown representability", *J. Amer. Math. Soc.* **9** (1996), no. 1, p. 205–236).

[6] *Editor's note.* Essentially all the conjectures stated in XIII and XIV have now been proved; see the footnotes in those sections for references and comments.

to publication, including the collaborators already named in this Introduction. In particular, I wish to thank Mlle Chardon for the good grace with which she carried out the thankless task of physically preparing the final manuscript for photo-offset reproduction.

Bures-sur-Yvette, April 1968

A. Grothendieck.

Exposé I

Global and Local Cohomological Invariants Relative to a Closed Subspace

1. The functors Γ_Z and $\underline{\Gamma}_Z$

Let X be a topological space, and let $\underline{C}_X$ be the category of sheaves of abelian groups on X. Let Φ be a family of supports in the sense of Cartan. The functor Γ_Φ on $\underline{C}_X$ is defined by Original French p. 6

$$\Gamma_\Phi(F) = \{f \in \Gamma(F) \mid \operatorname{supp}(f) \in \Phi\}. \tag{1}$$

If Z is a closed subset of X, by an abuse of notation we denote by Γ_Z the functor Γ_Φ, where Φ is the family of closed subsets of X contained in Z.[1] Thus

$$\Gamma_Z(F) = \{f \in \Gamma(F) \mid \operatorname{supp}(f) \subset Z\}. \tag{2}$$

We wish to extend this definition to the case in which Z is a *locally closed* subset of X, hence closed in a suitable open subset V of X. In that case we set

$$\Gamma_Z(F) = \Gamma_Z(F|_V). \tag{3}$$

We must verify that $\Gamma_Z(F)$ “does not depend” on the chosen open set. It is enough to show that, if V' is open and $V \supset V' \supset Z$, then the map $\rho^V_{V'} : F(V) \to F(V')$ maps $\Gamma_Z(F|_V)$ isomorphically onto $\Gamma_Z(F|_{V'})$. But

$$\Gamma_Z(F|_V) = \ker \rho^V_{V-Z}. \tag{4}$$

Consequently, if $f \in \Gamma_Z(F|_V)$ and $\rho^V_{V'}(f) = \rho^V_{V-Z}(f) = 0$, then $f = 0$, since $(V', V - Z)$ is a covering of V. Likewise, if $f' \in \Gamma_Z(F|_{V'})$, then $f' \in F(V')$ and $0 \in F(V - Z)$ define an element $f \in F(V)$ such that $\rho^V_{V'}(f) = f'$ and $f \in \Gamma_Z(F|_V)$. Thus $\rho^V_{V'}$ induces an isomorphism Original French p. 7

$$\Gamma_Z(F|_V) \longrightarrow \Gamma_Z(F|_{V'}).$$

Note that every open subset W of Z is induced by an open subset U of X in which W is closed. It follows that $W \mapsto \Gamma_W(F)$ defines a presheaf on Z; one verifies that it is a sheaf, denoted by $i^!(F)$, where $i : Z \to X$ is the canonical immersion. We obtain

$$\Gamma_Z(F) = \Gamma(i^!(F)). \tag{5}$$

[1] At this single occurrence the French printed and TeX sources underline Γ_Z. The defining equation (2) and the systematic distinction used thereafter identify the intended functor here as the global, non-underlined Γ_Z. The source anomaly and this explicit resolution are recorded in the tranche comparison ledger.

The sheaf $i^!(F)$ is a subsheaf of $i^*(F)$. Indeed, the canonical homomorphism

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

is injective on $\Gamma_{U\cap Z}(F|_U) \subset \Gamma(F|_U)$. In summary, we have the following result.

Proposition 1.1. *There exists a unique subsheaf $i^!(F)$ of $i^*(F)$ such that, for every open subset U of X for which $U \cap Z$ is closed in U, the map*

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

induces an isomorphism

$$\Gamma_{U\cap Z}(F|_U) \longrightarrow \Gamma(U \cap Z, i^!(F)).$$

If Z is open, then simply

$$i^!(F) = i^*(F) = F|_Z, \qquad \Gamma_Z(F) = \Gamma(Z, F). \tag{6}$$

Suppose again that Z is arbitrary. As U ranges through the open subsets of X, the assignment

$$U \longmapsto \Gamma_{U\cap Z}(F|_U) = \Gamma(U \cap Z, i^!(F))$$

is a sheaf on X, which we denote by $\underline{\Gamma}_Z(F)$. More precisely, the preceding formula, which expresses the fact that $i^!$ commutes with restriction to open subsets, gives an isomorphism

$$\underline{\Gamma}_Z(F) = i_*(i^!(F)). \tag{7}$$

By definition, for every open subset U of X, Original French p. 8

$$\Gamma(U, \underline{\Gamma}_Z(F)) = \Gamma_{U\cap Z}(F|_U). \tag{8}$$

Here there is a characteristic difference between the case in which Z is closed and the case in which Z is open. In the first case, formula (8) shows that $\underline{\Gamma}_Z(F)$ may be regarded as a subsheaf of F; hence there is a *canonical immersion*

$$\underline{\Gamma}_Z(F) \hookrightarrow F. \tag{8$'$}$$

When Z is open, on the other hand, equation (6) shows that the right-hand side of equation (8) is $\Gamma(U \cap Z, F)$ and therefore receives a homomorphism from $\Gamma(U, F)$. Thus there is a *canonical homomorphism* in the opposite direction:[2]

$$F \longrightarrow \underline{\Gamma}_Z(F). \tag{8$''$}$$

This is none other than the canonical homomorphism

$$F \longrightarrow i_*i^*(F),$$

in view of the isomorphism

$$\underline{\Gamma}_Z(F) \simeq i_*i^*(F) \tag{6 bis}$$

deduced from equations (6) and (7).

For variable F, the objects $\Gamma_Z(F)$, $\underline{\Gamma}_Z(F)$, and $i^!(F)$ may of course be regarded as functors in F, with values, respectively, in the category of abelian groups, the category of sheaves of abelian groups on X, and the category of sheaves of abelian groups on Z. It is sometimes convenient to interpret the functor

$$i^! : \underline{C}_X \longrightarrow \underline{C}_Z$$

as the right adjoint of the familiar functor

$$i_! : \underline{C}_Z \longrightarrow \underline{C}_X$$

defined by the following proposition.

Original French p. 9

Proposition 1.2. *Let G be a sheaf of abelian groups on Z. There exists a unique subsheaf of $i_*(G)$, denoted $i_!(G)$, such that for every open subset U of X the (identity) isomorphism*

$$\Gamma(U \cap Z, G) = \Gamma(U, i_*(G))$$

induces an isomorphism

$$\Gamma_{\Phi_{U\cap Z,U}}(U \cap Z, G) = \Gamma(U, i_!(G)),$$

where $\Phi_{U\cap Z,U}$ denotes the family of subsets of $U \cap Z$ that are closed in U.

The verification reduces to observing that the left-hand side is a sheaf as U varies: namely, for a section of $i_*(G)$ on U, viewed as a section of G on $U \cap Z$, the property of having support closed *in U* is *local on U*. The sheaf $i_!(G)$ just defined is also known as the *sheaf obtained from G by extension by zero* outside Z; see Godement [Godement]. In particular, if Z is closed, then

$$i_!(G) = i_*(G). \tag{9}$$

In general, however, the canonical injection $i_!(G) \to i_*(G)$ is not an isomorphism, as is already well known when Z is open. Evidently $i_!(G)$ depends functorially on G (and is in fact an exact functor of G). We therefore have:

Proposition 1.3. *There is an isomorphism of bifunctors in G and F, where G is a sheaf of abelian groups on Z and F is a sheaf of abelian groups on X:*

$$\mathrm{Hom}(i_!(G), F) = \mathrm{Hom}(G, i^!(F)). \tag{10}$$

To define such an isomorphism is equivalent to defining functorial homomorphisms

$$i_! i^!(F) \longrightarrow F, \qquad G \longrightarrow i^! i_!(G),$$

satisfying the familiar compatibility conditions (see, for example, Shih's exposé in the Cartan seminar on cohomological operations).

Recalling that $i_!$ is exact, and hence takes monomorphisms to monomorphisms, we conclude:

Original French p. 10

Corollary 1.4. *If F is injective, then $i^!(F)$ is injective; consequently $\underline{\Gamma}_Z(F) = i_* i^!(F)$ is also injective.*

Replacing X by a variable open subset U of X, Proposition 1.3 also gives:

Corollary 1.5. *There is an isomorphism, functorial in F and G,*

$$\underline{\mathrm{Hom}}(i_!(G), F) = i_*(\underline{\mathrm{Hom}}(G, i^!(F))). \tag{11}$$

Taking G to be the constant sheaf on Z defined by $\mathbf{Z}$, denoted $\mathbf{Z}_Z$, Proposition 1.3 and Corollary 1.5 specialize as follows.

Corollary 1.6. *There are isomorphisms, functorial in F,*

$$\begin{aligned} \Gamma_Z(F) &= \mathrm{Hom}(\mathbf{Z}_{Z,X}, F), \\ \underline{\Gamma}_Z(F) &= \underline{\mathrm{Hom}}(\mathbf{Z}_{Z,X}, F), \end{aligned} \tag{12}$$

[2] The original reference was (8).

where $\mathbf{Z}_{Z,X} = i_!(\mathbf{Z}_Z)$ *is the sheaf of abelian groups on* X *obtained from the constant sheaf* $\mathbf{Z}_Z$ *on* Z *by extension by zero outside* Z.

Remark 1.7. Suppose that X is a ringed space, equip Z with the sheaf of rings $\mathcal{O}_Z = i^{-1}(\mathcal{O}_X)$, and now let $\underline{\mathrm{C}}_X$ and $\underline{\mathrm{C}}_Z$ denote the categories of modules on X and Z, respectively. The preceding considerations extend word for word, with F a module on X and G a module on Z, and with statements 1.3–1.6 interpreted accordingly.

To conclude these generalities, let us examine what happens when the locally closed subset Z is changed. Let $Z' \subset Z$ be another locally closed subset, and let

$$j : Z' \longrightarrow Z, \qquad i' : Z' \longrightarrow X, \qquad i' = ij$$

be the canonical inclusions. There are functorial isomorphisms

Original French p. 11

$$(ij)^! = j^! i^!, \qquad (ij)_! = i_! j_!. \tag{13}$$

The first isomorphism in (13) defines a functorial isomorphism

$$\Gamma_{Z'}(F) = \Gamma(Z', (ij)^!(F)) \simeq \Gamma(Z', j^!(i^!(F))) = \Gamma_{Z'}(i^!(F)). \tag{14}$$

Suppose now that Z' is *closed* in Z, and let

$$Z'' = Z - Z'$$

be its complement in Z. This is open in Z and hence locally closed in X. Applying the canonical injection (8′) to $i^!(F)$ on Z with the subset Z', and using (14), gives an injective functorial canonical homomorphism

$$\Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F). \tag{15}$$

If Z' is replaced by Z'' in (14) and equation (8″) is used, one obtains a functorial canonical homomorphism

$$\Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F). \tag{15′}$$

Proposition 1.8. *Under the preceding hypotheses, the sequence of functorial homomorphisms*

$$0 \longrightarrow \Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F) \tag{16}$$

is exact. If F *is flasque, the sequence remains exact after a zero is added on the right.*

Proof. Replace X by an open subset V in which Z is closed. This reduces the assertion to the case in which Z, and hence Z', is closed. Then Z'' is closed in the open subset $X - Z'$, and there is a canonical injection

$$\Gamma_{Z''}(F) \longrightarrow \Gamma(X - Z', F).$$

The exactness of (16) then says simply that the sections of F with support in Z' are precisely those whose restriction to $X - Z'$ is zero.

If F is flasque, every element of $\Gamma_{Z''}(F)$, regarded as a section of F on $X - Z'$, extends to a section of F on X. The latter section plainly has support in Z, proving that the last homomorphism in (16) is then surjective. □ Original French p. 12

Corollary 1.9. *There is a functorial exact sequence*

$$0 \longrightarrow \underline{\Gamma}_{Z'}(F) \longrightarrow \underline{\Gamma}_Z(F) \longrightarrow \underline{\Gamma}_{Z''}(F), \tag{16 bis}$$

and, if F is flasque, the sequence remains exact after a zero is added on the right.

Proposition 1.8 may be interpreted, by means of Corollary 1.6, as a statement about the functors Hom and $\underline{\mathrm{Hom}}$, as follows. First note that if G is a sheaf of abelian groups on Z, inducing the sheaves $j^*(G)$ and $k^*(G)$ on Z' and Z'', respectively, where $j : Z' \to Z$ and $k : Z'' \to Z$ are the canonical injections, then there is a canonical exact sequence of sheaves on X:

$$0 \longrightarrow k^*(G)_X \longrightarrow G_X \longrightarrow j^*(G)_X \longrightarrow 0. \tag{17}$$

Here, to simplify notation, the subscript X denotes the sheaf on X obtained by extension by zero outside the space on which the sheaf in question is defined. Sequence (17) generalizes a familiar exact sequence for $Z = X$ (see Godement [Godement]); it follows from that sequence by writing it on Z and applying the functor $i_!$. Taking $G = \mathbf{Z}_Z$, we obtain in particular:

Proposition 1.10. *Under the preceding hypotheses, there is an exact sequence of sheaves of abelian groups on X:*

$$0 \longrightarrow \mathbf{Z}_{Z'',X} \longrightarrow \mathbf{Z}_{Z,X} \longrightarrow \mathbf{Z}_{Z',X} \longrightarrow 0. \tag{18}$$

The two exact sequences in Proposition 1.8 and Corollary 1.9 are exactly the sequences obtained from (18) *by applying, respectively, the functors* $\mathrm{Hom}(-, F)$ *and* $\underline{\mathrm{Hom}}(-, F)$.

This also gives an immediate proof that sequences (16) and (16 bis) remain exact after a zero is added on the right, provided that F is *injective.*

2. The functors $\mathrm{H}^*_Z(X, F)$ and $\underline{\mathrm{H}}^*_Z(F)$

Original French p. 13

Definition 2.1. *The derived functors, in the variable F, of the functors $\Gamma_Z(F)$ and $\underline{\Gamma}_Z(F)$ are denoted by $\mathrm{H}^*_Z(X, F)$ and $\underline{\mathrm{H}}^*_Z(F)$, respectively.*

They are cohomological functors with values, respectively, in the category of abelian groups and in the category of sheaves of abelian groups on X. When Z is closed, $\mathrm{H}^*_Z(X, F)$ is, by definition, simply $\mathrm{H}^*_\Phi(X, F)$, where Φ denotes the family of closed subsets of X contained in Z. When Z is open, we shall see that $\mathrm{H}^*_Z(X, F)$ is simply $\mathrm{H}^*(Z, F) = \mathrm{H}^*(Z, F|_Z)$, by the following proposition.

Proposition 2.2 (Excision theorem). *Let V be an open subset of X containing Z. There is an isomorphism of cohomological functors in F:*

$$\mathrm{H}^*_Z(X, F) \longrightarrow \mathrm{H}^*_Z(V, F|_V). \tag{19}$$

Indeed, there is a functorial isomorphism $\Gamma^X_Z \simeq \Gamma^V_Z j^!$, where $j \colon V \to X$ is the inclusion and $j^!$ is therefore the restriction functor (cf. equation (14)). This last functor is exact and, by Corollary 1.4, takes injectives to injectives; the isomorphism (19) follows immediately.

When Z is open, we may take $V = Z$, and obtain:

Corollary 2.3. *Suppose that Z is open. There is an isomorphism of cohomological functors*

$$\mathrm{H}^*_Z(X, F) = \mathrm{H}^*(Z, F). \tag{20}$$

The isomorphisms of Corollary 1.6 and the definitions give the following (cf. [Tôhoku]).

Proposition 2.3 bis. *There are isomorphisms of cohomological functors*

Original French p. 14

$$\mathrm{H}^*_Z(X, F) \simeq \mathrm{Ext}^*(X; \mathbf{Z}_{Z,X}, F), \tag{21}$$

$$\underline{\mathrm{H}}^*_Z(F) \simeq \underline{\mathrm{Ext}}^*(\mathbf{Z}_{Z,X}, F). \tag{21 bis}$$

Editorial note. This proposition is numbered 2.3 in the original edition; the corrected branch designates it 2.3 bis.

We may therefore apply the results of [Tôhoku] on the Ext of modules. First, we record the following interpretation of the sheaves $\underline{\mathrm{H}}^*_Z(F)$ in terms of the global groups $\mathrm{H}^*_Z(X, F)$.

Corollary 2.4. *$\underline{\mathrm{H}}^*_Z(F)$ is canonically isomorphic to the sheaf associated with the presheaf*

$$U \longmapsto \mathrm{H}^*_{Z\cap U}(U, F|_U).$$

In particular, using Corollary 2.3, we obtain:

Corollary 2.5. *Suppose that Z is open. There is an isomorphism of cohomological functors*

$$\underline{\mathrm{H}}^*_Z(F) = \mathrm{R}^* i_* i^*(F), \tag{22}$$

where $i\colon Z \to X$ is the inclusion.

The Ext spectral sequence gives the following important spectral sequence.

Theorem 2.6. *There is a spectral sequence, functorial in F, abutting to $\mathrm{H}^*_Z(X, F)$ and having initial term*

$$E_2^{p,q}(F) = \mathrm{H}^p(X, \underline{\mathrm{H}}^q_Z(F)). \tag{23}$$

Remarks 2.7. It follows immediately from Corollary 2.4 that the sheaves $\underline{\mathrm{H}}^q_Z(F)$ vanish on $X - \overline{Z}$, and also vanish in the interior of Z when $q \neq 0$; hence, for such q, $\underline{\mathrm{H}}^q_Z(F)$ is supported on the boundary of Z.

Consequently, the right-hand side of (23) may be interpreted as a cohomology group on $\overline{Z}$. We shall use Theorem 2.6 when Z is closed in X; in that case the right-hand side of (23) may be interpreted as a cohomology group computed on Z: Original French p. 15

$$E_2^{p,q}(F) = \mathrm{H}^p(Z, \underline{\mathrm{H}}^q_Z(F)). \tag{23 bis}$$

The corresponding equation was numbered (23) in the original edition.

When Z is open, the spectral sequence of Theorem 2.6 is simply the Leray spectral sequence for the continuous map $i\colon Z \to X$, in view of the interpretation in Corollary 2.5 used to compute its initial term.

We return to the exact sequence (18). It gives rise to an exact Ext sequence (cf. [Tôhoku]). *Editorial note.* The original edition cited Proposition 1.10 at this point; the corrected branch cites equation (18).

Theorem 2.8. *Let Z be a locally closed subset of X, let Z' be a closed subset of Z, and put $Z'' = Z - Z'$. There is a long exact sequence, functorial in F:*

$$\begin{aligned} 0 \longrightarrow \mathrm{H}^0_{Z'}(X,F) \longrightarrow \mathrm{H}^0_Z(X,F) \longrightarrow \mathrm{H}^0_{Z''}(X,F) \xrightarrow{\partial} \mathrm{H}^1_{Z'}(X,F) \longrightarrow \mathrm{H}^1_Z(X,F) \longrightarrow \cdots \\ \cdots \longrightarrow \mathrm{H}^i_{Z'}(X,F) \longrightarrow \mathrm{H}^i_Z(X,F) \longrightarrow \mathrm{H}^i_{Z''}(X,F) \xrightarrow{\partial} \mathrm{H}^{i+1}_{Z'}(X,F) \longrightarrow \cdots. \end{aligned} \tag{24}$$

Recall how this exact sequence is obtained. Let $C(F)$ be an injective resolution of F. Then the exact sequence (18)[3] gives the exact sequence

$$0 \longrightarrow \Gamma_{Z'}(C(F)) \longrightarrow \Gamma_Z(C(F)) \longrightarrow \Gamma_{Z''}(C(F)) \longrightarrow 0. \tag{25}$$

This is precisely the sequence defined in Proposition 1.8. Its cohomology long exact sequence is (24).

[3] See the preceding editorial note.

Source note. The corrected French TeX prints $\Gamma(C(F))$ in the middle term of (25). Equation (24), the construction from (18), and Proposition 1.8 require $\Gamma_Z(C(F))$; the support subscript is restored here and the emendation is not silent.

The most important case for us is the one in which Z is closed (and we may always reduce to it by replacing X with an open subset V in which Z is closed). Then Z' is closed, Z'' is closed in the open subset $X - Z'$, and we may write

$$\mathrm{H}^i_{Z''}(X, F) = \mathrm{H}^i_{Z''}(X - Z', F|_{X-Z'}). \tag{26}$$

This permits us to write the exact sequence (24) in terms of cohomology with support in a given closed subset. The most common case is $Z = X$. Writing $Z' = A$ for simplicity, we obtain: Original French p. 16

Corollary 2.9. *Let A be a closed subset of X. There is a long exact sequence, functorial in F:*

$$\begin{gathered} 0 \longrightarrow \mathrm{H}^0_A(X, F) \longrightarrow \mathrm{H}^0(X, F) \longrightarrow \mathrm{H}^0(X - A, F) \xrightarrow{\partial} \mathrm{H}^1_A(X, F) \longrightarrow \cdots \\ \cdots \longrightarrow \mathrm{H}^i_A(X, F) \longrightarrow \mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(X - A, F) \xrightarrow{\partial} \mathrm{H}^{i+1}_A(X, F) \longrightarrow \cdots . \end{gathered} \tag{27}$$

This exact sequence shows that $\mathrm{H}^i_A(X, F)$ plays the role of a relative cohomology group of X modulo $X - A$, with coefficients in F. It is in this capacity that it arises naturally in applications. By "sheafifying" (24) and (27), or by arguing directly, and taking into account that the sheaf associated with $U \mapsto \mathrm{H}^i(U, F)$ is zero for $i > 0$, we obtain:

Corollary 2.10. *Under the hypotheses of Theorem 2.8 there is a long exact sequence, functorial in F:*

$$\cdots \longrightarrow \underline{\mathrm{H}}^i_{Z'}(F) \longrightarrow \underline{\mathrm{H}}^i_Z(F) \longrightarrow \underline{\mathrm{H}}^i_{Z''}(F) \xrightarrow{\partial} \underline{\mathrm{H}}^{i+1}_{Z'}(F) \longrightarrow \cdots . \tag{24 bis}$$

Corollary 2.11. *Let A be a closed subset of X. There is an exact sequence, functorial in F:*

$$0 \longrightarrow \underline{\mathrm{H}}^0_A(F) \longrightarrow F \longrightarrow f_*(F|_{X-A}) \xrightarrow{\partial} \underline{\mathrm{H}}^1_A(F) \longrightarrow 0, \tag{28}$$

and, for $i \geq 2$, canonical isomorphisms

$$\underline{\mathrm{H}}^i_A(F) = \underline{\mathrm{H}}^{i-1}_{X-A}(F) = \mathrm{R}^{i-1} f_*(F|_{X-A}), \tag{29}$$

where $f\colon X - A \to X$ is the inclusion.

Thus $\underline{\mathrm{H}}^0_A(F)$ and $\underline{\mathrm{H}}^1_A(F)$ are defined, respectively, as the kernel and cokernel of the canonical homomorphism

$$F \longrightarrow f_* f^*(F) = f_*(F|_{X-A}),$$

and $\underline{\mathrm{H}}^i_A(F)$ for $i \geq 2$ is expressed in terms of the derived functors of f_*.

Source note. The French display (28) omits the argument (F) from $\underline{\mathrm{H}}^1_A$. The explanatory sentence immediately following the display supplies it; it is restored here and recorded as an emendation.

Corollary 2.12. *Let F be a sheaf of abelian groups on X. If F is flasque, then for every locally closed subset Z of X and every integer $i \neq 0$,* Original French p. 17

$$\mathrm{H}^i_Z(X, F) = 0, \qquad \underline{\mathrm{H}}^i_Z(F) = 0.$$

Conversely, if $\mathrm{H}^1_Z(X, F) = 0$ for every closed subset Z of X, then F is flasque.

Suppose that F is flasque. It induces a flasque sheaf on every open subset. To prove that $\mathrm{H}^i_Z(X, F) = 0$ for $i > 0$, we may therefore assume that Z is closed; the assertion then follows

from the exact sequence (27). For every locally closed Z, "sheafifying"—that is, applying Corollary 2.4—gives $\underline{\mathrm{H}}^i_Z(F) = 0$ for $i > 0$. Conversely, suppose that $\mathrm{H}^1_Z(X, F) = 0$ for every closed Z. Then (27) shows that $\mathrm{H}^0(X, F) \to \mathrm{H}^0(X - Z, F)$ is surjective for every such Z, which means precisely that F is flasque.
Editorial note. At both corresponding occurrences in the French proof, the original edition cited Corollary 2.9; the corrected branch specifies equation (27).

Combining Theorems 2.6 and 2.8, we deduce:

Proposition 2.13. *Let F be a sheaf of abelian groups on X, let Z be a closed subset of X, put $U = X - Z$, and let N be an integer. The following conditions are equivalent:*

(i) $\underline{\mathrm{H}}^i_Z(F) = 0$ *for* $i \leq N$.

(ii) *For every open subset V of X, the canonical homomorphism*

$$\mathrm{H}^i(V, F) \longrightarrow \mathrm{H}^i(V \cap U, F)$$

is

a) *bijective for $i < N$;*

b) *injective for $i = N$.*

If $N > 0$, it is enough in (ii) *to require* a).

To prove (i) $\Rightarrow$ (ii), the local nature of the sheaves $\underline{\mathrm{H}}^i_Z(F)$ reduces us to proving the following.

Corollary 2.14. *If condition* (i) *of Proposition 2.13 holds, then*

$$\mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F)$$

is bijective for $i < N$ and injective for $i = N$.

Indeed, by the exact sequence (27), this is equivalent to $\mathrm{H}^i_Z(X, F) = 0$ for $i \leq N$, and this relation is an immediate consequence of the spectral sequence (23 bis). Original French p. 18
Editorial note. At both corresponding occurrences in the French text, the original edition cited Theorem 2.6; the corrected branch specifies equation (23 bis).

Conversely, condition (ii) of Proposition 2.13 says that, for every open subset V of X,

$$\mathrm{H}^i_{Z \cap V}(V, F|_V) = 0 \qquad \text{for } i \leq N,$$

which implies (i) by Corollary 2.4. Moreover, if $N > 0$, condition b) is superfluous. Indeed, if $N = 1$, condition a) together with (28) gives $\underline{\mathrm{H}}^i_Z(F) = 0$ for $i \leq N$. If $N > 1$, condition a) for $i = N - 1 > 0$, together with (29), gives $\underline{\mathrm{H}}^i_Z(F) = 0$ for $i \leq N$.

Taking Corollary 2.11 into account, this also proves condition (i) of Proposition 2.13.

Remark. Let $Y \to X$ be a closed immersion, and suppose that locally it has the form $\{0\} \times Y \subset R^n \times Y$. Suppose that F is a locally constant sheaf on X. Then

$$\underline{\mathrm{H}}^i_Y(F) \simeq \begin{cases} 0, & i \neq n, \\ F \otimes \underline{\mathrm{T}}_{Y,X}, & i = n, \end{cases} \qquad \underline{\mathrm{T}}_{Y,X} \simeq \underline{\mathrm{H}}^n_Y(\mathbf{Z}_X), \tag{30}$$

where $\underline{\mathrm{T}}_{Y,X}$ is the extension to X of a sheaf on Y locally isomorphic to $\mathbf{Z}_Y$; it is called the *normal orientation sheaf* of Y in X.

Source note. The corrected French prose prints lowercase x where the ambient space must be X, and its sentence defining $\underline{\mathrm{T}}_{Y,X}$ is grammatically incomplete. The reading above preserves the display and states its evident mathematical syntax explicitly.

Using the spectral sequence (23 bis), we obtain in this case

$$\mathrm{H}^i_Y(X, F) \simeq \mathrm{H}^{i-n}(Y, F \otimes \underline{\mathrm{T}}_{Y,X}), \tag{31}$$

and recover the *Gysin homomorphism*

$$\mathrm{H}^j(Y, F \otimes \underline{\mathrm{T}}_{Y,X}) \longrightarrow \mathrm{H}^{j+n}(X, F). \tag{32}$$

Bibliography

[Godement] R. Godement, *Topologie algébrique et théorie des faisceaux*, Act. Scient. et Ind., vol. 1252, Hermann, Paris.

[Tôhoku] A. Grothendieck, "Sur quelques points d'algèbre homologique," *Tôhoku Math. J.* **9** (1957), 119–221.

Exposé II

Application to Quasi-coherent Sheaves on Preschemes

Proposition 1. *Let X be a prescheme and let Z be a locally closed subset of the form $Z = U - V$, where U and V are open subsets of X such that $V \subset U$ and the canonical immersions $U \to X$ and $V \to X$ are quasi-compact. Then, for every quasi-coherent $\mathcal{O}_X$-module F, the sheaves $\underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent.* Original French p. 19

By Exposé I, equation (24), there is an exact sequence of relative cohomology sheaves

$$\longrightarrow \underline{\mathrm{H}}^{i-1}_U(F) \longrightarrow \underline{\mathrm{H}}^i_V(F) \longrightarrow \underline{\mathrm{H}}^i_Z(F) \longrightarrow \underline{\mathrm{H}}^i_U(F) \longrightarrow \underline{\mathrm{H}}^{i+1}_V(F) \longrightarrow .$$

By EGA III, 1.4.17, for the sheaves $\underline{\mathrm{H}}^i_Z(F)$ to be quasi-coherent it therefore suffices that the sheaves $\underline{\mathrm{H}}^i_U(F)$ and $\underline{\mathrm{H}}^i_V(F)$ be quasi-coherent. We may thus suppose that Z is open and that the canonical immersion $j\colon Z \to X$ is quasi-compact.

Since Z is open, Exposé I, equation (22), gives a canonical isomorphism

$$\underline{\mathrm{H}}^i_Z(F) \simeq \mathrm{R}^i j_*(F|_Z).$$

But j is separated (EGA I, 5.5.1) and quasi-compact; hence, by EGA III, 1.4.10, the sheaves $\mathrm{R}^i j_*(F|_Z) = \underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent. This completes the proof.

Corollary 2. *Let Z be a closed subset of X such that the canonical immersion $X - Z \to X$ is quasi-compact. Then the $\mathcal{O}_X$-modules $\underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent.*

Corollary 3. *If X is locally noetherian, then, for every locally closed subset Z of X and every quasi-coherent $\mathcal{O}_X$-module F, the sheaves $\underline{\mathrm{H}}^i_Z(F)$ are quasi-coherent.*

This follows immediately from Corollary 2 and EGA I, 6.6.4.

Corollary 4. *Suppose that X is the spectrum of a ring A. Let U be a quasi-compact open subset of X, put $Y = X - U$, and let F be a quasi-coherent $\mathcal{O}_X$-module. There is an isomorphism of cohomological functors in F:* Original French p. 20

$$\underline{\mathrm{H}}^i_Y(F) = \big(\widetilde{\mathrm{H}^i_Y(X,F)}\big). \tag{4.1}$$

There is moreover an exact sequence, functorial in F,

$$0 \longrightarrow \mathrm{H}^0_Y(X,F) \longrightarrow \mathrm{H}^0(X,F) \longrightarrow \mathrm{H}^0(U,F) \longrightarrow \mathrm{H}^1_Y(X,F) \longrightarrow 0, \tag{4.2}$$

and there are isomorphisms, functorial in F,

$$\mathrm{H}^i_Y(X,F) \simeq \mathrm{H}^{i-1}(U,F), \qquad i \geq 2. \tag{4.3}$$

By Corollary 2, the sheaves $\underline{\mathrm{H}}^i_Y(F)$ are quasi-coherent. Since X is affine, it follows that $\mathrm{H}^p(X, \underline{\mathrm{H}}^i_Y(F)) = 0$ for $p > 0$. The spectral sequence of Exposé I, equation (23), therefore degenerates, and

$$\mathrm{H}^i_Y(X, F) = \Gamma(\underline{\mathrm{H}}^i_Y(F)).$$

Equation (4.1) now follows from EGA I, 1.1.3.7; equations (4.2) and (4.3) follow from the cohomology exact sequence of Exposé I, equation (27), and from the vanishing $\mathrm{H}^i(X, F) = 0$ for $i > 0$, since X is affine.
Editorial note. The corrected branch replaces the original citation of Proposition 1 by Corollary 2 in the first sentence of this proof and regularizes both Exposé I equation references; those corrected references are used above.

Under the hypotheses of Corollary 4,[1] U is a finite union of affine open subsets X_f. Consequently, one can find an ideal I, generated by finitely many elements f_α and defining Y; write $\mathbf{f} = (f_\alpha)$. With the notation of EGA III, 1,[2] we have the following result.

Proposition 5. *Suppose that X is the spectrum of a ring A. Let $\mathbf{f} = (f_\alpha)$ be a finite family of elements of A, let Y be the closed subset of X that they define, let M be an A-module, and let F be the associated sheaf. There are isomorphisms of ∂-functors in M:*

$$\mathrm{H}^i((\mathbf{f}), M) \simeq \mathrm{H}^i_Y(X, F). \tag{5.1}$$

We shall also write $\mathrm{H}^i_J(M) = \mathrm{H}^i_Y(X, F)$ when Y is the closed subset of $X = \operatorname{Spec} A$ defined by an ideal J of A.

For $i = 0$ and $i = 1$, use the exact sequences (4.2) and EGA III, 1.4.3.2; for $i \geq 2$, use (4.3) and EGA III, 1.4.3.1. This gives isomorphisms functorial in M. One verifies that, up to a sign depending only on i, they are compatible with the boundary operator; hence the isomorphism of ∂-functors in (5.1) exists. Original French p. 21
Editorial note. For the case $i \geq 2$, the original edition cited EGA III, 1.4.3.2; the corrected branch cites 1.4.3.1.

Now let X be a prescheme, let Y be a closed subset of X, and let $f\colon Y \to X$ be the inclusion. Let I be a quasi-coherent ideal defining Y in X, and let F be a sheaf on X.

We have seen that there are isomorphisms of ∂-functors in F

$$\mathrm{Ext}^i_{\mathcal{O}_X}(X; f_*f^{-1}(\mathcal{O}_X), F) \longrightarrow \mathrm{H}^i_Y(X, F), \tag{$*$}$$
$$\underline{\mathrm{Ext}}^i_{\mathcal{O}_X}(f_*f^{-1}(\mathcal{O}_X), F) \longrightarrow \underline{\mathrm{H}}^i_Y(F). \tag{$**$}$$

Source note. The corrected French source calls these arrows isomorphisms but displays plain right arrows in both $(*)$ and $(**)$. The arrows have been preserved exactly; no unmarked replacement by $\xrightarrow{\sim}$ has been made.

Let n, m be integers such that $m \geq n \geq 0$. Denote by $i_{n,m}$ the canonical map

$$\mathcal{O}_{Y_m} = \mathcal{O}_X/I^{m+1} \longrightarrow \mathcal{O}_X/I^{n+1} = \mathcal{O}_{Y_n},$$

and by j_n the map $f_*f^{-1}(\mathcal{O}_X) \to \mathcal{O}_{Y_n}$. The pairs $(\mathcal{O}_{Y_n}, i_{n,m})$ form a projective system, and the maps j_n are compatible with the $i_{n,m}$.

Applying the functor $\mathrm{Ext}^i_{\mathcal{O}_X}(X; \cdot, F)$ yields a morphism

$$\varphi'\colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; \mathcal{O}_{Y_n}, F) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(X; f_*f^{-1}(\mathcal{O}_X), F).$$

[1]The corrected French source adds: "For consistency and clarity, only equations have been numbered in parentheses."

[2]Recall that $\mathrm{H}^\bullet(\mathbf{f}, M)$ is the Koszul cohomology $\mathrm{H}^\bullet(\mathrm{Hom}(K_\bullet(\mathbf{f}), M))$ of $\mathbf{f}$ with values in M (EGA III, 1.1.2), and that $\mathrm{H}^\bullet((\mathbf{f}), M)$ is the direct limit (ibid., 1.1.6.5) $\varinjlim_n \mathrm{H}^\bullet(\mathbf{f}^n, M)$, whose transition morphisms are induced by the natural morphisms $K_\bullet(\mathbf{f}^{n+1}) \to K_\bullet(\mathbf{f}^n)$ (ibid., 1.1.6).

It is readily seen to be a morphism of cohomological functors in F. Consequently, the morphism

$$\varphi\colon \varinjlim_n \operatorname{Ext}^i_{\mathcal{O}_X}(X; \mathcal{O}_{Y_n}, F) \longrightarrow \mathrm{H}^i_Y(X, F),$$

obtained by composing φ' with $(*)$, is itself a morphism of cohomological functors in F.

We define similarly Original French p. 22

$$\underline{\varphi}\colon \varinjlim_n \underline{\operatorname{Ext}}^i_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F) \longrightarrow \underline{\mathrm{H}}^i_Y(F).$$

Our goal is the following theorem.

Theorem 6. a) *Let X be a locally noetherian prescheme, let Y be a closed subset of X defined by a coherent ideal I, and let F be a quasi-coherent $\mathcal{O}_X$-module. Then $\boldsymbol{\varphi}$ is an isomorphism.*

b) *If X is noetherian, then φ is an isomorphism.*

Source note. The corrected source defines the sheaf-level morphism immediately above as $\underline{\varphi}$ but denotes it by the bold symbol $\boldsymbol{\varphi}$ in Theorem 6(a) and Lemma 7. Both graphic forms are retained rather than silently normalized.

Theorem 6 will follow from Theorem 6(a) and the following lemma.

Lemma 7. *If the underlying topological space of X is noetherian and $\boldsymbol{\varphi}$ is an isomorphism, then φ is also an isomorphism.*

We first prove Lemma 7. There is a spectral sequence

$$\mathrm{H}^p(X, \underline{\mathrm{H}}^q_Y(F)) \Longrightarrow \mathrm{H}^*_Y(X, F). \tag{7.1}$$

There is, on the other hand, a direct system of spectral sequences

$$\mathrm{H}^p(X, \underline{\operatorname{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)) \Longrightarrow \operatorname{Ext}^*_{\mathcal{O}_X}(X; \mathcal{O}_{Y_n}, F). \tag{7.2_n}$$

Editorial note. In both abutments above, the original edition printed a superscript m; the corrected branch replaces it by the superscript $*$, which is preserved here.

It follows from the definitions of φ and $\underline{\varphi}$ that these morphisms are associated with a homomorphism Φ of spectral sequences, from the direct limit of (7.2_n) to (7.1). If the underlying space of X is noetherian, then by Godement, 4.12.1,[3]

$$\varinjlim_n \mathrm{H}^p(X, \underline{\operatorname{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)) \xrightarrow{\sim} \mathrm{H}^p\left(X, \varinjlim_n \underline{\operatorname{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)\right).$$

Thus Φ_2 may be written as a morphism

$$\mathrm{H}^p\left(X, \varinjlim_n \underline{\operatorname{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_n}, F)\right) \longrightarrow \mathrm{H}^p(X, \underline{\mathrm{H}}^q_Y(F)),$$

which is precisely the morphism deduced from $\underline{\varphi}$.

If $\underline{\varphi}$ is an isomorphism, then so is Φ_2, and hence so is φ by EGA 0_{III}, 11.1.5. This proves Original French p. 23
Lemma 7.

We now prove Theorem 6(a). This is a local question on X. By Corollary 4 and EGA I, 1.3.9 and 1.3.12, we may suppose that X is the spectrum of a ring A. Write M for the A-module whose associated sheaf is F. It is therefore enough to prove that, under the hypotheses of Theorem 6(a), the canonical homomorphism

$$\varinjlim_n \operatorname{Ext}^i_A(A/I^n, M) \longrightarrow \mathrm{H}^i_Y(X, M) \tag{7.3}$$

[3] Cf. the first bibliographic reference at the end of Exposé I.

is an isomorphism.
Source note. The corrected French text first uses M in display (7.3) without introducing it. The affine correspondence with the previously given quasi-coherent sheaf F is made explicit above.

Let f_α be finitely many elements of A that generate I, and put $\mathbf{f} = (f_\alpha)$. The sequence of ideals $(\mathbf{f}^n)$ is decreasing and cofinal with the sequence of ideals I^n, so that (7.3) is equivalent to a morphism of ∂-functors in M:

$$\varinjlim_n \operatorname{Ext}^i_A(A/(\mathbf{f}^n), M) \longrightarrow \mathrm{H}^i_Y(X, M). \tag{7.4}$$

On the other hand, there are canonical isomorphisms

$$\begin{aligned} \varinjlim_n \operatorname{Hom}_A(A/(\mathbf{f}^n), M) &\simeq \varinjlim_n (m \in M \mid (\mathbf{f}^n)m = 0) \\ &\simeq \mathrm{H}^0((\mathbf{f}), M). \end{aligned} \tag{7.5}$$

Since $\varinjlim_n \operatorname{Ext}^i_A(A/(\mathbf{f}^n), M)$ is a universal ∂-functor in M, there is a unique morphism of ∂-functors in M

$$\varinjlim_n \operatorname{Ext}^i_A(A/(\mathbf{f}^n), M) \longrightarrow \mathrm{H}^i((\mathbf{f}), M) \tag{7.6}$$

that agrees in degree zero with (7.5).

The comparison obtained from (7.3) and (5.1) is a morphism of ∂-functors in M that agrees with (7.6) in degree zero; it therefore agrees with (7.6) in every degree. Theorem 6(a) is consequently an immediate consequence of the following lemma.
Source note. The corrected French sentence calls the comparison just used "the composite of (7.3) and (5.1)." As the displayed arrows in (7.3) and (5.1) have the same target rather than composable source and target, the intended comparison requires identifying that target through the isomorphism (5.1), equivalently using its inverse. The mathematical comparison is stated explicitly above; the source's directional anomaly is recorded here rather than silently concealed.

Lemma 8. *Let A be a noetherian ring, let I be an ideal generated by a finite family $\mathbf{f} = (f_\alpha)$ of elements, and let M be an A-module. Then the homomorphisms* (7.6) *are isomorphisms.*

Lemma 9. *Let A be a ring, let $\mathbf{f} = (f_\alpha)$ be a finite family of elements of A, let I be the ideal generated by $\mathbf{f}$, and let $i > 0$ be an integer. The following conditions are equivalent:* Original French p. 24

a) *The homomorphism* (7.6) *is an isomorphism for every M.*

b) $\mathrm{H}^i((\mathbf{f}), M) = 0$ *for every injective M.*

c) *The projective system* $(\mathrm{H}_i(\mathbf{f}^n, A)) = \mathrm{H}_{i,n}$ *is essentially zero; that is, for every n there exists $n' > n$ such that $\mathrm{H}_{i,n'} \to \mathrm{H}_{i,n}$ is zero.*

Source note. Lemma 9(c), and later Lemma 11, print their systems with the index $n \in \mathbf{Z}$, whereas the proofs explicitly take $n > 0$ and use ordinary positive powers. The printed indexing is retained; this translation does not introduce negative powers.

Condition a) trivially implies b).

Condition b) implies a). Indeed, b) implies that $M \mapsto \mathrm{H}^i((\mathbf{f}), M)$ is a universal cohomological functor, so (7.6) is a morphism of universal cohomological functors. It is an isomorphism in degree zero and hence in every degree.

Condition c) implies b). Indeed, if M is injective, then for every n

$$\mathrm{H}^i(\mathbf{f}^n, M) = \operatorname{Hom}(\mathrm{H}_i(\mathbf{f}^n, A), M) = \operatorname{Hom}(\mathrm{H}_{i,n}, M).$$

Thus c) implies that, for every i, the direct system $(\mathrm{H}^i(\mathbf{f}^n, M))_{n\in\mathbf{Z}}$ is essentially zero, whence b).

Condition b) implies c). Indeed, fix $n > 0$, and let j be a monomorphism from $\mathrm{H}_{i,n}$ into an injective module M. For $n' \geq n$, let $j_{n'} \in \mathrm{Hom}(\mathrm{H}_{i,n'}, M)$ be the composite of j with the transition homomorphism $t_{n',n}\colon \mathrm{H}_{i,n'} \to \mathrm{H}_{i,n}$. The $j_{n'}$ define an element of $\mathrm{H}^i((\mathbf{f}), M)$, which is zero by hypothesis. There is therefore an n_0 such that $j_{n'} = 0$ if $n' > n_0$. Since j is a monomorphism, $j_{n'} = 0$ implies $t_{n',n} = 0$, which proves c).
Editorial note. In the last implication, the original edition shortened the transition-map symbol to $t_{n'}$; the corrected branch restores $t_{n',n}$, as retained above.

Corollary 10. *Suppose that the underlying space of $X = \mathrm{Spec}(A)$ is noetherian. The preceding conditions hold for every finite family of elements of A and every $i > 0$ (or, equivalently, for $i = 1$) if and only if, for every injective A-module M, the sheaf F associated with M is flasque.*

The condition is necessary. Indeed, let $\mathbf{f} = (f_\alpha)$ be a finite family of elements of A, let Y be the closed subset defined by $\mathbf{f}$, and put $U = X - Y$. We then have the exact sequence

$$\mathrm{H}^0(X, F) \longrightarrow \mathrm{H}^0(U, F) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0,$$

and Lemma 9(b) shows that $\mathrm{H}^0(X, F) \to \mathrm{H}^0(U, F)$ is surjective.

The condition is sufficient by (5.1) and the fact that, for every closed subset Y of X and every flasque sheaf F on X, $\mathrm{H}^i_Y(X, F) = 0$ for $i > 0$. Original French p. 25

Lemma 11. *Under the hypotheses of Lemma 9, for every noetherian A-module N and every $i > 0$, the projective system $(\mathrm{H}_{i,n}(N))_{n\in\mathbf{Z}}$, where $\mathrm{H}_{i,n}(N) = \mathrm{H}_i(\mathbf{f}^n, N)$, is essentially zero.*

Proof. We argue by induction on the number m of elements of $\mathbf{f}$.

If $m = 1$, the family $\mathbf{f}$ consists of a single element, say f. Then $\mathrm{H}_{i,n}(N) = 0$ for $i > 1$, and $\mathrm{H}_{1,n}(N)$ is canonically isomorphic to the annihilator $N(n)$ of f^n in N. For $n' \geq n$, the transition homomorphism $N(n') \to N(n)$ is multiplication by $f^{n'-n}$. The $N(n)$ form an increasing sequence of submodules of N. Since N is noetherian, there is an n_0 such that $N(n) = N(n_0)$ for $n \geq n_0$. Hence all the $N(n)$ are annihilated by f^{n_0}, and every transition homomorphism $N(n') \to N(n)$ is zero when $n' \geq n + n_0$. The lemma is thus proved for $m = 1$.

Now suppose that $m > 1$ and that the lemma has been proved for all integers $m' < m$. Put $\mathbf{g} = (f_1, \ldots, f_{m-1})$ and $\mathbf{h} = f_m$.

For every $n > 0$, EGA III, 1.1.4.1 gives an exact sequence

$$0 \longrightarrow \mathrm{H}_0(\mathbf{h}^n, \mathrm{H}_i(\mathbf{g}^n, N)) \longrightarrow \mathrm{H}_i(\mathbf{f}^n, N) \longrightarrow \mathrm{H}_1(\mathbf{h}^n, \mathrm{H}_{i-1}(\mathbf{g}^n, N)) \longrightarrow 0,$$

and, as n varies, a projective system of exact sequences. The induction hypothesis shows that, for $i > 0$, the modules $\mathrm{H}_i(\mathbf{g}^n, N)$ form an essentially zero projective system. The same is therefore true of $\mathrm{H}_0(\mathbf{h}^n, \mathrm{H}_i(\mathbf{g}^n, N))$, which is identified with a quotient of $\mathrm{H}_i(\mathbf{g}^n, N)$. For the terms on the right, factor the transition morphisms from n' to n as

$$\begin{aligned}\mathrm{H}_1(\mathbf{h}^{n'}, \mathrm{H}_{i-1}(\mathbf{g}^{n'}, N)) &\longrightarrow \mathrm{H}_1(\mathbf{h}^{n'}, \mathrm{H}_{i-1}(\mathbf{g}^{n}, N))\\ &\longrightarrow \mathrm{H}_1(\mathbf{h}^{n}, \mathrm{H}_{i-1}(\mathbf{g}^{n}, N)).\end{aligned}$$

Since $\mathrm{H}_{i-1}(\mathbf{g}^n, N)$ is a noetherian module, the case $m = 1$ shows that, for fixed n, there exists $n' > n$ such that the second arrow is zero. One therefore sees that, in this projective system of exact sequences, the two outer projective systems are essentially zero. The middle projective system is therefore essentially zero as well. ☐ Original French p. 26

We have thus proved Lemma 11, hence Lemma 8, and consequently Theorem 6.

Remark. Theorem 6 can also be obtained by proving the condition of Corollary 10 with the aid of the structure theorems for injective modules over a noetherian ring (Matlis, Gabriel).

Exposé III

Cohomological Invariants and Depth

1. Review

We shall state a few definitions and results which the reader may find, for example, in Chapter I of the course given by J.-P. Serre at the Collège de France in 1957–58.[1] Original French p. 27

Definition 1.1. *Let A be a ring (commutative with identity, as throughout what follows), and let M be an A-module (unital, as throughout what follows). We call:*

- *the* annihilator of M*, denoted* $\operatorname{Ann} M$*, the set of* $a \in A$ *such that* $am = 0$ *for every* $m \in M$*;*
- *the* support of M*, denoted* $\operatorname{Supp} M$*, the set of prime ideals* $\mathfrak{p}$ *of* A *such that the localization* $M_{\mathfrak{p}}$ *is nonzero;*
- *the* "assassin of M," *or the* "set of prime ideals associated with M," *denoted* $\operatorname{Ass} M$*, the set of prime ideals* $\mathfrak{p}$ *for which there exists a nonzero element of* M *whose annihilator is* $\mathfrak{p}$*.*

If $\mathfrak{a}$ *is an ideal of* A*, we denote by* $\mathfrak{r}(\mathfrak{a})$ *the radical of* $\mathfrak{a}$ *in* A*, that is, the set of elements of* A *some power of which lies in* $\mathfrak{a}$*.*

The following results hold under the assumption that *A is noetherian and M is finitely generated.*

Proposition 1.1.

(i) $\operatorname{Ass} M$ *is a finite set.*

(ii) *An element of A annihilates a nonzero element of M if and only if it belongs to one of the prime ideals associated with M.* Original French p. 28

(iii) *The radical of the annihilator of M, $\mathfrak{r}(\operatorname{Ann} M)$, is the intersection of the prime ideals associated with M that are minimal with respect to inclusion in $\operatorname{Ass} M$.*

Proposition 1.2. *Let $\mathfrak{p}$ be a prime ideal of A. The following assertions are equivalent:*

(i) $\mathfrak{p} \in \operatorname{Supp} M$.

[1] *Editorial note.* The reissue of Serre's text (J.-P. Serre, *Algèbre locale. Multiplicités*, course at the Collège de France, 1957–58, written up by Pierre Gabriel, second edition, Lecture Notes in Mathematics, vol. 11, Springer-Verlag, 1965) no longer contains proofs of these statements. One may consult N. Bourbaki, *Algèbre commutative*, Masson, as Serre himself suggests.

(ii) *There exists* $\mathfrak{q} \in \operatorname{Ass} M$ *such that* $\mathfrak{q} \subset \mathfrak{p}$.

(iii) $\mathfrak{p} \supset \operatorname{Ann} M$.

(iii bis) $\mathfrak{p} \supset \mathfrak{r}(\operatorname{Ann} M)$.

Proposition 1.3. *Let* N *be a finitely generated* A*-module. Then*

$$\operatorname{Ass} \operatorname{Hom}_A(N, M) = \operatorname{Supp} N \cap \operatorname{Ass} M.$$

2. Depth

Throughout this section, A denotes a commutative ring, I an ideal of A, and M and N two A-modules. We denote by X the prime spectrum of A (its structure sheaf will not be used in this section), and by Y the closed set defined by I, namely $Y = \operatorname{Supp}(A/I) = \{\mathfrak{p} \in X \mid \mathfrak{p} \supset I\}$. Original French p. 28

Lemma 2.1. *Suppose that* A *is noetherian, that* M *and* N *are finitely generated, and that* $\operatorname{Supp} N = Y$. *Then the following assertions are equivalent:*

(i) $\operatorname{Hom}_A(N, M) = 0$.

(ii) $\operatorname{Supp} N \cap \operatorname{Ass} M = \varnothing$.

(iii) *The ideal* I *is not a zero divisor on* M*, meaning that, for every* $m \in M$*, the equality* $Im = 0$ *implies* $m = 0$.

(iv) *There exists an* M*-regular element in* I. *(An element* $a \in A$ *is called* M*-regular if multiplication by* a *on* M *is injective.)*

(v) *For every* $\mathfrak{p} \in Y$*, the maximal ideal* $\mathfrak{p}A_\mathfrak{p}$ *of the local ring* $A_\mathfrak{p}$ *is not associated with* $M_\mathfrak{p}$*; in symbols,* $\mathfrak{p}A_\mathfrak{p} \notin \operatorname{Ass} M_\mathfrak{p}$.

Proof. The equivalence (i)⇔(ii) follows because $\operatorname{Ass} \operatorname{Hom}_A(N, M) = \varnothing$ is equivalent to (ii) by Proposition 1.3, and to (i) by an immediate consequence of Proposition 1.2.

The implication (iii)⇒(ii) follows by contradiction. If $\mathfrak{p} \in \operatorname{Supp} N \cap \operatorname{Ass} M$, then $\mathfrak{p} \supset I$ and there is an $m \in M$ whose annihilator is $\mathfrak{p}$; hence $Im \subset \mathfrak{p}m = 0$, contrary to (iii). Original French p. 29

The implication (iv)⇒(iii) is immediate.

Conditions (ii) and (iv) are equivalent because $\operatorname{Supp} N = Y$, so (ii) says that I is contained in no prime ideal associated with M. Since those associated ideals are prime and finite in number, this is equivalent to saying that I is not contained in their union. By Proposition 1.1(ii), that union is precisely the set of elements of A which are not M-regular.

To prove (i)⇒(v), let $\mathfrak{p} \in Y$. From $\operatorname{Hom}_A(N, M) = 0$ and the formula

$$\big(\operatorname{Hom}_A(N, M)\big)_\mathfrak{p} = \operatorname{Hom}_{A_\mathfrak{p}}(N_\mathfrak{p}, M_\mathfrak{p})$$

one obtains $\operatorname{Hom}_{A_\mathfrak{p}}(N_\mathfrak{p}, M_\mathfrak{p}) = 0$. Thus, by Proposition 1.3,

$$\operatorname{Supp} N_\mathfrak{p} \cap \operatorname{Ass} M_\mathfrak{p} = \varnothing.$$

But $\mathfrak{p}A_\mathfrak{p} \in \operatorname{Supp} N_\mathfrak{p}$, so $\mathfrak{p}A_\mathfrak{p} \notin \operatorname{Ass} M_\mathfrak{p}$.

Finally, suppose (v), and let $\mathfrak{p} \in \operatorname{Ass} M$. There is an $m \in M$ with annihilator $\mathfrak{p}$. Its canonical image in $M_\mathfrak{p}$ is nonzero, and its annihilator contains $\mathfrak{p}A_\mathfrak{p}$ and hence equals it. Thus $\mathfrak{p}A_\mathfrak{p} \in \operatorname{Ass} M_\mathfrak{p}$, so (v) gives $\mathfrak{p} \notin Y$. This proves (i). □

We shall now work with these conditions after replacing the functor Hom by its derived functors.

Theorem 2.2. *Let A be a commutative ring, I an ideal of A, M an A-module, and n an integer.* Original French p. 30

(a) *If there is a sequence $f_1, \dots, f_{n+1}$ of elements of I which is M-regular (that is, f_1 is M-regular and f_{i+1} is regular on $M/(f_1, \dots, f_i)M$ for $i \leq n$), then, for every A-module N annihilated by a power of I,*

$$\operatorname{Ext}^i_A(N, M) = 0 \qquad \text{for } i \leq n.$$

(b) *If, in addition, A is noetherian, M is finitely generated, and there is a finitely generated A-module N with $\operatorname{Supp} N = V(I)$ and $\operatorname{Ext}^i_A(N, M) = 0$ for $i \leq n$, then there exists an M-regular sequence $f_1, \dots, f_{n+1}$ of elements of I.*

We first prove (a), by induction. If $n < 0$, the assertion is empty. Suppose $n \geq 0$ and that (a) has been proved for every $n' < n$. By hypothesis there is an M-regular element $f_1 \in I$. Denote by f_1^i multiplication by f_1 on $\operatorname{Ext}^i_A(N, M)$ and by f_1^M multiplication by f_1 on M. The sequence

$$0 \longrightarrow M \xrightarrow{f_1^M} M \longrightarrow M/f_1M \longrightarrow 0 \tag{III.1}$$

is exact, and therefore so is

$$\operatorname{Ext}^{i-1}_A(N, M/f_1M) \xrightarrow{\delta} \operatorname{Ext}^i_A(N, M) \xrightarrow{f_1^i} \operatorname{Ext}^i_A(N, M).$$

By hypothesis, $I^r N = 0$ for some r, so f_1^0 is nilpotent. Since Ext^i is a universal cohomological functor, the same is true of f_1^i for every i. Moreover, M/f_1M has a regular sequence of length n. The induction hypothesis therefore gives

$$\operatorname{Ext}^{i-1}_A(N, M/f_1M) = 0 \qquad \text{for } i \leq n.$$

[2] Consequently, for $i \leq n$, f_1^i is both injective and nilpotent, so $\operatorname{Ext}^i_A(N, M) = 0$.

We now prove (b), again by induction. If $n < 0$, the assertion is empty. If $n = 0$, it follows from Lemma 2.1, (i)⇒(iv). If $n > 0$, the case $n = 0$ gives an M-regular element $f_1 \in I$. From (III.1) one obtains the exact sequence Original French p. 31

$$\operatorname{Ext}^{i-1}_A(N, M) \longrightarrow \operatorname{Ext}^{i-1}_A(N, M/f_1M) \longrightarrow \operatorname{Ext}^i_A(N, M). \tag{III.2}$$

The hypotheses of (b) therefore hold for M/f_1M and $n-1$. By induction there is a regular sequence of n elements of I on M/f_1M, and hence an M-regular sequence of $n+1$ elements of I beginning with f_1.

This theorem suggests the following generalization of the classical definition of the depth of a finitely generated module over a noetherian ring.

Definition 2.3. *Let A be a commutative ring with identity, M an A-module, and I an ideal of A. The I-*depth *of M, denoted* $\operatorname{prof}_I M$, *is the supremum in $\mathbb{N} \cup \{+\infty\}$ of the natural numbers n such that, for every finitely generated A-module N annihilated by a power of I,*

$$\operatorname{Ext}^i_A(N, M) = 0 \qquad \text{for every } i < n.$$

[2] *Source correction.* The French authority prints $\operatorname{Ext}^{i-1}_A(N, M) = 0$ for $i \leq n-1$. The quotient module and the bound are both defective: applying the induction hypothesis to the length n sequence on M/f_1M gives the displayed formula, which is exactly what is needed below.

It follows from the preceding theorem that, if n is the supremum of the lengths of M-regular sequences of elements of I, then $n \leq \operatorname{prof}_I M$. More precisely:

Proposition 2.4. *Let A be a commutative ring, I an ideal of A, M an A-module, and $n \in \mathbb{N}$. Consider the assertions:*

(1) $n \leq \operatorname{prof}_I M$.

(2) *For every finitely generated A-module N annihilated by a power of I, one has*

$$\operatorname{Ext}^i_A(N, M) = 0 \qquad \textit{for } i < n.$$

(3) *There exists a finitely generated A-module N such that* $\operatorname{Supp} N = V(I)$, *and such that* $\operatorname{Ext}^i_A(N, M) = 0$ *whenever $i < n$.* Original French p. 32

(4) *There is an M-regular sequence of length n consisting of elements of I.*

The following logical implications hold:

$$\begin{array}{ccccc} (1) & \Longleftrightarrow & (2) & \Longleftarrow & (4) \\ & & \Downarrow & & \\ & & (3) & & \end{array}$$

If A is noetherian and M is finitely generated, all four conditions are equivalent.

Proof. Conditions (1) and (2) are equivalent by definition, and (2)⇒(3) follows by taking $N = A/I$. The implication (4)⇒(2) is Theorem 2.2(a). Finally, when A is noetherian and M is finitely generated, (3)⇒(4) is Theorem 2.2(b). □

From now until the end of the section, we assume that A is noetherian and M is finitely generated.

Corollary 2.5. *If $f \in I$ is M-regular, then*

$$\operatorname{prof}_I M = \operatorname{prof}_I(M/fM) + 1.$$

Indeed, if $n \leq \operatorname{prof}_I(M/fM)$, there is an (M/fM)-regular sequence $f_1, \ldots, f_n$ of elements of I. Hence $f, f_1, \ldots, f_n$ is M-regular, so $n + 1 \leq \operatorname{prof}_I M$ and $\operatorname{prof}_I M \geq \operatorname{prof}_I(M/fM) + 1$. Conversely, for every finitely generated A-module N annihilated by a power of I, the exact sequence (III.2) shows that if $i < \operatorname{prof}_I M$, then $\operatorname{Ext}^{i-1}_A(N, M/fM) = 0$.[3] Thus $\operatorname{prof}_I M - 1 \leq \operatorname{prof}_I(M/fM)$.

Corollary 2.6. *Every finite M-regular sequence consisting of elements of I can be extended to a maximal M-regular sequence, whose length is necessarily the I-depth of M.*

Remark 2.7. *One can scarcely refrain from saying that the greater the depth of an A-module, the more beautiful it is. A module whose support does not meet $V(I)$ is among the most beautiful: indeed,* $\operatorname{prof}_I M$ *is finite if and only if* $\operatorname{Supp} M \cap V(I) \neq \varnothing$. Original French p. 33

Remark 2.8. *If A is a semi-local ring, let $\mathfrak{r}(A)$ be its radical and $k = A/\mathfrak{r}(A)$ its residue ring. The relevant notion of depth is obtained by taking $I = \mathfrak{r}(A)$; accordingly, we write simply* $\operatorname{prof} M$ *for the $\mathfrak{r}(A)$-depth of an A-module M. This is the notion of "homological codimension"*

[3] *Source correction.* The French authority prints $i \leq \operatorname{prof}_I M$. In the displayed exact sequence, vanishing of both adjacent terms is guaranteed for $i < \operatorname{prof}_I M$; this is precisely the range needed for the following depth inequality.

(cf. Serre, op. cit.*), formerly denoted* $\operatorname{codh}_A M$*, defined as the infimum of the integers* i *such that* $\operatorname{Ext}^i_A(k, M) \neq 0$*. Indeed,* $\operatorname{Supp} k = V(\mathfrak{r}(A))$*.*

Proposition 2.9. *If* A *is noetherian and* M *is finitely generated, then*

$$\operatorname{prof}_I M = \inf_{\mathfrak{p} \in V(I)} \operatorname{prof} M_{\mathfrak{p}}.$$

Corollary 2.10. *If* A *is a noetherian semi-local ring and* M *is a finitely generated* A*-module, then*

$$\operatorname{prof} M = \inf_{\mathfrak{m}} \operatorname{prof} M_{\mathfrak{m}},$$

where $\mathfrak{m}$ *ranges over the maximal ideals of* A*.*

The corollary follows immediately from Proposition 2.9, since the prime ideals containing the radical are precisely the maximal ideals.

We now prove Proposition 2.9. Let $f \in I$. If f is M-regular and $\mathfrak{p} \supset I$, then the image g of f in $A_{\mathfrak{p}}$ belongs to the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ and is $M_{\mathfrak{p}}$-regular, as follows from the exact sequence

$$0 \longrightarrow M_{\mathfrak{p}} \xrightarrow{g'} M_{\mathfrak{p}} \longrightarrow (M/fM)_{\mathfrak{p}} \longrightarrow 0, \tag{III.3}$$

where g' denotes multiplication by g on $M_{\mathfrak{p}}$. This also shows that $(M/fM)_{\mathfrak{p}} \simeq M_{\mathfrak{p}}/gM_{\mathfrak{p}}$. Applying Corollary 2.5 to M and to $M_{\mathfrak{p}}$ gives, by induction, $\operatorname{prof}_I M \leq \nu(M)$, where

Original French p. 34

$$\nu(M) = \inf_{\mathfrak{p} \in V(I)} \operatorname{prof} M_{\mathfrak{p}}.$$

More precisely, induction gives $\nu(M) = \nu(M/fM) + 1$ whenever f is M-regular. It remains to prove that, if $\nu(M) \geq 1$, there is an M-regular element in I. Applying Lemma 2.1 to $M_{\mathfrak{p}}$, $A_{\mathfrak{p}}$, and $\mathfrak{p}A_{\mathfrak{p}}$ for every $\mathfrak{p} \in V(I)$ shows that $\mathfrak{p}A_{\mathfrak{p}} \notin \operatorname{Ass} M_{\mathfrak{p}}$. Applying the same lemma to A, M, and I gives the conclusion.

Proposition 2.11. *Let* $u : A \to B$ *be a homomorphism of noetherian rings, let* I *be an ideal of* A*, and let* M *be a finitely generated* A*-module. Put* $I_B = I \otimes_A B$ *and* $M_B = M \otimes_A B$*. If* B *is flat over* A*, then*

$$\operatorname{prof}_{I_B} M_B \geq \operatorname{prof}_I M;$$

if B *is faithfully flat over* A*, equality holds.*

Indeed, put $N = A/I$. Flatness gives $N \otimes_A B = B/I_B$; set $N_B = N \otimes_A B$. Again by flatness and the noetherian hypotheses,

$$\operatorname{Ext}^i_B(N_B, M_B) = \operatorname{Ext}^i_A(N, M) \otimes_A B.$$

Thus $\operatorname{Ext}^i_A(N, M) = 0$ implies $\operatorname{Ext}^i_B(N_B, M_B) = 0$, and the converse holds when B is faithfully flat over A.

3. Depth and Topological Properties

Lemma 3.1. *Let* X *be a topological space,* Y *a closed subspace, and* F *a sheaf of abelian groups on* X*. Put* $U = X - Y$*. If* n *is an integer, the following conditions are equivalent:* Original French p. 34

(i) $\mathcal{H}^i_Y(X, F) = 0$ *for* $i < n$*.*

Original French p. 35

(ii) *For every open subset V of X, the group homomorphism*

$$\mathrm{H}^i(V, F) \longrightarrow \mathrm{H}^i(V \cap U, F)$$

is bijective for $i < n - 1$ and injective for $i = n - 1$.

(iii) *For every open subset V of X,*

$$\mathrm{H}^i_{Y \cap V}(V, F|_V) = 0 \qquad \textit{for } i < n.$$

Proof. First, (ii) and (iii) are equivalent. Indeed, let V be an open subset of X, and put $X' = V$, $Y' = Y \cap V$, $F' = F|_V$, and $U' = X' - Y'$. Since Y' is closed in X', there is an exact sequence

$$\mathrm{H}^i_{Y'}(X', F') \longrightarrow \mathrm{H}^i(X', F') \xrightarrow{\rho_i} \mathrm{H}^i(U', F') \longrightarrow \mathrm{H}^{i+1}_{Y'}(X', F').$$

If the two outer terms vanish, ρ_i is bijective; if the term on the left vanishes, ρ_i is injective. Thus (iii) implies (ii). Conversely, if $i < n$, then $\mathrm{H}^i_{Y'}(X', F')$ vanishes because ρ_i is injective and ρ_{i-1} is surjective.

Next, (i) implies (iii). The local-to-global spectral sequence gives

$$\mathrm{H}^p(X, \mathcal{H}^q_Y(X, F)) \Longrightarrow \mathrm{H}^{p+q}_Y(X, F).$$

By hypothesis, $\mathcal{H}^q_Y(X, F) = 0$ for $q < n$; hence $\mathrm{H}^{p+q}_Y(X, F) = 0$ whenever $p + q < n$.

Finally, (iii) implies (i), since (iii) says that the presheaf

$$V \longmapsto \mathrm{H}^i_{Y \cap V}(V, F|_V)$$

is zero. Its associated sheaf is therefore zero; because Y is closed, that associated sheaf is $\mathcal{H}^i_Y(X, F)$. □

Remark 3.2. *The equivalence of (i) and (ii) was proved in Proposition I.2.13. As was observed there, if $n \geq 2$, the condition that ρ_{n-1} be injective may be omitted.*[4] Original French p. 36

Proposition 3.3. *Let X be a locally noetherian prescheme, Y a closed subprescheme of X, and F a coherent $\mathcal{O}_X$-module. The conditions of Lemma 3.1 are equivalent to each of the following conditions:*

(iv) *For every $x \in Y$, one has* $\operatorname{prof} F_x \geq n$.

(v) *For every coherent $\mathcal{O}_X$-module G on X whose support is contained in Y,*

$$\mathcal{E}xt^i_{\mathcal{O}_X}(G, F) = 0 \qquad \textit{for } i < n.$$

(vi) *There exists a coherent $\mathcal{O}_X$-module G whose support equals Y and such that*

$$\mathcal{E}xt^i_{\mathcal{O}_X}(G, F) = 0 \qquad \textit{for } i < n.$$

If X is affine, Proposition 2.4 supplies what is needed to establish the equivalence of the three conditions in Proposition 3.3. These conditions are local except for (v)⇒(vi); for that implication one may take $G = \mathcal{O}_Y$ and invoke Proposition 2.4 again. It therefore suffices to prove (i)⇒(vi) and (v)⇒(i).

Let J be the ideal of Y; it is a coherent sheaf of ideals. Put $\mathcal{O}_{Y_m} = \mathcal{O}_X / J^{m+1}$. This is a coherent $\mathcal{O}_X$-module whose support equals Y, and Theorem II.6(b) gives

$$\mathcal{H}^i_Y(X, F) = \varinjlim_m \mathcal{E}xt^i_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F).$$

Thus (v) implies (i). Moreover, the transition morphisms in the projective system of the $\mathcal{O}_{Y_m}$ are epimorphisms. Original French p. 37

If the functor $\mathcal{E}xt^i$ is left exact in its first argument—at least when that argument lies in the category of coherent $\mathcal{O}_X$-modules with support contained in Y—then the transition morphisms of the inductive system obtained by applying $\mathcal{E}xt^i$ to the $\mathcal{O}_{Y_m}$ are injective. Condition (i) makes the limit zero, and therefore makes $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F)$ zero for every m.

We argue by induction. The assertion is trivial for $n < 0$. Suppose that (i)⇒(vi) has been proved in degrees below q. Then (i)⇒(v), so the long exact $\mathcal{E}xt$ sequence shows that $\mathcal{E}xt^q$ is left exact in its first argument. Hence $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F) = 0$ for every m. Thus (i)⇒(vi) holds through degree q. □

Example 3.4. *Let A be a noetherian local ring, $\mathfrak{m}$ its maximal ideal, M a finitely generated A-module, and n an integer. Put $X = \operatorname{Spec}(A)$, $Y = \{\mathfrak{m}\}$, and $U = X - Y$. Let F be the sheaf associated with M. The following conditions are equivalent:* Original French p. 37

(1) $\operatorname{prof} M \geq n$.

(2) *The natural homomorphism*

$$\mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F)$$

is injective if $i = n - 1$ and bijective if $i < n - 1$.

(3) $\operatorname{Ext}^i_A(k, M) = 0$ *if $i < n$, where $k = A/\mathfrak{m}$.*

(4) $\mathrm{H}^i_Y(X, F) = 0$ *if $i < n$.*

Taking Remark 3.2 into account, one obtains the following corollary. Original French p. 38

Corollary 3.5. *Let X be a locally noetherian prescheme, Y a closed subprescheme of X, and F a coherent $\mathcal{O}_X$-module. The following conditions are equivalent:* Original French p. 38

(1) *For every $x \in Y$,* $\operatorname{prof} F_x \geq 2$.

(2) *For every open subset V of X, the natural homomorphism*

$$\mathrm{H}^0(V, F) \longrightarrow \mathrm{H}^0(V \cap (X - Y), F)$$

is bijective.

Theorem 3.6 (Hartshorne). *Let X be a locally noetherian prescheme and Y a closed subprescheme of X. Suppose that* $\operatorname{prof} \mathcal{O}_{X,x} \geq 2$ *for every $x \in Y$. Then the natural map*

$$\pi_0(X) \longrightarrow \pi_0(X - Y)$$

is bijective.

Proof. Since X is locally noetherian, it is locally connected. It therefore suffices to prove that X is connected if and only if $X - Y$ is connected. A locally ringed space $(X, \mathcal{O}_X)$ is connected if and only if $\mathrm{H}^0(X, \mathcal{O}_X)$ is not a direct product of two nonzero rings. By Corollary 3.5, applied with $F = \mathcal{O}_X$, the hypothesis implies that

$$\mathrm{H}^0(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^0(X - Y, \mathcal{O}_X)$$

is an isomorphism, and the conclusion follows. □

[4] The original edition repeated a proof here, but that proof was not entirely correct.

Corollary 3.7. *Let X be a locally noetherian prescheme. Let d be an integer such that $\dim \mathcal{O}_{X,x} \geq d$ implies $\operatorname{prof} \mathcal{O}_{X,x} \geq 2$. If X is connected, then it is connected in codimension $d-1$: namely, if X' and X'' are two irreducible components of X, there is a sequence of irreducible components of X,*

$$X' = X_0, X_1, \ldots, X_n = X'',$$

such that, for every i with $0 \leq i < n$, the codimension in X of $X_i \cap X_{i+1}$ is at most $d-1$.

First note that if X is Cohen–Macaulay, then $d = 2$ has the property just invoked. Recall also that the codimension of Y in X is defined as the infimum of the dimensions of the local rings of X at the points of Y. Original French p. 39

Proof (reduction). Let $\mathcal{F}$ be the collection of closed subsets of X whose codimension in X is at least d. This is an antifilter of closed subsets of X. A closed subset Y belongs to $\mathcal{F}$ if and only if, for every $y \in Y$, there is an open neighborhood V of y such that $Y \cap V$ has codimension at least d in V.[5] Finally, if X is connected and $Y \in \mathcal{F}$, then $X - Y$ is connected by Hartshorne's theorem. The corollary therefore follows from the next lemma, which is purely topological.

Lemma 3.8. *Let X be a connected, locally noetherian topological space, and let $\mathcal{F}$ be an antifilter of closed subsets of X. Suppose that every closed subset $Y \subset X$ which belongs locally to $\mathcal{F}$—that is, for every $x \in X$ there are an open neighborhood V of x in X and a $Y' \in \mathcal{F}$ such that $V \cap Y = V \cap Y'$—belongs to $\mathcal{F}$. The following conditions are equivalent:*

(i) *For every $Y \in \mathcal{F}$, the complement $X - Y$ is connected.*

(ii) *If X' and X'' are distinct irreducible components, there is a sequence $X_0, X_1, \ldots, X_n$ of irreducible components of X such that $X' = X_0$, $X'' = X_n$, and*

$$X_i \cap X_{i+1} \notin \mathcal{F} \qquad (0 \leq i < n).$$

Source correction. The corrected French branch prints $1 \leq i < n$ in (ii), omitting the first adjacent pair. Both the chain's indexing and the proof below require $0 \leq i < n$; the target displays that range rather than silently reproducing the defect.

Proof. Assume (ii), and let $Y \in \mathcal{F}$. We must prove that $U = X - Y$ is connected. If U' and U'' are two irreducible components of U, there are irreducible components X' and X'' of X such that $X' \cap U = U'$ and $X'' \cap U = U''$. Choose a sequence $X_0, \ldots, X_n$ having the property in (ii), and put $U_i = X_i \cap U$ for $0 \leq i \leq n$. The U_i are irreducible components of U, and $U_i \cap U_{i+1}$ is nonempty for $0 \leq i < n$. Otherwise $X_i \cap X_{i+1} \subset Y$ would belong to $\mathcal{F}$, contrary to the choice of the X_i. Hence U is connected. Original French p. 40

Conversely, assume (i). Let

$$Y = \bigcup_{X', X''} (X' \cap X''),$$

where the union is over distinct irreducible components X', X'' of X for which $X' \cap X'' \in \mathcal{F}$. This family of intersections is locally finite because X is locally noetherian; its members are closed, so Y is closed. Moreover, Y belongs locally to $\mathcal{F}$, and hence $Y \in \mathcal{F}$. Thus $U = X - Y$ is connected.

Let X' and X'' be distinct irreducible components of X, and let U' and U'' be their traces on U; by the construction of Y, these traces are nonempty. They are irreducible components

[5] Source correction: the French TeX says "an open neighborhood V of X"; the quantified point y and the local condition require "of y."

of U. Since U is connected and locally noetherian, there is a sequence $U_0, \dots, U_n$ of irreducible components of U such that $U_0 = U'$, $U_n = U''$, and, for $0 \le i < n$,

$$U_i \cap U_{i+1} \neq \varnothing \quad \text{and} \quad U_i \cap U_{i+1} \neq U_i.$$

Let $X_0, \dots, X_n$ be the irreducible components of X satisfying $X_i \cap U = U_i$. If $X_i \cap X_{i+1} \in \mathcal{F}$, then by the construction of Y one would have either $U_i \cap U_{i+1} = \varnothing$ or $U_i = U_{i+1}$, contradicting the choice of the U_i. ☐

Corollary 3.9. *Let A be a noetherian local ring. Suppose that, for every prime ideal $\mathfrak{p}$ of A,*

$$(\dim A_{\mathfrak{p}} \ge 2) \Longrightarrow (\operatorname{prof} A_{\mathfrak{p}} \ge 2).$$

Suppose also that A satisfies the chain condition.[6] *Then every minimal prime ideal $\mathfrak{p}$ of A satisfies* $\dim(A/\mathfrak{p}) = \dim A$. *Equivalently, all irreducible components of* $\operatorname{Spec} A$ *have the same dimension, namely* $\dim A$.

Indeed, put $X = \operatorname{Spec} A$. If X' and X'' are two irreducible components of X, Corollary 3.7 joins them by a chain with the stated properties. It is therefore enough to prove that two successive components have the same dimension, which follows from the second hypothesis.

Example 3.10. *Let X be the union of two complementary linear subspaces of respective dimensions 2 and 3 in a vector space of dimension 5. More precisely, let $X = \operatorname{Spec} A$, where* Original French p. 41

$$A = B/(\mathfrak{p} \cap \mathfrak{q}), \qquad B = k[X_1, \dots, X_5],$$

$\mathfrak{p} = (X_1, X_2, X_3)$, *and* $\mathfrak{q} = (X_4, X_5)$. *Removing the intersection point x of the two linear subspaces disconnects X. Consequently, the depth of $\mathcal{O}_{X,x}$ is 1, since it cannot be at least 2 by Theorem 3.6. The failure of the preceding corollary's equidimensionality conclusion gives another reason.*

More generally, let X be the union of two linear subspaces of dimensions $p, q \ge 2$ in a vector space of dimension $p + q$. Under no embedding of X into a regular scheme is X even set-theoretically a complete intersection at the origin. Indeed, after possibly modifying its scheme structure without changing the underlying topological space near the origin, X would then be Cohen–Macaulay and hence of depth at least 2 at the origin, which it is not.

Remark 3.11. *Let X be a locally noetherian prescheme, Y a closed subprescheme of X, and F an $\mathcal{O}_X$-module. Depth is a purely topological notion, expressed in terms of the vanishing of $\mathcal{H}^i_Y(X, F)$ for $i < n$. One also wishes to study these sheaves for a fixed i, or for $i > n$. The following result addresses this question.*

Lemma 3.12. *Let m be an integer. The equality $\mathcal{H}^i_Y(X, F) = 0$ for every $i > m$ and every coherent F holds if and only if it holds for $F = \mathcal{O}_X$.*

By passage to a direct limit, it then holds for every quasi-coherent sheaf. For example, if Y can be locally described by m equations—or, as one says, if Y is locally a set-theoretic complete intersection (as happens, for example, when X and Y are nonsingular)—the computation of $\mathcal{H}^i_Y(X, F)$ by the Koszul complex shows that these sheaves vanish for $i > m$. This fact was used implicitly in Example 3.10. The cohomological condition is nevertheless insufficient, as the following example shows.

Example 3.13. *Let $X = \operatorname{Spec} A$, where A is a normal noetherian local ring of dimension* Original French p. 42

[6] Cf. EGA 0_{IV}, 14.3.2.

2, and let Y be a curve in X. The complement $X - Y$ can be shown to be affine. Hence, for $i > 1$, the supported-cohomology exact sequence gives

$$\mathrm{H}^i_Y(X, \mathcal{O}_X) \simeq \mathrm{H}^{i-1}(X - Y, \mathcal{O}_X) = 0.$$

Consequently the corresponding local-cohomology sheaf $\mathcal{H}^i_Y(\mathcal{O}_X)$ vanishes. Nevertheless, one can construct a curve which is not described by a single equation.

Source correction. The corrected French branch and its rendered PDF write a sheaf-local-cohomology term with support $X - Y$ in the middle of the displayed isomorphism. The stated reason is instead the vanishing of $\mathrm{H}^{i-1}(X - Y, \mathcal{O}_X)$ because $X - Y$ is affine; the target displays the supported/global exact-sequence formula that this reason requires and then states the sheaf consequence separately.

We shall seek[7] conditions ensuring that $\mathcal{H}^i_Y(X, F)$ is coherent for a fixed i. In general it is not: for example, consider $\mathrm{H}^n_{\mathfrak{m}}(A)$ when A is a noetherian local ring of dimension $n > 0$. If A is a discrete valuation ring with fraction field K, then

$$\mathrm{H}^1_{\mathfrak{m}}(A) \simeq K/A,$$

which is not a finitely generated A-module.

To illuminate the question, it is equivalent to the following problem. Let $f : U \to X$ be an open immersion and G a coherent sheaf on U. Find criteria ensuring that the higher direct images $\mathrm{R}^i f_* G$ are coherent sheaves on X for a fixed i. Such conditions are necessary for the use of formal geometry encountered in Exposé IX and the subsequent exposés.

[7] Cf. Exposé VIII.

Exposé IV

Dualizing Modules and Functors

1. Generalities on Functors of Modules

Fix a commutative noetherian ring A. Let $\mathcal{C}$ be the category of finitely generated A-modules, $\mathcal{C}'$ the category of all A-modules, and $\mathbf{Ab}$ the category of abelian groups. This section studies certain properties of additive functors Original French p. 43

$$T : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$

For $M \in \mathrm{Ob}\,\mathcal{C}$, the group $T(M)$ has a canonical A-module structure. If $f_M : M \to M$ denotes multiplication by $f \in A$, then f acts on $T(M)$ as $T(f_M)$. In other words, T factors as

$$\mathcal{C}^{\mathrm{op}} \xrightarrow{T_0} \mathcal{C}' \longrightarrow \mathbf{Ab},$$

where the second arrow is the forgetful functor. We henceforth regard $T(M)$ as endowed with this A-module structure.

Compose the isomorphism $M \xrightarrow{\sim} \mathrm{Hom}_A(A, M)$ with the map

$$\mathrm{Hom}_A(A, M) \longrightarrow \mathrm{Hom}_A(T(M), T(A))$$

induced by T. This gives the following mutually adjoint forms of the same pairing:

$$\begin{aligned} M &\longrightarrow \mathrm{Hom}_A(T(M), T(A)), \\ M \times T(M) &\longrightarrow T(A), \\ T(M) &\longrightarrow \mathrm{Hom}_A(M, T(A)). \end{aligned}$$

They define a natural morphism of contravariant functors

$$\varphi_T : T \longrightarrow \mathrm{Hom}_A(-, T(A)).$$

Proposition 1.1. *The following two properties are equivalent:* Original French p. 44

i) *φ_T is an isomorphism of functors.*

ii) *T is left exact.*

The implication (i)⇒(ii) is immediate. For the converse, use the following fact: if $u : F \to F'$ is a morphism between additive left-exact functors from $\mathcal{C}^{\mathrm{op}}$ to $\mathbf{Ab}$ and $u(A)$ is an isomorphism, then u is an isomorphism. Here one uses that A is noetherian, so every finitely generated A-module is finitely presented.

Remark 1.2. *In particular, the representable functors $T : \mathcal{C}'^{\mathrm{op}} \to \mathbf{Ab}$ are precisely those which commute with arbitrary inverse limits in $\mathcal{C}'^{\mathrm{op}}$ (indexed by a preordered set, not necessarily filtered).*

Let $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Ab})_{\mathrm{lex}}$ denote the full subcategory of additive functors that are left exact. We have proved an equivalence of categories

$$\mathcal{C}' \xrightarrow{\sim} \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Ab})_{\mathrm{lex}},$$

with mutually quasi-inverse functors

$$H \longmapsto \mathrm{Hom}_A(-, H), \qquad T \longmapsto T(A).$$

Let J be an ideal of A, put $Y = V(J) \subset \operatorname{Spec} A$, and let $\mathcal{C}_Y$ be the full subcategory of $\mathcal{C}$ consisting of finitely generated A-modules M with $\operatorname{Supp} M \subset Y$. Then

$$\mathcal{C}_Y = \bigcup_n \mathcal{C}^{(n)},$$

where $\mathcal{C}^{(n)}$ is the full subcategory of modules satisfying $J^n M = 0$.

Proposition 1.3. *With this notation, let $T : \mathcal{C}_Y^{\mathrm{op}} \to \mathbf{Ab}$ be an additive functor, and put* Original French p. 45

$$H = \varinjlim_n T(A/J^n).$$

There is a natural morphism

$$\varphi_T : T \longrightarrow \mathrm{Hom}_A(-, H),$$

and the following conditions are equivalent:

i) *φ_T is an isomorphism.*

ii) *T is left exact.*

Source correction. The proposition line prints $T : \mathcal{C}_Y \to \mathbf{Ab}$, while its target, its restrictions, and the concluding category equivalence are contravariant. The target restores the required opposite category.

Proof. First define φ_T. For $M \in \operatorname{Ob} \mathcal{C}_Y$, choose n with $J^n M = 0$. Then M is an A/J^n-module. If T_n is the restriction of T to $\mathcal{C}^{(n)\mathrm{op}}$ and $H_n = T(A/J^n)$, the construction of Proposition 1.1 gives

$$T_n \longrightarrow \mathrm{Hom}_A(-, H_n).$$

Composing with $H_n \to H$ gives, for every such M,

$$T(M) = T_n(M) \longrightarrow \mathrm{Hom}_A(M, H),$$

and these maps define φ_T.

It is clear that (i) implies (ii). Conversely, suppose T is left exact and let $M \in \operatorname{Ob} \mathcal{C}^{(n)}$. Proposition 1.1 applied over A/J^n gives

$$T_n(M) \xrightarrow{\sim} \mathrm{Hom}_A(M, H_n).$$

For every $n' \geq n$, Proposition 1.1 applied over $A/J^{n'}$ gives a compatible isomorphism with $H_{n'}$. Hence

$$T(M) \simeq \varinjlim_{n' \geq n} \mathrm{Hom}_A(M, H_{n'}).$$

Because this is a filtered direct limit and M is finitely presented,

$$\varinjlim_{n' \geq n} \operatorname{Hom}_A(M, H_{n'}) \xrightarrow{\sim} \operatorname{Hom}_A\left(M, \varinjlim_{n' \geq n} H_{n'}\right) = \operatorname{Hom}_A(M, H).$$

Consequently, if $\mathcal{C}'_Y$ denotes the category of all A-modules supported in Y, without a finite-generation condition, there is a natural equivalence

$$\mathcal{C}'_Y \xrightarrow{\sim} \operatorname{Fun}(\mathcal{C}_Y^{\mathrm{op}}, \mathbf{Ab})_{\mathrm{lex}}.$$

□

Application

Retain the preceding notation, and let Original French p. 46

$$T^* : \mathcal{C}_Y^{\mathrm{op}} \longrightarrow \mathbf{Ab}$$

be an exact ∂-functor. For every $i \in \mathbb{Z}$, put

$$H_n^i = T^i(A/J^n), \qquad H^i = \varinjlim_n H_n^i.$$

Theorem 1.4. *Let $n \in \mathbb{Z}$. Suppose there is an $i_0 \in \mathbb{Z}$ such that $T^i = 0$ for every $i < i_0$. Then the following three conditions are equivalent:*

i) $T^i = 0$ *for every* $i < n$.

ii) $H^i = 0$ *for every* $i < n$.

iii) *There exists an* $M_0 \in \mathcal{C}_Y$ *such that* $\operatorname{Supp} M_0 = Y$ *and* $T^i(M_0) = 0$ *for every* $i < n$.

Proof. Clearly (i) implies (ii) and (iii), taking $M_0 = A/J$. We prove (ii)⇒(i) by induction on n. It is true for $n = i_0$; suppose it has been proved through n. If $H^i = 0$ for every $i < n+1$, the induction hypothesis gives $T^i = 0$ for $i < n$. The vanishing of T^{n-1} makes T^n left exact, so Proposition 1.3 gives

$$T^n \xrightarrow{\sim} \operatorname{Hom}_A(-, H^n) = 0.$$

It remains to prove (iii)⇒(ii), again by induction. The assertion is true for $n = i_0$. Suppose it has been proved through n, and let $M_0 \in \mathcal{C}_Y$ satisfy $\operatorname{Supp} M_0 = Y$ and $T^i(M_0) = 0$ for every $i < n+1$. The induction hypothesis gives $H^i = 0$ for $i < n$; it remains to show $H^n = 0$. The preceding implication gives $T^{n-1} = 0$, and therefore

$$T^n \xrightarrow{\sim} \operatorname{Hom}_A(-, H^n).$$

Moreover,

$$\operatorname{Ass} \operatorname{Hom}_A(M_0, H^n) = \operatorname{Supp} M_0 \cap \operatorname{Ass} H^n = \operatorname{Ass} H^n,$$

because

$$\operatorname{Ass} H^n \subset \operatorname{Supp} H^n \subset Y = \operatorname{Supp} M_0.$$

Thus

$$T^n(M_0) = 0 \iff \operatorname{Ass} H^n = \varnothing \iff H^n = 0,$$

which completes the proof. □

2. Characterization of Exact Functors

The ring A remains commutative and noetherian. Retain the notation of Proposition 1.3: Original French p. 47

$$Y = V(J), \qquad T : \mathcal{C}_Y^{\mathrm{op}} \longrightarrow \mathbf{Ab}, \qquad H = \varinjlim_n T(A/J^n).$$

Assume that T is left exact, so that

$$T(M) \xrightarrow{\sim} \operatorname{Hom}_A(M, H).$$

Proposition 2.1. *The following properties are equivalent:*

i) *T is exact.*

ii) *H is injective in $\mathcal{C}'$, the category of all A-modules.*

Proof. Only (i)⇒(ii) needs proof. Suppose the restriction of $\operatorname{Hom}_A(-, H)$ to $\mathcal{C}_Y^{\mathrm{op}}$ is exact. Since A is noetherian, it is enough to prove the following extension property: whenever $N \subset M$ are finitely generated A-modules, every homomorphism $f : N \to H$ extends to a homomorphism $\bar{f} : M \to H$.

By the definition of H and finite generation of N, there is an integer n such that $J^n f(N) = 0$. Give M and N their J-adic topologies. By Krull's theorem, the J-adic topology on N is equivalent to the topology induced from the J-adic topology on M. Hence there is a $V = J^k M$ such that

$$U = V \cap N \subset J^n N.$$

Thus $U \subset \ker f$, and f factors as

$$\begin{tikzcd} N \arrow[r, two heads] \arrow[dr, "f"'] & N/U \arrow[d, "u"] \\ & H. \end{tikzcd}$$

Both N/U and M/V belong to $\mathcal{C}_Y$. Exactness therefore extends u to a map $\bar{u} : M/V \to H$:

$$\begin{tikzcd} N/U \arrow[r, hook] \arrow[dr, "u"'] & M/V \arrow[d, "\bar{u}"] \\ & H. \end{tikzcd}$$

The composite $M \to M/V \xrightarrow{\bar{u}} H$ is the required extension $\bar{f}$. $\square$

Original French p. 48

Retain the notation $Y = V(J)$ and let $\mathcal{C}_Y$ denote the category of finitely generated A-modules whose support is contained in Y.

Corollary 2.2. *Let K be an injective A-module. Then the submodule* Original French p. 48

$$H_J^0(K) = \{x \in K : J^n x = 0 \textit{ for some } n \geq 0\}$$

of elements annihilated by a suitable power of J is injective.

Proof. By Proposition 2.1, it is enough to verify that the restriction to $\mathcal{C}_Y^{\mathrm{op}}$ of $\operatorname{Hom}_A(-, H_J^0(K))$ is exact. Let $M \in \mathcal{C}_Y$. There is an integer k such that $J^k M = 0$, and the inclusion

$$\operatorname{Hom}_A(M, H_J^0(K)) \longrightarrow \operatorname{Hom}_A(M, K)$$

is an isomorphism: every map $M \to K$ has image annihilated by J^k and therefore lands in $H_J^0(K)$. The assertion follows because $\operatorname{Hom}_A(-, K)$ is exact. $\square$

3. The Case Where T Is Left Exact and Every $T(M)$ Is Finitely Generated

As above, let

$$T : \mathcal{C}_Y^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$ Original French p. 48

From now on, assume that T is left exact and factors through $\mathcal{C}_Y$:

$$\begin{tikzcd} \mathcal{C}_Y^{\mathrm{op}} \arrow[rr, "T"] \arrow[dr, "\widetilde{T}"'] & & \mathbf{Ab} \\ & \mathcal{C}_Y \arrow[ur, "\text{forget}"'] & \end{tikzcd}$$

Thus $T(T(M)) = (T \circ T)(M)$ is defined. Under the representation $T(M) \simeq \mathrm{Hom}_A(M, H)$, the canonical evaluation map

$$M \longrightarrow \mathrm{Hom}_A(\mathrm{Hom}_A(M, H), H)$$

defines a natural morphism

$$\eta_M : M \longrightarrow (T \circ T)(M).$$ Original French p. 49

Proposition 3.1. *Assume, as before, that A is noetherian, and assume in addition that A/J is artinian. The following conditions are equivalent:*

i) *T is left exact and, for every $M \in \mathcal{C}_Y$, the module $T(M)$ is finitely generated and $\eta_M : M \to (T \circ T)(M)$ is an isomorphism.*

ii) *T is exact and, for every residue field $k = A/\mathfrak{m}$ of a maximal ideal $\mathfrak{m} \supset J$, one has $T(k) \simeq k$.*

iii) *There is an isomorphism $T \simeq \mathrm{Hom}_A(-, H)$ with H injective and, for every k as in* (ii)*, one has $\mathrm{Hom}_A(k, H) \simeq k$.*

iv) *T is exact and $\mathrm{length}_A T(M) = \mathrm{length}_A M$ for every $M \in \mathcal{C}_Y$.*

Proof. The equivalence of *(ii)* and *(iii)* follows from Proposition 2.1.

(ii)⇒(iv). If $M \in \mathcal{C}_Y$, then M is an A/J^n-module for some n; since A/J^n is artinian, M has finite length. We argue by induction on $\mathrm{length}_A M$. When $\mathrm{length}_A M = 1$, the module M is one of the residue fields occurring in *(ii)*, so the assertion holds. If $\mathrm{length}_A M > 1$, choose a nonzero proper submodule $M' \subset M$ and put $M'' = M/M'$. Exactness of T turns

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

into

$$0 \longrightarrow T(M'') \longrightarrow T(M) \longrightarrow T(M') \longrightarrow 0.$$

Both end terms have smaller length, and therefore

$$\begin{aligned} \mathrm{length}_A T(M) &= \mathrm{length}_A T(M') + \mathrm{length}_A T(M'') \\ &= \mathrm{length}_A M' + \mathrm{length}_A M'' = \mathrm{length}_A M. \end{aligned}$$

(ii)⇒(i). By *(iv)*, $T(M)$ has finite length and hence is finitely generated. It remains to prove that every η_M is an isomorphism. Again use induction on $\mathrm{length}_A M$. For a residue field, $T(k)$ and $(T \circ T)(k)$ both have length one; evaluation at $1 \in k$ shows that η_k is nonzero, hence it is an isomorphism. For an exact sequence as above, naturality gives the commutative diagram with exact rows

Original French p. 50

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r] \arrow[d, "\eta_{M'}"'] & M \arrow[r] \arrow[d, "\eta_M"'] & M'' \arrow[r] \arrow[d, "\eta_{M''}"'] & 0 \\
0 \arrow[r] & (T\circ T)(M') \arrow[r] & (T\circ T)(M) \arrow[r] & (T\circ T)(M'') \arrow[r] & 0.
\end{tikzcd}$$

The outer vertical arrows are isomorphisms by induction, hence so is η_M.

(i)⇒(ii). Let

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be exact in $\mathcal{C}_Y$, and let Q be the cokernel of $T(M) \to T(M')$. Left exactness gives the exact sequence

$$0 \longrightarrow T(M'') \longrightarrow T(M) \longrightarrow T(M') \longrightarrow Q \longrightarrow 0.$$

Applying T at the left end and using naturality of evaluation yields

$$\begin{tikzcd}
0 \arrow[r] & T(Q) \arrow[r] & (T\circ T)(M') \arrow[r] & (T\circ T)(M) \\
& & M' \arrow[u, "\eta_{M'}", "\sim"'] \arrow[r, hook] & M. \arrow[u, "\sim", "\eta_M"']
\end{tikzcd}$$

Thus $T(Q) = 0$, and condition *(i)* gives $Q \simeq (T \circ T)(Q) = 0$. Therefore $T(M) \to T(M')$ is surjective and T is exact.

Now let $k = A/\mathfrak{m}$ with $\mathfrak{m} \supset J$. The A-module $T(k)$ is a finite-dimensional k-vector space. It is nonzero because $k \simeq (T \circ T)(k)$. Choose a decomposition $T(k) \simeq k \oplus V$. Additivity and biduality give

$$\begin{aligned}(T \circ T)(k) &\simeq T(k) \oplus T(V) \\ &\simeq k \oplus V \oplus T(V) \simeq k,\end{aligned}$$

so comparison of finite lengths forces $V = 0$. Hence $T(k) \simeq k$.

(iv)⇒(iii). Proposition 2.1 supplies an injective module H and an isomorphism

$$T \simeq \operatorname{Hom}_A(-, H).$$

It remains only to check the residue fields. For $k = A/\mathfrak{m}$ as above, $T(k)$ has length one, hence is isomorphic to a residue field $k' = A/\mathfrak{m}'$. Moreover

$$\operatorname{Supp} k' = \operatorname{Supp} \operatorname{Hom}_A(k, H) \subseteq \operatorname{Supp} k = \{\mathfrak{m}\}.$$

Thus $\mathfrak{m}' = \mathfrak{m}$ and $k' \simeq k$, as required. □

Remark 3.2. *One can show that condition* (iv) *is equivalent to the condition* Original French p. 51

(iv)′ *For every $M \in \operatorname{Ob} \mathcal{C}_Y$, one has*

$$\operatorname{length} T(M) = \operatorname{length} M.$$

4. Dualizing Module. Dualizing Functor

Definition 4.1. *Let A be a noetherian local ring and let $\mathfrak{m}$ be its maximal ideal. A* dualizing functor *for A is a functor*

$$T : \mathcal{C}_{\mathfrak{m}}^{\mathrm{op}} \longrightarrow \mathbf{Ab},$$

where $\mathcal{C}_{\mathfrak{m}}$ denotes $\mathcal{C}_Y$ when $Y = V(\mathfrak{m})$, that satisfies the equivalent conditions of Proposition 3.1. An A-module I is said to be dualizing for A *if the functor*

$$M \longmapsto \operatorname{Hom}_A(M, I)$$

is dualizing.

Definition 4.1 can be generalized without assuming that A is local.

Definition 4.2. *Let A be a noetherian ring, and let $\overline{\mathcal{C}}$ be the full subcategory of $\mathcal{C}$ consisting of the A-modules of finite length. A* dualizing functor *is an exact A-linear functor*

$$T : \overline{\mathcal{C}}^{\mathrm{op}} \longrightarrow \overline{\mathcal{C}}$$

such that the morphism of functors

$$\mathrm{id} \longrightarrow T \circ T$$

is an isomorphism.

We shall prove an existence theorem and also prove that the module I representing such a functor is locally artinian. We shall further show that, for every maximal ideal $\mathfrak{m}$ of A, the $\mathfrak{m}$-primary component of the socle of I has length one.

Proposition 4.3. *Let A and B be noetherian local rings with maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$, and suppose that B is a finite A-algebra. If I is a dualizing module for A, then $\mathrm{Hom}_A(B, I)$ is a dualizing module for B.* Original French p. 51

Proof. Let Original French p. 52

$$R : \mathcal{C}_{\mathfrak{m}_B} \longrightarrow \mathcal{C}_{\mathfrak{m}_A}$$

be the restriction-of-scalars functor; it is exact. Let T be a dualizing functor for A,

$$T : \mathcal{C}_{\mathfrak{m}_A}^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$

It is exact, and for every $M \in \mathrm{Ob}\, \mathcal{C}_{\mathfrak{m}_A}$ the natural morphism

$$M \longrightarrow (T \circ T)(M)$$

is an isomorphism. Consequently, the contravariant composite $T \circ R$ is a dualizing functor for B. If I represents T, the classical formula

$$\mathrm{Hom}_A(M, I) \simeq \mathrm{Hom}_B(M, \mathrm{Hom}_A(B, I)),$$

valid for every B-module M, shows that $\mathrm{Hom}_A(B, I)$ is a dualizing module for B. ☐

Corollary 4.4. *Let A be a noetherian local ring, let $\mathfrak{a}$ be an ideal of A, and put $B = A/\mathfrak{a}$. If I is a dualizing module for A, then the annihilator of $\mathfrak{a}$ in I is a dualizing module for B.*

Lemma 4.5. *Let A be a noetherian local ring and let I be a locally artinian A-module. There is a canonical isomorphism*

$$I \xrightarrow{\sim} \widehat{I} = I \otimes_A \widehat{A}.$$

Proof. Let I_n be the annihilator of $\mathfrak{m}^n$ in I, where $\mathfrak{m}$ is the maximal ideal of A. To say that I is locally artinian means that I is the direct limit of the I_n and that each I_n has finite length. Since tensor product commutes with direct limits, we are reduced to the case in which I is artinian. In that case some power $\mathfrak{m}^k$ of the maximal ideal annihilates I. Thus, for $p \geq k$,

$$I \xrightarrow{\sim} I \otimes_A A/\mathfrak{m}^p,$$

and hence

$$I \xrightarrow{\sim} I \otimes_A \widehat{A},$$

because A is noetherian and I is finitely generated.

It follows that restriction of scalars from $\widehat{A}$ to A and extension of scalars induce mutually quasi-inverse *equivalences* between the category of locally artinian $\widehat{A}$-modules and the category of locally artinian A-modules. ☐ Original French p. 53

Proposition 4.6. *Let A be a noetherian local ring, let $\widehat{A}$ be its completion, let I be a dualizing module for A (respectively, for $\widehat{A}$), and let J be the completed module*[1] *of I (respectively, the A-module obtained by restriction of scalars). Then J is a dualizing module for $\widehat{A}$ (respectively, for A). Moreover, the underlying abelian groups of I and J are isomorphic.* Original French p. 53

Proof. One merely observes that the equivalence between the category of locally artinian A-modules and the category of locally artinian $\widehat{A}$-modules induces an isomorphism between the bifunctors $\mathrm{Hom}_A(-,-)$ and $\mathrm{Hom}_{\widehat{A}}(-,-)$, and that the characterization of a dualizing functor or a dualizing module involves only these bifunctors. □

Theorem 4.7. *Let A be a noetherian local ring.*

a) *A dualizing module I always exists.*

b) *Any two dualizing modules are isomorphic (by a noncanonical isomorphism).*

c) *An A-module I is dualizing if and only if it is an injective envelope of the residue field k of A.*

Remark 4.8. *Proposition 4.6 reduces the question to the case of a complete noetherian local ring. By a structure theorem of Cohen,*[2] *such a ring is a quotient of a regular ring. Proposition 4.3 then permits us to assume that A is regular. As we shall see later, this observation gives an explicit calculation of the dualizing module.*[3] *We shall nevertheless prove Theorem 4.7 by other means.*

Recollections

Before proving the theorem, we recall several facts about injective envelopes; see Gabriel, *Des Catégories Abéliennes*, thesis, Paris, 1961, Chapter II, §5. Original French p. 54

Let $\mathcal{C}$ be an abelian category in which direct limits exist and are exact[4] (for example, the category of modules). Every object M embeds in an injective object, and an *injective envelope* of M means an injective object containing M that is minimal among such objects. The following properties hold:

i) Every object M has an injective envelope I.

ii) If I and J are two injective envelopes of M, there is an isomorphism $I \simeq J$ (generally not unique) that induces the identity on M.

iii) The object I is an *essential extension* of M: if $P \subset I$ and $P \cap M = \{0\}$, then $P = \{0\}$. Conversely, if I is injective and is an essential extension of M, then I is an injective envelope of M.

---

[1] Here "the completed module" means $\widehat{I} = I \otimes_A \widehat{A}$ (cf. Lemma 4.5): this is I endowed with its canonical $\widehat{A}$-module structure, not its $\mathfrak{m}$-adic completion. For example, if p is prime and $A = \widehat{A} = \mathbb{Z}_p$ is the ring of p-adic integers, then the injective envelope of $k = \mathbb{Z}/p\mathbb{Z}$ is the discrete $\mathbb{Z}_p$-module $\mathbb{Q}_p/\mathbb{Z}_p$, whose p-adic completion is zero.

[2] See I. S. Cohen, "On the structure and ideal theory of complete local rings," *Trans. Amer. Math. Soc.* **59** (1946), 54–106.

[3] This was the method followed by Grothendieck in 1957. The method by injective envelopes used below appears to be due to K. Morita, "Duality for modules and its applications to the theory of rings with minimum conditions," *Sc. Rep. Tokyo Kyoiku Daigaku* **6** (1958/59), 83–142. Morita's work is independent of Grothendieck's, considerably predates this seminar, and is not restricted to commutative base rings.

[4] More precisely, small filtered direct limits are assumed exact; one should also assume the existence of a generator. See [Tôhoku]. For the category of modules, which suffices here, one may instead consult Chapter 10 of Bourbaki's *Algèbre*.

Granting these facts, it plainly suffices to prove part *c)* of Theorem 4.7.

Proof. Suppose first that I is a dualizing module for A. Then I is injective and $\mathrm{Hom}_A(k, I) \simeq k$. Composing this isomorphism with the inclusion

$$\mathrm{Hom}_A(k, I) \hookrightarrow \mathrm{Hom}_A(A, I) \simeq I$$

gives an inclusion

$$k \hookrightarrow I.$$

We show that I is an injective envelope of k. Let J be an injective module such that

$$k \subset J \subset I.$$

Since J is injective, there is an injective A-submodule J' of I such that $I = J \oplus J'$. We claim that $\mathrm{Hom}_A(k, J') = 0$. Indeed,

$$\mathrm{Hom}_A(k, I) \simeq \mathrm{Hom}_A(k, J) \oplus \mathrm{Hom}_A(k, J').$$

The first summand is a nonzero vector subspace of $\mathrm{Hom}_A(k, I) \simeq k$ (it contains the inclusion $k \subset J$). Consequently $\mathrm{Hom}_A(k, J) \simeq k$ and $\mathrm{Hom}_A(k, J') = 0$. Original French p. 55

Induction on length now gives $\mathrm{Hom}_A(M, J') = 0$ for every finite-length A-module M. The module I is the direct limit of the modules $\mathrm{Hom}_A(A/\mathfrak{m}^n, I)$ (cf. Proposition 1.3), which have finite length by hypothesis. Hence the projection $I \to J'$ vanishes, and therefore $J' = 0$.

Conversely, let I be an injective envelope of k. By Propositions 2.1 and 3.1*(ii)*, to prove that I is dualizing it suffices to show that $V = \mathrm{Hom}_A(k, I)$ is isomorphic to k. There are inclusions

$$k \subset V \subset I.$$

As a k-vector space, V decomposes as $k \oplus V'$, where V' is a vector subspace of I with $V' \cap k = 0$. Since I is an essential extension of k, we have $V' = 0$, and thus $V = k$. □

Corollary 4.9. *Let A be a noetherian local ring. Every dualizing module for A is locally artinian.*

Proof. Let I be a dualizing module; it is an injective envelope of k. Using the notation and result of Corollary 2.2, we have

$$k \subset H^0_{\mathfrak{m}}(I) \subset I,$$

and $H^0_{\mathfrak{m}}(I)$ is injective. It follows that $I = H^0_{\mathfrak{m}}(I)$, and hence that I is locally artinian.[5] □

5. Consequences of the theory of dualizing modules

The functor Original French p. 56

$$T = \mathrm{Hom}_A(-, I) \colon \mathcal{C}_{\mathfrak{m}} \longrightarrow \mathcal{C}_{\mathfrak{m}}$$

is an anti-equivalence. Indeed, $T \circ T$ is isomorphic to the identity functor, and the argument is formal from that point on.

One deduces the usual properties of the notion of orthogonality:

Let $M^* = \mathrm{Hom}_A(M, I) = T(M)$, and let $N \subset M$ be a submodule. The *orthogonal of N* is the submodule N' of M^* consisting of those elements of M^* that vanish on N. In this way one obtains a bijection between the set of submodules of M and the set of submodules of M^*, reversing the order relation.

In particular:

[5] As already observed, one may alternatively note directly that I is the direct limit of the modules $\mathrm{Hom}_A(A/\mathfrak{m}^n, I)$.

- $\operatorname{length}_M N = \operatorname{colength}_{M^*} N'$.
- Cyclic modules, that is, those for which $M/\mathfrak{m}M$ has dimension 0 or 1, correspond under the duality to modules whose socle has length 0 or 1.
- If A is artinian, the ideals of A correspond to the submodules of I.

And so on.

Let A be a noetherian local ring. Write $\mathrm{D}A$ for the category of A-modules M such that, for every $n \in \mathbb{N}$,

$$M^{(n)} = M/\mathfrak{m}^{n+1}M$$

has finite length and such that

$$M = \varprojlim_n M^{(n)},$$

and let $\widehat{A}$ be the completion of A. Restriction of scalars and completion are quasi-inverse equivalences between $\mathrm{D}A$ and $\mathrm{D}\widehat{A}$; up to isomorphism, they are compatible with taking the underlying abelian groups of the modules in question. Write $\mathrm{C}A$ for the category of locally artinian A-modules with finite-dimensional socle.

Proposition 5.1. *Let A be a noetherian local ring, and let I be a dualizing module for A. The functors* Original French p. 57

$$\operatorname{Hom}_A(-, I) \colon (\mathrm{C}A)^{\mathrm{op}} \longrightarrow \mathrm{D}A$$

and

$$\operatorname{Hom}_{\widehat{A}}(-, I) \colon \mathrm{D}A \longrightarrow (\mathrm{C}A)^{\mathrm{op}}$$

are equivalences of categories, quasi-inverse to one another.

Moreover, transporting these functors by the equivalences of categories between $\mathrm{D}A$ and $\mathrm{D}\widehat{A}$ on the one hand, and between $\mathrm{C}A$ and $\mathrm{C}\widehat{A}$ on the other, yields the functor $\operatorname{Hom}_{\widehat{A}}(-, I)$.

Proof. Let $X \in \operatorname{Ob}(\mathrm{C}A)$. By definition,

$$X = \varinjlim_{k \in \mathbb{N}} X_k, \qquad X_k = \operatorname{Hom}_A(A/\mathfrak{m}^{k+1}, X),$$

and therefore

$$\operatorname{Hom}_A(X, I) = \varprojlim_k \operatorname{Hom}_A(X_k, I).$$

Thus

$$Y = \operatorname{Hom}_A(X, I) = \varprojlim_k \operatorname{Hom}_A(X_k, I)$$

is a finitely generated $\widehat{A}$-module, as follows from *EGA* 0_{I}, 7.2.9. In this connection, observe that $\mathrm{D}\widehat{A}$ is also the category of finitely generated $\widehat{A}$-modules, or, equivalently, that $\mathrm{D}A$ is the category of A-modules that are complete and finitely generated over $\widehat{A}$.[6]

Now let Y be such a module, and let $f \colon Y \to I$ be a $\widehat{A}$-homomorphism. The image of f is a finitely generated submodule, hence is annihilated by $\mathfrak{m}^k$ for some k: indeed, every $x \in I$ is annihilated by some power of $\mathfrak{m}$. Consequently f factors through $Y/\mathfrak{m}^kY$, and it follows that

$$\begin{aligned} \operatorname{Hom}_{\widehat{A}}(Y, I) &= \varinjlim_k \operatorname{Hom}_{\widehat{A}}(Y^{(k)}, I), \qquad \text{where } Y^{(k)} = Y/\mathfrak{m}^{k+1}Y, \\ &= \varinjlim_k (Y^{(k)})^*. \end{aligned}$$

[6] The printed French and the corrected TeX write $Y = \varprojlim_k X_k$. But the X_k were introduced as an increasing direct system, whereas the immediately preceding display gives $\operatorname{Hom}_A(X, I) = \varprojlim_k \operatorname{Hom}_A(X_k, I)$. The target therefore restores this latter module as Y, which is also the object required by the remainder of the proof.

This object belongs to $\mathrm{Ob}(CA)$. It follows immediately that the two functors in the statement are quasi-inverse to one another. □

It follows from the preceding discussion that replacing A by $\widehat{A}$ changes neither the categories or functors under consideration nor the underlying abelian groups of the modules involved. Proposition 5.1 may therefore be restated as follows.

Original French p. 58

The restriction of $\mathrm{Hom}_{\widehat{A}}(-, I)$ to the category of finitely generated $\widehat{A}$-modules takes its values in the category of locally artinian $\widehat{A}$-modules having finite-dimensional socle. It admits a quasi-inverse, namely the restriction of $\mathrm{Hom}_{\widehat{A}}(-, I)$ in the opposite direction. On the intersection of these two categories the two functors coincide (of course) and define an autoduality of the category of finite-length $\widehat{A}$-modules.

Example 5.2 (Macaulay). *Let A be a local ring with residue field k. Let k_0 be a subfield of A such that k is finite over k_0, and put $[k : k_0] = d$. Every finite-length A-module may be regarded as a finite-dimensional k_0-vector space of dimension*

$$d \operatorname{length}(M).$$

The functor

$$T(M) = \mathrm{Hom}_{k_0}(M, k_0)$$

is exact and preserves length, hence is dualizing for A. Its associated dualizing module is

$$A' = \varinjlim_n \mathrm{Hom}_{k_0}(A/\mathfrak{m}^n, k_0),$$

the topological dual of A endowed with the $\mathfrak{m}$-adic topology.

Example 5.3. *Let A be a regular noetherian local ring of dimension n, let $\mathfrak{m}$ be its maximal ideal, and let k be its residue field. There is a regular system of parameters $(x_1, x_2, \ldots, x_n)$ that generates $\mathfrak{m}$ and is an A-regular sequence. The Koszul complex therefore gives*

$$\mathrm{Ext}^i_A(k, A) = 0 \quad \text{if } i \neq n,$$
$$\mathrm{Ext}^n_A(k, A) \simeq k.$$

Since the depth of A is n, one has $\mathrm{Ext}^i_A(M, A) = 0$ for $i < n$ whenever M is annihilated by a power of $\mathfrak{m}$. One also has $\mathrm{Ext}^i_A(M, A) = 0$ for $i > n$, because the global cohomological dimension of A equals n. Thus $\mathrm{Ext}^n_A(-, A)$ is exact, and $\mathrm{Ext}^n_A(k, A) \simeq k$. It follows that:

Original French p. 59

Proposition 5.4. *If A is a regular noetherian local ring of dimension n, the functor*

$$M \longmapsto \mathrm{Ext}^n_A(M, A)$$

is dualizing. The associated dualizing module is

$$I = \varinjlim_r \mathrm{Ext}^n_A(A/\mathfrak{m}^r, A).$$

It is isomorphic to $H^n_{\mathfrak{m}}(A)$ (Exposé II, Theorem 6).[7]

Remark 5.5. *If A satisfies the hypotheses of both preceding examples, the two resulting dualizing modules are isomorphic. Suppose, for example, that A is regular of dimension n,*

Original French p. 59

[7]For a ring A, an ideal $J \subset A$, an A-module M, and $i \in \mathbb{Z}$, set $H^i_J(M) = H^i_Y(X, \widetilde{M})$, where $X = \mathrm{Spec}(A)$ and $Y = V(J)$.

complete, and of equal characteristic. There is then a coefficient field, say K. After choosing a system of parameters $(x_1, \ldots, x_n)$ of A, one can construct an isomorphism

$$A \simeq K[[T_1, \ldots, T_n]],$$

and hence, as we shall see, an explicit isomorphism between the two dualizing modules

$$v : H^n_{\mathfrak{m}}(A) \longrightarrow A'.$$

This isomorphism has an intrinsic interpretation in terms of the module

$$\Omega^n = \Omega^n(A/K)$$

of completed relative differentials of top degree. Indeed, Ω^n has a basis consisting of

$$dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n.$$

There is therefore an isomorphism

$$u : H^n_{\mathfrak{m}}(\Omega^n) \longrightarrow H^n_{\mathfrak{m}}(A).$$

A remarkable fact is that the composite

$$vu = w : H^n_{\mathfrak{m}}(\Omega^n) \longrightarrow A'$$

is independent of the chosen system of parameters and commutes with change of the base field.

To construct v, compute $H^n_{\mathfrak{m}}(A)$ from the Koszul complex associated with the x_i. One obtains Original French p. 60

$$H^n_{\mathfrak{m}}(A) = \varinjlim_r A/(x_1^r, \ldots, x_n^r).$$

The transition morphisms are as follows. Put

$$I_r = A/(x_1^r, \ldots, x_n^r),$$

and let $e^r_{a_1,\ldots,a_n}$ be the image of $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ in I_r. The elements $e^r_{a_1,\ldots,a_n}$, for $0 \leq a_i < r$, form a basis of I_r.

For an integer s, the transition morphism

$$t_{r,r+s} : I_r \longrightarrow I_{r+s}$$

is multiplication by $x_1^s x_2^s \cdots x_n^s$; consequently,

$$t_{r,r+s}(e^r_{a_1,\ldots,a_n}) = e^{r+s}_{a_1+s,\ldots,a_n+s}.$$ [8]

Giving an A-homomorphism $w : M \to A'$ is equivalent to giving a K-linear form $w' : M \to K$ that is continuous on every finitely generated submodule. For $M = H^n_{\mathfrak{m}}(\Omega^n)$, defining w is therefore equivalent to defining a linear form

$$\rho : H^n_{\mathfrak{m}}(\Omega^n) \longrightarrow K,$$

called the residue form.[9] *To construct ρ, it is enough to define compatible forms $\rho_r : I_r \to K$ by*

$$\rho_r(e^r_{a_1,\ldots,a_n}) = \begin{cases} 1, & \text{if } a_i = r-1 \text{ for } 1 \leq i \leq n, \\ 0, & \text{otherwise.} \end{cases}$$

[8] The printed French and the corrected TeX write $u_{r,r+s}$ in this display, immediately after defining the transition map as $t_{r,r+s}$. The target emends the isolated u to t; the domain, codomain, and multiplication rule make the reference unambiguous.

[9] For a fuller treatment of residues, see R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics 20, Springer, 1966.

Exposé V

Local Duality and Structure of the $H^i(M)$

1. Complexes of homomorphisms

1.1

Let $F^\bullet$ and $G^\bullet$ be two graded modules. Write Original French p. 61

$$\mathrm{Hom}^\bullet(F^\bullet, G^\bullet) \tag{1}$$

for the graded module of homomorphisms of graded modules from $F^\bullet$ to $G^\bullet$. Thus

$$\mathrm{Hom}^s(F^\bullet, G^\bullet) = \prod_k \mathrm{Hom}(F^k, G^{k+s}). \tag{2}$$

Suppose that $F^\bullet$ (respectively $G^\bullet$) is a complex, and let d_1 (respectively d_2) be its differential. For $h \in \mathrm{Hom}^s(F^\bullet, G^\bullet)$, set[1]

$$d(h) = h \circ d_1 + (-1)^{s+1} d_2 \circ h. \tag{3}$$

One checks immediately that $d \circ d = 0$; hence $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$, equipped with d, is a complex. The cohomology group of this complex is denoted by

$$\underline{\mathrm{H}}^\bullet(F^\bullet, G^\bullet). \tag{4}$$

If $G^\bullet$ is injective in every degree, then

$$F^\bullet \longmapsto \underline{\mathrm{H}}^\bullet(F^\bullet, G^\bullet)$$

is an exact ∂-functor. Likewise, for arbitrary $F^\bullet$,

$$G^\bullet \longmapsto \underline{\mathrm{H}}^\bullet(F^\bullet, G^\bullet)$$

is an exact δ-functor on the category of complexes $G^\bullet$ that are injective in every degree.

Remark 1.2. The cycles of $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ that commute or anticommute with the differentials, according to the degree. The boundaries of $\mathrm{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ that are homotopic to zero.

[1] *Editor's note.* The original sign convention was different, but it is incompatible with the convention of Exposé VIII. The latter seems more reasonable because in that case the degree-zero cohomology is the set of homotopy classes of morphisms from $F^\bullet$ to $G^\bullet$. The calculations below have been modified accordingly.

Let A be a ring, let M (respectively N) be an A-module, and let $R(M)$ (respectively $R(N)$) be an injective resolution of M (respectively N). There is then a canonical isomorphism[2] Original French p. 62

$$\underline{\mathrm{H}}^s(R(M), R(N)) \simeq \operatorname{Ext}^s(M, N). \tag{1.3}$$

Indeed, let $i\colon M \to R(M)$ be the canonical augmentation, and let $h \in \operatorname{Hom}^s(R(M), R(N))$. Denote by t_s the map

$$h \longmapsto h_0 \circ i$$

from $\operatorname{Hom}^s(R(M), R(N))$ to $\operatorname{Hom}(M, R(N)^s)$. The family $(t_s)_{s\geq 0}$ defines a homomorphism of (ordinary) complexes[3]

$$t\colon \operatorname{Hom}^\bullet(R(M), R(N)) \longrightarrow \operatorname{Hom}^\bullet(M, R(N));$$

that is,

$$(dh)_0 \circ i = d_2 \circ h_0 \circ i.$$

One checks easily that, on passing to cohomology, t induces an isomorphism. It follows in particular that

$$\underline{\mathrm{H}}^\bullet(R(M), R(N))$$

does not "depend" on the chosen injective resolution $R(M)$ (respectively $R(N)$) of M (respectively N).

To every exact sequence of A-modules Original French p. 62

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{5}$$

one associates an exact sequence of injective resolutions

$$0 \longrightarrow R(M') \longrightarrow R(M) \longrightarrow R(M'') \longrightarrow 0. \tag{6}$$

One checks that the isomorphism *(1.3)* commutes with the homomorphisms

$$\underline{\mathrm{H}}^s(R(M'), R(N)) \longrightarrow \underline{\mathrm{H}}^{s+1}(R(M''), R(N)), \tag{8}$$

$$\operatorname{Ext}^s(R(M'), R(N)) \longrightarrow \operatorname{Ext}^{s+1}(R(M''), R(N)), \tag{9}$$

deduced from *(6)* and *(5)*.[4]

Let P be a third A-module, and let $R(P)$ be an injective resolution of P. Then composition of graded morphisms gives a pairing Original French p. 63

$$\operatorname{Hom}^i(R(N), R(M)) \times \operatorname{Hom}^j(R(M), R(P)) \longrightarrow \operatorname{Hom}^{i+j}(R(N), R(P)), \tag{10}$$

which defines a pairing

$$\underline{\mathrm{H}}^i(R(N), R(M)) \times \underline{\mathrm{H}}^j(R(M), R(P)) \longrightarrow \underline{\mathrm{H}}^{i+j}(R(N), R(P)), \tag{11}$$

and hence a homomorphism of functors in M:

$$\underline{\mathrm{H}}^i(R(N), R(M)) \longrightarrow \operatorname{Hom}\Big(\underline{\mathrm{H}}^j(R(M), R(P)), \underline{\mathrm{H}}^{i+j}(R(N), R(P))\Big). \tag{1.4}$$

We shall see that *(1.4)* is a homomorphism of δ-functors in M. The exact sequences *(5)* and *(6)* give a commutative diagram:

[2] *Editor's note.* The strange original numbering has been preserved.

[3] Here M also denotes the complex $M[0]$ consisting of M placed in degree zero.

[4] *Source note.* The corrected French TeX and the direct compiled source page both use resolution arguments in *(9)*: $R(M')$ and $R(N)$ in the source term, and $R(M'')$ and $R(N)$ in the target term. This bounded translation reproduces that literal reading; it does not silently replace those arguments by M', M'', and N.

$$\begin{array}{ccc}
\mathrm{Hom}^i(R(N),R(M')) & \longrightarrow & \mathrm{Hom}\Big(\mathrm{Hom}^j(R(M'),R(P)),\mathrm{Hom}^{i+j}(R(N),R(P))\Big) \\
\Big\downarrow & & \Big\downarrow{\scriptstyle \mathrm{Hom}(q,\mathrm{id})} \\
\mathrm{Hom}^i(R(N),R(M)) & \longrightarrow & \mathrm{Hom}\Big(\mathrm{Hom}^j(R(M),R(P)),\mathrm{Hom}^{i+j}(R(N),R(P))\Big) \\
\Big\downarrow{\scriptstyle p} & & \Big\downarrow \\
\mathrm{Hom}^i(R(N),R(M'')) & \longrightarrow & \mathrm{Hom}\Big(\mathrm{Hom}^j(R(M''),R(P)),\mathrm{Hom}^{i+j}(R(N),R(P))\Big) .
\end{array}$$

Let $h \in \mathrm{Hom}^i(R(N),R(M''))$ (respectively $g \in \mathrm{Hom}^j(R(M'),R(P))$) be a cycle, and let $h' \in \mathrm{Hom}^i(R(N),R(M))$ (respectively $g' \in \mathrm{Hom}^j(R(M),R(P))$) be such that $p(h') = h$ (respectively $q(g') = g$). To say that *(1.4)* is a homomorphism of δ-functors in M is to say that

$$g \circ dh' - dg' \circ h \tag{12}$$

is a coboundary in $\mathrm{Hom}^\bullet(R(N),R(P))$.

Now

$$dh' = h' \circ d_1 + (-1)^{i+1} d_2 \circ h',$$
$$dg' = g' \circ d_2 + (-1)^{j+1} d_3 \circ g',$$

with the obvious notation. Thus *(12)* becomes

$$g \circ h' \circ d_1 + (-1)^{i+1} g \circ d_2 \circ h' - g' \circ d_2 \circ h - (-1)^{j+1} d_3 \circ g' \circ h.$$

On the other hand, since h and g are cycles, Original French p. 64

$$g \circ d_2 = (-1)^j d_3 \circ g,$$
$$d_2 \circ h = (-1)^i h \circ d_1.$$

Consequently *(12)* is finally written as

$$d\Big(g \circ h' + (-1)^{i+1} g' \circ h\Big),$$

which completes the proof.

Thus *(1.3)* and *(1.4)* give a homomorphism of δ-functors in M:

$$\mathrm{Ext}^i(N,M) \longrightarrow \mathrm{Hom}\Big(\mathrm{Ext}^j(M,P),\mathrm{Ext}^{i+j}(N,P)\Big). \tag{1.5}$$

2. The local duality theorem for a regular local ring

Let A be a regular local ring of dimension r, let $\mathfrak{m}$ be the maximal ideal of A, and let M be a finitely generated A-module. Set Original French p. 64

$$H^i(M) = H^i_{\mathfrak{m}}(M) \qquad (\text{so } H^i(M) = \varinjlim_k \mathrm{Ext}^i(A/\mathfrak{m}^k, M)).$$

We saw in *IV, 5.4* that $I = H^r(A)$ is a dualizing module for A; denote the associated dualizing functor by D. In *(1.5)*, put $N = A/\mathfrak{m}^k$ and $P = A$. We then obtain a homomorphism of δ-functors in M,

$$\varphi_k \colon \mathrm{Ext}^i(A/\mathfrak{m}^k, M) \longrightarrow \mathrm{Hom}\Big(\mathrm{Ext}^{r-i}(M,A),\mathrm{Ext}^r(A/\mathfrak{m}^k,A)\Big). \tag{13}$$

Passing to the direct limit over k gives a homomorphism of δ-functors

$$\varphi\colon H^i(M) \longrightarrow D\Big(\mathrm{Ext}^{r-i}(M,A)\Big)\,. \tag{14}$$

Theorem 2.1 (Local duality theorem). The preceding functorial homomorphism φ is an isomorphism.

Proof. If $i > r$, the right-hand side of *(14)* is trivially zero, and the left-hand side is zero because Original French p. 65

$$H^i(M) = \varinjlim_k \mathrm{Ext}^i(A/\mathfrak{m}^k, M),$$

and each $\mathrm{Ext}^i(A/\mathfrak{m}^k, M)$ is zero (syzygy theorem).

If $i = r$, the preceding discussion shows that the two functors in M, $H^r(M)$ and $D(\mathrm{Hom}(M,A))$, are right exact. Since A is noetherian and M is finitely generated, it is enough to verify the isomorphism for $M = A$, where it is immediate.

To show that φ is a functorial isomorphism, it now suffices to proceed by descending induction on i and observe that every finitely generated module admits a finite presentation, while for $i < r$ both sides of *(14)* vanish when M is finitely generated and free. This is clear for the right-hand side. Since H^i commutes with finite direct sums, for the left-hand side it is enough to show that $H^i(A) = 0$ for $i < r$. This follows from *III, 3.4*, since $\mathrm{prof}(A) = r$. ☐

3. Application to the structure of the $H^i(M)$

Theorem 3.1. *Let A be a Noetherian local ring, let D be a dualizing functor for A, and let $M \neq 0$ be a finitely generated A-module of dimension n. Then:* Original French p. 65

(i) $H^i(M) = 0$ *if* $i < 0$ *or if* $i > n$.

(ii) $D(\widehat{H^i(M)})$ *is a finitely generated* $\widehat{A}$*-module of dimension at most* i.

(iii) $H^n(M) \neq 0$*; moreover, if A is complete, then* $D(H^n(M))$ *has dimension n and*

$$\mathrm{Ass}(D(H^n(M))) = \{\, \mathfrak{p} \in \mathrm{Ass}(M) \mid \dim(A/\mathfrak{p}) = n \,\}.$$

Proof. Let I be the dualizing module associated with D. We know that $\widehat{I}$ is a dualizing module for $\widehat{A}$. On the other hand,

$$\begin{aligned} \widehat{H^i(M)} &= H^i(\widehat{M}),\\ D(\widehat{H^i(M)}) &= \mathrm{Hom}\big(H^i(\widehat{M}), \widehat{I}\big),\\ \dim \widehat{M} &= \dim M. \end{aligned}$$

Consequently, we may suppose that A is complete. By Cohen's theorem, every complete local ring is a quotient of a regular local ring. To reduce to that case, we need the following lemma. Original French p. 66

Lemma 3.2. *Let X (respectively Y) be a ringed space, let X' (respectively Y') be a closed subspace of X (respectively Y), and let $f\colon X \to Y$ be a morphism of ringed spaces such that $f^{-1}(Y') = X'$. Let F be an $\mathcal{O}_X$-module. Denote by A (respectively B) the ring $\Gamma(\mathcal{O}_X)$ (respectively $\Gamma(\mathcal{O}_Y)$), and by $\overline{f}\colon B \to A$ the ring homomorphism corresponding to f. There is a spectral sequence of B-modules with initial term* Original French p. 66

$$E_2^{p,q} = H^p_{Y'}(Y, R^q f_*(F)), \tag{15}$$

converging to the B-module $H^\bullet_{X'}(X,F)_{[\overline{f}]}$.

Proof. Let $\mathcal{O}_{Y,Y'}$ be the sheaf $\mathcal{O}_Y|_{Y'}$, extended by zero outside Y' (see Exposé I). There is an isomorphism of B-modules

$$\operatorname{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)) \simeq \operatorname{Hom}(f^*(\mathcal{O}_{Y,Y'}), F)_{[\bar{f}]}. \tag{16}$$

Now

$$f^*(\mathcal{O}_{Y,Y'}) = \mathcal{O}_{X,X'}. \tag{17}$$

Moreover, if G is an injective $\mathcal{O}_X$-module, then $f_*(G)$ is an injective $\mathcal{O}_Y$-module, at least when f is flat; one can easily reduce to this case by replacing $\mathcal{O}_X$, and so forth, by the constant sheaves of rings $\mathbb{Z}$. Hence the spectral sequence of the composite functor

$$F \longmapsto \operatorname{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)),$$

whose initial term is

$$E_2^{p,q} = \operatorname{Ext}^p(Y; \mathcal{O}_{Y,Y'}, R^q f_*(F)),$$

converges, in view of (16) and (17), to

$$\operatorname{Ext}^\bullet(X; \mathcal{O}_{X,X'}, F)_{[\bar{f}]}.$$

The lemma then follows from Exposé I, Proposition 2.3 bis. □ Original French p. 67

Now let $f\colon B \to A$ be a surjective homomorphism of local rings. Let Original French p. 67

$$f\colon \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(B)$$

be the corresponding morphism of affine schemes. Set $X = \operatorname{Spec}(A)$ (respectively $X' = \{\mathfrak{m}_A\}$) and $Y = \operatorname{Spec}(B)$ (respectively $Y' = \{\mathfrak{m}_B\}$), and let M be an A-module and $\widetilde{M}$ the corresponding $\mathcal{O}_X$-module. Since $R^q f_*(\widetilde{M}) = 0$ for $q > 0$, the spectral sequence (15) degenerates, and Lemma 3.2 gives an isomorphism of B-modules

$$H^n_{\{\mathfrak{m}_B\}}(Y, f_*(\widetilde{M})) \simeq H^n_{\{\mathfrak{m}_A\}}(X, \widetilde{M})_{[f]}, \tag{18}$$

hence an isomorphism of B-modules

$$H^n_{\mathfrak{m}_B}(M_{[f]}) \simeq H^n_{\mathfrak{m}_A}(M)_{[f]}. \tag{19}$$

On the other hand, if D_A (respectively D_B) is the dualizing functor for A (respectively B), then

$$D_A(M)_{[f]} \simeq D_B(M_{[f]}). \tag{20}$$

Finally, since there is an isomorphism of rings

$$B/\operatorname{Ann} M_{[f]} \simeq A/\operatorname{Ann} M, \tag{21}$$

it follows that the change of base rings under consideration changes nothing. We may therefore suppose that A is regular of dimension r.

By (2.1), we have

$$D(H^i(M)) = \operatorname{Ext}^{r-i}(M, A). \tag{22}$$

Original French p. 67

We shall prove the equivalence of the following properties:

Original French p. 68

(a) $\dim \operatorname{Ext}^j(M, A) \leq r - j$;

(b) for every $\mathfrak{p} \in X = \operatorname{Spec}(A)$ such that $\dim A_{\mathfrak{p}} < j$, one has $\operatorname{Ext}^j(M, A)_{\mathfrak{p}} = 0$;

(c) $\operatorname{codim}(\operatorname{Supp}(\operatorname{Ext}^j(M, A)), X) \geq j$.

To prove (a) $\Longrightarrow$ (b), let $\mathfrak{p} \in X$ with $\dim A_{\mathfrak{p}} < j$. Then $\dim A/\mathfrak{p} > r - j$, and hence, by (a),

$$\operatorname{Ann}(\operatorname{Ext}^j(M, A)) \not\subset \mathfrak{p},$$

which implies that $\operatorname{Ext}^j(M, A)_{\mathfrak{p}} = 0$. Let $\mathfrak{p} \in \operatorname{Supp}(\operatorname{Ext}^j(M, A))$. Then $\operatorname{Ext}^j(M, A)_{\mathfrak{p}} \neq 0$, so by (b) we have $\dim A_{\mathfrak{p}} \geq j$. Therefore

$$\operatorname{codim}(\operatorname{Supp}(\operatorname{Ext}^j(M, A)), X) = \inf\{\dim A_{\mathfrak{p}} \mid \mathfrak{p} \in \operatorname{Supp}(\operatorname{Ext}^j(M, A))\} \geq j,$$

that is, (b) $\Longrightarrow$ (c). Finally, (c) implies (a) trivially.
Let us now prove the theorem.

Original French p. 68

(i) Let $x = (x_1, \ldots, x_r)$ be a system of parameters for A such that $x_i \in \operatorname{Ann} M$ for $i = 1, \ldots, r - n$. Let $K^\bullet((x^k), M)$ be the Koszul complex. It is easy to see that the map

$$K^i((x^k), M) \longrightarrow K^i((x^{k'}), M)$$

for $k < k'$ is zero if $i > n$. It follows that

$$H^i(M) = \varinjlim H^i((x^k), M) = 0$$

if $i > n$. On the other hand, it is trivial that $H^i(M) = 0$ if $i < 0$; hence (i) is proved.

(ii) Since A is regular, $\dim A_{\mathfrak{p}} < j$ implies that the global homological dimension of $A_{\mathfrak{p}}$ is strictly less than j, and hence

$$\operatorname{Ext}^j(M, A)_{\mathfrak{p}} = \operatorname{Ext}^j_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, A_{\mathfrak{p}}) = 0.$$

Thus we have proved (b), and consequently (a). Assertion (ii) then follows from (22) and (a).

(iii) There exists a $\mathfrak{p} \in \operatorname{Supp}(M)$ such that $\dim A_{\mathfrak{p}} = r - n$ and such that $\operatorname{Supp}(M_{\mathfrak{p}}) = \{\mathfrak{m}A_{\mathfrak{p}}\}$. Since $A_{\mathfrak{p}}$ is regular if A is, we have $\operatorname{depth} A_{\mathfrak{p}} = r - n$, and therefore

$$\operatorname{Ext}^{r-n}_A(M, A)_{\mathfrak{p}} = \operatorname{Ext}^{r-n}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, A_{\mathfrak{p}}) \neq 0. \tag{23}$$

Taking (22) into account, this implies, on the one hand,

Original French p. 69

$$H^n(M) \neq 0,$$

and, on the other hand,

$$\dim D(H^n(M)) \geq n,$$

hence, by (ii),

$$\dim D(H^n(M)) = n.$$

Now let $Y = \operatorname{Supp}(M)$. By (i), we know that $D(H^n(M')) = \operatorname{Ext}^{r-n}(M', A)$ is a left exact functor in M' on the category $(\mathcal{C}_Y)^\circ$. Hence there exists an A-module H and an isomorphism of functors in M':

$$\operatorname{Ext}^{r-n}(M', A) = \operatorname{Hom}(M', H).$$

Let Y_i, $i = 1, \ldots, k$, be the irreducible components of Y of maximum dimension. We shall see that the assertion $\operatorname{Ext}^{r-n}(M', A) \neq 0$ is equivalent to the assertion that there exists an i such that $\operatorname{Supp} M' \supset Y_i$. Indeed, if $\operatorname{Supp} M' \supset Y_i$, then $\dim(M') = n$, and hence $\operatorname{Ext}^{r-n}(M', A) \neq 0$.

If $\operatorname{Supp} M' \not\supset Y_i$ for every $i = 1, \dots, k$, then $\dim M' < n$, and

$$D\big(H^n(M')\big) = \operatorname{Ext}^{r-n}(M', A) = 0.$$

Since $\operatorname{Ass}\big(\operatorname{Ext}^{r-n}(M, A)\big) = \operatorname{Supp} M \cap \operatorname{Ass}(H)$, we see that the last assertion of (iii) follows from the following lemma:

Lemma 3.3. *Let $X = \operatorname{Spec}(A)$, let Y be a closed subset of X, let*

$$T\colon (\mathcal{C}_Y)^\circ \longrightarrow \mathbf{Ab}$$

be a left exact functor, and let Y_i, $i = 1, \dots, k$, be a family of irreducible components of Y such that the assertion $T(M) = 0$ is equivalent to the assertion that, for every i, $\operatorname{Supp} M \not\supset Y_i$. Then T is representable by a module H such that

$$\operatorname{Ass}(H) = \bigcup_{i=1}^{k} \{y_i\},$$

where y_i is the generic point of Y_i, for $i = 1, \dots, k$.

Proof. For $y \in Y$, construct an A-module $M(y)$ such that Original French p. 70

$$\operatorname{Supp}(M(y)) = \overline{\{y\}}.$$

Suppose that $y \neq y_i$ for every $i = 1, \dots, k$. Then $Y_i \not\subset \operatorname{Supp}\big(M(y)\big)$ for every $i = 1, \dots, k$, and hence $T\big(M(y)\big) = 0$. It follows that

$$\operatorname{Ass}(T(M(y))) = \operatorname{Supp}(M(y)) \cap \operatorname{Ass}(H) = \varnothing,$$

and therefore $y \notin \operatorname{Ass}(H)$. If $y = y_i$, then $Y_i \subset \operatorname{Supp}\big(M(y)\big)$, so $T\big(M(y)\big) \neq 0$, and hence

$$\operatorname{Ass}(T(M(y))) = \operatorname{Supp}(M(y)) \cap \operatorname{Ass}(H) \neq \varnothing.$$

By the first part of the proof, this implies $y \in \operatorname{Ass}(H)$, which proves the lemma. ☐

Example 3.4. *Let A be a noetherian ring, let $X = \operatorname{Spec}(A)$, and let Y be a closed subset of X such that $X - Y$ is affine. Then, for every irreducible component Y_α of Y, one has*

$$\operatorname{codim}(Y_\alpha, X) \leq 1.$$

Proof. Consider X as a prescheme over X. Let $y_\alpha \in Y_\alpha$ be a generic point, and consider the morphism

$$\operatorname{Spec}(\mathcal{O}_{X,y_\alpha}) \longrightarrow X.$$

The affine scheme obtained by base-changing X to $\operatorname{Spec}(\mathcal{O}_{X,y_\alpha})$ is canonically isomorphic to $\operatorname{Spec}(\mathcal{O}_{X,y_\alpha})$.

By (EGA I 3.2.7), if y_0 is the unique closed point of

$$Y_0 = \operatorname{Spec}(\mathcal{O}_{X,y_\alpha}),$$

then $Y_0 - y_0$ is affine. By (EGA III 1.3.1), one obtains

$$\mathrm{H}^i(Y_0 - y_0, \mathcal{O}_{Y_0}) = 0 \qquad \text{if } i > 0.$$

Hence, by (I 2.9),

$$\mathrm{H}^{i-1}(\mathcal{O}_{X,y_\alpha}) = \mathrm{H}^i_{\{y_0\}}(Y_0, \mathcal{O}_{Y_0}) = 0 \qquad \text{if } i \geq 2.$$ French source PDF p. 62 / running 54

By 3.1(iii), it follows that

$$\dim \mathcal{O}_{X,y_\alpha} \leq 1,$$

and therefore

$$\operatorname{codim}(Y_\alpha, X) = \inf_{y \in Y_\alpha} \dim \mathcal{O}_{X,y} \leq 1.$$

□

Let A be a noetherian local ring, let $\mathfrak{m}$ be its maximal ideal, and let M be a finitely generated A-module. Suppose that A is a quotient of a regular local ring. Set $X = \operatorname{Spec}(A)$ and, for every $x \in X$, set $\mathfrak{m}_x = \mathfrak{m}A_x$. Original French p. 71

Proposition. The following two conditions are equivalent:

(a) $\mathrm{H}^i(M)$ has finite length;

(b) for every $x \in X - \{\mathfrak{m}\}$,

$$\mathrm{H}_{\mathfrak{m}_x}^{i-\dim \overline{\{x\}}}(M_x) = 0.$$

Proof. In view of (3.2), we may suppose that A is regular. By (2.1), we have

$$\mathrm{H}^i(M) = \mathrm{D}\big(\operatorname{Ext}^{r-i}(M, A)\big),$$

where $r = \dim A$. By (IV 4.7), condition (a) is equivalent[5] to

$$\operatorname{Ext}^{r-i}(M, A) \text{ has finite length.} \tag{24}$$

Condition (24) is equivalent to

$$\text{for every } x \in X - \{\mathfrak{m}\}, \qquad \operatorname{Ext}_A^{r-i}(M, A)_x = 0. \tag{25}$$

On the other hand, A_x is regular of dimension $r - \dim \overline{\{x\}}$. Hence, by (2.1),

$$\begin{aligned} \mathrm{H}_{\mathfrak{m}_x}^{i-\dim \overline{\{x\}}}(M_x) &= \mathrm{D}\Big(\operatorname{Ext}_{A_x}^{(r-\dim \overline{\{x\}})-(i-\dim \overline{\{x\}})}(M_x, A_x)\Big) \\ &= \mathrm{D}\Big(\operatorname{Ext}_{A_x}^{r-i}(M_x, A_x)\Big). \end{aligned} \tag{26}$$

Since M is finitely generated, one has

$$\operatorname{Ext}_A^{r-i}(M, A)_x = \operatorname{Ext}_{A_x}^{r-i}(M_x, A_x),$$

which proves the proposition.

Corollary. For $\mathrm{H}^i(M)$ to have finite length for $i \leq n$, it is necessary and sufficient that

$$\operatorname{depth}(M_x) > n - \dim \overline{\{x\}}$$

for every $x \in X - \{\mathfrak{m}\}$.

Proof. This follows from (3.5) and (III 3.1).

[5]Indeed, one must show that, if E is a finitely generated A-module and $\mathrm{D}(E)$ has finite length, then E has finite length. Let K (respectively Q) be the kernel (respectively cokernel) of the canonical morphism $\epsilon\colon E \to \mathrm{DD}(E)$. The canonical morphism $\gamma\colon \mathrm{D}(E) \to \mathrm{DDD}(E)$ satisfies $\mathrm{D}(\epsilon) \circ \gamma = \mathrm{id}_{\mathrm{D}(E)}$. Since $\mathrm{D}(E)$ has finite length, γ is an isomorphism, and hence so is $\mathrm{D}(\epsilon)$. Since D is exact, it follows that $\mathrm{D}(K)$ and $\mathrm{D}(Q)$ are zero. It is enough to prove that an A-module whose dual is zero is itself zero, for then $E = \mathrm{DD}(E)$ has finite length, just as $\mathrm{D}(E)$ does. Indeed, let M_0 be a finitely generated submodule of such a module M. Since D is exact, $\mathrm{D}(M_0)$ is a quotient of $\mathrm{D}(M)$ and is therefore zero. Again by exactness, $\mathrm{D}(M_0/\mathfrak{m}_A M_0) = 0$; by biduality, the finite-length module $M_0/\mathfrak{m}_A M_0$ is therefore zero. Nakayama's lemma then gives $M_0 = 0$, and finally $M = 0$.

Exposé VI

The Functors $\mathrm{Ext}^{\bullet}_Z(X; F, G)$ and $\mathcal{E}xt^{\bullet}_Z(F, G)$

1. Generalities

1.1

Let $(X, \mathcal{O}_X)$ be a ringed space and let Z be a locally closed subset of X. Let F and G be $\mathcal{O}_X$-modules. We shall denote by $\mathrm{Ext}^i_Z(X; F, G)$ (respectively $\mathcal{E}xt^i_Z(F, G)$) the ith derived functor of

$$G \longmapsto \Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G)) \quad \text{(respectively } G \longmapsto \underline{\Gamma}_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G))\text{).}$$

Lemma 1.2. *The sheaf $\mathcal{E}xt^i_Z(F, G)$ is canonically isomorphic to the sheaf associated with the presheaf*

$$U \longmapsto \mathrm{Ext}^i_{Z \cap U}(U; F|_U, G|_U).$$

This follows from ([Tôhoku], 3.7.2) and from the fact that

$$\Gamma(U; \underline{\Gamma}_Z(\mathcal{H}om_{\mathcal{O}_X}(F, G))) \simeq \Gamma_{Z\cap U}\Big(\mathcal{H}om_{\mathcal{O}_X|_U}(F|_U, G|_U)\Big)$$

canonically.

Theorem 1.3 (Excision theorem). *Let V be an open subset of X containing Z. Then there is an isomorphism of cohomological functors*

$$\mathrm{Ext}^{\bullet}_X(X; F, G) \simeq \mathrm{Ext}^{\bullet}_V(V; F|_V, G|_V). \tag{1.3.1}$$

Indeed, if $G^{\bullet}$ is an injective resolution of G, then $G^{\bullet}|_V$ is an injective resolution of $G|_V$. The theorem follows immediately.

1.4

Let $\mathcal{O}_{X,Z}$ be the $\mathcal{O}_X$-module defined by the following conditions ([Godement], 2.9.2):

$$\mathcal{O}_{X,Z}|_{X-Z} = 0 \qquad \text{and} \qquad \mathcal{O}_{X,Z}|_Z = \mathcal{O}_X|_Z.$$

We have seen that, for every $\mathcal{O}_X$-module H, there is a functorial isomorphism

$$\Gamma_Z(H) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, H).$$

Consequently, there are isomorphisms functorial in F and G:

$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, \mathcal{H}om_{\mathcal{O}_X}(F,G)), \tag{1.4.1}$$
$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G), \tag{1.4.2}$$
$$\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G)) \simeq \operatorname{Hom}_{\mathcal{O}_X}(F, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, G)) = \operatorname{Hom}_{\mathcal{O}_X}(F, \underline{\Gamma}_Z(G)). \tag{1.4.3}$$

In particular, it follows from (1.4.2) that there is a ∂-functorial isomorphism in F and G,

$$\theta \colon \operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G) \xrightarrow{\sim} \operatorname{Ext}^i_Z(X; F, G).$$

1.5

By definition, the functor $G \longmapsto \Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G))$ is the composite of $G \longmapsto \mathcal{H}om_{\mathcal{O}_X}(F,G)$ and Γ_Z. Since Γ_Z is left exact (I 1.9), since $\mathcal{H}om_{\mathcal{O}_X}(F,G)$ is flasque when G is injective, and since Γ_Z is exact on flasque sheaves (I 2.12), it follows from ([Tôhoku], 2.4.1) that there is a spectral functor abutting to $\operatorname{Ext}^\bullet_Z(X; F, G)$ whose initial term is

$$H^p_Z(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F,G)).$$

On the other hand, it follows from (1.4.3) that $\Gamma_Z(\mathcal{H}om_{\mathcal{O}_X}(F,G))$ is the composite of $G \longmapsto \underline{\Gamma}_Z(G)$ and $H \longmapsto \operatorname{Hom}_{\mathcal{O}_X}(F,H)$. Since $\underline{\Gamma}_Z$ takes injectives to injectives (I 1.4), it follows from ([Tôhoku], 2.4.1) that there is a spectral functor abutting to $\operatorname{Ext}^\bullet_Z(X; F, G)$ whose initial term is

$$\operatorname{Ext}^p_{\mathcal{O}_X}(X; F, \mathcal{H}^q_Z(G)).$$

Finally, it follows from (1.4.2) and the Ext spectral sequence that there is a spectral functor abutting to $\operatorname{Ext}^\bullet_Z(X; F, G)$ whose initial term is $H^p(X; \mathcal{E}xt^\bullet_Z(F,G))$. Hence:

Theorem 1.6. *There are three spectral functors abutting to* $\operatorname{Ext}^\bullet_Z(X; F, G)$ *whose initial terms are respectively*

$$H^p_Z(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F,G)), \tag{1.6.1}$$
$$H^p(X, \mathcal{E}xt^q_Z(F,G)), \tag{1.6.2}$$
$$\operatorname{Ext}^p_{\mathcal{O}_X}(X; F, \mathcal{H}^q_Z(G)). \tag{1.6.3}$$

1.7

Let Z' be a closed subset of Z, and set $Z'' = Z - Z'$. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{X,Z''} \longrightarrow \mathcal{O}_{X,Z} \longrightarrow \mathcal{O}_{X,Z'} \longrightarrow 0. \tag{1.7.1}$$

It generalizes the exact sequence of ([Godement], 2.9.3). This exact sequence splits locally; hence, for every $\mathcal{O}_X$-module F, there is another exact sequence

$$0 \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z''} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X,Z'} \longrightarrow 0. \tag{1.7.2}$$

Now let G be an $\mathcal{O}_X$-module. Applying the functor $\operatorname{Hom}_{\mathcal{O}_X}(\bullet, G)$ to (1.7.2), one obtains from (1.4.2) and the long exact sequence of Ext the following theorem.

Theorem 1.8. *Let Z be a locally closed subset of X, let Z' be a closed subset of Z, and set $Z'' = Z - Z'$. There is an exact sequence, functorial in F and G,*

$$0 \longrightarrow \operatorname{Hom}_{Z'}(F,G) \longrightarrow \operatorname{Hom}_Z(F,G) \longrightarrow \operatorname{Hom}_{Z''}(F,G) \longrightarrow \operatorname{Ext}^1_{Z'}(F,G) \longrightarrow \cdots$$
$$\cdots \longrightarrow \operatorname{Ext}^i_Z(F,G) \longrightarrow \operatorname{Ext}^i_{Z''}(F,G) \longrightarrow \operatorname{Ext}^{i+1}_{Z'}(F,G) \longrightarrow \cdots.$$

Corollary 1.9. *Let Y be a closed subset of X, and set $U = X - Y$. There is an exact sequence, functorial in F and G,*

$$0 \longrightarrow \mathrm{Hom}_Y(F,G) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(F,G) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X|U}(F|_U, G|_U) \longrightarrow \mathrm{Ext}^1_Y(F,G) \longrightarrow \cdots$$
$$\cdots \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(F,G) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X|U}(F|_U, G|_U) \longrightarrow \mathrm{Ext}^{i+1}_Y(F,G) \longrightarrow \cdots.$$

This corollary is an immediate consequence of Theorems (1.3) and (1.8).

2. Applications to Quasi-Coherent Sheaves on Preschemes

Proposition 2.1. *Let X be a locally Noetherian prescheme. For every locally closed subset Z of X, every coherent $\mathcal{O}_X$-module F, and every quasi-coherent $\mathcal{O}_X$-module G, the sheaves $\mathcal{E}xt^i_Z(F,G)$ are quasi-coherent.*

As in (1.6.3), one shows that the sheaves $\mathcal{E}xt^i_Z(F,G)$ are obtained as the abutment of a spectral sequence whose initial term is

$$\mathcal{E}xt^p_{\mathcal{O}_X}(F, \mathcal{H}^q_Z(G)).$$

By (II, Corollary 3), the $\mathcal{H}^q_Z(G)$ are quasi-coherent, and therefore so are the $\mathcal{E}xt^p_{\mathcal{O}_X}(F, \mathcal{H}^q_Z(G))$, since F is coherent. The proposition follows immediately.

2.2

Let Y now be a closed subprescheme of X, and let $\mathcal{I}$ be an ideal defining Y. Let m and n be integers such that $m \geq n \geq 0$. We denote by $i_{n,m}$ the canonical map

$$\mathcal{O}_{Y_m} = \mathcal{O}_X/\mathcal{I}^{m+1} \longrightarrow \mathcal{O}_X/\mathcal{I}^{n+1} = \mathcal{O}_{Y_n}$$

and by j_n the map $\mathcal{O}_{X,Y} \to \mathcal{O}_{Y_n}$. The $(\mathcal{O}_{Y_n}, i_{n,m})$ form a projective system, and the j_n are compatible with the $i_{n,m}$.

Applying the functor $\mathrm{Ext}^i_{\mathcal{O}_X}(F \otimes \bullet, G)$, we obtain a morphism

$$\varphi' \colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{X,Y}, G);$$

this is a morphism of cohomological functors in G. The morphism

$$\varphi \colon \varinjlim_n \mathrm{Ext}^i_{\mathcal{O}_X}(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathrm{Ext}^i_Y(X; F, G),$$

which is the composite $\theta \circ \varphi'$ (cf. (1.4)), is therefore also a morphism of cohomological functors in G.

In the same way one defines

$$\underline{\varphi} \colon \varinjlim_n \mathcal{E}xt^i_{\mathcal{O}_X}(F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \mathcal{E}xt^i_Y(X; F, G).$$

Theorem 2.3. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X defined by a coherent ideal $\mathcal{I}$, let F be a coherent $\mathcal{O}_X$-module, and let G be a quasi-coherent $\mathcal{O}_X$-module. Then:*

a) $\underline{\varphi}$ *is an isomorphism.*

b) If X is Noetherian, φ is an isomorphism.

Proof of Theorem 2.3

The proof of b) is almost word for word the same as that of (II 6 b)), using the spectral sequence (1.6.2), so we do not reproduce it.

For the proof of a), by (2.1) we may assume that X is affine with ring A, that F (respectively G) is defined by an A-module M (respectively N), and that $\mathcal{I}$ is defined by an ideal I. It is enough to prove that the homomorphism

$$\varinjlim_n \mathrm{Ext}^i_A(M/I^nM, N) \longrightarrow \mathrm{Ext}^i_Y(X, F, G) \tag{2.3.1}$$

induced by $\underline{\varphi}$ is an isomorphism.

Indeed, for $i = 0$, the two sides of (2.3.1) may be canonically identified with the submodule of $\mathrm{Hom}_A(M, N)$ consisting of the elements of $\mathrm{Hom}_A(M, N)$ annihilated by a power of I. One then sees that the homomorphism (2.3.1) is simply the identity map.

The functor

$$N \longmapsto \varinjlim_n \mathrm{Ext}^\bullet_A(M/I^nM, N)$$

is a universal ∂-functor. We shall show that the same holds for the functor $N \longmapsto \mathrm{Ext}^\bullet_Y(M, N)$. Indeed, if N is an injective module, then by (II, 9 and 11), $\mathcal{H}^q_Y(N) = 0$ for $q \neq 0$; and by (IV.2.2), $H^0_Y(N)$ is injective.

It follows that

$$\mathrm{Ext}^p_{\mathcal{O}_X}(X; M, \mathcal{H}^q_Y(N)) = 0 \qquad \text{if } p + q \neq 0.$$

Thus, by (1.6.3), $\mathrm{Ext}^i_Y(M, N) = 0$ for $i \neq 0$ when N is injective. This completes the proof.

Bibliography

The same references as those listed at the end of Exposé I, cited respectively as [Tôhoku] and [Godement].

Exposé VII

Vanishing Criteria and Coherence Conditions for the Sheaves $\mathcal{E}xt^i_Y(F,G)$

1. Study for $i < n$

We first prove a lemma.

Lemma 1.1. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X, and let G be a quasi-coherent $\mathcal{O}_X$-module. Suppose that, for every coherent $\mathcal{O}_X$-module F whose support is contained in Y, one has*

$$\mathcal{E}xt^{n-1}(F,G) = 0.$$

Then, for every coherent $\mathcal{O}_X$-module F and every closed subset Z of X such that $Y \supset$ $\operatorname{Supp} F \cap Z$, one has an isomorphism

$$\mathcal{E}xt^n_Z(F,G) \cong \mathcal{H}om(F, \mathcal{H}^n_Y(G)).$$

Proof of Lemma 1.1

We first observe that

$$\mathcal{E}xt^i_Z(F,G) = \mathcal{E}xt^i_{Z\cap \operatorname{Supp} F}(F,G).$$

(This is immediate; cf. Exposé VI.) We first prove the assertion for $Z = X$, so that $\operatorname{Supp} F \subset Y$. On the category of coherent $\mathcal{O}_X$-modules whose support is contained in Y, the functor

$$F \longmapsto \mathcal{E}xt^n(F,G)$$

is left exact. By (IV, 1.3), it is represented by

$$I = \varinjlim_k \mathcal{E}xt^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G),$$

where $\mathcal{I}$ is a defining ideal of Y. By (II, 6), one has

$$\mathcal{H}^n_Y(G) \cong \varinjlim_k \mathcal{E}xt^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G).$$

This proves the assertion when $Z = X$. Again by (VI, 2.3),

$$\mathcal{E}xt^n_Z(F,G) \cong \varinjlim_k \mathcal{E}xt^n(F/\mathcal{J}^{k+1}F, G),$$

where $\mathcal{J}$ is a defining ideal of Z. The support of $F/\mathcal{J}^{k+1}F$ is contained in Y whenever $Z \cap \operatorname{Supp} F \subset Y$. What we have just proved therefore gives

$$\mathcal{E}xt^n_Z(F, G) \cong \varinjlim_k \mathcal{H}om(F/\mathcal{J}^{k+1}F, \mathcal{H}^n_Y(G)).$$

It remains to show that the natural homomorphism

$$\varinjlim_k \mathcal{H}om(F/\mathcal{J}^{k+1}F, \mathcal{H}^n_Y(G)) \longrightarrow \mathcal{H}om(F, \mathcal{H}^n_Y(G))$$

is an isomorphism when $Z \cap \operatorname{Supp} F \subset Y$. Since X admits a cover by Noetherian affine open subsets, we reduce to the case where X is affine Noetherian. Then $F(X)$ is a finitely generated $\mathcal{O}_X(X)$-module and $\operatorname{Supp} \mathcal{H}^n_Y(G) \subset Y$. Hence every homomorphism

$$u : F(X) \longrightarrow \mathcal{H}^n_Y(G)(X)$$

is annihilated by a power of $\mathcal{I}$, and therefore by a power of $\mathcal{J}$. This proves the lemma.

Proposition 1.2. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X, let G be a quasi-coherent $\mathcal{O}_X$-module, and let n be an integer. For any closed subsets Z and S of X such that $Z \cap S = Y$, the following conditions are equivalent:*

(i) $\mathcal{H}^i_Y(G) = 0$ *for* $i < n$;

(ii) there exists a coherent $\mathcal{O}_X$-module F with support S such that

$$\mathcal{E}xt^i_Z(F, G) = 0 \qquad \textit{for } i < n;$$

(iii) for every coherent $\mathcal{O}_X$-module F whose support is contained in S (that is, $\operatorname{Supp} F \cap Z = \operatorname{Supp} F \cap Y$*), one has*

$$\mathcal{E}xt^i_Z(F, G) = 0 \qquad \textit{for } i < n;$$

(iv) for every coherent $\mathcal{O}_X$-module F, one has

$$\mathcal{E}xt^i_Y(F, G) = 0 \qquad \textit{for } i < n.$$

Moreover, if these conditions hold, then for every coherent $\mathcal{O}_X$-module F and every closed subset Z' of X such that $Z' \cap \operatorname{Supp} F = Y \cap \operatorname{Supp} F$, one has isomorphisms

$$\mathcal{E}xt^n_Z(F, G) \cong \mathcal{E}xt^n_Y(F, G) \cong \mathcal{H}om(F, \mathcal{H}^n_Y(G)).$$

Source note. The quantified subset in the preceding sentence is Z', but both the corrected French TeX and the direct compiled page print Z in the first term of the final display. The proof immediately below uses Z'. Both source controls confirm these facts; Z' is probably intended, but the edition decision remains unresolved. This reader therefore preserves printed Z and does not silently emend the authority.

Proof of Proposition 1.2: induction

We argue by induction. The proposition is immediate for $n < 0$. Suppose that it has been proved for $n < q$. If one of the conditions holds for $n = q$ and for two subsets Z and S as in the statement, then the induction hypothesis gives, for every closed subset Z' of X and every coherent $\mathcal{O}_X$-module F such that $Z' \cap \operatorname{Supp} F = Y \cap \operatorname{Supp} F$, isomorphisms

$$\mathcal{E}xt^{q-1}_{Z'}(F, G) \cong \mathcal{H}om(F, \mathcal{H}^{q-1}_Y(G)) \cong \mathcal{E}xt^{q-1}_Y(F, G). \tag{1.1}$$

Consequently:

(i)⇒(iv) take $Z' = Y$ in (1.1);

(iv)⇒(iii) take $Z' = Z$ in (1.1);

(iii)⇒(ii) take $F = \mathcal{O}_S$;

(ii)⇒(i) take $Z' = Z$ in (1.1). This gives $\mathcal{H}om(F, \mathcal{H}_Y^{q-1}(G)) = 0$. Moreover,

$$\operatorname{Supp} \mathcal{H}_Y^{q-1}(G) \subset Y = Z \cap S \subset S = \operatorname{Supp} F,$$

so the following lemma applies.

Lemma 1.3. *Let X be a prescheme, let P be a coherent $\mathcal{O}_X$-module, and let H be a quasi-coherent $\mathcal{O}_X$-module such that*

$$\mathcal{H}om(P, H) = 0 \qquad \textit{and} \qquad \operatorname{Supp} P \supset \operatorname{Supp} H.$$

Then $H = 0$.

Proof of Lemma 1.3

It suffices to prove the lemma when X is affine, since the affine open subsets form a basis for the topology of X and the hypotheses are preserved under restriction to an open subset. In that case, we are reduced to a problem about A-modules, where $X = \operatorname{Spec}(A)$. We apply the formula (valid under the sole hypothesis that M is of finite type):

$$\operatorname{Ass} \operatorname{Hom}_A(P, H) = \operatorname{Supp} P \cap \operatorname{Ass} H.$$

We know that

$$\operatorname{Ass} H \subset \operatorname{Supp} H \subset \operatorname{Supp} P \qquad \text{and} \qquad \operatorname{Ass} \operatorname{Hom}_A(P, H) = \varnothing;$$

hence $\operatorname{Ass} H = \varnothing$, and therefore $H = 0$.
Source note. The corrected French TeX and direct PDF print M in the finite-type parenthetical. Since the displayed formula involves P and no M is introduced, P is the high-confidence intended reading; no silent emendation is made while the French-authority disposition is pending.

Close of the proof of Proposition 1.2

To complete the proof of the proposition, it remains to observe that condition (iv) permits the application of Lemma 1.1.

Corollary 1.4. *Let G be a coherent Cohen–Macaulay $\mathcal{O}_X$-module, and let $n \in \mathbb{Z}$. The conditions of Proposition 1.2 are equivalent to:*

$$\text{(v)} \qquad \operatorname{codim}(Y \cap \operatorname{Supp} G, \operatorname{Supp} G) \geq n.$$

Proof

Recall first that an $\mathcal{O}_X$-module is said to be Cohen–Macaulay if, for every $x \in X$, the stalk G_x is a Cohen–Macaulay $\mathcal{O}_{X,x}$-module; that is, for every $x \in S = \operatorname{Supp} G$,

$$\operatorname{depth} G_x = \dim G_x = \dim \mathcal{O}_{S,x}. \tag{1.2}$$

By Proposition III.3.3, condition (i) of Proposition 1.2 is equivalent to

$$\operatorname{depth}_Y G = \inf_{x \in Y} \operatorname{depth} G_x \geq n, \tag{1.3}$$

and hence also to

$$\operatorname{depth}_Y G = \inf_{x \in Y \cap S} \operatorname{depth} G_x \geq n,$$

since the depth of the zero module is infinite.

By definition,

$$\operatorname{codim}(Y \cap S, S) = \inf_{x \in S \cap Y} \dim \mathcal{O}_{S,x};$$

the conclusion follows by applying formula (1.2).

We shall now prove a result that will allow us to deduce the coherence conditions we have in view from certain vanishing criteria.

Lemma 1.5. *Let X be a locally Noetherian prescheme. Let T^* be an exact contravariant ∂-functor defined on the category of coherent $\mathcal{O}_X$-modules and taking values in the category of $\mathcal{O}_X$-modules. Let Y be a closed subset of X, and let $i \in \mathbb{Z}$. Suppose that, for every coherent $\mathcal{O}_X$-module F whose support is contained in Y, the modules $T^i F$ and $T^{i-1} F$ are coherent. Let F be a coherent $\mathcal{O}_X$-module. Then $T^i F$ is coherent if and only if $T^i F''$ is coherent, where*

$$F'' = F / \Gamma_Y(F).$$

Indeed, $F' = \Gamma_Y(F)$ is coherent because X is locally Noetherian. The long exact cohomology sequence of T^* then gives

$$T^{i-1} F' \longrightarrow T^i F'' \longrightarrow T^i F \longrightarrow T^i F',$$

where the two outer terms are coherent; the conclusion follows.

Lemma 1.6. *If F and G are coherent and* $\operatorname{Supp} F \subset Y$*, then $\mathcal{E}xt^i_Y(F, G)$ is coherent.*

Indeed, $\mathcal{E}xt^i_Y(F, G)$ is isomorphic to $\mathcal{E}xt^i(F, G)$. More generally, this holds on any ringed space X: if Z is a closed subset containing $Y \cap \operatorname{Supp} F$, then $\mathcal{E}xt^i_Z(F, G)$ is isomorphic to $\mathcal{E}xt^i_Y(F, G)$ (cf. Exposé VI).

Proposition 1.7. *Suppose that F and G are coherent, and put*

$$\operatorname{Supp} F = S, \qquad S' = \overline{S \cap (X - Y)}.$$

Suppose that $\operatorname{depth} G_x \geq n$ *for every $x \in Y \cap S'$. Then $\mathcal{E}xt^i_Y(F, G)$ is coherent for $i < n$.*

Indeed, Lemma 1.6 allows us to apply Lemma 1.5 to

$$T^*(F) = \mathcal{E}xt^*_Y(F, G).$$

Putting $F'' = F/\Gamma_Y(F)$, we see that $\operatorname{Supp} F'' = S'$. By Proposition III.3.3, the hypothesis on the depth of G ensures that $\mathcal{H}^i_{Y \cap S'}(G) = 0$ for $i < n$; Proposition 1.2 then gives $T^i F'' = 0$ for $i < n$, and the conclusion follows from Lemma 1.5.

2. Study for $i > n$

Let X be a locally Noetherian *regular* prescheme; that is, suppose that all its local rings are regular. Let Y be a closed subset of X, and let F and G be two coherent $\mathcal{O}_X$-modules. Put

$$S = \operatorname{Supp} F, \qquad S' = \overline{S \cap (X - Y)},$$

and put

$$m = \sup_{x \in Y \cap S} \dim \mathcal{O}_{X,x},$$
$$n = \sup_{x \in Y \cap S'} \dim \mathcal{O}_{X,x}.$$

Then $n \leq m$.

Proposition 2.1. *In the situation described above:*

1. $\mathcal{E}xt^i_Y(F,G) = 0$ *if* $i > m$*;*
2. $\mathcal{E}xt^i_Y(F,G)$ *is coherent if* $i > n$.

Proof of Proposition 2.1

First note that $\mathcal{E}xt^i_Y(F,G)$ is coherent for every i when $\operatorname{Supp} F \subset Y$. Moreover, putting $F'' = F/\Gamma_Y(F)$ as above, we see that $\operatorname{Supp} F'' = S'$. Thus assertion (2) follows from assertion (1) and Lemma 1.5.[1]

To prove assertion (1), first observe that

$$\mathcal{E}xt^i_Y(F,G) \cong \varinjlim_k \mathcal{E}xt^i(F/\mathcal{J}^k F, G),$$

where $\mathcal{J}$ is the ideal defining Y. On the other hand, Theorem 4.2.2 of A. Grothendieck, "Sur quelques points d'algèbre homologique," *Tôhoku Mathematical Journal* **9** (1957), 119–221, shows that the $\mathcal{E}xt$ sheaves commute with the formation of stalks, at least when X is a locally Noetherian prescheme and the first argument is coherent. Since direct limits do likewise, for every $x \in X$ there are isomorphisms[2]

$$(\mathcal{E}xt^i_Y(F,G))_x \cong \varinjlim_k \operatorname{Ext}^i_{\mathcal{O}_{X,x}}((F/\mathcal{J}^k F)_x, G_x).$$

Since

$$\operatorname{Supp} \mathcal{E}xt^i_Y(F,G) \subset S \cap Y,$$

it is enough to note that $x \in Y \cap S$ implies $\dim \mathcal{O}_{X,x} \leq m$, and hence

$$\operatorname{Ext}^i_{\mathcal{O}_{X,x}}((F/\mathcal{J}^k F)_x, G_x) = 0 \qquad \text{if } i > m,$$

because the global cohomological dimension of a regular local ring is equal to its dimension.[3]

Let X be a locally Noetherian prescheme. For every subset P of X, set

$$D(P) = \{\dim \mathcal{O}_{X,p} \mid p \in P\}.$$

Lemma 2.2

If P is the underlying space of a connected subprescheme of X, then $D(P)$ is an interval.[4]

---

[1] The French TeX and printed page cite Lemma 1.3. The coherence reduction used here is Lemma 1.5; Lemma 1.3 is the earlier $\mathcal{H}om$/support lemma.

[2] The French TeX and printed stalk formula omit the superscript i on the local Ext term. The preceding direct-limit formula and the following vanishing display both require Ext^i.

[3] Cf. EGA 0_{IV}, 17.3.1.

[4] The French TeX and printed witness read "of A." The ambient prescheme introduced immediately before is X, and no object A occurs in this argument. The reader therefore uses X transparently.

Indeed, let a and b belong to $D(P)$, corresponding to points p and q of P. We show that there is a sequence of points of P,

$$p = p_1, \ldots, p_n = q,$$

such that, for $1 \leq i < n$,

$$|\dim \mathcal{O}_{X,p_i} - \dim \mathcal{O}_{X,p_{i+1}}| = 1.$$

It will follow that $D(P)$ contains the interval $[a, b]$.[5]

For this, observe that p and q can be joined by a chain of irreducible components of P in which two successive components meet. We are reduced to the case where p is the generic point of an irreducible component Q of P, and where $q \in Q$, hence $q \supset p$ as ideals of $\mathcal{O}_q$; there the assertion follows immediately from the definition of dimension.

[5]The French TeX and printed witness introduce the numerical endpoints as "a and B" and then print the interval $[p, q]$, although p and q are points of P. The reader normalizes the two numerical endpoints to a, b and the interval to $[a, b]$. The absolute-value bars are present in the current French TeX and printed witness and are preserved.

Proposition 2.3

Let X be a locally Noetherian regular prescheme, let Y be a closed subset of X, and let F be a coherent $\mathcal{O}_X$-module. Put

$$P = Y \cap \overline{\operatorname{Supp} F \cap (X - Y)}.$$

Let $n \in \mathbb{Z}$, and suppose that $n \notin D(P)$. Then $\mathcal{E}xt^n_Y(F, \mathcal{O}_X)$ is coherent.[6]

Proof

The conclusion is local, and the hypotheses are preserved under restriction to an open subset. Since P is closed, it is locally Noetherian and hence locally connected. We may therefore suppose that X is affine and Noetherian and that P is connected.

If $P = \varnothing$, the depth hypothesis of Proposition 1.7 with threshold $n+1$ is vacuous, and the conclusion follows. Assume henceforth that $P \neq \varnothing$. By Lemma 2.2, $D(P)$ is an interval. Since $n \notin D(P)$, exactly one of the following alternatives holds:[7]

$$n > d \quad \text{for every } d \in D(P), \qquad \text{or} \qquad n < d \quad \text{for every } d \in D(P).$$

In the first case, put $m = \sup D(P)$. Then $n > m$, and Proposition 2.1, applied with $G = \mathcal{O}_X$, shows that $\mathcal{E}xt^n_Y(F, \mathcal{O}_X)$ is coherent.

In the second case, for every $x \in P$ regularity gives

$$n < \dim \mathcal{O}_{X,x} = \operatorname{depth} \mathcal{O}_{X,x},$$

and hence $\operatorname{depth} \mathcal{O}_{X,x} \geq n+1$. Proposition 1.7 with threshold $n+1$ therefore gives coherence in every degree $i < n+1$, in particular in degree $i = n$.

[6] The French TeX and printed statement omit the closure and read $P = Y \cap \operatorname{Supp} F \cap (X - Y)$, which is empty literally. Section 2 defines $S' = \overline{\operatorname{Supp} F \cap (X - Y)}$, and Propositions 2.1 and 1.7 both use the locus $Y \cap S'$. The reader restores that uniquely supported closure.

[7] The French TeX and print write $D(P) = [a, b[$ and then treat only $n > b$ and $n < a$, leaving the allowed case $n = b$ uncovered. The print does not decide whether the intended one-character repair was a closed upper endpoint or a non-strict upper inequality. The reader states the order alternatives forced by $n \notin D(P)$ and Lemma 2.2 and therefore does not guess that glyph.

Exposé VIII

The Finiteness Theorem

1. A Biduality Spectral Sequence[1]

Let us state the result we wish to establish:

Proposition 1.1. *Let A be a Noetherian ring and let I be an ideal of A. Put $X = \operatorname{Spec}(A)$ and $Y = V(I)$. Let M be a finitely generated A-module of finite projective dimension. Let $\mathcal{F} = \widetilde{M}$ be the $\mathcal{O}_X$-module associated with M.*

1) *There is a spectral sequence*

$$\mathrm{H}_Y(X, \mathcal{F}) \Longleftarrow \operatorname{Ext}^p_Y(\operatorname{Ext}^{-q}(M, A), A).$$

2) *There is a spectral sequence*

$$\mathcal{H}_Y(X, \mathcal{F}) \Longleftarrow \mathcal{E}xt^p_Y(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

Of course, 2) follows from 1) by observing that, if M and N are two finitely generated A-modules and if we put $\mathcal{F} = \widetilde{M}$ and $\mathcal{G} = \widetilde{N}$, then there are isomorphisms

$$\begin{aligned} \mathcal{H}_Y(\mathcal{F}) &\simeq \mathrm{H}_Y\widetilde{(X, \mathcal{F})}, \\ \mathcal{E}xt_Y(\mathcal{F}, \mathcal{G}) &\simeq \operatorname{Ext}_Y\widetilde{(\mathcal{F}, \mathcal{G})}, \\ \mathcal{E}xt_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) &\simeq \operatorname{Ext}_A\widetilde{(M, N)}. \end{aligned}$$

Let $\mathcal{C}$ be the category of A-modules, and let $\mathbf{Ab}$ be the category of abelian groups. Let $\underline{\mathrm{F}}$ be the functor

$$\begin{aligned} \underline{\mathrm{F}} \colon \mathcal{C} &\longrightarrow \mathbf{Ab}, \\ M &\longmapsto \Gamma_Y(\widetilde{M}). \end{aligned}$$

We know from Exposé II that there is an isomorphism of ∂-functors

$$\mathrm{H}^*_Y(X, \widetilde{M}) \simeq \mathrm{R}^*\underline{\mathrm{F}}(M).$$

Furthermore, let Ext^*_Y denote the right derived functors, in the second argument, of

$$\underline{\mathrm{F}} \circ \operatorname{Hom} \colon \mathcal{C}^\circ \times \mathcal{C} \longrightarrow \mathbf{Ab}.$$

[1] A reader familiar with Verdier's language of derived categories will recognize the spectral sequence associated with a *biduality isomorphism.* Cf. SGA 6, I.

We know from Exposé VI that there is an isomorphism of ∂-functors

$$\mathrm{Ext}^*_Y(M, N) \simeq \mathrm{Ext}^*_Y(\widetilde{M}, \widetilde{N}).$$

Finally, let us record the following result from Exposé VI: if C is an injective A-module and N is a finitely generated A-module, the sheaf

$$\mathcal{H}om(\widetilde{N}, \widetilde{C}) \simeq \widetilde{\mathrm{Hom}(N, C)}$$

is flasque; hence

$$\mathrm{R}^1\underline{\mathrm{F}}(\mathrm{Hom}(N, C)) = 0.$$

It remains for us to prove the following result.

Lemma 1.2. — Let A be a Noetherian ring and let $\mathcal{C}$ be the category of A-modules. Let $\underline{\mathrm{F}}\colon \mathcal{C} \longrightarrow \mathbf{Ab}$ be a left exact additive functor such that, for every finitely generated A-module N and every injective A-module C, one has

$$\mathrm{R}^1\underline{\mathrm{F}}(\mathrm{Hom}(N, C)) = 0.$$

Let M be a finitely generated A-module of finite projective dimension. There is a spectral sequence

$$\mathrm{R}^*\underline{\mathrm{F}}(M) \Longleftarrow \mathrm{Ext}^p_{\underline{\mathrm{F}}}(\mathrm{Ext}^{-q}(M, A), A),$$

where $\mathrm{Ext}^p_{\underline{\mathrm{F}}}$ denotes the p-th right derived functor of $\underline{\mathrm{F}} \circ \mathrm{Hom}$.

We shall consider only complexes whose differential has degree $+1$. By the hypothesis on M, there exists a projective resolution of M of finite length,

$$u\colon L^\bullet \longrightarrow M,$$

where, moreover, the L^p are finitely generated modules and $L^p = 0$ if $p \notin [-n, 0]$. On the other hand, let $v\colon M \longrightarrow I^\bullet$ be an injective resolution of M. I claim that

$$v \circ u\colon L^\bullet \longrightarrow I^\bullet \tag{1.1}$$

is an injective resolution of $L^\bullet$. We must specify what this means.

Definition 1.3. — Let $X^\bullet$ be a complex of A-modules. An *injective resolution* of $X^\bullet$ is a homomorphism of complexes

$$x\colon X^\bullet \longrightarrow CX^\bullet$$

such that CX^p is injective for every $p \in \mathbb{Z}$, and x induces an isomorphism in homology.

Proposition 1.4. — Every left-bounded complex, i.e., every complex for which there exists $q \in \mathbf{Z}$ such that $X^p = 0$ for $p < q$, admits an injective resolution. Moreover, suppose that $u\colon X^\bullet \longrightarrow Y^\bullet$ is a homomorphism of complexes (both left-bounded), and that $x\colon X^\bullet \longrightarrow CX^\bullet$ and $y\colon Y^\bullet \longrightarrow CY^\bullet$ are injective resolutions of $X^\bullet$ and $Y^\bullet$. Then there exists a homomorphism of complexes

$$Cu\colon CX^\bullet \longrightarrow CY^\bullet,$$

unique up to homotopy, such that the diagram

$$\begin{array}{ccc} X^\bullet & \xrightarrow{x} & CX^\bullet \\ {\scriptstyle u}\big\downarrow & & \big\downarrow{\scriptstyle Cu} \\ Y^\bullet & \xrightarrow{y} & CY^\bullet \end{array}$$

commutes up to homotopy.

The proof is left to the reader.[2]

Let us recall a notation introduced in Exposé V.

Notation. — Let $X^\bullet$ and $Y^\bullet$ be two complexes. We denote by $\operatorname{Hom}^\bullet(X^\bullet, Y^\bullet)$ the *simple complex* whose component of degree n is

$$(\operatorname{Hom}^\bullet(X^\bullet, Y^\bullet))^n = \prod_{-p+q=n} \operatorname{Hom}(X^p, Y^q),$$

also denoted by $\operatorname{Hom}^n(X^\bullet, Y^\bullet)$, and whose differential is given by

$$\partial_n \colon \operatorname{Hom}^n(X^\bullet, Y^\bullet) \longrightarrow \operatorname{Hom}^{n+1}(X^\bullet, Y^\bullet),$$
$$\partial_n = d' + (-1)^{n+1} d'',$$

where d' and d'' are the differentials (of degree $+1$) induced by those of $X^\bullet$ and $Y^\bullet$.

Let $A^\bullet$ then be the complex defined by $A^p = 0$ if $p \neq 0$ and $A^0 = A$. Let

$$a \colon A^\bullet \longrightarrow CA^\bullet$$

be an injective resolution of $A^\bullet$. Consider the double complex

$$Q^{p,q} = \operatorname{Hom}(\operatorname{Hom}(L^{-q}, A), CA^p). \tag{1.2}$$

The first spectral sequence of the bicomplex $\underline{F}Q^{\bullet\bullet}$ will give the conclusion of Lemma 1.2.

Set

$$L'^\bullet = \operatorname{Hom}^\bullet(L^\bullet, A^\bullet), \tag{1.3}$$

and

$$P^\bullet = \operatorname{Hom}^\bullet(L'^\bullet, CA^\bullet). \tag{1.4}$$

One sees easily that $P^\bullet$ is the simple complex associated with $Q^{\bullet\bullet}$. Let us compute the abutment of the spectral sequence, that is, the homology of $\underline{F}P^\bullet$. To this end, using the fact that $L^\bullet$ is projective of finite type in every degree, one proves that

$$L^\bullet \cong \operatorname{Hom}^\bullet(L'^\bullet, A^\bullet).$$

From the homomorphism $a \colon A^\bullet \to CA^\bullet$, one obtains a homomorphism

$$b \colon \operatorname{Hom}^\bullet(L'^\bullet, A^\bullet) \longrightarrow \operatorname{Hom}^\bullet(L'^\bullet, CA^\bullet),$$

or equivalently a homomorphism

$$c \colon L^\bullet \longrightarrow P^\bullet. \tag{1.5}$$

This being said, it is easy to see, using the fact that $L'^\bullet$ is projective of finite type in every degree and bounded on the left, that (1.5) is an injective resolution of $L^\bullet$. Using Proposition 1.4, one concludes that $P^\bullet$ is homotopy equivalent to $I^\bullet$, where $I^\bullet$ is the injective resolution of M introduced above in (1.1). It follows that the *abutment* of the first spectral sequence of the double complex $\underline{F}Q^{\bullet\bullet}$, which is $H^*(\underline{F}P^\bullet)$, *is isomorphic* to $R^*\underline{F}(M)$.

The initial term of the first spectral sequence of the bicomplex $\underline{F}Q^{\bullet\bullet}$ is

$$E_2^{p,q} = {}'H^p({}''H^q(\underline{F}Q^{\bullet\bullet})).$$

For every $p \in \mathbf{Z}$, CA^p is injective. By the hypothesis on $\underline{F}$, the functor (restricted to the category of finitely generated modules)

$$N \longmapsto \underline{F}\operatorname{Hom}(N, CA^p)$$

[2] See also H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton Mathematical Series, vol. 19, Princeton University Press, 1956.

is exact. Hence one obtains isomorphisms

$$''H^q(\underline{\mathrm{F}}\,\mathrm{Hom}(L'^{\bullet}, CA^p)) \cong \underline{\mathrm{F}}\,\mathrm{Hom}(H^{-q}(L'^{\bullet}), CA^p).$$

By the definition of $\mathrm{Ext}^*_{\underline{\mathrm{F}}}$ as the right-derived functor of $\underline{\mathrm{F}} \circ \mathrm{Hom}$, one therefore obtains isomorphisms

$$E_2^{p,q} \cong \mathrm{Ext}^p_{\underline{\mathrm{F}}}(H^{-q}(L'^{\bullet}), A).$$

Source note. The French TeX and printed display have $H^q(L'^{\bullet})$ in the last formula. The preceding displayed isomorphism, the statement of Lemma 1.2, and the identity below require $H^{-q}(L'^{\bullet})$. This target therefore uses H^{-q} as a transparent editorial emendation. It is not a restoration from an independent original scan; the French authority remains unchanged pending editorial action.

Now

$$L'^{\bullet} = \mathrm{Hom}^{\bullet}(L^{\bullet}, A^{\bullet}),$$

where $L^{\bullet}$ is a projective resolution of M; hence there are isomorphisms

$$\mathrm{Ext}^q(M, A) \cong H^q(L'^{\bullet}).$$

Substituting $-q$ in this last identity gives the conclusion. □

VIII.2 The Finiteness Theorem

Theorem VIII.2.1. [(1)]. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Suppose that X is locally embeddable in a regular prescheme*[*]. *Let $i \in \mathbf{Z}$. Suppose that:*

a) *for every $x \in U = X - Y$, one has*

$$H^{i-c(x)}(\mathcal{F}_x) = 0,$$

where we have set[(2)]*:*

$$c(x) = \mathrm{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \tag{VIII.2.6}$$

Then:

b) *$\mathcal{H}^i_Y(\mathcal{F})$ is coherent.*

Corollary VIII.2.2. *Editor's note. Strictly speaking, this is a corollary of the proof that follows, not of the statement. The implication* c)⇒a) *is tautological. The converse is not, but follows from the proof. More precisely, as below, one covers X by open subsets embeddable in regular schemes; as explained below, this permits one to reduce to the case where $X = \mathrm{Spec}(A)$ is affine and regular and $\mathcal{F} = \widetilde{M}$, where M is an A-module of finite projective dimension. In this case it is shown that conditions* a) *and* c) *are equivalent to the dual conditions* a′) *and* c′)*. One then shows that* c′) *implies condition* d) *(see below), which in turn implies* a′)*. See the discussion following 2.4. Under the hypotheses of the preceding theorem, condition* a) *is equivalent to:*

[(1)] *Editor's note.* For an analogous statement, but in a somewhat more general situation, see Mme Raynaud (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental," *Ann. Sci. Éc. Norm. Sup.* (4) **7** (1974), pp. 29–52, Proposition II.2.1).

[*] This condition may be generalized to the hypothesis that a "dualizing complex," in the sense defined in R. Hartshorne, *Residues and Duality* (cited in the note to Exposé IV, p. 46), exists locally on X.

[(2)] *Editor's note.* As in Exposé V, $H^*(\mathcal{F}_x)$ denotes the local cohomology $H^*_{\mathfrak{m}_x}(\mathcal{F}_x)$.

c) *for every* $x \in U$ *such that* $c(x) = 1$, *one has* $H^{i-1}(\mathcal{F}_x) = 0$.

Corollary VIII.2.3. *Let* X *be a locally Noetherian prescheme that is locally embeddable in a regular prescheme, let* Y *be a closed subset of* X, *let* $\mathcal{F}$ *be a coherent* $\mathcal{O}_X$*-module, and let* n *be an integer. The following conditions are equivalent:*

(i) *for every* $x \in U$, *one has* $\operatorname{prof} \mathcal{F}_x > n - c(x)$;

(ii) *for every* $x \in U$ *such that* $c(x) = 1$, *one has* $\operatorname{prof} \mathcal{F}_x \geq n$;

(iii) *for every* $i \in \mathbb{Z}$, $\mathcal{H}^i_Y(\mathcal{F})$ *is coherent if* $i \leq n$;

(iv) $R^i i_*(\mathcal{F}|_U)$ *is coherent for* $i < n$[(4)].

Assume these results established when X is the spectrum of a regular Noetherian ring A and when $\mathcal{F}$ is the sheaf associated with an A-module of finite projective dimension.

First observe that, if $(X_j)_{j \in J}$ is an open covering of X by open subsets embeddable in a regular scheme, each of the conditions above is equivalent to the conjunction of the analogous conditions obtained by replacing X by X_j, Y by $Y_j = Y \cap X_j$, and $\mathcal{F}$ by $\mathcal{F}|_{X_j}$. Indeed, only the conditions involving $c(x)$ can present a difficulty. Let $x \in U$. If $x \in X_j$, set

$$c_j(x) = \operatorname{codim}(X_j \cap \overline{\{x\}} \cap Y, X_j \cap \overline{\{x\}}).$$

Necessarily $c_j(x) \geq c(x)$. Let $y \in \overline{\{x\}} \cap Y$ be a point that "realizes the codimension," that is, such that $c(x) = \dim \mathcal{O}_{\overline{\{x\}},y}$, and let X_j be a member of the covering such that $y \in X_j$. Then $x \in X_j$, and hence $c_j(x) = c(x)$. It follows that condition (a) for the X_j implies condition (a) for X. At this point we have only a partial converse: condition (a) for X implies condition (a) for those X_j such that $c(x) = c_j(x)$, which suffices for our purposes.[(5)]

Choose a covering of X by open subsets embeddable in a regular prescheme. Applying the preceding observation, we see that we may assume that X is closed in a regular X'. The reduction to X' is then immediate.

We may therefore assume that X is regular, and even affine, by covering X with affine open subsets. That we may assume that $\mathcal{F} = \widetilde{M}$, where M has finite projective dimension, will follow from the next lemma.

Lemma VIII.2.4. *Let* X *be a regular Noetherian prescheme. Let* $\mathcal{F}$ *be a coherent* $\mathcal{O}_X$*-module. The function that assigns to each* $x \in X$ *the projective dimension of* $\mathcal{F}_x$ *is bounded above.*

Indeed, let $x \in X$, and let U be an affine open neighborhood of x. Let $L^\bullet$ be a projective resolution of the module $\mathcal{F}(U)$, where the L^i are finitely generated. By hypothesis, the ring $\mathcal{O}_{X,x}$ is regular, so the projective dimension of $\mathcal{F}_x$ is finite; denote this integer by d. Set

$$K = \ker(L^{-d} \longrightarrow L^{-d+1}).$$

The module K_x is free, since d is the projective dimension of $\mathcal{F}_x$ ([M], Ch. VI, Prop. 2.1). By (EGA 0_{I}, 5.4.1 Errata), it follows that the $\mathcal{O}_U$-module $\widetilde{K}$ is free on a neighborhood U' of

(4) *Editor's note.* This condition appeared only in the body of the proof, but not in the statement of the corollary; since it is used in Section 3, it has been added.

(5) *Editor's note.* In fact, condition (a) for X implies condition (a) for every X_j, as asserted in the original text, but establishing this requires reading the proof below in detail. At this point the implication does not appear formal. For a property (a), (b), or (c), append a subscript J to denote its conjunction over the X_j. The proof *below* establishes $(c_J) \Rightarrow (a_J)$ (this is the chain of implications $(c') \Rightarrow (d) \Rightarrow (a')$). But tautologically $(a) \Rightarrow (c)$, and $(c) \iff (c_J)$; hence $(a) \Rightarrow (a_J)$.

x, with $U' \subset U$. Choose $f \in \mathcal{O}_X(U)$ such that $x \in D(f) \subset U'$. We then have a projective resolution of M_f, where $M = \mathcal{F}(U)$:

$$0 \longrightarrow K_f \longrightarrow (L^{-d})_f \longrightarrow \cdots \longrightarrow M_f \longrightarrow 0.$$

Source note. After choosing only f, French line 2729 and the compiled page print $M_{f'}$, whereas the displayed resolution ends in M_f. Moreover, the corrected kernel is a submodule of L^{-d}, but French line 2730 prints $(L^{d-1})_f$. The target uses M_f and $(L^{-d})_f$ as transparent editorial emendations forced by the chosen localization, the kernel definition, and the displayed resolution. These are not restorations from an independent original scan; the French authority remains unchanged pending editorial review.

This proves that the function under consideration is upper semicontinuous. Since X is quasi-compact, the conclusion follows.

We henceforth assume that X is affine, Noetherian, and regular, and that $\mathcal{F} = \widetilde{M}$, where M is a finitely generated A-module, necessarily of finite projective dimension. We shall proceed in several steps. First, we find a condition (d), equivalent to (a), and prove that it is also equivalent to (c). Then, using the spectral sequence of the preceding subsection, we prove $(d) \Rightarrow (b)$. It remains to prove (iii) $\Rightarrow$ (ii); indeed, (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii) follows immediately from $(a) \Leftrightarrow (c) \Rightarrow (b)$.

Let $x \in U$. By hypothesis, $\mathcal{O}_{X,x}$ is a regular local ring. If D denotes the dualizing functor for $\mathcal{O}_{X,x}$, it follows from (V, 2.1) that

$$D\, H^{i-c(x)}(\mathcal{F}_x) \simeq \operatorname{Ext}^{d(x)-i}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{O}_{X,x}),$$

where

$$d(x) = \dim \mathcal{O}_{X,x} + c(x) = \dim \mathcal{O}_{X,x} + \operatorname{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \tag{2.2}$$

Since X is Noetherian and $\mathcal{F}$ is coherent,

$$D\, H^{i-c(x)}(\mathcal{F}_x) \simeq (\mathcal{E}xt^{d(x)-i}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X))_x. \tag{2.3}$$

Moreover, a module vanishes if and only if its dual vanishes (cf. the editor's note (4) on p. 54). For every $q \in \mathbb{Z}$, set

$$\begin{cases} S_q = \operatorname{Supp} \mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X), \\ S'_q = S_q \cap U, \quad (U = X - Y), \\ Z_q = \overline{S'_q} \cap Y. \end{cases} \tag{2.4}$$

It follows from formula (2.3) that conditions (a) and (c) are respectively equivalent to:

(a$'$) for every $q \in \mathbb{Z}$ and every $x \in S'_q$, one has $q + i \neq d(x)$;

(c$'$) for every $q \in \mathbb{Z}$ and every $x \in S'_q$, if $c(x) = 1$, then $q + i \neq d(x)$.

Here is condition (d), promised above:

(d) for every $q \in \mathbb{Z}$ and every $y \in Z_q$, one has $q + i \neq \dim \mathcal{O}_{X,y}$.

These conditions are equivalent.

(a$'$) $\Rightarrow$ (c$'$) is recorded for completeness.

(d) $\Rightarrow$ (a$'$). Indeed, let $q \in \mathbb{Z}$ and $x \in S'_q$. Choose a point $y \in \overline{\{x\}} \cap Y$ that realizes the codimension,[6] that is, such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = \operatorname{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}) = c(x). \tag{2.5}$$

[6] *Editor's note.* Throughout what follows, closures of points are endowed with the reduced structure.

Since X is regular at y, it follows that

$$\dim \mathcal{O}_{X,y} = d(x) \qquad \text{(cf. (2.2))}. \tag{2.6}$$

But $y \in \overline{\{x\}}$, and hence $y \in Z_q$; the conclusion follows.

(c$'$) $\Rightarrow$ (d). Let $q \in \mathbb{Z}$ and $y \in Z_q$. Assume provisionally that there exists $x \in S'_q$ such that

$$y \in \overline{\{x\}} \qquad \text{and} \qquad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

(One also says that y is an immediate specialization of x.) It follows that $c(x) = 1$, since y realizes the codimension of $\overline{\{x\}} \cap Y$ in $\overline{\{x\}}$, while $x \notin Y$. Condition (c$'$) therefore gives

$$q + i \neq d(x).$$

Whence the conclusion, upon observing that $d(x) = \dim \mathcal{O}_{X,y}$ by (2.6). The assertion provisionally assumed above is the content of the following lemma.

Lemma 2.5. Let X be a locally Noetherian prescheme and Y a closed subset of X. Set $U = X - Y$, and suppose that U is dense in X. For every $y \in Y$, there exists $x \in U$ such that y is an immediate specialization of x, that is,

$$y \in \overline{\{x\}} \qquad \text{and} \qquad \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

We have applied the lemma with X equal to the prescheme $\overline{S'_q}$ and Y equal to the subset $Y \cap \overline{S'_q}$.

Proof of Lemma 2.5. There exists $x \in U$ such that $y \in \overline{\{x\}}$. Choose such an x for which $r = \dim \mathcal{O}_{\overline{\{x\}},y}$ is minimal. We must prove that $r = 1$. By the choice of x, every $z \in \operatorname{Spec}(\mathcal{O}_{\overline{\{x\}},y})$ with $z \neq x$ lies in Y. Consequently, $\{x\}$ is open in $\operatorname{Spec}(\mathcal{O}_{\overline{\{x\}},y})$, and the conclusion follows.□

The second step is to deduce (b) *from* (d).

Set

$$D(Z_q) = \{\dim \mathcal{O}_{X,y} \mid y \in Z_q\}.$$

By (d), for every $q \in \mathbb{Z}$ one has $q + i \notin D(Z_q)$. Applying Proposition VII.2.3, we see that

$$\mathcal{E}xt_Y^{q+i}(\mathcal{E}xt^q(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X) \quad \text{is coherent.}$$

The initial term of the spectral sequence of the preceding subsection is

$$E_2^{p,q} = \mathcal{E}xt_Y^p(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

It follows that $E_2^{p,q}$ is coherent for every $p, q \in \mathbb{Z}$ such that $p + q = i$. Now there are only finitely many pairs (p, q) such that $p + q = i$,[S] and this spectral sequence converges to $\mathcal{H}_Y^*(\mathcal{F})$; whence the conclusion.

It remains to prove that (iii) *implies* (ii). Let

$$i \colon U \longrightarrow X$$

denote the canonical immersion of U into X. In view of the exact homology sequence associated with the closed subset Y (I, 2.11), one sees that (iii) is *equivalent to*:

[S] *Source note.* Taken literally for arbitrary integers, this sentence is false. In context it must be read as concerning the terms that can contribute, in view of the finite-projective-dimension setup above. The French authority is unchanged.

(iv) $R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n$.

French printed p. 94 Indeed, there is an exact sequence

$$0 \longrightarrow \mathcal{H}^0_Y(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_U) \longrightarrow \mathcal{H}^1_Y(\mathcal{F}) \longrightarrow 0.$$

Now $\mathcal{H}^0_Y(\mathcal{F})$ is a quasi-coherent subsheaf of the coherent sheaf $\mathcal{F}$, and is therefore coherent. Hence $\mathcal{H}^1_Y(\mathcal{F})$ is coherent if and only if $i_*(\mathcal{F}|_U)$ is coherent. Moreover, for $p > 0$, the exact cohomology sequence associated with the closed subset Y reduces to isomorphisms

$$R^p i_*(\mathcal{F}|_U) \xrightarrow{\approx} \mathcal{H}^{p+1}_Y(\mathcal{F}).$$

We shall prove that (iv) *implies* (ii). To this end, recall (ii):

(ii) For every $x \in U$ such that $c(x) = 1$, one has

$$\operatorname{prof} \mathcal{F}_x \geq n.$$

We argue by induction on n.

If $n = 0$, both conditions are vacuous.

If $n = 1$, assume that $i_*(\mathcal{F}|_U)$ is coherent. We argue by contradiction and suppose that there exists $x \in U$ such that $c(x) = 1$ and

$$\operatorname{prof} \mathcal{F}_x = 0,$$

i.e. $x \in \operatorname{Ass} \mathcal{F}_x$. Let $y \in \overline{\{x\}} \cap Y$ be such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

Set

$$A = \mathcal{O}_{X,y} \qquad \text{and} \qquad X' = \operatorname{Spec}(A).$$

Perform the base change $v\colon X' \longrightarrow X$, which is flat:

$$\begin{array}{ccc} U' = X' \times_X U & \xrightarrow{v'} & U \\ {\scriptstyle i'}\big\downarrow & & \big\downarrow{\scriptstyle i} \\ X' & \xrightarrow{v} & X. \end{array} \tag{2.7}$$

The morphism i is separated, since it is an immersion, and is of finite type, since it is an open immersion and X is locally Noetherian. Since the base change is flat, EGA III, 1.4.15 gives an isomorphism

$$v^*(i_*(\mathcal{F}|_U)) \approx i'_*(v'^*(\mathcal{F}|_U)). \tag{2.8}$$

French printed p. 95 Denote by $\underline{x}$ (respectively, $\underline{y}$) the ideal of A corresponding to x (respectively, y).

Set $\mathcal{G} = v'^*(\mathcal{F}|_U)$. Then $\mathcal{G}$ is coherent and $\underline{x} \in \operatorname{Ass} \mathcal{G}$, so there exists a monomorphism

$$\mathcal{O}_{\overline{\{x\}}} \longrightarrow \mathcal{G};$$

consequently, $i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})$ is coherent. By the choice of y, $\dim(A/\underline{x}) = 1$, and therefore the support of $\mathcal{O}_{\overline{\{x\}}}$ is precisely

$$\overline{\{\underline{x}\}} = \{\underline{x}\} \cup \{\underline{y}\},$$

since $\overline{\{\underline{x}\}} = \mathrm{Spec}(A/\underline{x})$ as a scheme. It follows that

$$(\mathcal{O}_{\overline{\{x\}}}|_{U'})(U') = \mathrm{Frac}(A/\underline{x}),$$

the field of fractions of $A/\underline{x}$, and

$$i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})(X') = \mathrm{Frac}(A/\underline{x}).$$

But $\mathrm{Frac}(A/\underline{x})$ is not a finitely generated A-module, because $\underline{x}$ is different from the maximal ideal of A. This is a contradiction.

Suppose that $n > 1$, and assume the result established for every $n' < n$. By the induction hypothesis, for every $x \in U$ such that $c(x) = 1$, one has $x \notin \mathrm{Ass}\,\mathcal{F}_x$. Let x be such a point, and let $y \in \overline{\{x\}} \cap Y$ be an immediate specialization of x, i.e.

$$\dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

Perform the base change

$$v\colon \mathrm{Spec}(\mathcal{O}_{X,y}) \longrightarrow X,$$

retaining the notation of diagram (2.7). Applying EGA III, 1.4.15, one obtains isomorphisms

$$v^*(R^p i_*(\mathcal{F}|_U)) \simeq R^p i'_*(v'^*(\mathcal{F}|_U)), \qquad p \in \mathbb{Z}.$$

Thus we are reduced to the case in which X is the spectrum of a local ring A in which $\underline{x}$ is a prime ideal of dimension 1, i.e. $\dim(A/x) = 1$. Set

$$\mathcal{F}' = \underline{\Gamma}_Y(\mathcal{F}) \qquad \text{and} \qquad \mathcal{F}'' = \mathcal{F}/\mathcal{F}'.$$

We have $\mathcal{F}_x \simeq \mathcal{F}''_x$ and $y \notin \mathrm{Ass}\,\mathcal{F}''$. Moreover, $\mathcal{F}'|_U = 0$; hence the long exact sequence of the $R^p i_*$ gives isomorphisms

$$R^p i_*(\mathcal{F}|_U) \simeq R^p i_*(\mathcal{F}''|_U), \qquad p \in \mathbb{Z}.$$

French printed p. 96 Since $n > 1$, neither x nor y belongs to $\mathrm{Ass}\,\mathcal{F}''$. Now $\underline{x}$ and $\underline{y}$ are the only prime ideals of A containing $\underline{x}$. It follows from Exposé III, 2.1 that there exists an element $g \in \underline{x}$ that is M-regular, where we have set

$$\mathcal{F} = \widetilde{M}, \qquad M = \mathcal{F}(X).$$

Thus there is an exact sequence

$$0 \longrightarrow M \xrightarrow{g'} M \longrightarrow N \longrightarrow 0,$$

in which g' denotes multiplication by g on M. From the resulting long exact sequence one sees that

$$R^p i_*(\widetilde{N}|_U) \quad \text{is coherent for } p < n-1.$$

Therefore, by the induction hypothesis,

$$\mathrm{prof}(\widetilde{N})_x \geq n-1, \qquad \text{and hence} \qquad \mathrm{prof}\,\mathcal{F}_x \geq n.$$

□

3. Applications

From these results one deduces a coherence condition for the higher direct images of a coherent sheaf under a *morphism that is not proper.*

Theorem 3.1.—Let $f\colon X \to Y$ be a morphism of preschemes. Suppose that Y is *locally Noetherian* and that f is *proper.* Suppose that X is *locally embeddable in a regular prescheme.* Let $n \in \mathbb{Z}$. Let U be an open subset of X, and let $\mathcal{F}$ be a *coherent* $\mathcal{O}_U$-module. Suppose that, for every $x \in U$ such that

$$\operatorname{codim}(\overline{\{x\}} \cap (X \setminus U), \overline{\{x\}}) = 1,$$

one has $\operatorname{prof} \mathcal{F}_x \geq n$. *Then* the $\mathcal{O}_Y$-modules

$$R^p(f \circ g)_*(\mathcal{F})$$

are coherent for $p < n$, where g is the canonical immersion of U into X.

Indeed, there is a Leray spectral sequence whose abutment is $R^*(f \circ g)_*(\mathcal{F})$ and whose initial term is given by

$$E_2^{p,q} = R^p f_*(R^q g_*(\mathcal{F})).$$

Moreover, there exists a *coherent* $\mathcal{O}_X$-module $\mathcal{G}$ such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). It then follows from the preceding section that condition (iv) on p. 74 is satisfied [French printed p. 97], i.e. that $R^q g_*(\mathcal{G}|_U)$ is coherent for $q < n$. Applying the finiteness theorem of EGA III, 3.2.1 to f and the sheaves $R^q g_*(\mathcal{F})$, one finds that $E_2^{p,q}$ is coherent for $q < n$, whence the conclusion.

Proposition 3.2.—Let X be a locally Noetherian prescheme, locally embeddable in a regular prescheme. Let U be an open subset of X, and let $i\colon U \to X$ be the canonical immersion. Let $n \in \mathbb{Z}$. Finally, let $\mathcal{F}$ be a coherent Cohen–Macaulay $\mathcal{O}_U$-module (on U!). The following conditions are equivalent:

(a) $R^p i_*(\mathcal{F})$ is coherent for $p < n$;

(b) for every irreducible component S' of the closure $\overline{S}$ of the support S of $\mathcal{F}$, one has

$$\operatorname{codim}(S' \cap (X \setminus U), S') > n.$$

Proof, opening.—Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$-module such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). Applying Corollary 2.3 to $\mathcal{G}$, we find that condition (a) is equivalent to (c):

(c) for every $x \in S$, one has

$$\operatorname{prof} \mathcal{F}_x > n - c(x), \qquad c(x) = \operatorname{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$

Proof, (a)⇒(b).—Indeed, let S' be an irreducible component of $\overline{S}$, and let s be its generic point. Since $\mathcal{F}$ is Cohen–Macaulay, one has $\operatorname{prof} \mathcal{F}_s = \dim \mathcal{O}_{S,s} = 0$. Moreover, $\overline{\{s\}} = S'$, and the conclusion follows.

Proof, (b)⇒(a), *opening.*—Let $x \in S$, and let S' be an irreducible component of $\overline{S}$ such that $x \in S'$. Let s be the generic point of S'. Since $\mathcal{F}$ is Cohen–Macaulay, we know that

$$\operatorname{prof} \mathcal{F}_x = \dim \mathcal{O}_{\overline{\{s\}},x}.$$

Proof, (b)⇒(a), *case split.*—If $c(x) = +\infty$, there is nothing to prove. Otherwise, there exists $y \in Y \cap \overline{\{x\}}$ such that

$$c(x) = \dim \mathcal{O}_{\overline{\{x\}},y}.$$

Proof, (b)⇒(a), *conclusion.*—Now $\mathcal{O}_{X,y}$ is a quotient of a regular local ring by hypothesis; hence

$$\dim \mathcal{O}_{\overline{\{s\}},y} = \dim \mathcal{O}_{\overline{\{s\}},x} + \dim \mathcal{O}_{\overline{\{x\}},y} > n.$$

□

Bibliography

[M] H. Cartan & S. Eilenberg – *Homological Algebra*, Princeton Math. Series vol. 19, Princeton University Press, 1956.

Exposé IX

Algebraic Geometry and Formal Geometry

The purpose of this exposé is to generalize Theorems 3.4.2 and 4.1.5 of *EGA III* to the case of a morphism that is not proper.

1. The Comparison Theorem

Let $f\colon X \to X'$ be a morphism of preschemes that is *separated* and of *finite type*. Suppose that X' is locally noetherian. Let Y' be a closed subset of X', and let $Y = f^{-1}(Y')$. Let $\widehat{X}$ and $\widehat{X'}$ be the formal completions of X and X' along Y and Y', respectively. Let $\widehat{f}$ be the morphism induced by f upon passing to the completions.

$$\begin{tikzcd} X \arrow[d, "f"'] & Y \arrow[l] \arrow[d] \\ X' & Y' \arrow[l] \end{tikzcd} \quad , \qquad \begin{tikzcd} \widehat{X} \arrow[r, "j"] \arrow[d, "\widehat{f}"'] & X \arrow[d, "f"] \\ \widehat{X'} \arrow[r, "i"] & X'. \end{tikzcd} \tag{1.1}$$

We denote by j (resp. i) the morphism from $\widehat{X}$ to X (resp. from $\widehat{X'}$ to X'). It is known that i and j are *flat*.

Let $\mathcal{I}'$ be an ideal of definition of Y', and let $\mathcal{J} = f^*(\mathcal{I}') \cdot \mathcal{O}_X$; this is an ideal of definition of Y. Hence

$$\widehat{X'} = \left(Y', \varprojlim_{k\in\mathbb{N}} \mathcal{O}_{X'}/\mathcal{I}'^{k+1}\right), \qquad \widehat{X} = \left(Y, \varprojlim_{k\in\mathbb{N}} \mathcal{O}_X/\mathcal{J}^{k+1}\right). \tag{1.2}$$

For every $k \in \mathbb{N}$, set

$$Y'_k = \left(Y', \mathcal{O}_{X'}/(\mathcal{I}')^{k+1}\right), \qquad Y_k = \left(Y, \mathcal{O}_X/\mathcal{J}^{k+1}\right). \tag{1.3}$$

Let F be a coherent $\mathcal{O}_X$-module. For every $k \in \mathbb{N}$, set

$$F_k = F/\mathcal{J}^{k+1}F, \qquad \widehat{F} = j^*(F) \simeq \varprojlim F_k. \tag{1.4}$$

If we set

$$R^i f_*(F)^\wedge = \varprojlim_{k\in\mathbb{N}} \left(R^i f_*(F) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{Y'_k}\right), \qquad i \in \mathbb{Z}, \tag{1.5}$$

there is a natural homomorphism

$$r_i : i^*(R^i f_*(F)) \longrightarrow R^i f_*(F)^\wedge, \tag{1.6}$$

which is an *isomorphism* when $R^i f_*(F)$ is *coherent*.

As explained in *EGA III*, 4.1.1, there is a commutative diagram

$$\begin{array}{ccc} i^*(R^i f_*(F)) & \xrightarrow{\quad \rho_i \quad} & R^i \widehat{f}_*(\widehat{F}) \\ \Big\downarrow r_i & & \Big\downarrow \psi_i \\ R^i f_*(F)^\wedge & \xrightarrow{\quad \varphi_i \quad} & \varprojlim_{k \in \mathbb{N}} R^i f_*(F_k). \end{array} \tag{1.7}$$

In the cited passage one obtains a commutative diagram because $R^i f_*(F)$ is known to be coherent, and the source and target of (1.6) are identified. In our case, $R^i f_*(F)$ will be coherent only for certain values of i, for which we shall study (1.7).

Consider the graded ring

$$\mathcal{S} = \bigoplus_{k \in \mathbb{N}} \mathcal{I}'^k, \tag{1.8}$$

and the graded $\mathcal{S}$-module

$$\mathcal{H}^i = \bigoplus_{k \in \mathbb{N}} R^i f_*(\mathcal{J}^k F), \qquad i \in \mathbb{Z}, \tag{1.9}$$

whose $\mathcal{S}$-module structure is defined as follows.

The sheaf $R^i f_*(\mathcal{J}^k F)$ is associated with the presheaf that assigns to each affine open subset U' of X'

$$H^i\left(f^{-1}(U'), \, \mathcal{J}^k F\Big|_{f^{-1}(U')} \right). \tag{1.10}$$

Thus let U' be an affine open subset of X', set

$$U = f^{-1}(U'),$$

and let $x' \in \mathcal{I}'^m(U')$. Let x be the image of x' in $\mathcal{J}^m(U)$. Multiplication by x on $F|_U$ maps $\mathcal{J}^k F|_U$ into $\mathcal{J}^{k+m} F|_U$ and hence, by functoriality, gives a morphism

$$\mu^i_{x',k}(U') : H^i(U, \mathcal{J}^k F|_U) \longrightarrow H^i(U, \mathcal{J}^{k+m} F|_U), \tag{1.11}$$

defined for every $i \in \mathbb{Z}$ and every $k \in \mathbb{N}$. Upon passage to the associated sheaf, these morphisms give $\mathcal{H}^i$ its graded $\mathcal{S}$-module structure.

Theorem 1.1. Let n be an integer. Suppose that the graded $\mathcal{S}$-module $\mathcal{H}^i$ is finitely generated for $i = n-1$ and $i = n$. Then:

(0) r_n and r_{n-1} are isomorphisms, and $R^i \widehat{f}_*(\widehat{F})$ is coherent for $i = n-1$;

(1) for $i = n-1$, the maps ρ_i, φ_i, and ψ_i are topological isomorphisms. In particular, the filtration on $R^{n-1} f_*(F)$ defined by the kernels of the homomorphisms

$$R^{n-1} f_*(F) \longrightarrow R^{n-1} f_*(F_k) \tag{1.12}$$

is $\mathcal{I}'$-good;

(2) for $i = n$, the maps ρ_i, φ_i, and ψ_i are monomorphisms; moreover, the filtration on $R^n f_*(F)$ is $\mathcal{I}'$-good, and ψ_n is an isomorphism;

(3) for $i = n-2, n-1$, the projective system $R^i f_*(F_k)$ satisfies the uniform Mittag–Leffler condition; that is, there is a *fixed* integer $k \geq 0$ such that, for every $p \geq 0$ and every $p' \geq p + k$,

$$\operatorname{Im}\Big[R^i f_*(F_{p'}) \longrightarrow R^i f_*(F_p) \Big] = \operatorname{Im}\Big[R^i f_*(F_{p+k}) \longrightarrow R^i f_*(F_p) \Big].$$

Proceeding as in *EGA III*, 4.1.8, it is easy to *reduce to the case where X' is the spectrum of a noetherian ring A*. In this case, it is known that

$$R^i f_*(F) = \widetilde{H^i(X,F)} \qquad \text{(cf. (1.10))}. \tag{1.13}$$

Let I be the ideal of A such that $\widetilde{I} = \mathcal{I}'$, and set

$$S = \bigoplus_{k\in\mathbb{N}} I^k, \tag{1.14}$$

$$H^i = \bigoplus_{k\in\mathbb{N}} H^i(X, \mathcal{J}^k F), \qquad i \in \mathbb{Z}, \tag{1.15}$$

where H^i carries the graded S-module structure defined by (1.11), with $U' = X'$. [1]

The proof is modeled on that of *EGA III*, 4.1.5; let us give a summary.

We now work with φ_i and ψ_i, to which the following module homomorphisms correspond:

$$\begin{tikzcd} & H^i(\widehat{X},\widehat{F}) \arrow[d, "\psi_i"] \\ H^i(X,F)^\wedge \arrow[r, "\varphi_i"] & \varprojlim_{k\in\mathbb{N}} H^i(X,F_k). \end{tikzcd} \tag{1.16}$$

(a) Suppose only that H^i is a graded S-module of finite type. It follows that *the filtration on $H^i(X,F)$ defined by the modules*

$$R^i_k = \ker\big(H^i(X,F) \longrightarrow H^i(X,F_k)\big) \tag{1.17}$$

is I-good. Indeed, the exact cohomology sequence

$$H^i(X, \mathcal{J}^{k+1}F) \longrightarrow H^i(X,F) \longrightarrow H^i(X,F_k) \tag{1.18}$$

shows that the *graded S-module $\bigoplus_{k\in\mathbb{N}} R^i_k$ is a quotient* of the graded S-submodule

$$\bigoplus_{k\in\mathbb{N}} H^i(X, \mathcal{J}^{k+1}F)$$

of H^i, and hence *is of finite type*, since S is noetherian. This proves the first assertion.

Set

$$M^i = H^i(X,F), \qquad H^i_k = H^i(X,F_k). \tag{1.19}$$

There is a commutative diagram

$$\begin{tikzcd} H^i(X,F)^\wedge \arrow[r, "s_i"] \arrow[dr, "\varphi_i"'] & \varprojlim_k (M^i/R^i_k) \arrow[d, "t_i"] \\ & \varprojlim_k H^i_k, \end{tikzcd} \tag{1.20}$$

[1] The corrected French TeX and same-edition PDF print $U = X'$ here. In the construction preceding (1.11), however, $U = f^{-1}(U')$; specializing the affine open to the whole base therefore requires $U' = X'$. The French authority is left unchanged.

in which s_i is an *isomorphism*, since the filtration of $H^i(X,F)$ is I-good. Moreover, t_i is a *monomorphism*: the functor $\varprojlim$ is left exact and, for every $k \geq 0$, the natural morphism $M^i/R^i_k \longrightarrow H^i_k$ is a monomorphism by the definition of R^i_k.

To study the surjectivity of t_i, introduce

$$Q^i_k = \operatorname{coker}\bigl(H^i(X,F) \longrightarrow H^i(X,F_k)\bigr), \tag{1.21}$$

which gives a projective system of exact sequences

$$0 \longrightarrow R^i_k \longrightarrow M^i \longrightarrow H^i_k \longrightarrow Q^i_k \longrightarrow 0. \tag{1.22}$$

Using the exact cohomology sequence

$$H^i(X,F) \longrightarrow H^i(X,F_k) \longrightarrow H^{i+1}(X,\mathcal{J}^{k+1}F), \tag{1.23}$$

we see that the *graded S-module*

$$Q^i = \bigoplus_{k\in\mathbb{N}} Q^i_k \tag{1.24}$$

is a graded S-submodule of H^{i+1}. Moreover, for every $k \geq 0$,

$$I^{k+1}Q^i_k = 0, \tag{1.25}$$

because Q^i_k is the image of H^i_k.

(b) Suppose only that H^{i+1} is of finite type, and consider t_i alone, disregarding s_i. Since S is noetherian, Q^i is of finite type; as $I^{k+1}Q^i_k = 0$, there exist integers $r \geq 0$ and $k_0 \geq 0$ such that

$$I^r Q^i_k = 0 \qquad \text{for } k \geq k_0. \tag{1.26}$$

It follows that the *projective system $(Q^i_k)_{k\in\mathbb{N}}$ is essentially zero*, and hence that the *projective system $(H^i_k)_{k\in\mathbb{N}}$ satisfies the uniform Mittag–Leffler condition.* From the exact sequence (1.22) we obtain the exact sequence

$$0 \longrightarrow M^i/R^i_k \longrightarrow H^i_k \longrightarrow Q^i_k \longrightarrow 0, \tag{1.27}$$

and therefore the exact sequence

$$0 \longrightarrow \varprojlim_k M^i/R^i_k \xrightarrow{t_i} \varprojlim_k H^i_k \longrightarrow \varprojlim_k Q^i_k. \tag{1.28}$$

But the projective system $(Q^i_k)_{k\in\mathbb{N}}$ is essentially zero, so t_i is an *isomorphism.*

(c) Let us prove that if H^i is of finite type, then ψ_i is an isomorphism. It suffices to apply *EGA* 0_{III}, 13.3.1, taking the affine open subsets as a basis of open subsets of X. This is legitimate: by (b), the projective system $(H^{i-1}_k)_{k\in\mathbb{N}}$ satisfies the Mittag–Leffler condition.

The theorem follows formally from (a), (b), and (c). Observe that, in fact, at each step the proof uses finite generation of H^i for only one value of i.

Let us give some examples in which the hypothesis of Theorem 1.1 is satisfied.

Corollary 1.2. *Suppose that $\mathcal{I}'$ is generated by a section t' of $\mathcal{O}_{X'}$, and denote by t the corresponding section of $\mathcal{O}_X$. Let F be a coherent $\mathcal{O}_X$-module, and let n be an integer. Assume that:*

(i) *t is F-regular (that is, multiplication by t on F is a monomorphism);*

(ii) *$R^if_*(F)$ is coherent for $i = n-1$ and $i = n$.*

Then the hypothesis of Theorem 1.1 is satisfied.

Indeed, multiplication by t^k defines an isomorphism $F \xrightarrow{\sim} \mathcal{J}^k F$, from which it follows that

$$H^i \simeq R^i f_*(F) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{X'}[T], \tag{1.29}$$

where T is an indeterminate. This proves the result.

Corollary 1.3. *Suppose that $X' = \operatorname{Spec}(A)$, where A is a noetherian ring that is separated and complete for the I-adic topology. Suppose that the graded S-module H^i is of finite type for $i = n-1$ and $i = n$ (cf. (1.14) and (1.15)). Then the hypotheses of Theorem 1.1 are satisfied, and there is a commutative diagram of isomorphisms*

$$\begin{tikzcd} H^i(X,F) \arrow[rr, "\rho'_i"] \arrow[dr, "\varphi'_i"'] & & H^i(\widehat{X},\widehat{F}) \arrow[dl, "\psi_i"] \\ & \varprojlim_k H^i(X,F_k) & \end{tikzcd} \qquad \text{for } i = n-1. \tag{1.30}$$

Source-numbering note (SGA2-IX-EQ130-RENDERDEF-001). The corrected French TeX labels this diagram `eq:IX.1.30`, the following proof cites (1.30), and the next numbered formula is (1.31). The same-edition rendered PDF nevertheless prints (1.1), apparently because its subsection step resets the equation counter. The French files are left unchanged; the intended tag (1.30) is used here.

Editor's note (1)

In the same vein, see the article by W.-L. Chow, "Formal functions on homogeneous spaces," *Invent. Math.* **86** (1986), no. 1, pp. 115–130. The author proves the following result. Let X be an algebraic variety over a field, homogeneous under an algebraic group G, and let Z be a complete subvariety of X of dimension > 0. Suppose that Z generates X in the following sense: given $p \in Z$, let Γ_p be the set of elements of G that send p into Z. One says that Z generates X if the group generated by the identity component of Γ_p is all of G. In that case, every formal rational function on X along Z is algebraic; compare the results of Hironaka and Matsumura cited in editor's note (3), p. 138.

Along the lines of the techniques introduced by those authors, we also mention the elegant algebraization result due to D. Gieseker, "On two theorems of Griffiths about embeddings with ample normal bundle," *Amer. J. Math.* **99** (1977), no. 6, pp. 1137–1150, Theorems 4.1 and 4.2. Let X be a connected projective variety of dimension > 0, locally a complete intersection over an algebraically closed field. Suppose there are two embeddings of X into smooth projective varieties Y and W. If the formal completions of X in Y and W are equivalent, then there exists a scheme U containing X as a closed subscheme which embeds into both Y and W as an étale neighborhood of X. In other words, formal equivalence implies étale equivalence.

See also G. Faltings, "Formale Geometrie und homogene Räume," *Invent. Math.* **64** (1981), pp. 123–165.

Source-defect note (SGA2-IX-L3156-PUNCT-SRCDEF-001; SGA2-IX-L3157-FALTINGS-TITLE-SRCDEF-001). The French TeX and its same-edition PDF omit the sentence stop before "Autrement dit." They also print the Faltings title in the mixed and misspelled form "Formale Geometrie and homogene Raüme." The target restores the sentence boundary and the published German title; the French authority remains unchanged.

Proof. We simply note that $H^i(X,F)$ is of finite type and hence isomorphic to its completion. One obtains (1.30) by transcribing the diagram of modules (1.7) into the category of A-modules and replacing the left-hand vertical by $H^i(X,F)$.

Source-numbering note (SGA2-IX-EQ130-RENDERDEF-001). The corrected French TeX cites its label `eq:IX.1.30`; the next numbered formula is (1.31). The same-edition rendered PDF prints the referenced diagram as (1.1) because of an apparent counter reset. The French authority remains unchanged; the intended reference (1.30) is retained here.

Proposition 1.4. *Let A be a noetherian ring. Let $t \in A$, and suppose that A is separated and complete for the (tA)-adic topology. Set*

$$X' = \operatorname{Spec}(A), \qquad Y' = V(t), \qquad I = (tA). \tag{1.31}$$

Let T be a closed subset of X', and set

$$X = X' \setminus T, \qquad Y = Y' \cap X = Y' \setminus (Y' \cap T). \tag{1.32}$$

Let F be a coherent $\mathcal{O}_X$-module. Finally, set

$$T' = \{x \in X' \mid \operatorname{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) = 1\}. \tag{1.33}$$

Suppose that

a) t is F-regular;

b) $\operatorname{prof}_{T'}(F) \geq n+1$*;*

c) A is a quotient of a regular noetherian ring.

Then, in diagram (1.30), the morphisms ρ'_i, φ'_i, and ψ_i are isomorphisms for $i < n$ and monomorphisms for $i = n$. Moreover, ψ_n is an isomorphism.

Source-notation and numbering note (SGA2-IX-L3177-CLOSURE-BRACING-SRCDEF-001; SGA2-IX-EQ130-RENDERDEF-001). In (1.33) the French TeX and its same-edition PDF put braces around $\overline{x}$, whereas the defining codimension formula in Exposé VIII, Theorem 2.1, uses $\overline{\{x\}}$. The target makes that intended closure-of-the-point notation explicit; French remains unchanged. French line 3189 cites `eq:IX.1.30`; the target retains (1.30), rather than the same-edition PDF's accidental counter-reset display (1.1).

Proof. By (1.3) and (1.2), it suffices to prove that $R^i f_*(F)$ is coherent for $i \leq n$; this follows from the finiteness theorem, Exposé VIII, Theorem 2.1.

In particular:

Example 1.5. Proposition 1.4 applies when A is a local ring and t belongs to the radical $\mathfrak{r}(A)$ of A. We then take $T = \{\mathfrak{r}(A)\}$. In this case, for $n = 1$, we obtain the following statement:

If A is noetherian, separated and complete for the t-adic topology, and is a quotient of a regular ring (for example, if A is complete), if moreover t is F-regular and $\operatorname{prof} F_x \geq 2$ *for every* $x \in \operatorname{Spec}(A)$ *such that* $\dim(A/x) = 1$*, then the natural homomorphism*

$$\Gamma(X, F) \longrightarrow \Gamma(\widehat{X}, \widehat{F})$$

is an isomorphism.

Indeed, retaining the notation of Proposition 1.4, we have $T = \{\mathfrak{r}(A)\}$, and formula (1.33) says that

$$T' = \{x \in \operatorname{Spec}(A) \mid \dim(A/x) = 1\}.$$

Source-cross-reference note (SGA2-IX-EXAMPLE15-SRCDEF-001). French lines 3196 and 3202 use the key `eq:IX.1.4`, which formally names equation (1.4), although the grammar and the retained hypotheses/notation unambiguously refer to Proposition 1.4. The same-edition PDF displays only the number '1.4' and does not resolve the key. The target makes the semantic reference explicit; the French authority remains unchanged.

IX.2 The Existence Theorem

Let us first state EGA III, 3.4.2 in a slightly more general form.

Let $f : \mathfrak{X} \to \mathfrak{X}'$ be an adic morphism[(*)] of formal preschemes, with $\mathfrak{X}'$ noetherian. Let $\mathcal{I}'$ be an ideal of definition of $\mathfrak{X}'$; since f is adic, $f^*\mathcal{I}' = \mathcal{J}$ is[(2)] an ideal of definition of $\mathfrak{X}$.

For every $n \in \mathbb{N}$, set

$$\mathfrak{X}_n = (\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathcal{J}^{n+1}); \tag{IX.2.1}$$

this is an ordinary prescheme with the same underlying topological space as $\mathfrak{X}$.

Let $\mathfrak{F}$ be a coherent $\mathcal{O}_{\mathfrak{X}}$-module. For every $k \in \mathbb{N}$, the $\mathcal{O}_{\mathfrak{X}_k}$-modules

$$F_k = \mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F} \tag{2.2}$$

are coherent. For every i, there is a homomorphism

$$\psi_i : R^i f_*(\mathfrak{F}) \longrightarrow \varprojlim_k R^i f_*(F_k), \tag{2.3}$$

induced by functoriality from the natural homomorphism

$$\mathfrak{F} \longrightarrow F_k. \tag{2.4}$$

Set

$$\mathbf{S} = \operatorname{gr}_{\mathcal{I}'} \mathcal{O}_{\mathfrak{X}'} = \bigoplus_{k\in\mathbb{N}} \mathcal{I}'^k/\mathcal{I}'^{k+1}, \tag{2.5}$$

$$\operatorname{gr}_{\mathcal{J}}(\mathfrak{F}) = \bigoplus_{k\in\mathbb{N}} \mathcal{J}^k\mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F}, \tag{2.6}$$

$$\mathbf{K}^i = R^i f_*(\operatorname{gr}_{\mathcal{J}}(\mathfrak{F})) = \bigoplus_{k\in\mathbb{N}} R^i f_*(\mathcal{J}^k\mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F}). \tag{2.7}$$

It is clear that $\mathbf{K}^i$ is endowed with the structure of a graded $\mathbf{S}$-module.

Theorem 2.1. *Suppose that $\mathbf{K}^i$ is a graded $\mathbf{S}$-module of finite type for $i = n-1$, $i = n$, and $i = n+1$. Then:*

(i) $R^n f_*(\mathfrak{F})$ is coherent.

(ii) The homomorphism ψ_n in (2.3) is a topological isomorphism. The natural filtration on the right-hand side of (2.3) is $\mathcal{I}'$-good.

(iii) The projective system of the $R^n f_*(F_k)$ satisfies the uniform Mittag-Leffler condition.

Proof. The proof follows very easily from EGA 0_{III}, 13.7.7 (cf. EGA III, 3.4.2), provided that the text on page 78 is corrected as indicated in (EGA III 2, Err_{III} 24).

Theorem 2.2[(3)]. *Let A be a noetherian adic ring, and let I be an ideal of definition of A. Let T be a closed subset of $X' = \operatorname{Spec}(A)$. Suppose that I is generated by an element $t \in A$. Retain the notation of (1.31), (1.32), and (1.33). Let $\mathfrak{F}$ be a coherent $\mathcal{O}_{\widehat{X}}$-module. Set*

$$F_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}, \tag{2.8}$$

Source-render note [SGA2-IX-T22-EQ28-COUNTER-SRCDEF-001]. The admitted French counter order renders this display as (2.1), although its label and the surrounding sequence require (2.8). The English target uses (2.8); the French authority remains unchanged.

where $\mathcal{J} = t\mathcal{O}_{\widehat{X}}$ is an ideal of definition of $\widehat{X}$. Suppose that *A is a quotient of a regular noetherian ring and that:*

[(*)]This hypothesis is not essential; cf. Exposé XII, p. 118.

[(2)]*Editor's note.* By the very definition; cf. EGA I, 10.12.1.

(1) *t is $\mathfrak{F}$-regular;*

(2) $\operatorname{prof}_{T'} F_0 \geq 2$.

Then there exists a coherent $\mathcal{O}_X$-module F such that $\widehat{F} \simeq \mathfrak{F}$.

Editor's note (3). Numerous algebraization statements have been obtained since, without mentioning those cited below; see the articles by Faltings or by Mme Raynaud cited, respectively, in editor's note (22) on p. 155 and editor's note (7) on p. 203.

Source-defect note [SGA2-IX-L3265-ONT-GRAMMAR-SRCDEF-001]. The admitted French reads *on été obtenus*; the plural subject requires *ont été obtenus*. The English phrase "have been obtained" gives the intended plural sense; the French authority remains unchanged.

One has in mind in particular Artin's results (see especially Artin M., "Algebraization of formal moduli. I," in *Global Analysis (Papers in Honor of K. Kodaira)*, Univ. Tokyo Press, Tokyo, 1969, pp. 21–71), but also recent results on the algebraicity of leaves of foliations; see especially Bost J.-B., "Algebraic leaves of algebraic foliations over number fields," *Publ. Math. Inst. Hautes Études Sci.* **93** (2001), pp. 161–221, and Chambert-Loir A., "Théorèmes d'algébricité en géométrie diophantienne (d'après J.-B. Bost, Y. André, D. & G. Chudnovsky)," in *Séminaire Bourbaki, Vol. 2000/2001*, *Astérisque*, vol. 282, Société mathématique de France, Paris, 2002, Exp. 886, pp. 175–209, and the references cited there. In particular, the latter two articles contain discussions of the connection between algebraization questions and the theory of Diophantine approximation.

Proof. It suffices to prove that $\widehat{f}_*(\mathfrak{F})$ is a coherent $\mathcal{O}_{\widehat{X'}}$-module, where $\widehat{f}\colon \widehat{X} \to \widehat{X'}$ is the morphism of formal preschemes obtained from the inclusion of X into X' by completion with respect to t. Indeed, A is separated and complete for the t-adic topology, so there exists an A-module F' whose completion is isomorphic to $\widehat{f}_*(\mathfrak{F})$. Since X is open in X', we may take

$$F = \widetilde{F'}|_X.$$

Source-defect note [SGA2-IX-L3280-PUSHFORWARD-BASE-SRCDEF-001]. The French TeX and the same-edition reader print $\mathcal{O}_{\widehat{X}}$ in the first sentence. Since the pushforward along $\widehat{f}\colon \widehat{X} \to \widehat{X'}$ is a sheaf on the target, the English target uses the type-correct $\mathcal{O}_{\widehat{X'}}$. The French authority remains unchanged under manager decision EG-SGA2-IX-T22-SOURCE-DEFECT-ADJUDICATION-20260719-0001.

It remains to show that Theorem 2.1 applies to the morphism of formal preschemes $\widehat{f}$ and to $\mathfrak{F}$. By hypothesis (1), for every $k \in \mathbb{N}$ there is an isomorphism

$$\mathcal{J}^k\mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F} \to \mathfrak{F}/\mathcal{J}\mathfrak{F},$$

and hence the hypothesis of Theorem 2.1 is satisfied provided that

$$R^i f_*(F_0) \text{ is coherent for } i \leq 1.$$

This follows from (2) and the finiteness theorem of Exposé VIII, Theorem 2.1. This proves the result.

Source-notation observation [SGA2-IX-L3283-PLAIN-ARROW-OBS-001]. The corrected TeX and its same-edition reader use a plain $\to$ in the display while the preceding prose explicitly calls the map an isomorphism. This is an observation, not a confirmed defect. The target retains both features; no $\sim$ has been added over the arrow.

It remains to specialize this statement under the assumption that A is a local ring.

Corollary 2.3. *Let A be a noetherian local ring, and let $t \in \mathfrak{r}(A)$ be an element of the radical of A. Suppose that A is separated and complete for the t-adic topology and, moreover, is a quotient of a regular ring (for example, suppose that A is a complete noetherian local ring). Set*

$$X' = \operatorname{Spec} A, \qquad T = \{\mathfrak{r}(A)\}, \tag{2.9}$$

and retain the notation of (1.31), (1.32), and (1.33). Let $\mathfrak{F}$ be an $\mathcal{O}_{\widehat{X}}$-module. Suppose that

1. t is $\mathfrak{F}$-regular;
2. $\operatorname{prof}_{T'} F_0 \geq 2$, where $F_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}$ and $\mathcal{J} = t\mathcal{O}_{\widehat{X}}$.

Then there exists a coherent $\mathcal{O}_X$-module F such that $\widehat{F} \simeq \mathfrak{F}$.

Source-structure note (SGA2-IX-L3304-ENUM-STRUCTURE-SRCDEF-001). The French TeX closes its `enumerate` environment only after the conclusion, syntactically nesting "Alors il existe" inside item (2). The same-edition PDF preserves the source order but visually masks this nesting: the conclusion begins at the corollary margin, not at the item-text margin. The conclusion is consequent on both hypotheses, not part of hypothesis (2); the English therefore closes the list before "Then." The French authority remains byte-identical; this disclosed repair follows the manager's final adjudication.

Note that here T' is the set of prime ideals $\mathfrak{p}$ of A such that

$$\dim(A/\mathfrak{p}) = 1.$$

Exposé X

Application to the Fundamental Group

Throughout this exposé, X will denote a locally noetherian prescheme, Y a closed subset of X, U a variable open neighborhood of Y in X, and $\widehat{X}$ the formal completion of X along Y (EGA I, 10.8). For every prescheme Z, let $\mathbf{Et}(Z)$ denote the category of étale coverings of Z, and let $\mathbf{L}(Z)$ denote the category of coherent locally free modules on Z.

1. Comparison of $\mathbf{Et}(\widehat{X})$ and $\mathbf{Et}(Y)$

Let $\mathcal{I}$ be an ideal of definition of Y in X. For every $n \in \mathbb{N}$, set

$$Y_n = \left(Y, (\mathcal{O}_X/\mathcal{I}^{n+1})\Big|_Y\right).$$

The Y_n form a direct system of ordinary preschemes, or also of formal preschemes when the structure sheaves are endowed with the discrete topology. We know (EGA I, 10.6.2) that $\widehat{X}$ is the direct limit, in the category of formal preschemes, of the direct system (Y_n). We also know (EGA I, 10.13) that to give a formal $\widehat{X}$-prescheme R of *finite type* is to give a direct system of ordinary Y_n-preschemes R_n of *finite type*, such that

$$R_n \simeq R_{n+1} \times_{Y_{n+1}} Y_n.$$

Moreover, for R to be an étale covering of $\widehat{X}$, it is necessary and sufficient that, for every n, R_n be an étale covering of Y_n.

That said, it is easy to see that nilpotent elements make no difference to étale coverings (SGA 1, 8.3); in other words, the base-change functor

$$\mathbf{Et}(Y_{n+1}) \longrightarrow \mathbf{Et}(Y_n)$$

is an equivalence of categories for every $n \in \mathbb{N}$. Hence:

Proposition 1.1. *With the notation introduced above, the natural functor*

$$\mathbf{Et}(\widehat{X}) \longrightarrow \mathbf{Et}(Y)$$

is an equivalence of categories (cf. SGA 1, 8.4).

2. Comparison of $\mathbf{Et}(Y)$ and $\mathbf{Et}(U)$, for varying U

We shall introduce two conditions from which the comparison theorem announced above will follow easily. Let X be a locally noetherian prescheme, and let Y be a closed subset of X. We say that the pair (X, Y) satisfies the *Lefschetz condition*, written $\mathrm{Lef}(X, Y)$, if, for every open subset U of X containing Y and every locally free coherent sheaf E on U, the natural homomorphism

$$\Gamma(U, E) \longrightarrow \Gamma(\widehat{X}, \widehat{E})$$

is an isomorphism.

We say that the pair (X, Y) satisfies the *effective Lefschetz condition*, written $\mathrm{Leff}(X, Y)$, if $\mathrm{Lef}(X, Y)$ holds and, in addition, for every locally free coherent sheaf $\mathcal{E}$ on $\widehat{X}$, there exist an open neighborhood U of Y, a locally free coherent sheaf E on U, and an isomorphism

$$\widehat{E} \simeq \mathcal{E}.$$

These conditions hold in two important examples:

Example 2.1[1]. — Let A be a noetherian ring, and let $t \in \mathfrak{r}(A)$ be an A-regular element belonging to the radical $\mathfrak{r}(A)$ of A. Assume that A is a quotient of a regular local ring and that A is complete for the t-adic topology (for example, that A is complete for the $\mathfrak{r}(A)$-adic topology). Set

$$X' = \mathrm{Spec}(A), \qquad Y' = V(t),$$

and, further, set

$$x = \mathfrak{r}(A), \qquad X = X' \setminus \{x\}, \qquad Y = Y' \setminus \{x\}.$$

Thus X is open in X' and $Y = X \cap Y'$. Then:

(i) If, for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$ (i.e. for every closed point of X), we have $\operatorname{depth} A_{\mathfrak{p}} \geq 2$, then $\mathrm{Lef}(X, Y)$ holds;

(ii) if, moreover, for every prime ideal $\mathfrak{p}$ of A such that $t \in \mathfrak{p}$ and $\dim A/\mathfrak{p} = 1$ (i.e. for every closed point of Y), we have $\operatorname{depth} A_{\mathfrak{p}} \geq 3$, then $\mathrm{Leff}(X, Y)$ holds.

Let us first prove that, for every open neighborhood U of Y in X, the complement of U in X is a union of finitely many closed points (in X). Observe that U is open in X, hence in X', and therefore $Z' = X' \setminus U$ is closed. Let I be an ideal defining Z'; it is enough to prove that A/I has dimension 1. Since

$$Z' \cap Y' = \{x\},$$

the ring $A/(I + (t))$ is Artinian, and the conclusion follows from the "Hauptidealsatz."

The *first hypothesis is equivalent to*: "for every prime ideal $\mathfrak{p}$ of A with $\mathfrak{p} \neq \mathfrak{r}(A)$, one has

$$\operatorname{depth} A_{\mathfrak{p}} \geq 3 - \dim(A/\mathfrak{p}).$$

Indeed, A is a quotient of a regular ring, so we may apply VIII, 2.3 to the prescheme X', the closed subset $\{x\}$, and the coherent sheaf $\mathcal{O}_{X'}$, observing that $c(\mathfrak{p}) = \dim(A/\mathfrak{p})$ for $\mathfrak{p} \in X = X' \setminus \{x\}$ (since x is the closed point of X').

[1] *Editor's note.* One can slightly improve (i): see Mme Raynaud (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental," *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), pp. 29–52, Corollaries I.1.4 and I.5). Condition (ii) can be improved so as to dispense with the depth conditions along Y (see Theorem 3.3 of *loc. cit.* for a precise statement). The proof of this latter point is very technical; moreover, the article cited above gives only indications of proof and refers to an earlier, detailed version published in the *Bulletin de la Société mathématique de France*.

Source oddity SGA2-X-L3357-U-VS-X-SRCDEF-001. The French authority prints $\mathfrak{p} \in U = X' - \{x\}$, although U was just quantified as an arbitrary open neighborhood and line 3345 defines $X = X' - \{x\}$. The English therefore uses X here. The French file remains unchanged; the same-edition PDF confirms only the printed U, not the repair.

Let U be an open neighborhood of Y in X, and let E be a locally free $\mathcal{O}_U$-module. Set $Z = X \setminus U$, and let $u\colon U \to X$ be the canonical immersion. We shall first prove that $u_*(E)$ is a coherent $\mathcal{O}_X$-module, or, equivalently, that $\mathcal{H}^i_Z(E')$ is coherent for $i = 0, 1$, where E' is a coherent extension of E to X. For this, we apply Theorem VIII, 2.1 to the prescheme X, the closed subset Z, and the coherent sheaf E'. It is enough to verify that, for every point $p \in U$ such that $c(p) = 1$, one has $\operatorname{depth} E'_p \geq 1$, where we have set

$$c(p) = \operatorname{codim}\bigl(\overline{\{p\}} \cap Z, \overline{\{p\}}\bigr).$$

Now suppose $p \in U$ and $c(p) = 1$, and let $\mathfrak{p}$ denote the ideal of A corresponding to p. Then $\dim(A/\mathfrak{p}) = 2$, since the complement of U is a union of finitely many closed points and A is a quotient of a regular ring. Moreover, E is locally free, so for every $p \in \operatorname{Supp} E$ one has $\operatorname{depth} E_p = \operatorname{depth} \mathcal{O}_{U,p}$. Finally, if $p \in U$ and $c(p) = 1$, then

$$\operatorname{depth} E'_p = \operatorname{depth} E_p = \operatorname{depth} \mathcal{O}_{U,p} = \operatorname{depth} A_{\mathfrak{p}} \geq 3 - 2 = 1.$$

We must now prove that the natural homomorphism

$$\Gamma(U, E) \longrightarrow \Gamma(\widehat{X}, \widehat{E}) \tag{X.0.1}$$

is an isomorphism. Set $\overline{E} = u_*(E)$. The sheaf $\overline{E}$ is coherent and has depth at least 2 at every closed point of X. It follows that $R^i f_*(\overline{E})$ is coherent for $i = 0, 1$, where $f\colon X \to X'$ denotes the canonical immersion of X into $X' = \operatorname{Spec}(A)$ (Exposé VIII). Applying IX, 1.5, we conclude that

$$\Gamma(U,\overline{E}) \longrightarrow \Gamma(\widehat{X},\widehat{\overline{E}}) \tag{X.0.2}$$

is an isomorphism, since A is complete for the t-adic topology.

There is a commutative diagram

$$\begin{tikzcd}
\Gamma(X,\overline{E}) \arrow[rr, "\simeq"] \arrow[dr, "\simeq"'] & & \Gamma(U, E) \arrow[dl] \\
& \Gamma(\widehat{X}, \widehat{E}) &
\end{tikzcd}$$

and the conclusion follows.

Now let $\mathcal{E}$ be a locally free coherent sheaf on $\widehat{X}$. If one has proved that $\mathcal{E}$ is algebraizable, that is, isomorphic to the formal completion of a coherent $\mathcal{O}_X$-module E, it is easy to see that E is locally free in a neighborhood of Y, and hence to prove $\operatorname{Leff}(X, Y)$. Let $\widehat{X'}$ be the formal spectrum of A for the t-adic topology; it is identified with the formal completion of X' along Y'. Denote by f the canonical immersion of X into X', by f' the canonical immersion of Y into Y', and by $\widehat{f}$ the morphism obtained by passing to the completions. For $\mathcal{E}$ to be algebraizable, it suffices that $\widehat{f}_*(\mathcal{E})$ be a coherent $\mathcal{O}_{\widehat{X'}}$-module, since A is complete for the t-adic topology.

Source note [SGA2-X-L3384-PUSHFORWARD-BASE-SRCDEF-001]. The French authority prints $\mathcal{O}_{\widehat{X}}$ as the coefficient ring of the pushforward. Because $\widehat{f}\colon \widehat{X} \to \widehat{X'}$, the target uses $\mathcal{O}_{\widehat{X'}}$. Only this coefficient ring is corrected.

Set $\mathcal{I} = t\mathcal{O}_{\widehat{X}}$; this is an ideal of definition of $\widehat{X}$.

For every $n \geq 0$, set

$$\mathcal{E}_n = \mathcal{E}/\mathcal{I}^{n+1}\mathcal{E}.$$

At every closed point $y \in Y$, the depth of $\mathcal{E}_0$ is at least two; indeed, t is an A-regular element, and therefore
$$\operatorname{depth} \mathcal{O}_{Y,y} = \operatorname{depth} \mathcal{O}_{X,y} - 1 \geq 2.$$

Source note [SGA2-X-L3386-Y0-VS-Y-SRCDEF-001]. The French authority prints $\mathcal{O}_{Y_0,y}$, but Y_0 is undefined in this argument. Since $Y = V(t) \cap X$, the regular-element quotient gives $\mathcal{O}_{Y,y}$.

It follows that $\widehat{f}_*(\mathcal{E})$ is coherent (IX 2.3). Q.E.D.

Example 2.2 (This will make it possible to compare the fundamental group of a projective variety with that of a hyperplane section).

Let K be a field, and let X be a proper K-prescheme. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_X$-module. Let $t \in \Gamma(X, \mathcal{L})$ be an $\mathcal{O}_X$-regular element; this means that, for every open subset U and every isomorphism
$$u\colon \mathcal{L}|_U \xrightarrow{\sim} \mathcal{O}_U,$$
the section $u(t)$ is a non-zero-divisor in $\mathcal{O}_U$ (a condition that does not depend on u). Let $Y = V(t)$ be the subscheme of X defined by the equation $t = 0$[2]. Then:

(i) If, for every closed point x of X, one has $\operatorname{depth} \mathcal{O}_{X,x} \geq 2$, then $\operatorname{Lef}(X, Y)$ holds;

(ii) if, moreover, for every closed point $y \in Y$, one has $\operatorname{depth} \mathcal{O}_{X,y} \geq 3$, then $\operatorname{Leff}(X, Y)$ holds.

This example will be treated in detail in Exposé XII.

Let S be a prescheme. It is known (EGA II, 6.1.2) that the functor which assigns to every finite flat covering $r\colon R \to S$ the $\mathcal{O}_S$-algebra $r_*(\mathcal{O}_R)$ induces an equivalence between the category of finite flat coverings of S and the category of coherent locally free $\mathcal{O}_S$-algebras.

Source note [SGA2-X-L3402-OS-VS-OX-SRCDEF-001]. The French authority prints $\mathcal{O}_X$ in both coefficient-ring positions. Since S is arbitrary and $r\colon R \to S$, the direct image and the equivalent algebra category are over $\mathcal{O}_S$.

Let U be an open neighborhood of Y, and let $r\colon R \to U$ be a finite flat covering of U. Let $\widehat{R}$ be the finite flat covering of $\widehat{X}$ obtained from it by base change. Then
$$\widehat{r}_*(\mathcal{O}_{\widehat{R}}) \simeq \widehat{r_*(\mathcal{O}_R)}.$$

Assume now that $\operatorname{Lef}(X, Y)$ holds. This implies that, for every U, the inverse-image functor
$$\mathsf{L}(U) \longrightarrow \mathsf{L}(\widehat{X})$$
is fully faithful. Indeed, let E and F be two coherent locally free $\mathcal{O}_U$-modules; $\mathcal{H}om(E, F)$ is likewise coherent and locally free.

Source note [SGA2-X-L3408-DUPLICATE-HOM-SRCDEF-001]. The French authority prints the preceding sentence twice consecutively. The target states the identical assertion once.

By hypothesis, the natural map
$$\Gamma(U, \mathcal{H}om(E, F)) \longrightarrow \Gamma(\widehat{X}, \widehat{\mathcal{H}om(E, F)})$$
is an isomorphism, which proves the assertion, since $\mathcal{H}om$ commutes with formal completion when all the modules involved are locally free. Formal completion also commutes with tensor products. It follows that the functor which assigns to every coherent locally free $\mathcal{O}_U$-algebra $\mathcal{A}$ the $\mathcal{O}_{\widehat{X}}$-algebra $\widehat{\mathcal{A}}$ is fully faithful. More precisely, if E is a coherent locally free $\mathcal{O}_U$-module,

[2] *Editor's note.* Condition (ii) is superfluous; see the editor's note to Example 2.1 (original running p. 90).

commutative $\mathcal{O}_U$-algebra structures on E correspond bijectively to commutative $\mathcal{O}_{\widehat{X}}$-algebra structures on $\widehat{E}$.

Source note [SGA2-X-L3412-BIJECTION-MISSING-FIRST-SIDE-SRCDEF-001]. The French authority introduces a bijective correspondence but prints only its $\mathcal{O}_{\widehat{X}}$ side. The target restores the omitted $\mathcal{O}_U$-algebra structures on E, as required by the preceding fully faithful tensor-compatible completion functor.

Proposition 2.3. Let X be a locally noetherian prescheme, and let Y be a closed subset of X. Let $\widehat{X}$ be the formal completion of X along Y. For every open subset U of X containing Y, let L_U (respectively P_U, respectively E_U) denote the functor that assigns to every locally free coherent $\mathcal{O}_U$-module (respectively every finite flat covering of U, respectively every étale covering of U) its pullback along $\widehat{X} \to X$.

(i) If $\mathrm{Lef}(X, Y)$ holds, then, for every open neighborhood U of Y, the functors L_U, P_U, and E_U are fully faithful.

(ii) If $\mathrm{Leff}(X, Y)$ holds, then, for every locally free coherent $\mathcal{O}_{\widehat{X}}$-module $\mathcal{E}$, there are an open neighborhood U of Y and a locally free coherent $\mathcal{O}_U$-module E such that

$$L_U(E) \simeq \mathcal{E}.$$

The analogous assertions hold for finite flat coverings and for étale coverings, with P_U and E_U, respectively.

Proof. Part (i) has already been proved.

Part (ii) follows from (i) and the hypothesis, at least for L_U and P_U. Moreover, if R is an étale covering of $\widehat{X}$, there are an open neighborhood U of Y in X and a finite flat covering R' of U such that

$$\widehat{R'} \simeq R.$$

By restriction this gives a covering R'' of Y, which is étale by Proposition 1.1. Hence the restriction of R' to some neighborhood U' of Y is étale. $\square$

Corollary 2.4. Suppose that $\mathrm{Lef}(X, Y)$ holds. A finite flat covering R of an open neighborhood U of Y is connected if and only if $R \times_U \widehat{X}$ is connected. In particular, Y is connected if and only if U is connected, and this is also equivalent to X being connected.

Proof. Indeed, a locally ringed space $(X, \mathcal{O}_X)$ is connected if and only if $\Gamma(X, \mathcal{O}_X)$ is not a direct product of two nonzero rings. But, by $\mathrm{Lef}(X, Y)$, one has

$$\Gamma(U, r_*\mathcal{O}_R) \simeq \Gamma(\widehat{X}, \widehat{r}_*\mathcal{O}_{\widehat{R}}).$$

Corollary 2.5. Suppose that $\mathrm{Lef}(X, Y)$ holds. Then, for every open neighborhood U of Y, the functor

$$\mathbf{Et}(U) \longrightarrow \mathbf{Et}(Y)$$

is fully faithful. Suppose that $\mathrm{Leff}(X, Y)$ holds. Then, for every étale covering R of Y, there are an open neighborhood U of Y and a covering R' of U such that $R' \times_U Y \simeq R$.

Corollary 2.6[(3)]. Suppose that $\mathrm{Lef}(X, Y)$ holds and that Y is connected. Then every open neighborhood U of Y is connected, and the natural homomorphism $\pi_1(Y) \longrightarrow \pi_1(U)$ is surjective. Suppose, moreover, that $\mathrm{Leff}(X, Y)$ holds. Then the natural homomorphism

$$\pi_1(Y) \longrightarrow \varprojlim_U \pi_1(U)$$

is an isomorphism. (N.B. A "base point" in Y is understood to have been chosen and is also taken as a base point in X for the definition of the fundamental groups.)

All of this follows immediately from Propositions 1.1 and 2.3.

Editor's note (3)

Combined with Proposition 3.3 and criteria XII.2.4 and XII.3.4, this yields the following relative Lefschetz theorem. Let $f\colon X \to S$ be a projective and flat morphism of connected noetherian schemes, and let D be an effective relative Cartier divisor on X that is relatively ample. If, for every $s \in S$, the depth of X_s at every closed point is ≥ 2, then D is connected and, for every open subset U of X containing D, the map

$$i_U\colon \pi_1(D) \longrightarrow \pi_1(U)$$

is surjective. If, in addition, the depth of X_s at every closed point of D_s is ≥ 3 and the local rings of X at its closed points are pure—for example, are complete intersections (cf. X, 3.4)—then i_X is an isomorphism.

Cf. J.-B. Bost, "Lefschetz theorem for Arithmetic Surfaces," *Ann. Sci. Éc. Norm. Sup. (4)* **32** (1999), pp. 241–312, Theorems 1.1 and 2.1.

When X is merely a smooth projective surface that is geometrically connected over a field, D is always connected and

$$\pi_1(D) \longrightarrow \pi_1(U)$$

is surjective (where U is an open subset containing D), provided only that D is nef and has self-intersection > 0; cf. *loc. cit.*, Theorem 2.3, and also Theorem 2.4 for surfaces assumed only normal and complete.

In the case of an arithmetic surface (normal and quasi-projective) X over a ring of integers $\mathcal{O}_K$, Bost, improving results of Y. Ihara (Y. Ihara, "Horizontal divisors on arithmetic surfaces associated with Belyĭ uniformizations," in *The Grothendieck theory of dessins d'enfants (Luminy, 1993)*, London Math. Soc. Lect. Note Series, vol. 200, Cambridge Univ. Press, Cambridge, 1994, pp. 245–254, or *loc. cit.*, Corollary 7.2), proved that if a point $P \in X(\mathcal{O}_K)$, which plays the role of the divisor D in the geometric situation, satisfies certain positivity conditions, then the map

$$\pi_1(X) \longrightarrow \pi_1(\operatorname{Spec}\mathcal{O}_K)$$

induced by the projection is an isomorphism whose inverse is the map

$$\pi_1(\operatorname{Spec}\mathcal{O}_K) \longrightarrow \pi_1(X)$$

induced by P; cf. *loc. cit.*, Theorem 1.2.

Accepted source-normalization note (SGA2-X-L3456-AU-DESSUS-NORMALIZATION-001@1). The corrected French control selects au-dessus from `\sisi{au dessus}{au-dessus}`; the target renders this as "over." This is an explicit source normalization, not a source defect.

3. Comparison of $\pi_1(X)$ and $\pi_1(U)$

Definition 3.1. Let X be a prescheme and Z a closed subset of X. Set $U = X \setminus Z$. We say that the pair (X, Z) is *pure* if, for every open subset V of X, the functor

$$\begin{aligned} \mathbf{Et}(V) &\longrightarrow \mathbf{Et}(V \cap U), \\ V' &\longmapsto V' \times_V (V \cap U) \end{aligned}$$

is an equivalence of categories*.

*For a notion that is more satisfactory in certain respects, cf. the commentary XIV, 1.6(d).

Accepted source-normalization note (SGA2-X-L3464-MAPSTO-NORMALIZATION-001@1). The corrected French control selects `\mto` from `\sisi{\rightsquigarrow}{\mto}`; the target renders the object map with $\longmapsto$. This is an explicit source normalization, not a source defect.

Definition 3.2. Let A be a noetherian local ring. Set $X = \operatorname{Spec} A$. Let $\mathfrak{r}(A)$ be the radical of A, and let $x = \mathfrak{r}(A)$ be the closed point of X. We say that A is *pure* if the pair $(X, \{x\})$ is pure.

We leave to the reader the task of not proving the following proposition:

Proposition 3.3. Let X be a locally noetherian prescheme and let Z be a closed subset of X. The pair (X, Z) is pure if and only if, for every $z \in Z$, the local ring $\mathcal{O}_{X,z}$ is pure**.

Source-defect note (SGA2-X-L3477-FERME-VS-FERMEE-SRCDEF-001). French line 3477 reads *une partie fermé de X*; the masculine adjective *fermé* disagrees with the feminine noun *partie*. The target renders the intended sense as "a closed subset of X." The admitted French control is unchanged.

That said, the following theorem is the main result of this section:

Theorem 3.4 (Purity theorem)(4).

(i) A regular noetherian local ring of dimension ≥ 2 is pure (Zariski–Nagata purity theorem).

(ii) A noetherian local ring of dimension ≥ 3 that is a complete intersection is pure.

Recall that a local ring A is said to be a *complete intersection* if and only if there exist a *regular* noetherian local ring B and a B-regular sequence $(t_1, \ldots, t_k)$ of elements of the radical $\mathfrak{r}(B)$ of B such that
$$A \simeq B/(t_1, \ldots, t_k).$$

In this connection, let us note that it would be less ambiguous to say that A is an *absolute* complete intersection, in contrast to the situation already encountered, in which X is a locally noetherian prescheme (not necessarily regular) and Y is a closed subset of X that is said to be "locally a set-theoretic complete intersection *in* X."

Let us first prove a few lemmas.

Lemma 3.5. — *Let X be a locally noetherian prescheme, and let U be an open subset of X. Set $Z = X \setminus U$. Let $i\colon U \to X$ be the canonical immersion of U into X. The following conditions are equivalent:*

(i) *For every open subset V of X, if $V' = V \cap U$, the restriction functor $F \longmapsto F|_{V'}$ from the category of locally free coherent $\mathcal{O}_V$-modules to the category of locally free coherent $\mathcal{O}_{V'}$-modules is fully faithful;*

(ii) *the natural homomorphism $\mathcal{O}_X \to i_*\mathcal{O}_U$ is an isomorphism;*

(iii) *for every $z \in Z$, one has* $\operatorname{prof} \mathcal{O}_{X,z} \geq 2$.

We have already seen (III 3.3) the equivalence of (ii) and (iii). Let us show that (ii) implies (i). Let F and G be two locally free coherent $\mathcal{O}_V$-Modules. Then $\mathcal{H}om(F, G)$ is also locally free and coherent, so
$$\mathcal{H}om(F, G) \longrightarrow i_*(\mathcal{H}om(F_{|V'}, G_{|V'}))$$

** Compare the noncommutative case of XIV, 1.8, whose proof is essentially the same as that of Proposition 3.3.

(4) N.D.E.: For the history of the methods employed, see Grothendieck's letter of October 1, 1961 to Serre, *Correspondance Grothendieck-Serre*, edited by Pierre Colmez and Jean-Pierre Serre, Documents Mathématiques, vol. 2, Société Mathématique de France, Paris, 2001.

is an isomorphism, and therefore

$$\mathrm{Hom}(F, G) \simeq \mathrm{Hom}(F_{|V'}, G_{|V'}).$$

Conversely, take $F = G = \mathcal{O}_X$ and apply (i) to every open subset V of X.

Here is a useful "descent lemma":

Lemma 3.6. — *Let X be a locally noetherian prescheme, and let Z be a closed subset of X. Set $U = X \setminus Z$. Suppose that the homomorphism $\mathcal{O}_X \to i_*\mathcal{O}_U$ is an isomorphism. Let $f\colon X_1 \to X$ be a faithfully flat and quasi-compact morphism. Set $Z_1 = f^{-1}(Z)$. If the pair (X_1, Z_1) is pure, then so is (X, Z).*

Proof of Lemma 3.6. Observe that the hypothesis $\mathcal{O}_X \simeq i_*\mathcal{O}_U$ is preserved under flat base extension, since i is quasi-compact and, in this case, formation of the direct image commutes with inverse image. This hypothesis implies that the functor

$$\mathbf{Et}(V) \longrightarrow \mathbf{Et}(U \cap V)$$

defined by

$$V' \longmapsto V' \times_V (V \cap U)$$

is fully faithful, as follows from the interpretation of an étale covering in terms of a coherent locally free algebra. It remains to prove effectivity. One may, for example, form the fiber square X_2 and the fiber cube X_3 of X_1 over X, and observe that a faithfully flat and quasi-compact morphism is a morphism of universal effective descent for the fibered category of étale coverings over the category of preschemes. The conclusion then follows formally[(*)].

Remark 3.7. — *We proved along the way that if $\mathcal{O}_X \to i_*\mathcal{O}_U$ is an isomorphism, then X is connected if and only if U is connected, and then $\pi_1(U) \to \pi_1(X)$ is surjective.*

Corollary 3.8. — *Let A be a noetherian local ring. Assume that* $\operatorname{depth} A \geq 2$. *If $\widehat{A}$ is pure, then A is pure.*

Proof. This follows from Lemma 3.5 and Lemma 3.6.

The following lemma is the essential point in the proof of the purity theorem:

Lemma 3.9. — *Let A be a noetherian local ring, and let $t \in \mathfrak{r}(A)$ be an A-regular element. Suppose that A is complete for the t-adic topology and is moreover a quotient of a regular local ring (for example, if A is complete). Set $B = A/tA$.*

(i) *If, for every prime ideal $\mathfrak{p}$ of A such that* $\dim A/\mathfrak{p} = 1$, *one has* $\operatorname{depth} A_{\mathfrak{p}} \geq 2$, *then the purity of B implies the purity of A.*

(ii) *If, for every prime ideal $\mathfrak{p}$ of A such that* $\dim A/\mathfrak{p} = 1$, *one has* $\operatorname{depth} A_{\mathfrak{p}} \geq 2$, *if $A_{\mathfrak{p}}$ is pure whenever $t \notin \mathfrak{p}$, and if*[(5)] $\operatorname{depth} A_{\mathfrak{p}} \geq 3$ *whenever $t \in \mathfrak{p}$, then the purity of A implies the purity of B.*

> ***Source note [SGA2-X-L3551-MISSING-EST-SRCDEF-001].*** *The French authority prints* si $A_{\mathfrak{p}}$ pur lorsque $t \notin \mathfrak{p}$, *omitting the finite copula* est. *The English supplies "is" in the unambiguous intended condition. The French authority remains unchanged.*

(*) Cf. J. Giraud, *Méthode de la descente*, Mémoire no. 2 du *Bulletin de la Société Mathématique de France* (1964).

(5) N.D.E.: This last condition can be improved; cf. the editor's note (1) on p. 90.

Let $X' = \mathrm{Spec}(A)$ and let $Y' = V(t)$, which we identify with the spectrum of B. Let $x = \mathfrak{r}(A)$, and set $X = X' \setminus \{x\}$ and $Y = Y' \setminus \{x\} = X \cap Y'$. Denote by $\widehat{X'}$ the formal spectrum of A for the t-adic topology; it is identified with the formal completion of X' along Y'.

Since A is complete for the t-adic topology, observe that $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(\widehat{X'})$ is an equivalence of categories. Likewise, $\mathbf{Et}(\widehat{X'}) \longrightarrow \mathbf{Et}(Y')$ is an equivalence by Proposition 1.1; hence $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(Y')$ is an equivalence of categories.

We prove (i). Consider the diagrams

$$\begin{tikzcd} X' & X \arrow[l] \\ Y' \arrow[u] & Y \arrow[l] \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Et}(X') \arrow[r, "a"] \arrow[d, "c"'] & \mathbf{Et}(X) \arrow[d, "b"] \\ \mathbf{Et}(Y') \arrow[r, "d"] & \mathbf{Et}(Y) \end{tikzcd}$$

We have just seen that c is an equivalence; d is also an equivalence by the hypothesis that B is pure; and finally b is fully faithful, as was seen in Example 2.1; cf. Proposition 2.3(i).

We prove (ii). This time suppose that A is pure, so a is an equivalence; likewise c. Let us show that b is an equivalence. By Example 2.1, we know that $\mathrm{Leff}(X, Y)$ holds, so b is already fully faithful; let us prove that it is essentially surjective. Apply Proposition 2.3(ii), noting that if U is an open neighborhood of Y in X, then $X \setminus U$ is a union of finitely many closed points. The pair $(X, X \setminus U)$ is therefore pure by Proposition 3.3, since at any such point $\mathfrak{p}$ one has $\mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}$, which is pure by hypothesis. This proves the assertion.

Proof of the purity theorem. We first prove (i) by induction on the dimension. Let A be a noetherian local ring of dimension 2. Set $X' = \mathrm{Spec}(A)$, $x = \mathfrak{r}(A)$, and $X = X' \setminus \{x\}$. We have $\operatorname{depth} A = 2$. We may therefore apply Lemma 3.5 to the pair $(X', \{x\})$, and hence $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(X)$ is fully faithful.

Now let $r\colon R \longrightarrow X$ be an étale covering defined by the coherent, locally free, étale $\mathcal{O}_X$-algebra $\mathcal{A} = r_*(\mathcal{O}_R)$. Let $i\colon X \longrightarrow X'$ denote the canonical immersion. I claim that $i_*(\mathcal{A}) = \mathcal{B}$ is a coherent $\mathcal{O}_{X'}$-algebra. Indeed, it suffices to apply the "finiteness theorem," VIII, 2.3. I also claim that this algebra has depth at least 2 at x: it is the direct image of an $\mathcal{O}_X$-module, with $X = X' \setminus \{x\}$. Since A is a regular ring of dimension 2, we have

$$\operatorname{pd} \mathcal{B} + \operatorname{depth} \mathcal{B} = \dim A = 2,$$

where $\operatorname{pd} \mathcal{B}$ denotes the projective dimension of $\mathcal{B}$. Thus $\operatorname{pd} \mathcal{B} = 0$, so $\mathcal{B}$ is projective and therefore free. Consequently, $\mathcal{B}$ defines a finite flat covering of $X' = \mathrm{Spec}(A)$. The set of points of X' at which this covering is not étale is a closed subset of X' whose defining ideal is principal, namely the discriminant ideal of $\mathcal{B}/A$. By construction this closed subset is contained in $x = \mathfrak{r}(A)$; it is therefore empty because $\dim A = 2$.

Let A now be a regular noetherian local ring with $\dim A = n \geq 3$. Suppose that (i) has been proved for rings of dimension less than n. To prove that A is pure, we may assume by Corollary 3.8 that A is complete. Choose $t \in \mathfrak{r}(A)$ whose image in $\mathfrak{r}(A)/\mathfrak{r}(A)^2$ is nonzero. Then $B = A/tA$ is a regular noetherian local ring of dimension $n-1$, and is therefore pure because $n - 1 \geq 2$. The conclusion follows from Lemma 3.9(i), which is applicable because A is complete.

Let us prove (ii). Let A be a noetherian local ring of dimension at least 3. Suppose that there exist a regular noetherian local ring B and a B-sequence $(t_1, \ldots, t_k)$ such that

$$A \simeq B/(t_1, \ldots, t_k).$$

We prove that A is pure by induction on k. If $k = 0$, this follows from (i). Suppose that $k \geq 1$ and that the result has been proved for $k' < k$. By Corollary 3.8, we may assume that A, and hence also B, is complete. Set

$$C = B/(t_1, \ldots, t_{k-1}).$$

Then $A \simeq C/t_kC$, and t_k is C-regular. By the induction hypothesis, C is pure; it therefore suffices to verify that Lemma 3.9(ii) is applicable. In the notation of that lemma, its A and B are here C and A, respectively. We have $\dim C \geq 4$; hence, for every prime ideal $\mathfrak{p}$ of C such that $\dim(C/\mathfrak{p}) = 1$, one has $\operatorname{depth} C_{\mathfrak{p}} \geq 3$. Moreover, $C_{\mathfrak{p}}$ is a complete intersection with $k' \leq k-1$, and is therefore pure by the induction hypothesis.

Theorem 3.10. — *Let X be a locally noetherian prescheme, and let Y be a closed subset of X. Suppose that* $\mathrm{Leff}(X, Y)$ *holds (cf. Examples 2.1 and 2.2). Suppose further that, for every open neighborhood U of Y and every $x \in X \setminus U$, the local ring $\mathcal{O}_{X,x}$ is either regular of dimension ≥ 2 or a complete intersection of dimension ≥ 3. Then*

$$\pi_0(Y) \longrightarrow \pi_0(X)$$

is a bijection, and, if X is connected,

$$\pi_1(Y) \longrightarrow \pi_1(X)$$

is an isomorphism.

There is nothing more to prove. Observe that in the two cited Examples 2.1 and 2.2, the complement of U is a union of finitely many closed points; it follows that the hypothesis on the dimension of $\mathcal{O}_{X,x}$ is not farcical.

Exposé XI

Application to the Picard Group

This exposé follows the pattern of the preceding one, but this time the result of no. 1 is weaker.

Throughout this exposé, X will denote a locally noetherian prescheme; $\mathcal{I}$, a quasi-coherent ideal of $\mathcal{O}_X$ (so $Y = V(\mathcal{I})$ is a closed subset of X); U, a variable open neighborhood of Y in X; and $\widehat{X}$, the formal completion of X along Y.

For every ringed space $(Z, \mathcal{O}_Z)$, let $\mathbf{P}(Z)$ denote the category of invertible $\mathcal{O}_Z$-modules, that is, modules locally free of rank 1, and let $\operatorname{Pic}(Z)$ denote the group of isomorphism classes of invertible modules on Z.

XI.1 Comparison of $\operatorname{Pic}(\widehat{X})$ and $\operatorname{Pic}(Y)$

For every $n \in \mathbf{N}$, set

$$X_n = (Y, \mathcal{O}_X/\mathcal{I}^{n+1}) \qquad \text{and} \qquad P_n = \mathcal{I}^{n+1}/\mathcal{I}^{n+2}.$$

The sequence of sheaves of abelian groups on Y

$$0 \longrightarrow P_n \xrightarrow{u} \mathcal{O}^*_{X_{n+1}} \xrightarrow{v} \mathcal{O}^*_{X_n} \longrightarrow 1 \tag{XI.1.1}$$

is *exact*. More precisely, the group structure on P_n is the additive structure, $u(x) = 1 + x$ for every $x \in P_n$, and v is the homomorphism induced by the injection $\mathcal{I}^{n+2} \longrightarrow \mathcal{I}^{n+1}$. To see that v is surjective, observe that, for every $y \in Y$, $\mathcal{O}_{X_n,y}$ is a local ring, the quotient of $\mathcal{O}_{X_{n+1},y}$ by a nilpotent ideal; the rest is equally trivial. From (XI.1.1) we deduce an exact cohomology sequence:

$$\mathrm{H}^1(Y, P_n) \xrightarrow{u^1} \mathrm{H}^1(Y, \mathcal{O}^*_{X_{n+1}}) \xrightarrow{v^1} \mathrm{H}^1(Y, \mathcal{O}^*_{X_n}) \xrightarrow{d} \mathrm{H}^2(Y, P_n). \tag{$*$}$$

Moreover, for every $n \in \mathbf{N}$, we know how to identify $\operatorname{Pic}(X_n)$ with $\mathrm{H}^1(Y, \mathcal{O}^*_{X_n})$. Furthermore, if E is an invertible $\mathcal{O}_{X_{n+1}}$-module corresponding to a cohomology class $c(E)$, the cohomology 6
class corresponding to the pullback of E to X_n is equal to $v^1(c(E))$. Hence the following proposition:

Proposition XI.1.1. *Retain the notation introduced above. Let $p \in \mathbf{N}$. The map*

$$\operatorname{Pic}(\widehat{X}) \longrightarrow \operatorname{Pic}(X_n)$$

has the following properties:

Source note [SGA2-XI-L3609-PIC-YN-VS-XN-SRCDEF-001]. *The French prints Y_n. Exposé XI locally defines and uses X_n, while Exposé X used Y_n for the same infinitesimal thickening. We therefore read X_n here.*

(i) it is injective for $n \geq p$ *if* $\mathrm{H}^1(Y, P_n) = 0$ *for every* $n \geq p$*;*

(ii) it is an isomorphism for $n \geq p$ *if* $\mathrm{H}^i(Y, P_n) = 0$ *for every* $n \geq p$ *and* $i = 1, 2$.

Of course, the exact sequence $(*)$ contains more information than the proposition above. The reader will have noticed that nothing has been said about the functor $\mathbf{P}(\widehat{X}) \longrightarrow \mathbf{P}(Y)$. Given two invertible $\mathcal{O}_{\widehat{X}}$-modules E, F, $H = \mathcal{H}om(E, F)$ is also invertible. If reduction modulo $\mathcal{I}^{n+1}$ is denoted by a subscript n, we obtain an exact sequence:

$$0 \longrightarrow H_0 \otimes P_n \longrightarrow \mathcal{H}om(E_{n+1}, F_{n+1}) \longrightarrow \mathcal{H}om(E_n, F_n) \longrightarrow 0.$$

Hence there is an exact cohomology sequence that we shall not write down and whose interpretation is evident; this observation can be used to study the functor $\mathbf{P}$.

XI.2 Comparison of $\mathrm{Pic}(X)$ and $\mathrm{Pic}(\widehat{X})$

The reader will find in Exposé X, no. 2, the proof of the following:

Proposition XI.2.1. *Suppose that* $\mathrm{Lef}(X, Y)$ *holds; then, for every open neighborhood* U *of* Y *in* X*, the functor*

$$\mathbf{P}(U) \to \mathbf{P}(\hat{X}) \tag{XI.2.2}$$

is fully faithful, and the map

$$\mathrm{Pic}(U) \to \mathrm{Pic}(\hat{X}) \tag{XI.2.3}$$

is therefore injective. If $\mathrm{Leff}(X, Y)$ *holds, the map (XI.2.4) is an isomorphism:* **7**

$$\varinjlim_{U} \mathrm{Pic}(U) \to \mathrm{Pic}(\hat{X}). \tag{XI.2.4}$$

Corollary XI.2.2. *Suppose that* $\mathrm{Lef}(X, Y)$ *holds and that, for every integer* $n \geq p$*, one has* $\mathrm{H}^1(Y, P_n) = 0$*; then, for every open subset* $U \supset Y$*, the maps*

$$\mathrm{Pic}(X) \to \mathrm{Pic}(U) \to \mathrm{Pic}(X_n)$$

are injective if $n \geq p$*. If* $\mathrm{Leff}(X, Y)$ *holds and if, moreover, for every integer* $n \geq p$*, one has* $\mathrm{H}^i(Y, P_n) = 0$ *for* $i = 1$ *and* $i = 2$*, then the map*

$$\varinjlim_{U} \mathrm{Pic}(U) \to \mathrm{Pic}(X_n)$$

is an isomorphism for $n \geq p$*.*

XI.3 Comparison of $\mathbf{P}(X)$ and $\mathbf{P}(U)$

A definition:

[**.** *remark]Definition 3.1*$^{(*)}$ *Let* X *be a prescheme and let* Z *be a closed subset of* X*. Put* $U = X - Z$*. We say that* X *is* parafactorial *at the points of* Z *if, for every open subset* V *of* X*, the functor* $\mathbf{P}(V) \to \mathbf{P}(V \cap U)$ *is an equivalence of categories. We also say that the pair* (X, Z) *is parafactorial.*

Recall that $\mathbf{P}(Z)$ denotes the category of locally free modules of rank 1 on Z.

Definition XI.3.2. A noetherian local ring is said to be *parafactorial* if the pair $(\mathrm{Spec}(A), \{\mathfrak{r}(A)\})$ is parafactorial.

One proves the following proposition, which shows that the notion is "pointwise":

Proposition XI.3.3. *Suppose that X is locally noetherian. The pair (X, Z) is parafactorial if and only if, for every $z \in Z$, the local ring $\mathcal{O}_{X,z}$ is parafactorial.* 8

Note that "parafactorial"includes "fully faithful". As in Lemma 3.5 of Exposé X, one proves the following:

Lemma XI.3.4. *If X is a locally noetherian prescheme and if $Z = X - U$ is a closed subset of X, the following conditions are equivalent:*

(i) for every open subset V of X, the functor $\mathbf{P}(V) \to \mathbf{P}(V \cap U)$ is fully faithful;

(ii) the homomorphism $\mathcal{O}_X \to i_*(\mathcal{O}_U)$ is an isomorphism;

(iii) for every $z \in Z$, one has $\operatorname{prof}(\mathcal{O}_{X,z}) \geq 2$.

Thus "parafactorial"means that the conditions of XI.3.4 are satisfied and that, for every open subset V of X, the homomorphism $\operatorname{Pic}(V) \to \operatorname{Pic}(V \cap U)$ is surjective. In particular, if X is the spectrum of a noetherian local ring, one obtains:

Proposition XI.3.5. *Let A be a noetherian local ring; it is parafactorial if and only if $\operatorname{prof} A \geq 2$ and $\operatorname{Pic}(X' - \{x\}) = 0$, where $X' = \operatorname{Spec}(A)$ and x is the unique closed point of X'.*

Notice that a local ring of dimension ≤ 1 is never parafactorial, since its depth is ≤ 1. Thus *factorial does not imply parafactorial*; the implication does, however, hold for noetherian local rings of dimension ≥ 2, as we shall see below.

Lemma XI.3.6. *Let X be a locally noetherian prescheme and let Z be a closed subset of X. Let $f\colon X_1 \to X$ be a faithfully flat and quasi-compact morphism. Put $Z_1 = f^{-1}(Z)$. If (X_1, Z_1) is parafactorial, then (X, Z) is parafactorial.*

First note that, if $i\colon (X - Z) \to X$ denotes the canonical immersion of $U = X - Z$ 9
into X, the formation of the direct image under i of a quasi-coherent $\mathcal{O}_U$-module commutes with base change by f, since f is flat. It is therefore equivalent to assume the equivalent conditions of Lemma XI.3.5 for (X, Z) or for (X_1, Z_1), since f is a morphism of descent for the category of quasi-coherent sheaves. It remains to prove that, for every open subset V of X, $\operatorname{Pic}(V) \to \operatorname{Pic}(V \cap U)$ is surjective. We make the base change $V \to X$, which changes nothing (*sic*), and we are reduced to $V = X$. We then note that, if $\mathcal{L}$ is an invertible $\mathcal{O}_U$-module and if $\mathcal{L}$ admits a locally free extension, this extension is isomorphic to $i_*(\mathcal{L})$, by what we have just seen. It remains to prove that $i_*(\mathcal{L})$ is invertible. Using once again the fact that direct image under i commutes with flat base change, and that "locally free of rank 1"is a property that descends under faithfully flat and quasi-compact morphisms, we are done.

Corollary XI.3.7. *Let A be a noetherian local ring; if $\hat{A}$ is parafactorial, then A is parafactorial.*

One should not believe that if A is parafactorial, then $\hat{A}$ is parafactorial[*].

Before turning to the statement of the main theorem of this no. , let us make the connection with divisor theory and the notion of a factorial ring[*].

[*]For a more detailed study of the notion of parafactoriality, and for the proof of XI.3.3, cf. EGA IV 21.13, 21.14.

[*]*Editor's note.* For a precise study of the relationship between the factoriality of A and that of its completion, see (Heitmann R., "Characterization of completions of unique factorization domains", *Trans. Amer. Math. Soc.* **337** (1993), no. 1, pp. 379–387).

[*]For the generalities that follow, cf. also EGA IV 21.

Let X be a *noetherian* and *normal* prescheme. Let $Z^1(X)$ be the free abelian group generated by the $x \in X$ such that $\dim \mathcal{O}_{X,x} = 1$. The local ring at such a point is a discrete valuation ring. We denote the corresponding normalized valuation by v_x. Let $K(X)$ be the ring of rational functions on X, and let

$$p\colon K(X)^* \to Z^1(X)$$

be the map that associates to every $f \in K(X)^*$ the codimension-one cycle:

$$(f) = \sum_{x\in X,\, \dim \mathcal{O}_{X,x}=1} v_x(f)\cdot x.$$

The image of p is denoted by $P(X)$, and its elements are called *principal divisors*$^{(*)}$. We set **10**

$$\mathrm{Cl}(X) = Z^1(X)/P(X).$$

Let $Z'^1(X)$ be the subgroup of $Z^1(X)$ whose elements are the *locally principal divisors*. We know that

$$\mathrm{Pic}(X) \simeq Z'^1(X)/P(X),$$

and consequently $\mathrm{Pic}(X)$ is identified with a subgroup of $\mathrm{Cl}(X)$.

Note that if U is a dense open subset of X, then $K(X) \to K(U)$ is an isomorphism, and that if $\mathrm{codim}(X-U, X) \geq 2$, i.e. if every $x \in X$ such that $\dim \mathcal{O}_{X,x} \leq 1$ belongs to U, the homomorphism $Z^1(X) \to Z^1(U)$, and consequently $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$, is also an isomorphism. Finally, if every $x \in U$ is factorial, i.e. if $\mathcal{O}_{X,x}$ is factorial, then $Z^1(U) = Z'^1(U)$, and hence $\mathrm{Pic}(U) \simeq \mathrm{Cl}(U)$.

Proposition XI.3.8. *Let X be a noetherian and normal prescheme. Let $(U_i)_{i\in I}$ be a family of open subsets of X such that:*

a) the U_i form a filter base$^{(*)}$*;*

b) if we set $Y_i = X - U_i$, then $\mathrm{codim}(Y_i, X) \geq 2$ *for every i,*

c) if $x \in U_i$ for every $i \in I$, then $\mathcal{O}_{X,x}$ is factorial.

Then there is an isomorphism:

$$\varinjlim_{i\in I} \mathrm{Pic}(U_i) \xrightarrow{\approx} \mathrm{Cl}(X).$$

Note that b) implies that every $x \in X$ such that $\dim \mathcal{O}_{X,x} \leq 1$ belongs to U_i for every i. Thus the U_i are dense, and moreover the homomorphism $Z^1(U_i) \to Z^1(X)$ is an isomorphism, as is $K(U_i) \to K(X)$. Thus $\mathrm{Pic}(U) \subset \mathrm{Cl}(U_i) \simeq \mathrm{Cl}(X)$. To prove the desired assertion, it is therefore enough to show that every $D \in Z^1(X)$ belongs to $Z'^1(U_i)$ for a suitable i. It is enough to do this for irreducible positive "divisors". Thus let $x \in X$ be such that $\dim \mathcal{O}_{X,x} = 1$. It **11**
is enough to prove that there exists an $i \in I$ such that $\overline{\{x\}}$ is locally principal at the points of U_i. Let $\mathcal{I}$ be the largest ideal of definition of the closed subset $\overline{\{x\}}$. The set of points in a neighborhood of which $\mathcal{I}$ is free is an open subset U. Now $U \supset \bigcap_{i\in I} U_i$ by c). If we set $Y = X - U$, then $Y \subset \bigcup_{i\in I} Y_i$ with $Y_i = X - U_i$; now Y is closed, and therefore has only *finitely* many generic points, so it is contained in the union of finitely many Y_i, and hence in Y_j for some $j \in I$, since the U_i form a filter base. Thus $U \supset U_j$. □

(*) In accordance with the terminology of EGA IV 21, we now prefer to reserve the term "divisors"for "locally principal divisors"or "Cartier divisors".

(*) *Editor's note.* i.e. a decreasing filtered family.

Corollary XI.3.9. *Let X be a noetherian and normal prescheme, and let Y be a closed subset of codimension ≥ 2. Suppose that, for every $p \in X - Y$, $\mathcal{O}_{X,p}$ is factorial. Then*

$$\mathrm{Pic}(X - Y) \to \mathrm{Cl}(X - Y) \to \mathrm{Cl}(X)$$

are isomorphisms.

Corollary XI.3.10. *Let A be a noetherian normal local ring. Set $X' = \mathrm{Spec}(A)$ and $x = \mathfrak{r}(A)$. For A to be* factorial*, it is necessary and sufficient that* $\mathrm{Pic}(X' - \{x\}) = 0$ *and that* $\mathfrak{p} \in X' - \{x\}$ *imply that $A_{\mathfrak{p}}$ is factorial.*

Indeed, for A to be factorial, it is necessary and sufficient that $\mathrm{Cl}(X') = 0^{(*)}$.

Corollary XI.3.11. *Let A be a noetherian local ring of dimension ≥ 2. Set $X' = \mathrm{Spec}(A)$, and let $x = \mathfrak{r}(A)$. Set $X = X' - \{x\}$. The following conditions are equivalent:*

(i) A is factorial;

(ii) a) *for every $y \in X$, $\mathcal{O}_{X,p}$ is factorial, and*

b) *A is parafactorial, i.e.* $\operatorname{prof} A \geq 2$ *and* $\mathrm{Pic}(X) = 0$.

Before proving this corollary, let us state the **12**

Serre's normality criterion 3.11$^{(*)}$. *Let A be a noetherian local ring. For A to be normal, it is necessary and sufficient that*

(i) for every prime ideal $\mathfrak{p}$ of A such that $\dim A_{\mathfrak{p}} \leq 1$*, $A_{\mathfrak{p}}$ is normal,*

(ii) for every prime ideal $\mathfrak{p}$ of A such that $\dim A_{\mathfrak{p}} \geq 2$*, one has* $\operatorname{prof} A_{\mathfrak{p}} \geq 2$.

Let us prove XI.3.11.

(i) $\Longrightarrow$ (ii). Since a localization of a factorial ring is again factorial, we have (ii) a). Moreover, A is normal, and hence $\operatorname{prof} A \geq 2$, since $\dim A \geq 2$ (XI.3.12 (ii)). Finally, A is parafactorial; indeed, $\mathrm{Pic}(X) \simeq \mathrm{Cl}(X') = 0$ (cf. XI.3.10).

(ii) $\Longrightarrow$ (i). Let us first prove that A is *normal* by applying Serre's criterion. Since $\dim A \geq 2$, condition (i) of the criterion is among the hypotheses. Moreover, for every $\mathfrak{p} \in X$, $A_{\mathfrak{p}}$ is factorial, hence normal, and therefore of depth ≥ 2, at least if $\dim A_{\mathfrak{p}} \geq 2$. Finally, $\operatorname{prof} A \geq 2$ by (ii) b). It remains to apply XI.3.10.

Let us summarize the foregoing:

Proposition XI.3.13. *Let X be a locally noetherian prescheme, and let $\mathcal{I}$ be a quasi-coherent ideal of X. Let $Y = V(\mathcal{I})$. Let $p \in \mathbf{N}$. Suppose that:*

1) one has $\mathrm{Leff}(X, Y)$ *(*Exposé *X);*

2) $\mathrm{H}^i(X, \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ *if $i = 1$ or 2 and if $n \geq p$;*

3) for every open neighborhood U of Y in X and every $x \in X - U$, the ring $\mathcal{O}_{X,x}$ is parafactorial.

Then, for every $n \geq p$, and every open neighborhood U of Y, the homomorphisms

$$\begin{tikzcd} \mathrm{Pic}(X) \arrow[rr] \arrow[dr] & & \mathrm{Pic}(X_n) \\ & \mathrm{Pic}(U) \arrow[ur] & \end{tikzcd}$$

are isomorphisms.

We know some parafactorial rings: 13

Theorem XI.3.14. *(i) (Auslander–Buchsbaum)*(*) *A regular noetherian local ring is factorial (and hence parafactorial if its dimension is* ≥ 2*).*

(ii) A noetherian local ring of dimension ≥ 4 *that is a* complete intersection *is parafactorial.*

Corollary XI.3.15 (Conjecture of Samuel(*))**.** *A noetherian local ring A that is a complete intersection and is factorial in codimension* ≥ 3 *(i.e.* $\dim A_{\mathfrak{p}} \leq 3$ *implies that* $A_{\mathfrak{p}}$ *is factorial) is factorial.*

Proof of the corollary

We argue by induction on the dimension of A.

If $\dim A \leq 3$, then A is factorial by hypothesis.

If $\dim A > 3$, then by the induction hypothesis, noting that a localization of a complete intersection is again a complete intersection, all localizations of A other than A are factorial. By Theorem XI.3.14 (ii), A is parafactorial, and therefore factorial by XI.3.11.

Let us prove XI.3.14 (i) (following Kaplansky)(*).

Let A be a regular noetherian local ring, and set $\dim A = n$. If $n = 0$ or 1, the result is known. Suppose $n \geq 2$; we argue by induction on n, and suppose $n \geq 2$ and the theorem proved for rings of dimension $< n$. Set $X' = \operatorname{Spec}(A)$ and $X = X' - \{x\}$, where $x = \mathfrak{r}(A)$. The localizations of A other than A are regular and of dimension $< n$, and hence factorial. Moreover, $\operatorname{prof} A = \dim A \geq 2$. It is therefore enough to prove that $\operatorname{Pic}(X) = 0$ (cor. XI.3.11). Thus let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-Module; we know that it can be extended to a coherent $\mathcal{O}_{X'}$-Module $\mathcal{L}'$. There exists a resolution of $\mathcal{L}'$ by free $\mathcal{O}_X$-Modules:

$$0 \longleftarrow \mathcal{L}' \longleftarrow \mathcal{L}'_1 \longleftarrow \cdots \longleftarrow \mathcal{L}'_n \longleftarrow 0,$$

because the cohomological dimension of A is finite. By restriction to X' we obtain a finite 14
free resolution. It is therefore enough to prove the following lemma:

Lemma XI.3.16. *Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{L}$ be a locally free $\mathcal{O}_X$-Module that admits a finite resolution by* finite-type free *modules. Then* $\det(\mathcal{L}) \simeq \mathcal{O}_X$*.*

Recall that $\det(\mathcal{L})$ is defined as the highest exterior power of $\mathcal{L}$. In the case under consideration, $\det(\mathcal{L}) \simeq \mathcal{L}$, because $\mathcal{L}$ is invertible, and hence the lemma gives the conclusion. Let us prove this lemma. Let

$$0 \longleftarrow \mathcal{L}_0 \longleftarrow \mathcal{L}_1 \longleftarrow \mathcal{L}_2 \longleftarrow \cdots \longleftarrow \mathcal{L}_n \longleftarrow 0,$$

be the asserted exact sequence, where $\mathcal{L}_0 = \mathcal{L}$. Since everything is locally free, we have:

$$\bigotimes_{0 \leq i \leq n} (\det(\mathcal{L}_i))^{(-1)^i} \simeq \mathcal{O}_X$$

(*) *Editor's note.* see Bourbaki, *Algèbre commutative* VII.1.4, corollary to Theorem 2, and VII.3.2, Theorem 1

(*) Cf. EGA IV 5.8.6.

(*) *Editor's note.* compare with the following purity result, due to Gabber. Let X be the spectrum of a regular local ring A of dimension 3, let a have nonzero differential, i.e. $a \in \mathfrak{m} - \mathfrak{m}^2$, and let U be the complement of $V(a)$. Then a vector bundle on U is free (for a simple proof, see (Swan R.G., "A simple proof of Gabber's theorem on projective modules over a localized local ring" *Proc. Amer. Math. Soc.* **103** (1988), no. 4, p. 1025–1030). The rank-1 case is a special case of Theorem XI.3.14. For purity results concerning vector bundles of arbitrary rank, whether in the analytic or algebraic setting, see (Gabber O., "On purity theorems for vector bundles", *Internat. Math. Res. Notices* (2002), no. 15, p. 783–788).

(*) *Editor's note.* for a proof in the same spirit, but more elementary, see Call F. & Lyubeznik G., "A simple proof of Grothendieck's theorem on the parafactoriality of local rings", in *Commutative algebra: syzygies, multiplicities, and birational algebra (South Hadley, MA, 1992)*, Contemp. Math., vol. 159, American Mathematical Society, Providence, RI, 1994, p. 15–18.

(*) This is the proof reproduced in EGA IV 21.11.1.

and all the $\mathcal{L}_i$, $i > 0$, are free, and hence so are their determinants, and consequently so is the determinant of $\mathcal{L}_0 = \mathcal{L}$. □

It remains for us to prove part (ii) of the theorem. First, let us prove a lemma that will allow us to proceed by induction:

Lemma XI.3.17. *Let A be a noetherian local ring that is a quotient of a regular ring. Let $t \in \mathfrak{r}(A)$ be an A-regular element. Suppose that A is complete for the t-adic topology. Set $X' = \operatorname{Spec}(A)$, $Y' = V(t) \simeq \operatorname{Spec}(B)$, $B = A/tA$, $X = X' - \{x\}$, $Y = Y' - \{x\}$, and $x = \mathfrak{r}(A)$. Suppose that:*

a) for every $y \in X$ closed in X, one has $\operatorname{prof} \mathcal{O}_{X,y} \geq 2$,

b) $\operatorname{prof} A/tA \geq 3$,

then the map $\operatorname{Pic}(X) \to \operatorname{Pic}(Y)$ is injective. In particular, if B is parafactorial, then A is also parafactorial. 15

We know that a) implies $\operatorname{Lef}(X, Y)$ by X 2.1. If we prove that $\mathrm{H}^1(Y, P_n) = 0$ for every $n \geq 0$, then we will know by (XI.2.2) that $\operatorname{Pic}(X) \to \operatorname{Pic}(Y)$ is injective. If, moreover, B is parafactorial, then we will know that $\operatorname{Pic}(Y) = 0$ (XI.3.5), and hence $\operatorname{Pic}(X) = 0$; moreover, $\operatorname{prof}(A) = 3 + 1 \geq 2$, since t is A-regular, and thus A will be parafactorial by XI.3.5.

Let $\mathcal{I} = \widetilde{(tA)}$ be the $\mathcal{O}_{X'}$-Module associated with the ideal tA. In no. XI.1, we set $P_n = (\mathcal{I}^{n+1}/\mathcal{I}^{n+2})_{|Y}$ for every $n \geq 0$. Since t is A-regular, $P_n \simeq \mathcal{O}_Y$. *It therefore remains for us to prove that* $\mathrm{H}^1(Y, \mathcal{O}_Y) = 0$. Now $Y = Y' - \{x\}$ is an open subset of Y', and hence we have an exact sequence (I (27)):

$$\mathrm{H}^1(Y', \mathcal{O}_{Y'}) \to \mathrm{H}^1(Y, \mathcal{O}_Y) \to \mathrm{H}^2_x(Y', \mathcal{O}_{Y'}),$$

whose right-hand term is zero by hypothesis b), and whose left-hand term is zero because Y' is affine. □

Lemma XI.3.18. *Keeping the hypotheses of XI.3.17, suppose in addition that:*

c) for every y closed in Y, one has $\operatorname{prof} \mathcal{O}_{X,y} \geq 3$,

d) $\operatorname{prof} A/tA \geq 4$ (stronger than b)),

e) for every y closed in X with $y \in Y$, the ring $\mathcal{O}_{X,y}$ is parafactorial.

Then the map $\operatorname{Pic}(X) \to \operatorname{Pic}(Y)$ is an isomorphism; in particular, for A to be parafactorial, it is necessary and sufficient that B be parafactorial.

We know (X 2.1) that a) and c) imply $\operatorname{Leff}(X, Y)$. Moreover, by the argument just given, d) implies that $\mathrm{H}^1(Y, P_n) = 0$ for every $n \geq 0$ and $i = 1$ or $i = 2$. Moreover, for every open neighborhood U of Y in X, the complement of U in X consists of finitely many closed points. By e) and Theorem XI.3.13, we deduce that $\operatorname{Pic}(X) \to \operatorname{Pic}(Y)$ is an isomorphism. On the other hand, $\operatorname{prof} A \geq \operatorname{prof} B \geq 2$; by the criterion XI.3.5, we deduce that A is parafactorial if and only if B is. 16

Let us now prove XI.3.14 (ii). Let R be a regular noetherian local ring. Let $(t_1, \ldots, t_k)$ be an R-sequence. Set $B = R/(t_1, \ldots, t_k)$ and suppose that $\dim B \geq 4$. We must prove that B is parafactorial. We argue by induction on k. If $k = 0$, then B is regular, and hence factorial by XI.3.14 (i), and therefore parafactorial by XI.3.11. Suppose $k \geq 1$ and the theorem proved for $k' < k$. Set $A = R/(t_1, \ldots, t_{k-1})$, so that $B = A/t_k A$. We may suppose that B is complete by XI.3.7. By the induction hypothesis, A is parafactorial. Let us prove that Lemma XI.3.18 applies. We have assumed that B is complete, and hence so is A, and therefore A is complete for the t-adic topology. If $x \in X$ and x is closed in X, then A_x is a complete intersection of dimension ≥ 4, with $k' < k$. By the induction hypothesis, A_x is parafactorial, and moreover has depth ≥ 4. This gives a), c), and e). Moreover, $\dim A \geq 5$, which gives d). □

Theorem XI.3.19. *Let X be a locally noetherian prescheme, and let $\mathcal{I}$ be a coherent sheaf of ideals of X. Set $Y = V(\mathcal{I})$. Let n be an integer. Suppose that:*

(i) one has $\operatorname{Leff}(X, Y)$ *(cf. examples X 2.1 and X 2.2),*

(ii) for every $p \geq n$, one has $\mathrm{H}^i(Y, \mathcal{I}^{p+1}/\mathcal{I}^{p+2}) = 0$ for $i = 1$ and $i = 2$,

(iii) for every open subset $U \supset Y$ and every $x \in X - U$, the ring $\mathcal{O}_{X,x}$ is regular of dimension ≥ 2 or a complete intersection of dimension ≥ 4.

Then, for every open subset $U \supset Y$ and every integer $p \geq n$, the homomorphisms

$$\operatorname{Pic}(X) \to \operatorname{Pic}(U) \to \operatorname{Pic}(Y_p)$$

are isomorphisms, where Y_p denotes the prescheme $(Y, \mathcal{O}_X/\mathcal{I}^{p+1})$.

It is enough to combine XI.3.13 and XI.3.14.

Exposé XII

Applications to projective algebraic schemes

XII.1 Projective duality theorem and finiteness theorem(*)

The following theorem, essentially contained in FAC(*) (except that, at the time, Serre did not have at his disposal the language of Ext for sheaves of modules(*)), is the global analogue of the local duality theorem (Exposé IV), which was modeled on it. 17

Theorem XII.1.1. *Let k be a field, $X = \mathbf{P}^r_k$ projective space of dimension r over k, and F a variable coherent module on X; then there is an isomorphism of ∂-functors in F:*

$$\mathrm{H}^i(X, F)' \xleftarrow{\sim} \mathrm{Ext}^{r-i}(X; F, \Omega^r_{X/k}), \tag{1}$$

where we set

$$\Omega^r_{X/k} = \mathcal{O}_{\mathbf{P}^r_k}(-r-1). \tag{2}$$

Remark. This module is of course also the module of relative differentials of degree r of X over k. In this form, the theorem remains valid if X is a proper smooth scheme over k (for the projective case, see A. Grothendieck, "*Théorèmes de dualité pour les faisceaux algébriques cohérents*", Séminaire Bourbaki, May 1957)(*). When F is locally free, one recovers Serre's duality theorem $\mathrm{H}^i(X, F)' \xrightarrow{\sim} \mathrm{H}^{r-i}(\mathcal{H}om_{\mathcal{O}_X}(F, \Omega^r_{X/k}))$. Theorem XII.1.1 (which also recovers the case of a projective X over k, as in *loc. cit.*) will suffice for our purposes.

The homomorphism (1) is deduced from the Yoneda pairing 18

$$\mathrm{H}^i(X, F) \times \mathrm{Ext}^{r-i}(X; F, \Omega^r_{X/k}) \to \mathrm{H}^r(X, \Omega^r_{X/k}), \tag{3}$$

and from a well-known isomorphism (cf. FAC, or EGA III 2.1.12):

$$\mathrm{H}^r(X, \Omega^r_{X/k}) = \mathrm{H}^r(\mathbf{P}^r_k, \mathcal{O}_{\mathbf{P}^r_k}(-r-1)) \xrightarrow{\sim} k. \tag{4}$$

To show that (1) is an isomorphism, one proceeds as in the case of the local duality theorem, noting that $\mathrm{H}^r(X, F)$, as a functor of F, is right exact (since $\mathrm{H}^n(X, F) = 0$ for $n > r$), and

(*)The present exposé, written in January 1963, is considerably more detailed than was the oral presentation in June 1962.

(*)J.-P. Serre, "Faisceaux algébriques cohérents", *Ann. of Math.* **61** (1955), p. 197-278.

(*)*Editor's note.* Readers interested in the history of mathematics will profitably consult the letter that Grothendieck wrote to Serre on 15 December 1955 and Serre's reply of 22 December of the same year; see *Correspondance Grothendieck-Serre*, edited by Pierre Colmez and Jean-Pierre Serre, Documents Mathématiques, vol. 2, Société Mathématique de France, Paris, 2001.

(*)For a more general duality theorem, cf. the Hartshorne seminar cited at the end of Exp. IV.

that every coherent module is isomorphic to a quotient of a direct sum of modules of the form $\mathcal{O}(-m)$, with m large. By descending induction on i, this reduces us to checking the assertion for a sheaf of the form $\mathcal{O}(-m)$, where it is contained in the well-known explicit computations (FAC, or EGA III 2.1.12). We may moreover assume $-m \leq -r-1$, in which case $\mathrm{H}^i(X, \mathcal{O}(-m)) = 0$ for $i \neq r$.

Corollary XII.1.2. *For a given coherent F, and for m sufficiently large, there is a canonical isomorphism*

$$\mathrm{H}^i(X, F(-m))' \xrightarrow{\sim} \mathrm{H}^0(X, \mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \Omega^r_{X/k})(m)) \tag{5}$$

(where $'$ denotes the vector-space dual).

Indeed, on projective space X, for every pair of coherent sheaves F, G and for n sufficiently large, there is a canonical isomorphism:

$$\mathrm{Ext}^n(X; F(-m), G) \simeq \mathrm{Ext}^n(X; F, G(m)) \xrightarrow{\sim} \mathrm{H}^0(X, \mathcal{E}xt^n_{\mathcal{O}_X}(F, G)(m)), \tag{6}$$

(the isomorphism between the first two terms being trivially valid for every m), as follows from the global Ext spectral sequence

$$\mathrm{H}^p(X, \mathcal{E}xt^q_{\mathcal{O}_X}(F, G(m))) \Longrightarrow \mathrm{Ext}^\bullet(X; F, G(m)),$$

which degenerates for m sufficiently large, by virtue of the fact that **19**

$$\mathcal{E}xt^q_{\mathcal{O}_X}(F, G(m)) \simeq \mathcal{E}xt^q_{\mathcal{O}_X}(F, G)(m),$$

and that the $\mathcal{E}xt^q_{\mathcal{O}_X}(F, G)$ are coherent sheaves. Thus (5) follows from (6) and (1).

Corollary XII.1.3. *For given i, F, the following conditions are equivalent:*

(i) $\mathrm{H}^i(X, F(-m)) = 0$ *for m large.*

(i bis) $\mathrm{H}^i(X, F(\cdot)) = \bigoplus_{m \in \mathbf{Z}} \mathrm{H}^i(X, F(-m))$ *is a finite-type S-module, where $S = k[t_o, \dots, t_r]$.*

(ii) $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \Omega^r_{X/k}) = 0$.

(ii bis) $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \mathcal{O}_X) = 0$.

(iii) $\mathrm{H}^i_x(F_x) = 0$ *for every closed point x of X.*

(iv) $\mathrm{H}^{i+1}_x(\widetilde{F}_x) = 0$ *for every closed point x of the punctured affine cone $\widetilde{X} = \mathrm{Spec}(S) - \mathrm{Spec}(k)$ over X, where $\widetilde{F}$ denotes the inverse image of F under the canonical morphism $\widetilde{X} \to X$.*

Proof. (i) $\iff$ (i bis), since the submodule of $\mathrm{H}^i(X, F(\cdot))$ that is the sum of the homogeneous components of degree $\geq \nu$ is of finite type over S (in fact, for $i \neq 0$, it is even of finite type over k), (cf. FAC or EGA III 2.2.1 and 2.3.2).

(i) $\iff$ (ii) by Corollary XII.1.2.

(ii) $\iff$ (ii bis), because $\Omega^r_{X/k}$ is locally isomorphic to $\mathcal{O}_X$.

(ii bis) $\iff$ (iii) by the local duality theorem for $\mathcal{O}_{X,x}$ (which is a regular local ring of **20**
dimension r), according to which the "dual"of $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \mathcal{O}_X)_x$ is identified with $\mathrm{H}^i_x(F_x)$ (V 2.1).

(ii bis) is equivalent to the analogous relation

$$\mathcal{E}xt^{r-i}_{\mathcal{O}_{\widetilde{X}}}(\widetilde{F}, \mathcal{O}_{\widetilde{X}}) = 0$$

(because $\widetilde{X} \to X$ is faithfully flat, so the inverse image of $\mathcal{E}xt^{r-i}_{\mathcal{O}_X}(F, \mathcal{O}_X)$ is isomorphic to $\mathcal{E}xt^{r-i}_{\mathcal{O}_{\widetilde{X}}}(\widetilde{F}, \mathcal{O}_{\widetilde{X}})$), and this last relation is equivalent to (iv) by the local duality theorem for the local ring $\mathcal{O}_{X,x}$, which is regular of dimension $r+1$.

In particular, applying this to all $i \leq n$, we find:

Corollary XII.1.4. *Equivalent conditions for given n, F:*

(i) $\mathrm{H}^i(X, F(-m)) = 0$ *for $i \leq n$ and m large.*

(i bis) $\mathrm{H}^i(X, F(\cdot))$ *is a finite-type S-module for $i \leq n$.*

(ii) $\mathrm{prof}(F_x) > n$ *for every closed point x of X.*

(iii) $\mathrm{prof}(\widetilde{F}_x) > n+1$ *for every closed point x of $\widetilde{X}$.*

The point of Corollaries XII.1.3 and XII.1.4 is to express a global condition (i) or (i bis) in terms of local conditions, namely the vanishing of local invariants such as $\mathrm{H}^i_x(X, F_x)$ or $\mathrm{H}^i_x(X, \widetilde{F}_x)$, or an inequality involving depth. In this form, these results remain trivially valid for an arbitrary projective scheme X, and a very ample invertible sheaf $\mathcal{O}_X(1)$ on X, as one sees by means of a suitable projective immersion $X \to \mathbf{P}^r_k$. (Of course, conditions XII.1.3 (ii) and XII.1.3 (ii bis) are no longer equivalent to the others in this general case, unless, for example, one assumes that X is regular.) One can moreover generalize to the case of a projective morphism $X \to S$ as follows:

Proposition XII.1.5. *Let $f : X \to S$ be a projective morphism, with S noetherian, let $\mathcal{O}_X(1)$ be an invertible module on X that is very ample relative to S, let F be a coherent module on X, flat over S, let s be a point of S, let X_s be the fiber of X at s (regarded as a projective scheme over $k(s)$), let F_s be the sheaf induced on X_s by F, and finally let i be an integer. Suppose that, for every closed point x of X_s, one has $\mathrm{H}^i_x(F_{s,x}) = 0$ (for example, $\mathrm{prof}(F_{s,x}) > i$). Then there exists an open neighborhood U of s such that the same condition holds for every $s' \in U$. Moreover, for such a U, one has* **21**

$$\mathrm{R}^i f_*(F(-m)) = 0 \text{ for } m \text{ large},$$

and if $\mathcal{S}$ is a quasi-coherent graded algebra on S, generated by $\mathcal{S}^1$, and defining X together with $\mathcal{O}_X(1)$ as $X = \mathrm{Proj}(\mathcal{S})$, $\mathcal{O}_X(1) = \mathrm{Proj}(\mathcal{S}(1))$, then the S-module

$$\mathrm{R}^i f_*(F(\cdot)) = \bigoplus_{m \in \mathbf{Z}} \mathrm{R}^i f_*(F(m))$$

is of finite type over U.

Embed X in an $X' = \mathbf{P}^r_S$ in such a way that $\mathcal{O}_X(1)$ is induced by $\mathcal{O}_{X'}(1)$ (this is possible provided that S is replaced by an affine neighborhood of s). Put [(*)] for every integer j, and every $t \in S$:

$$\underline{E}^j(t) = \mathcal{E}xt^j_{\mathcal{O}_{X'_t}}(F_{t'}, \mathcal{O}_{X'_t}(-r-1)) \tag{7}$$

Thus $\underline{E}^j(t)$ is a coherent module on X_t; I claim that, as t varies, the family of these modules is “constructible”in the following sense: for every $t \in S$ there exists a nonempty open subset V of the closure $\overline{t}$, endowed with the induced reduced structure, and a coherent module $\underline{E}^j(V)$ on $X_V = X \times_S V$, *flat* over V, such that, for every $t' \in V$, $\underline{E}^j(t)$ is isomorphic to the module induced by $\underline{E}^j(V)$ on X_t. To verify this assertion, putting $Z = \overline{t}$ with its induced structure, we consider the coherent modules **22**

$$\underline{E}^j(Z) = \mathcal{E}xt^j_{\mathcal{O}_{X'_Z}}(F_Z, \mathcal{O}_{X'_Z}(-r-1))$$

[(*)]The first part of XII.1.5 can also be obtained immediately by applying the purely local statement EGA IV 12.3.4 to the preceding $\underline{E}^j$, thereby bypassing most of the proof that follows.

(where the subscript Z still means that one takes the induced object over Z), and we take for V a nonempty open subset of Z such that the modules $\underline{E}^j(Z)$ are flat over V: this is possible, since one verifies immediately that $\underline{E}^j(Z) = 0$ for j outside the interval $[0, r]$, and one can apply SGA 1 IV 6.11. We then take $\underline{E}^j(V) = \underline{E}^j(Z)|_{X_V}$, and one readily verifies that it has the required property.

It follows from the preceding remark that there exists a finite partition of S formed by sets V_α of the form $V = V(t)$ as above (noetherian induction), and, applying Serre's theorem EGA III 2.2.1 to the $\underline{E}^j(V_i)$, one sees that there exists an integer m_0 such that

$$\mathrm{R}^i f_{V_\alpha *}(\underline{E}^j(V_\alpha)) = 0 \text{ for } i \neq 0, m \geq m_0, \text{ for } j,$$

from which it follows, using the flatness of $\underline{E}^j(V_\alpha)$ over V_α and straightforward Künneth-type relations (cf. EGA III par. 7), that

$$\mathrm{H}^i(X_t, \underline{E}^j(t)(m)) = 0 \text{ for } i \neq 0, m \geq m_0, \text{ for } j,$$

for every $t \in V_\alpha$, and hence for every t, since the V_α cover S. From this and the spectral sequence for global Ext it follows, by virtue of XII.1.1, as in the proof of XII.1.2, that there is an isomorphism

$$\mathrm{H}^i(X_t, F_t(-m))' \xleftarrow{\sim} \mathrm{H}^0(X_t, \underline{E}^{r-i}(t)(m)) \text{ for } m \geq m_0, \tag{8}$$

for every integer i, and every $t \in S$.

Now use the hypothesis on F_s, which can be written

$$\underline{E}^{r-i}(s) = 0, \tag{9}$$

and which, by virtue of (8), is equivalent to

$$\mathrm{H}^i(X_s, F_s(-m)) = 0 \text{ for } m \geq m_0. \tag{10}$$

Since F, and hence $F(-m)$, is flat over S, it follows from the Künneth-type relations already invoked that (for a given m) the same relation (10) holds when s is replaced by a t near s, in **23**
particular by every generization t of s. By virtue of (8), we shall therefore have, for such a generization,

$$\underline{E}^{r-i}(t) = 0, \tag{11}$$

and the set of $t \in s$ for which this relation holds is plainly constructible (since it induces an open subset on each V_α); as it contains the generizations of s, it contains an open neighborhood U of s. This proves the first assertion of XII.1.5. Moreover, for $t \in U$, it follows from (11) and (8) that

$$\mathrm{H}^i(X_t, F_t(-m) = 0 \text{ for } m \geq m_0, \ t \in U, \tag{12}$$

which, by virtue of the Künneth-type relations, implies (and in fact is much stronger than)

$$\mathrm{R}^i f_*(F(-m)) = 0 \text{ on } U, \text{ for } m \geq m_0. \tag{13}$$

This proves the second assertion of XII.1.5. Finally, the last assertion follows immediately by proceeding as at the beginning of the proof of XII.1.3.

[. *remark]Remark 1.6*(*) *The proof simplifies considerably (eliminating all consideration of constructibility) when one assumes from the outset that the hypothesis made for $s \in S$ holds for every $s' \in S$. In fact, when one assumes that F_s has depth $> i$ at the closed points of X,*

(*)This remark is made more precise by footnote 101.

there is a general statement, local in nature on X, *which says that the same condition holds for all the* X_t, *provided that* X *is replaced by a suitable open neighborhood of the fiber* X_s *(in other words, a certain subset of* X*, defined by conditions on the modules induced by* F *on the fibers, is open, cf.* EGA *IV). Since* f *is proper here, this neighborhood can therefore be taken to be of the form* $f^{-1}(U)$*, which recovers the first assertion of XII.1.5 without laborious dévissage. In this general case, one can still prove by the method of loc. cit. that the first assertion of XII.1.5 (proved here globally, using the fact that* X *is projective over* S*) again follows from a purely local statement on* X *(which the reader may formulate explicitly if useful).*

XII.2 Lefschetz theory for a projective morphism: Grauert comparison theorem

It is the following theorem: **24**

Theorem XII.2.1. *Let* $f : X \to S$ *be a projective morphism, with* S *Noetherian, let* $\mathcal{O}_X(1)$ *be an invertible module on* X*, relatively ample over* S*, let* Y *be the prescheme of zeros of a section* t *of* $\mathcal{O}_X(1)$*, let* J *be the ideal defining* Y*, let* X_n *be the closed subprescheme of* X *defined by* J^{n+1}*, let* $\widehat{X}$ *be the formal completion of* X *along* Y*, let* $\widehat{f} : \widehat{X} \to S$ *be the composite morphism* $\widehat{X} \to X \to S$*, and let* F *be a coherent module on* X*, flat over* S*. Assume in addition that, for every* $s \in S$*, the module* F_s *induced on the fiber* X_s *has depth* $> n$ *at the points of that fiber, and that* t *is* F*-regular. Under these conditions:*

(i) The canonical homomorphism

$$\mathrm{R}^i f_*(F) \to \mathrm{R}^i \widehat{f}_*(\widehat{F})$$

is an isomorphism for $i < n$*, and a monomorphism for* $i = n$*.*

(ii) The canonical homomorphism

$$\mathrm{R}^i \widehat{f}_*(\widehat{F}) \to \varprojlim_m \mathrm{R}^i f_*(F_m)$$

is an isomorphism for $i \leq n$*.*

Proof. We immediately reduce to the case where S is affine, and then to proving the

Corollary XII.2.2. *Under the conditions of XII.2.1, assume in addition that* S *is affine. Then:*

(i) The canonical homomorphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(\widehat{X}, \widehat{F})$$

is an isomorphism for $i < n$*, and a monomorphism for* $i = n$*.* **25**

(ii) The canonical homomorphism

$$\mathrm{H}^i(\widehat{X}, \widehat{F}) \to \varprojlim_m \mathrm{H}^i(X_m, F_m)$$

is an isomorphism for $i \leq n$*.*

After replacing $\mathcal{O}_X(1)$ by a tensor power, and t by a power of t, we may assume that $\mathcal{O}_X(1)$ is very ample relative to S. On the other hand, since t, and hence t^m, is F-regular, multiplication by t^m, regarded as a homomorphism from $F(-m)$ to F, is injective; therefore, for every $m \geq 0$, there is an exact sequence:

$$0 \to F(-m) \xrightarrow{t^m} F \to F_m \to 0, \tag{14}$$

and hence an exact cohomology sequence

$$\mathrm{H}^i(X, F(-m)) \to \mathrm{H}^i(X, F) \to \mathrm{H}^i(X, F_m) \to \mathrm{H}^{i+1}(X, F(-m)).$$

But by XII.1.5, we have $\mathrm{H}^i(X, F(-m)) = 0$ for $i \leq n$ and m sufficiently large, which proves the

Lemma XII.2.3. *For m large, the canonical homomorphism*

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(X, F_m)$$

is bijective if $i < n$, and injective if $i = n$.

This shows that, for $i < n$, the inverse system $(\mathrm{H}^i(X_m, F_m))_{m \geq 0}$ is essentially constant, and *a fortiori* satisfies the Mittag-Leffler condition; hence (in view of $\widehat{F} = \varprojlim F_m$), we obtain (ii) from EGA 0_{III} 13.3. On the other hand, (i) follows trivially from this, taking XII.2.3 into account.

Corollary 2.4(*). *Let $f : X \to S$ be a projective flat morphism, with S locally Noetherian, let $\mathcal{O}_X(1)$ be an invertible module on X, relatively ample over S, let t be a section of this module that is $\mathcal{O}_X$-regular, let Y be the closed subprescheme of zeros of t, and let $\widehat{X}$ be the formal completion of X along Y. Assume that, for every $s \in S$, X_s has depth ≥ 1 (resp. depth ≥ 2) at its closed points. Then, for every open neighborhood U of Y, the functor* 26

$$F \longmapsto \widehat{F}$$

*from the category of locally free coherent modules on U to the category of locally free coherent modules on $\widehat{X}$ is faithful (resp. fully faithful, i.e. the Lefschetz condition (*Lef*) of X 2 holds).*

For two locally free Modules F and G on U, introduce the Module

$$H = \mathcal{H}om_{\mathcal{O}_U}(F, G)$$

we are reduced to proving that the canonical homomorphism

$$\mathrm{H}^0(U, H) \to \mathrm{H}^0(\widehat{U}, \widehat{H}) \tag{15}$$

is injective (resp. bijective). Now the Modules H_t have depth ≥ 1 (resp. ≥ 2) at the closed points of X_t, and hence we can apply XII.2.1, which gives the conclusion XII.2.4 in the case where $U = X$. In the case of an arbitrary U, note that the question is local on S, so we may suppose that S is affine. Then every coherent Module on X is a quotient of a coherent locally free Module (since $\mathcal{O}_X(1)$ is an invertible relatively ample Module on X). Since the dual Module $H' = \mathcal{H}om(H, \mathcal{O}_U)$ extends to a coherent Module on X, which is therefore isomorphic to a cokernel of a homomorphism of locally free Modules on X, it follows by transposition that one can find a homomorphism

$$u' : L'^0 \to L'^1$$

of locally free Modules on X, inducing a homomorphism 27

$$u : L^0 \to L^1$$

of locally free Modules on U, such that there is an exact sequence

$$0 \to H \to L^0 \xrightarrow{u} L^1.$$

Using the five lemma (which becomes the three lemma), and the left exactness of the functor H^0, we are reduced to proving that (15) is injective (resp. bijective) when H is replaced there by L^0, L^1, which brings us back to the case where H is induced by a locally free Module H' on X. Moreover, in the non-parenthetical case this reduction is even unnecessary, since the kernel of (15) consists in any event of the sections of H on U that vanish in some suitable open neighborhood V of Y; but the restriction homomorphism $\mathrm{H}^0(U, H) \to \mathrm{H}^0(V, H)$ is injective, since H has depth ≥ 1 at the points of every closed subset Z of X not meeting Y (cf. the lemma below). In the parenthetical case, we are reduced to proving that

$$H^0(X, H') \to \mathrm{H}^0(U, H')$$

is bijective, which follows from the fact that H' has depth ≥ 2 at all the points of a closed subset $Z = X - U$ of X not meeting Y. It therefore remains only to prove the

Lemma XII.2.5. *Let F be a coherent Module on X, flat over S, such that for every $s \in S$, F_s has depth $\geq n$ at every closed point of X_s. Then, for every closed subset Z of X not meeting Y, F has depth $\geq n$ at all the points of Z.*

Indeed, for every $x \in X$, setting $s = f(x)$, we have

$$\mathrm{prof}(F_x) \geq \mathrm{prof}(F_{s,x}), \tag{16}$$

as can be seen by lifting arbitrarily a maximal $F_{s,x}$-regular sequence of elements of $\mathfrak{r}(\mathcal{O}_{X_s,x})$, 28
which gives an F_x-regular sequence by SGA 1 IV 5.7. Now if x belongs to a Z as in Lemma XII.2.5, then x is necessarily closed in X_s; in other words, Z is *finite* over S. Indeed, Z (endowed with a structure induced from X) is projective over S as a closed subprescheme of X, which is projective, and Z is affine over S as a closed subprescheme of $X - Y$, which is affine.

Remark XII.2.6. Suppose that for every $s \in S$, the section t_s of $\mathcal{O}_{X_s}(1)$ induced by t is $\mathcal{O}_{X_s}$-regular (which implies by SGA 1 IV 5.7 that t is $\mathcal{O}_X$-regular). Then the hypotheses made are stable under extension of the base $S' \to S$ (S' locally noetherian). Thus the conclusion remains valid after every base change.

XII.3 Lefschetz theory for a projective morphism: existence theorem

Theorem 3.1(*). *Let $f : X \to S$ be a projective morphism, with S noetherian, let $\mathcal{O}_X(1)$ be an invertible Module on X that is ample relative to S, let X_0 be the closed subprescheme*

(*) *Editor's note.* See Corollary I.1.4 of the article by Mme Raynaud (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental", *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52).

(*) *Editor's note.* For a version without the flatness hypothesis, see (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental", *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, Theorem II.3.3).

of zeros of a section t of $\mathcal{O}_X(1)$, let $\widehat{X}$ be the formal completion of X along X_0, let $\mathfrak{F}$ be a coherent Module on $\widehat{X}$, and let F_0 be the Module that it induces on X_0. Assume moreover:

a) $\mathfrak{F}$ is flat over S.

b) For every $s \in S$, the section t_s induced by t on X_s is $\mathfrak{F}_s$-regular (which implies that F_0 is also flat over S, cf. SGA 1 IV 5.7*).*

c) For every $s \in S$, F_{0s} has depth ≥ 2 at the closed points of X_{0s}.

Assume moreover that S admits an ample invertible sheaf. Under these conditions, there exists a coherent Module F on X, and an isomorphism from its formal completion $\widehat{F}$ to $\mathfrak{F}$.

This statement will follow from the following one:

Corollary XII.3.2. *Under conditions A)), B)), C)) above, the following hold:*

(i) The Module $\widehat{f}_*(\mathfrak{F})$ on S is coherent; hence, for every n, the Module $\widehat{f}_*(\mathfrak{F}(n))$ on S is coherent. **29**

(ii) For n large, the canonical homomorphism $\widehat{f}^*\widehat{f}_*(\mathfrak{F}(n)) \to \mathfrak{F}(n)$ is surjective.

Assume the corollary for the moment, and let us prove XII.3.1. By virtue of the last hypothesis made in XII.3.1, we may reduce to the case $X = \mathbf{P}^r_S$, after replacing $\mathcal{O}_X(1)$ and t by a suitable power. I claim that we may further suppose that $t_s \neq 0$ for every s. Otherwise, indeed, $\mathfrak{F}_s = 0$ by b), or equivalently, by Nakayama, $F_{0s} = 0$, i.e. s does not belong to the image of $\operatorname{Supp} F_0$ under the morphism $f_0 : X_0 \to S$ induced by f. Now this image S' is open by virtue of a), b), since F_0 is flat over S, and it is clear that it suffices to prove the conclusion of XII.3.1 in the situation obtained by restricting over S', since the coherent Module F' on $X|_{S'}$ thereby obtained will be the restriction of a coherent Module F on X, which will have the required property. We may therefore suppose that, in addition to hypotheses a), b), c), the following hypotheses also hold:

a′) $\mathcal{O}_X$ is flat over S.

b′) For every $s \in S$, the section t_s is $\mathcal{O}_{X_s}$-regular.

c′) For every $s \in S$, $\mathcal{O}_{X_{0s}}$ has depth ≥ 2 at the closed points of X_{0s}. (It suffices to choose $X = \mathbf{P}^r_S$ with $r \geq 3$, which is permissible.)

Now XII.3.2 implies that one can find an epimorphism

$$\widehat{L} \to \mathfrak{F} \to 0, \tag{17}$$

where L is a Module on X of the form $f_*(G)(-n)$, with G a coherent locally free Module on S: indeed, for n large, it suffices to express the coherent Module $\widehat{f}_*(\mathfrak{F})$ on S as a quotient of such a G. On the other hand, hypotheses a), b), c) on f, t imply that $\widehat{L}$ satisfies the same conditions a), b), c) as $\mathfrak{F}$. It follows easily that the same is true of the kernel of (17), to which the same argument may therefore be applied, so that $\mathfrak{F}$ is represented as the cokernel of a homomorphism

$$\widehat{L'} \to \widehat{L}, \tag{18}$$

where L, L' are locally free Modules on X. Now, by virtue of a′), the second part of c′), and either XII.2.1 or XII.2.4, the homomorphism (18) comes from a homomorphism $L' \to L$ of Modules on X. It now suffices to take for F the cokernel of $L' \to L$, and we are done.

It remains to prove XII.3.2. This had been done in the seminar by a somewhat laborious expedient, consisting in interpreting everything in terms of cohomology on the punctured projective cone of X relative to S, in order to reduce to Theorem 2.1. A more direct and more satisfactory method (although substantially identical) now seems to me to be the following. It consists in noting that in IX, no. IX.2 (and with the notation of that exposé), the hypothesis that the morphism $f : \mathcal{X} \to \mathcal{X}'$ is adic is not used anywhere in the proof of 2.1, via EGA 0_{III} 13.7.7; it suffices instead to suppose that $\mathcal{X}$ is also adic, and to choose two ideals of definition $\mathcal{J}$ for $\mathcal{X}'$, $\mathcal{I}$ for $\mathcal{X}$, such that $\mathcal{J}\mathcal{O}_{\mathcal{X}} \subset \mathcal{I}$, and to define $\mathcal{S} = \mathrm{gr}_{\mathcal{J}}(\mathcal{O}_{\mathcal{X}'})$, and consider $\mathrm{gr}_{\mathcal{I}}(\mathfrak{F})$. In any event, 2.1 can be applied directly to the morphism $\widehat{f} : \widehat{X} \to S$ considered in the present subsection, where one simply takes $\mathcal{J} = 0$. Thus, to verify that $\widehat{f}_*(\mathfrak{F})$ is coherent, it suffices, by virtue of *loc. cit.*, to verify that $\mathrm{R}^i f_{0*}(\mathrm{gr}_{\mathcal{I}}(\mathfrak{F}))$ is coherent on S for $i = 0, 1$; for this, note that by virtue of a) and b), the Module in question is none other than $\bigoplus_{m\geq 0} \mathrm{R}^i f_{0*}(F_0(-m))$, which is indeed coherent by virtue of hypothesis c) and XII.1.5.

This proves XII.3.2 (i). For XII.3.2 (ii), we shall need the

Lemma XII.3.3. *Under the conditions of A)), B)), C)) of XII.3.1, set* **30**

$$G_m = \widehat{f}_*(\mathfrak{F}(\cdot)_m) = \bigoplus_n \widehat{f}_*(\mathfrak{F}_m(n))$$

Then the projective system (G_m) satisfies the Mittag-Leffler condition.

We may suppose that S is affine, with ring A. Let $\mathcal{S}$ be a finitely generated graded A-algebra in positive degrees, and let $t' \in \mathcal{S}_1$, such that X embeds in $\mathrm{Proj}(\mathcal{S})$, with $\mathcal{O}_X(1)$ induced by $\mathcal{O}(1)$ and the section t the image of t'. Endow $\mathcal{S}$ with the $\mathcal{J}$-adic filtration, where $\mathcal{J} = t'\mathcal{S}$, and consider the projective system of the $\mathfrak{F}(\cdot)_m$ in the category of abelian sheaves on X_0. We are again under the preliminary conditions of EGA 0_{III} 13.7.7(*), and moreover $\mathrm{H}^i(X_0, \mathrm{gr}(\mathfrak{F}(\cdot))$ is a finite-type Module over $\mathrm{gr}_{\mathcal{J}}(\mathcal{S})$ for $i = 0, 1$. Indeed, since t' is $\mathfrak{F}$-regular, one sees at once that, as a Module over $(\mathcal{S}/t'\mathcal{S})[T]$ (of which $\mathrm{gr}_{\mathcal{J}}(\mathcal{S})$ is a quotient), the Module in question identifies with $\mathrm{H}^i(X_0, F_0(\cdot)) \otimes_{\mathcal{S}/t'\mathcal{S}} (\mathcal{S}/t'\mathcal{S})[T]$; but by XII.1.5, $\mathrm{H}^i(X_0, F_0(\cdot))$ is of finite type over $\mathcal{S}$, and hence over $\mathcal{S}/t'\mathcal{S}$, for $i = 0, 1$, which proves our assertion. Consequently, we are under the conditions for applying 0_{III} 13.7.7 with $n = 1$, which proves XII.3.3.

This point being established (and still supposing S affine, as we may for proving XII.3.2 (ii)), let m_0 be such that $m \geq m_0$ implies $\Im(G_m \to G_0) = \Im(G_{m_0} \to G_0)$, so that both sides are also equal to $\Im(\varprojlim G_m \to G_0) = \Im(\widehat{f}_*(\mathfrak{F}(\cdot)) \to f_*\mathfrak{F}_0(\cdot))$. Now note that for large n, $\mathfrak{F}_{m_0}(n)$ is generated by its sections, and therefore $\mathfrak{F}_0(n)$ is generated by sections that lift to $\mathfrak{F}_{m_0}$, and hence (by the choice of m_0) that lift to $\mathfrak{F}$. Thus the sections of $\mathfrak{F}$ generate $\mathfrak{F}_0$, and hence also $\mathfrak{F}$ by Nakayama. This proves XII.3.2 (ii), and therefore XII.3.1.

Corollary XII.3.4. *Let $f : X \to S$ be a projective flat morphism, with S locally noetherian, let $\mathcal{O}_X(1)$ be an invertible Module on X, relatively ample over S, and let t be a section of this Module such that for every $s \in S$, the induced section t_s on the fiber X_s is $\mathcal{O}_{X_s}$-regular; let X_0* **31**
*be the closed subprescheme of zeros of t, and let $\widehat{X}$ be the formal completion of X along X_0. Suppose that for every $s \in S$, X_{0s} has depth ≥ 2 at its closed points (i.e. X_s has depth ≥ 3 at the closed points of X_{0s}), and X_s has depth ≥ 2 at its closed points. Under these conditions, the pair (X, X_0) satisfies the effective Lefschetz condition (*Leff*) of paragraph 2 of Exposé* X, *i.e. :*

a) For every open neighborhood U of X_0 in X, the functor

$$F \longmapsto \widehat{F}$$

from the category of coherent locally free Modules on U to the category of coherent locally free Modules on $\widehat{X}$ is fully faithful.

(*)Corrected as indicated in IX p. 78.

b) For every coherent locally free Module $\mathfrak{F}$ on $\widehat{X}$, there exist an open neighborhood U of X_0 and a coherent locally free Module F on U such that $\mathfrak{F}$ is isomorphic to $\widehat{F}$.

Indeed, a) has already been noted in XII.2.4 under weaker conditions. For b), we apply XII.3.1, which gives the conclusion, at least if S is noetherian and admits an absolutely ample invertible Module, in particular if S is affine. Indeed, if F is a coherent module on X such that $\widehat{F}$ is isomorphic to $\mathfrak{F}$, and hence locally free, it follows that F is locally free on a neighborhood U of X_0, and $F|_U$ will satisfy the required condition. But now note that by XII.2.5, for such an F, its direct image under the immersion $U \to X$ is coherent, and is moreover independent of the chosen solution (U, F) (taking account of the fact that two solutions coincide in a neighborhood of X_0, by a)). More precisely, one can find a coherent Module F on X and an isomorphism $\widehat{F} \xrightarrow{\sim} \mathfrak{F}$, such that F has depth ≥ 2 at all the points of X that are closed in their fiber, and this determines F up to unique isomorphism. By this uniqueness property, the solutions of the problem obtained over the affine open subsets of S glue together, giving a coherent F on all of X and an isomorphism $\widehat{F} \simeq \mathfrak{F}$. Restricting F to the open subset U of points at which it is free gives what was required.

Thanks to XII.2.4 and XII.3.4, in the setting of a projective algebraic scheme and a **32**
"hyperplane section"thereof, one can apply the general facts established in Exposés X and XI concerning conditions (Lef) and (Leff). Thus:

Corollary XII.3.5. *Let X be a projective algebraic scheme equipped with an ample invertible module $\mathcal{O}_X(1)$, let t be a section of this module that is $\mathcal{O}_X$-regular, and let X_0 be the closed subscheme of zeros of t. Assume that X has depth ≥ 2 at its closed points (resp. and depth ≥ 3 at the closed points of X_0). Then $\pi_0(X_0) \to \pi_0(X)$ is bijective; in particular, X is connected if and only if X_0 is connected, and, after choosing a geometric base point in X_0, $\pi_1(X_0) \to \pi_1(X)$ is surjective. More generally, for every open $U \supset X_0$, the homomorphism $\pi_1(X_0) \to \pi_1(U)$ is surjective (resp. the homomorphism $\pi_1(X_0) \to \varprojlim_U \pi_1(U)$ is bijective). In the parenthesized case, if one assumes in addition that the local ring at every closed point of X not in X_0 is pure (3.2) (for example, regular, or merely a complete intersection), then $\pi_1(X_0) \to \pi_1(X)$ is an isomorphism.*

Apply 2.4 and XII.3.4. Note that, in the parenthesized case, the hypothesis that X have depth ≥ 3 at the closed points of X_0 implies that every irreducible component of X having dimension $\neq 0$ has dimension ≥ 3 (as one sees by noting that such a component necessarily meets X_0, and by considering a closed point of the intersection).

Remark. When X is normal and has dimension ≥ 2 at all its points, it has depth ≥ 2 at its closed points, and the non-parenthesized hypotheses of XII.3.5 hold. In this case there is a more elementary proof of the surjectivity of $\pi_1(X_0) \to \pi_1(X)$ using Bertini's theorem (cf. SGA 1 X.2.10). If one assumes in addition that X_0 is normal and that X has dimension ≥ 3 at all its points, then the parenthesized hypotheses of XII.3.5 hold. In this case, XII.3.5 was established by Grauert (indeed, thanks to the normality hypothesis, one can then dispense with the existence theorem XII.3.1 by various devices) . This proof of Grauert's was the **33**
starting point of the "Lefschetz theory"that is the subject of the present seminar.

Corollary XII.3.6. *Let X, $\mathcal{O}_X(1)$, t, and X_0 be as in XII.3.5. Assume that X has depth ≥ 2 at its closed points, and that*

$$\mathrm{H}^i(X_0, \mathcal{O}_{X_0}(-n)) = 0$$

for $n > 0$ and for $i = 1$ (resp. for $i = 1$ and $i = 2$), which, by XII.1.4, implies that X_0 has depth ≥ 2 (resp. ≥ 3) at its closed points, i.e. that X has depth ≥ 3 (resp. ≥ 4) at the closed points of X_0. Under these conditions, for every open neighborhood U of X_0, $\mathrm{Pic}(U) \to \mathrm{Pic}(X_0)$ is injective; in particular, $\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$ is injective (resp. $\varprojlim_U \mathrm{Pic}(U) \to \mathrm{Pic}(X_0)$ is *bijective). In the parenthesized case, if one assumes in addition that the local ring of X at every closed point not in X_0 is parafactorial (XI.3.1) (for example, regular, or more generally a complete intersection), then* $\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$ *is bijective.*

Apply XI XI.3.13 and XI.3.14, noting that the parenthesized hypothesis implies that the irreducible components of X having dimension $\neq 0$ have dimension ≥ 4. In particular, applying this when X is a global complete intersection of dimension ≥ 4 in projective space, one obtains:

Corollary XII.3.7. *Let X be an algebraic scheme of dimension ≥ 3 that is a complete intersection in the scheme $\mathbf{P}^r_k$. Then* $\mathrm{Pic}(X)$ *is the free group generated by the class of the sheaf* $\mathcal{O}_X(1)$.

Proceed by induction on the number of hypersurfaces whose intersection is X, applying XII.3.6 and noting that, for a complete intersection X of dimension ≥ 3, one has $\mathrm{H}^i(X, \mathcal{O}_X(n)) = 0$ for $i = 1, 2$ and every n.

Remark XII.3.8. In the case where X is a nonsingular hypersurface, XII.3.7 is due to Andreotti. The result XII.3.7 can also be expressed (when X is nonsingular) by saying that the homogeneous coordinate ring of X is factorial, and in this form it is contained in XI XI.3.14 (ii). We also note that Serre had given a proof of XII.3.7 in the nonsingular case by transcendental methods, using a specialization argument to reduce to characteristic 0, where the Lefschetz theorem is available in its classical form. Of course, the fact that the purely algebraic proof given here makes it possible to dispense with nonsingularity hypotheses in the statement of the Lefschetz theorem invites one to reconsider it in the classical case as well. Cf. the following exposé, which proposes conjectures in this direction. 34

In Corollaries XII.3.5 and XII.3.7, we have worked over a base field, whereas the key theorems XII.2.4 and XII.3.4 are valid over an arbitrary base. To generalize Corollaries XII.3.5 and XII.3.6 to a general S, we must give workable criteria for a point of X (flat over S) to have a "pure"resp. parafactorial local ring. This will be the subject of the following n°.

XII.4 Formal completion and normal flatness

Theorem XII.4.1. *Let X be a locally Noetherian prescheme, locally embeddable in a regular scheme, Y a closed subset of X, $U = X - Y$, X_0 a closed subprescheme of X defined by an ideal $\mathcal{J}$, $\widehat{X}$ the formal completion of X along X_0, U_0 the trace of X_0 on U, $\widehat{U}$ the formal completion of U along U_0, $i : U \to X$ and $\widehat{i} : \widehat{U} \to \widehat{X}$ the canonical immersions, and n an integer. Assume that*

a) X is normally flat along X_0 at the points of $Y \cap X_0$, i.e. at these points the Modules $\mathcal{J}^n/\mathcal{J}^{n+1}$ over X_0 are flat, i.e. locally free.

b) For every $x \in Y \cap X_0$, one has $\operatorname{prof} \mathcal{O}_{X_0,x} \geq n + 2$.

Under these conditions, the following assertions hold: 35

1. *Let F be a coherent Module on U, and suppose that:*

   c) *For every $x \in Y - Y \cap X_0$, one has* $\operatorname{prof} \mathcal{O}_{X,x} \geq n + 2$.

   d) *F is free at the points of U_0, and has depth $\geq n + 1$ at all the points of U where it is not free.*

   Then the graded Module

$$\bigoplus_{m \geq 0} \mathrm{R}^p i_*(\mathcal{J}^m F)$$

   over $\bigoplus_{m \geq 0} \mathcal{J}^m$ is of finite type for $p \leq n$.

2. Let $\mathfrak{F}$ be a coherent Module on $\widehat{U}$; then the graded Module

$$\bigoplus_{m\geq 0} \mathrm{R}^p i_*(\mathcal{J}^m\mathfrak{F}/\mathcal{J}^{m+1}\mathfrak{F})$$

over $\bigoplus_{m\geq 0}\mathcal{J}^m/\mathcal{J}^{m+1} = \mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X)$ is of finite type for $p \leq n$.

Proof. 1) Let $X' = \mathrm{Spec}(\bigoplus_{m\geq 0}\mathcal{J}^m)$; the base change $f : X' \to X$ then defines $U' = X' - Y'$, X'_0, $U'_0 = X'_0 U'$, and immersions $i' : U' \to X'$, $i'_0 : U'_0 \to X'_0$. Thus we have a Cartesian square

$$\begin{tikzcd} X' \arrow[d, "f"'] & U' \arrow[l, "i'"'] \arrow[d, "g"] \\ X & U \arrow[l, "i"'] \end{tikzcd}$$

and we have

$$\bigoplus_{m\geq 0} \mathrm{R}^p i_*(\mathcal{J}^m F) = \mathrm{R}^p i_*\Big(\bigoplus_{m\geq 0}\mathcal{J}^m F\Big) = \mathrm{R}^p i_*(g_*(F')), \tag{19}$$

where $F' = g^*F$, so that there is indeed a canonical isomorphism **36**

$$g_*(F') \xrightarrow{\sim} \bigoplus_{m\geq 0}\mathcal{J}^m F,$$

since this is true at the points of U_0, because F is free there by d), and also at the points of U_0, because there one has $\mathcal{J}^m = \mathcal{O}_U$, (so that in both cases, $\mathcal{J}^m \otimes_{\mathcal{O}_U} F \to \mathcal{J}^m F$ is an isomorphism).

On the other hand, since f, and hence g, are affine, one has

$$\mathrm{R}^p i_*(g_*(F')) = \mathrm{R}^p (ig)_*(F') = \mathrm{R}^p (fi')_*(F') = f_*(R^p i'_*(F')) \tag{20}$$

and therefore, comparing (19) and (20), we see that assertion 1 is equivalent to the following: $R^p i'_*(F')$ is a Module of finite type, i.e. coherent, on X', for every $p \leq n$. Now, since X is locally embeddable in a regular scheme, the same is true of X', which is of finite type over X, and we may apply the coherence criterion VIII VIII.2.3 to a coherent extension F'' of F': we wish to express that $\mathcal{H}^p_Y(F'')$ is coherent for $p \leq n+1$, and this is also equivalent to saying that, for every $x' \in U'$ such that

$$\mathrm{codim}(\overline{x'} \cap Y', \overline{x'}) = 1, \tag{20 bis}$$

one has

$$\mathrm{prof}\, F'_{x'} \geq n+1. \tag{21}$$

Now this condition is satisfied at the points x' where F' is not free, because for such an x' one has $x' \notin U'_0$ by d), and therefore g is an isomorphism there; again by d), F has depth $\geq n+1$ at $g(x')$, and hence F' has depth $\geq n+1$ at x'. It is therefore enough to verify condition (21) at those $x' \in U'$ satisfying (21) at which F' is free. For this, it is enough to prove that one has **37**

$$\mathrm{prof}\, \mathcal{O}_{X',x'} \geq n+1 \tag{21 bis}$$

at these points; *a fortiori* it is enough to establish this relation at all the points x of U' satisfying (20 bis). Now, again by criterion VIII.2.3 in Exposé VIII, this is equivalent to the assertion that the Modules

$$\mathcal{H}^p_{Y'}(\mathcal{O}_{X'}) \quad \text{for } p \leq n+1$$

are coherent. In fact, we shall prove that they are even zero, or, equivalently by Exposé III, that one has

$$\mathrm{prof}\, \mathcal{O}_{X',x'} \geq n+2 \quad \text{for every } x' \in Y'. \tag{22}$$

For this, we distinguish two cases. If $x' \notin X'_0$, then f is an immersion at x', and we must verify that F has depth $\geq n+2$ at the image $x = f(x')$, which is precisely condition c). If, on the other hand, $x' \in X'_0$, i.e. $x = f(x') \in X_0$ and hence $x \in Y \cap X_0$, we apply conditions a) and b) by means of the

Lemma XII.4.2. *Let X be a locally Noetherian prescheme, X_0 a closed subprescheme of X defined by an ideal $\mathcal{J}$, $X' = \mathrm{Spec}(\bigoplus_{m\geq 0}\mathcal{J}^m)$, $X'_0 = \mathrm{Spec}(\bigoplus_{m\geq 0}\mathcal{J}^m/\mathcal{J}^{m+1}) = X' \times_X X_0$, and x a point of X_0 at which X is normally flat along X_0, i.e. such that $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X) = \bigoplus_{m\geq 0}\mathcal{J}^m/\mathcal{J}^{m+1}$ is flat there as a Module over X_0. Then, for every sequence of elements f_i $(1 \leq i \leq m)$ of $\mathcal{O}_{X,x}$ whose images in $\mathcal{O}_{X_0,x}$ form an $\mathcal{O}_{X_0,x}$-regular sequence, and every $x' \in X'$ above x, the images of the f_i in $\mathcal{O}_{X',x'}$ (respectively in $\mathcal{O}_{X'_0,x'}$) likewise form an $\mathcal{O}_{X',x'}$-regular sequence (respectively an $\mathcal{O}_{X'_0,x'}$-regular sequence); in particular, one has*

$$\mathrm{prof}\,\mathcal{O}_{X',x'} \geq \mathrm{prof}\,\mathcal{O}_{X_0,x}, \quad \mathrm{prof}\,\mathcal{O}_{X'_0,x'} \geq \mathrm{prof}\,\mathcal{O}_{X_0,x}. \tag{23}$$

To prove it, we may suppose that X is local with closed point x, and hence affine with 38
ring $A = \mathcal{O}_{X,x}$, with $\mathcal{J}$ defined by the Ideal J; it is enough to prove that, for every sequence f_i $(1 \leq i \leq m)$ of elements of A whose images in A/J form an A/J-regular sequence, the f_i likewise form a $(\bigoplus_{m\geq 0} J^m)$-regular sequence and a $(\bigoplus_{m\geq 0} J^m/J^{m+1})$-regular sequence, i.e. for every m, it forms a J^m-regular sequence and a J^m/J^{m+1}-regular sequence. The second assertion is trivial, since J^m/J^{m+1} is a free module over A/J. The first follows by considering the J-adic filtration of J^m and noting that the sequence of the f_i is regular on the graded module associated with J^m for that filtration.

This proves XII.4.2, and consequently XII.4.1 1).

Let us prove XII.4.1 2). For this, use the Cartesian square

$$\begin{tikzcd}
X'_0 \arrow[d, "f_0"'] & U'_0 \arrow[l, "i'_0"'] \arrow[d, "g_0"] \\
X_0 & U_0 \arrow[l, "i_0"']
\end{tikzcd}$$

and, proceeding as at the beginning of the proof of 1), we find that

$$\bigoplus_{m\geq 0} \mathrm{R}^p i_*(\mathcal{J}^m\mathfrak{F}/\mathcal{J}^{m+1}\mathfrak{F}) \simeq f_{0*}(\mathrm{R}^p i'_{0*}(\mathfrak{F}'_0)), \tag{24}$$

where $\mathfrak{F}_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}$ and $\mathfrak{F}'_0 = g_0^*(\mathfrak{F}_0)$, (using the fact that $\mathfrak{F}$ is locally free). Thus the conclusion of 2) is equivalent to saying that, for $p \leq n$, $\mathrm{R}^p i'_{0*}(\mathfrak{F}'_0)$ is a coherent Module.

Here again, taking into account that $\mathfrak{F}'_0$ is locally free, criterion VIII VIII.2.3 allows us to reduce to proving the same assertion after replacing $\mathfrak{F}'_0$ by $\mathcal{O}_{U'_0}$, i.e. to proving that the Modules 39

$$\mathcal{H}^p_{Y'_0}(\mathcal{O}_{X'_0}) \quad \text{for } p \leq n+1 \quad (\text{where } Y'_0 = Y' \cap X'_0 = X'_0 - U'_0)$$

are coherent. Once again, we prove that they are in fact zero, i.e., that one has

$$\mathrm{prof}\,\mathcal{O}_{X'_0,x'} \geq n+2 \quad \text{for every } x' \in Y'_0. \tag{25}$$

This does indeed follow from conditions a) and b), in view of XII.4.2. This completes the proof of XII.4.1.

Remark XII.4.3. It follows immediately, by descent, that the hypothesis that X be locally embeddable in a regular scheme can be replaced by the following weaker hypothesis: there exists a faithfully flat and quasi-compact morphism $\overline{X} \to X$ such that $\overline{X}$ is locally embeddable in a regular scheme.

Theorem XII.4.1 enables us to apply the results of Exposé IX (the comparison and existence theorems). We shall be interested in particular in the

Corollary XII.4.4. *Assume that conditions a), b), and c) of Theorem XII.4.1 hold, with $n = 1$ and $X = \mathrm{Spec}(A)$, where A is separated and complete for the J-adic topology. Then*

1. *The functor $F \longmapsto \widehat{F}$ from the category of locally free coherent modules on U to the category of locally free coherent modules on $\widehat{U}$ is fully faithful.*

2. *For every locally free coherent module $\mathfrak{F}$ on $\widehat{U}$, there exists a coherent module F on U and an isomorphism $\widehat{F} \xrightarrow{\sim} \mathfrak{F}$.*

*In particular, if, for every $x \in U$ whose closure in U does not meet U_0, i.e. such that $\overline{x} \cap X_0 \subset Y$, one has $\operatorname{prof} \mathcal{O}_{U,x} \geq 2$, then the pair (U, U_0) satisfies the effective Lefschetz condition (*Leff*) of Exposé X.*

(For the last assertion, one proceeds as in X 2.1.)

Special case of XII.4.4:

Corollary XII.4.5. *Let A be a Noetherian ring, let J be an ideal of A contained in the radical, and let $A_0 = A/J$. Assume* **40**

(i) $\operatorname{prof} A_0 \geq 3$.

(ii) $\operatorname{gr}_J(A)$ *is a free A_0-module.*

(iii) A is complete for the J-adic topology.

Let $X = \operatorname{Spec}(A)$, $X_0 = \operatorname{Spec}(A_0) = V(J)$, let a be the closed point of X, let $U = X - \{a\}$, $U_0 = X_0 - \{a\}$, and let $\widehat{U}$ be the formal completion of U along U_0. Then the functor $F \longmapsto \widehat{F}$ from the category of locally free coherent modules on U to the category of locally free coherent modules on $\widehat{U}$ is fully faithful. Moreover, for every locally free coherent module $\mathfrak{F}$ on $\widehat{U}$, there exists a coherent module (not necessarily locally free!) F on U, and an isomorphism $\widehat{F} \simeq \mathfrak{F}$.

Note that, thanks to XII.4.3, we did not have to assume that A is a quotient of a regular ring, because the completion of A for the $\mathfrak{r}(A)$-adic topology satisfies this condition in any case.

Proceeding as in Exposés X and XI, we deduce from XII.4.5

Corollary XII.4.6. *Under the conditions of XII.4.5, one has the following:*

a) U and U_0 are connected (III 3.1).

Choosing a geometric base point in U_0, the homomorphism

$$\pi_1(U_0) \to \pi_1(U)$$

is surjective.

b) The homomorphism

$$\operatorname{Pic}(U) \to \operatorname{Pic}(U_0)$$

is injective.

To prove b), in view of XII.4.5, this amounts to checking that every isomorphism $L'_0 \xrightarrow{\sim} L_0$ lifts to an isomorphism $\widehat{L'} \xrightarrow{\sim} \widehat{L}$. For this, one lifts step by step to isomorphisms $L'_n \xrightarrow{\sim} L_n$; the obstructions lie in $\mathrm{H}^1(U_0, \mathcal{J}^n/\mathcal{J}^{n+1})$, and these modules are zero because J^n/J^{n+1} is free and $\operatorname{prof} A_0 \geq 3$. **41**

We are now in a position to prove the

Theorem XII.4.7. *Let A be a noetherian local ring, let J be an ideal of A contained in its radical, and let $A_0 = A/J$. Suppose that* *(i)* $\operatorname{prof} A_0 \geq 3$.

(ii) $\operatorname{gr}_J(A)$ *is a free module over* A_0.

Then, if A_0 *is "pure"(X 3.1) (resp. parafactorial (XI XI.3.1)), the same holds for* A.

Proof. By descent, we may suppose that we also have

(iii) A is complete for the J-adic topology.

Indeed, by (i) and (ii), we have $\operatorname{prof}(A) \geq 3$, and hence $\operatorname{prof}(\widehat{A}) \geq 3$, where $\widehat{A}$ is the completion of A for the J-adic topology, and we apply X 3.6 and XI XI.3.6. We are therefore under the conditions of XII.4.5. Since $\operatorname{prof}(A) \geq 3 \geq 2$, to say that A is parafactorial means simply that $\operatorname{Pic}(U) = 0$, and by XII.4.6 b), it is enough for this that $\operatorname{Pic}(U_0) = 0$, i.e. that A_0 be parafactorial. To prove that A is "pure"if A_0 is, we must prove that if V is an étale covering of U, defined by an algebra $\mathcal{B}$ on U, then $\mathrm{H}^0(U, \mathcal{B})$ is a finite étale algebra over A. Now, since A_0 is pure, the same holds for the A_n (which differ from it only by nilpotent elements), and therefore, for every n, $B_n = \mathrm{H}^0(U, \mathcal{B}_n)$ is an étale algebra over A/J^{n+1}; these algebras of course fit together, so that $\varprojlim B_n$ is an étale algebra over A. By XII.4.5, this algebra is none other than $\mathrm{H}^0(U, \mathcal{B})$, which proves our assertion.

Corollary XII.4.8. *Let* $f : X \to Y$ *be a flat morphism of locally noetherian preschemes, let* $x \in X$*, and let* $y = f(x)$*. Suppose that* $\mathcal{O}_{X_y,x}$ *is a "pure"(resp. parafactorial) local ring of depth* ≥ 3*. Then the same holds for* $\mathcal{O}_{X,x}$*.* 42

This is the result of the kind promised at the end of the preceding no. , for generalizing Corollaries XII.3.5 and following. Thus, using XII.3.4, we obtain the

Corollary XII.4.9. *Let* $f : X \to S$ *be a projective flat morphism, with* S *locally Noetherian, let* $\mathcal{O}_X(1)$ *be an invertible module on* X*, ample relative to* S*, let* t *be a section of* $\mathcal{O}_X(1)$ *such that, for every* $s \in S$*, the section* t_s *induced on* X_s *is* $\mathcal{O}_{X_s}$*-regular, let* X_0 *be the closed subscheme of zeros of* t*, and let* X_m *be the closed sub-scheme of zeros of* t^{m+1}*. Assume that, for every* $s \in S$*,* X_s *has depth* ≥ 3 *at all its closed points. Then:*

a) If the local rings at the closed points of $X_s - X_{0,s}$ $(s \in S)$ *are "pure"—for example, if they are complete intersections—then the functor* $X' \longmapsto X'_0 = X' \times_X X_0$ *from the category of finite ètale coverings of* X *to the category of finite ètale coverings of* X_0 *is an equivalence of categories; in particular, after choosing a geometric base point in* X_0*, the homomorphism*

$$\pi_1(X_0) \to \pi_1(X)$$

is an isomorphism (*)

(*) *Editor's note.* Let us point out the striking connectedness result obtained subsequently by Fulton and Hansen, in the case where $S = \operatorname{Spec}(k)$ (k an algebraically closed field). Let $g : X \to \mathbf{P}^m_k \times \mathbf{P}^m_k$ be such that $\dim g(X) > m$; then the inverse image of the diagonal is connected. Among other things, this makes it possible to generalize Corollary XII.4.9 when f is the structural morphism of projective space $\mathbf{P}^m_k$ over $S = \operatorname{Spec}(k)$: more precisely, an irreducible subvariety X of $\mathbf{P}^m_k$ of dimension $> m/2$ has trivial fundamental group! (cf. Fulton W. & Hansen J., "A connectedness theorem for projective varieties, with applications to intersections and singularities of mappings", *Ann. of Math. (2)* **110** (1979), no. 1, p. 159-166). For generalizations to Grassmannians or abelian varieties, see Debarre O., "Théorèmes de connexité pour les produits d'espaces projectifs et les grassmanniennes", *Amer. J. Math.* **118** (1996), no. 6, p. 1347-1367 and "Théorèmes de connexité et variétés abéliennes", *Amer. J. Math.* **117** (1995), no. 3, p. 787–805. The triviality result for the fundamental group of X above was obtained independently by Faltings, who moreover proves that $\operatorname{Pic}(X)$ has no torsion prime to the characteristic of k, using methods of algebraization of formal vector bundles, more in the spirit of Grothendieck's techniques, cf. (Faltings G., "Algebraization of some formal vector bundles" *Ann. of Math. (2)* **110** (1979), no. 3, p. 501–514).

b) If the local rings at the closed points of $X_s - X_{0,s}$ ($s \in S$) are "parafactorial"—for example, regular, or complete intersections of dimension ≥ 4—then, for every integer m such that $\mathrm{R}^i f_{0*}(\mathcal{O}_{X_0}(-n)) = 0$ for $n > m$ and $i = 1, 2$, the map $\mathrm{Pic}(X) \to \mathrm{Pic}(X_m)$ is bijective.

Moreover, if S is Noetherian and the $X_{0,s}$ have depth ≥ 3 at their closed points, such m exist (cf. XII.1.5).

Remark XII.4.10. Under the conditions of the last assertion of XII.4.9 b), we saw in XII.1.5 that there exists an m such that $n > m$ even implies $\mathrm{H}^i(X_0, \mathcal{O}_{X_0}(-n)) = 0$ for $i = 1, 2$, and indeed for $i \leq 2$). This condition is stronger than $\mathrm{R}^i f_*(\mathcal{O}_{X_0}(-n)) = 0$ for $i = 1, 2$, and it has the further advantage of being stable under base change. The same is true of the depth hypotheses made in XII.4.9, and also of a hypothesis of the type "the X_s are locally complete intersections". It follows under these conditions that XII.4.9 b) also implies that the morphism of functors **43**

$$\mathbf{Pic}_{X/S} \to \mathbf{Pic}_{X_n/S}$$

in $\mathbf{Sch}_{/S}$ is an isomorphism, and hence so is the morphism of relative Picard schemes, when these exist:

$$\mathrm{Pic}_{X/S} \to \mathrm{Pic}_{X_n/S}\,.$$

Even when S is the spectrum of an algebraically closed field, this statement is considerably more precise than the statement that merely says that $\mathrm{Pic}(X) \to \mathrm{Pic}(X_n)$ is bijective.

One may ask whether one can always take $n = 0$ in the preceding conclusions (thus assuming that the $X_{0,s}$ have depth ≥ 3 at their closed points). When X_0 is smooth over S and the residual characteristics of S are zero, this is indeed so, by the Kodaira "vanishing theorem"(proved by transcendental methods, using a Kähler metric), which implies that, for every connected smooth projective scheme of dimension n over a field k of characteristic zero and every ample invertible module L on X, one has $\mathrm{H}^i(X, L^{-1}) = 0$ for $i \neq n$. It is not known[(*)] at present whether this theorem can be replaced by a generalization in characteristic $p > 0$, and whether the smoothness hypothesis can be replaced by a hypothesis of a more general nature (involving depth, or of "complete intersection"type, and so on).

XII.5 Universal finiteness conditions for a nonproper morphism

Recall for reference the **44**

Proposition XII.5.1. *Let $f : X \to S$ be a proper morphism of preschemes, with S locally noetherian, let U be an open subset of X, let $g : U \to X$ be the canonical immersion, let $h = fg : U \to S$, and let F be a Module on U. Suppose that the Modules $\mathrm{R}^i g_*(F)$ are coherent for $i \leq n$ (a hypothesis local in nature on X, which in practice is checked using the criterion VIII 2.3). Then $\mathrm{R}^i h_*(F)$ is coherent for $i \leq n$.*

[(*)] *Editor's note.* As Raynaud observed, the decomposition result for the de Rham complex of Deligne and Illusie readily implies the vanishing of the group $H^i(X, L^{-1})$ (with L ample on X, projective and smooth over k of characteristic $p > 0$) for $i < \inf(p, \dim(X))$, provided that X can be lifted to a flat scheme over $W_2(k)$ (cf. Deligne P. & Illusie L., "Relèvements modulo p^2 et décomposition du complexe de de Rham", *Invent. Math.* **89** (1987), no. 2, p. 247–270); this gives a purely algebraic proof of Kodaira's result for projective varieties in characteristic zero. If X cannot be lifted, it is well known that the "vanishing theorem"is false; cf. the example in (Raynaud M., "Contre-exemple au "vanishing theorem" en caractéristique $p > 0$", in *C.P. Ramanujam—a tribute*, Tata Inst. Fund. Res. Studies in Math., vol. 8, Springer, Berlin-New York, 1978, p. 273–278); see also the very elegant examples in (Haboush W. & Lauritzen N., "Varieties of unseparated flags", in *Linear algebraic groups and their representations (Los Angeles, CA, 1992)*, Contemp. Math., vol. 153, American Mathematical Society, Providence, RI, 1993, p. 35–57), simplified in (Lauritzen N. & Rao A.P., "Elementary counterexamples to Kodaira vanishing in prime characteristic", *Proc. Indian Acad. Sci. Math. Sci.* **107** (1997), no. 1, p. 21–25). On the other hand, I know of no example in which the map $\mathrm{Pic}(X_{n+1}) \to \mathrm{Pic}(X_n)$ is not surjective for $n > 1$ in positive characteristic, where X_n denotes a thickened hyperplane section of a smooth projective X as above.

This follows at once from the Leray spectral sequence

$$\mathrm{E}_2^{p,q} = \mathrm{R}^p f_*(\mathrm{R}^q g_*(F)) \Longrightarrow \mathrm{R}^* h_*(F), \tag{26}$$

and from the fact that the higher direct images under f of a coherent Module on X are coherent (EGA III 3.2.1).

Proposition XII.5.2. *Let S be a locally noetherian prescheme, let $\mathcal{S}$ be a quasi-coherent graded Algebra of finite type over S, generated by $\mathcal{S}_1$, let X be a subprescheme of* $\mathrm{Proj}(\mathcal{S})$*, let $\mathcal{O}_X(1)$ be the invertible Module on X induced by* $\mathrm{Proj}(\mathcal{S}(1))$ *and very ample relative to S, let U be an open subset of X, let $g : U \to X$ be the canonical immersion, let $h = fg : U \to S$, and let F be a quasi-coherent Module on U, giving twisted Modules $F(m) = F \otimes \mathcal{O}_X(m)$) ($m \in \mathbf{Z}$); let n be an integer and m_0 an integer. The following conditions are equivalent:*

(i) $\mathrm{R}^i g_*(F)$ *is coherent for $i \leq n$.*

(ii) $\bigoplus_{m>m_0} \mathrm{R}^i h_*(F(m))$ *is a finite-type $\mathcal{S}$-Module for $i \leq n$.*

Proof. Replacing F by $F(m)$ in the spectral sequence above, we obtain a spectral sequence of graded $\mathcal{S}$-Modules 45

$$\mathrm{E}_2^{p,q} = \bigoplus_{m \geq m_0} \mathrm{R}^p f_*(\mathrm{R}^q g_*(F(m)) \Longrightarrow \bigoplus_{m \geq m_0} \mathrm{R}^* h_*(F(m)).$$

Since we have

$$\mathrm{R}^q g_*(F(m)) \simeq \mathrm{R}^q g_*(F)(m),$$

we see that if the $\mathrm{R}^i g_*(F)$ are coherent, then $\mathrm{E}_2^{p,q}$ is of finite type over $\mathcal{S}$ for $q \leq n$, by part a) of Lemma XII.5.3 below, which implies that the abutment is of finite type over $\mathcal{S}$ in degree $i \leq n$. This proves (i) $\Longrightarrow$ (ii). Moreover, reasoning in the abelian category of graded $\mathcal{S}$-Modules modulo the thick subcategory $\mathcal{C}$ of those that are quasi-coherent of finite type, we obtain from the preceding spectral sequence

$$\bigoplus_{m \geq m_0} \mathrm{R}^{n+1} h_*(F(m)) \simeq \bigoplus_{m \geq m_0} f_*(\mathrm{R}^{n+1} g_*(F)(m)) \mod \mathcal{C},$$

which proves that if the left-hand side is a finite-type $\mathcal{S}$-Module, then $\mathrm{R}^{n+1} g_*(F)$ is coherent, by part b) of Lemma XII.5.3. This proves the implication (ii) $\Longrightarrow$ (i) by induction on n. It remains to prove:

Lemma XII.5.3. *Let S, $\mathcal{S}$, X, f be as in XII.5.2, let G be a quasi-coherent Module on X, and let m_0 be an integer. Then*

a) If G is coherent, then for every integer i, the graded Module

$$\bigoplus_{m \geq m_0} \mathrm{R}^i f_*(G(m))$$

over $\mathcal{S}$ is of finite type.

b) Conversely, suppose that the Module $\bigoplus_{m \geq m_0} \mathrm{R}^i f_*(G(m))$ over $\mathcal{S}$ is of finite type; then G is coherent.

Proof of XII.5.3. For a), the case $i = 0$ is given in EGA III 2.3.2, and the case $i > 0$ in EGA III 2.2.1 (i)(ii), which says that the $\mathrm{R}^i f_* G(m)$ are coherent and zero for large m (if we suppose S 46
noetherian, as we may). For b), note that G is isomorphic to $\mathbf{Proj}(\bigoplus_{m \geq m_0} f_*(G(m))$ (EGA II 3.4.4 and 3.4.2), which proves that G is coherent if $\bigoplus_{m \geq m_0} f_*(G(m))$ is of finite type over $\mathcal{S}$, by *loc. cit.* 3.4.4.

Corollary XII.5.4 (S noetherian)**.** *Suppose that* $\mathrm{R}^i g_*(F)$ *is coherent for* $i \leq n$*. Then, for* $i \leq n+1$ *and large* m*, there is a canonical isomorphism:*

$$\mathrm{R}^i h_*(F(m)) \simeq f_*(\mathrm{R}^i g_*(G)(m)).$$

Indeed, the spectral sequence (26) for $F(m)$ then degenerates in degree $\leq n$, by EGA III 2.2.1 (ii), which gives the result at once (and in fact recovers the implication (ii) $\Longrightarrow$ (i) of 5.2).

Corollary XII.5.5. *Under the preliminary conditions of XII.5.2, with* S *noetherian, the following conditions are equivalent:*

(i) $\bigoplus_{m \geq m_0} h_*(F(m)$ *is of finite type over* S*, and* $\mathrm{R}^i h_*(F(m)) = 0$ *for* $0 < i \leq n$ *and large* m*.*

(ii) $g_*(F)$ *is coherent, and* $\mathrm{R}^i g_*(F) = 0$ *for* $0 < i \leq n$*.*

(ii bis) $g_*(F)$ *is coherent, and* $\operatorname{prof}_Y g_*(F) > n+1$*.*

The equivalence of (ii) and (ii bis) is contained in III 3.3. Moreover, by XII.5.2, conditions (i) and (ii) both imply that the $\mathrm{R}^i g_*(F)$ ($i \leq n$) are coherent. The equivalence of (i) and (ii) then follows from XII.5.4, taking into account the fact that for every coherent Module G on X, one has $G = 0$ if and only if $f_*(G(m)) = 0$ for large m, for example by EGA III 2.2.1 (iii).

Remark XII.5.6. The criteria XII.5.2 and XII.5.5 can be interpreted by saying that the **47**
"simultaneous finiteness condition"XII.5.2 (ii) is expressed by local regularity properties (in terms of depth by VIII VIII.2.1) of F at the points of U near $Y = X - U$, whereas the "asymptotic vanishing condition"XII.5.5 (i) is distinctly stronger in nature and is expressed by local regularity conditions on $g_*(F)$ at the points of Y itself. It would be interesting, in order to generalize Lefschetz-type theorems for projective morphisms to quasi-projective morphisms, to find local criteria on X that are necessary and sufficient for the $\mathcal{S}$-Modules $\bigoplus_{m \geq 0} \mathrm{R}^i h_*(F(m))$ for $i \leq n$ to be of finite type. When S is the spectrum of a field (and doubtless more generally if it is the spectrum of an artinian ring) and $Y = X - U$ is finite, one can show that it is necessary and sufficient that the following conditions hold:

1°) $\operatorname{prof} F_x > n$ for every closed point x of U (compare XII.1.4).

2°) $\mathrm{R}^i g_*(F)$ is coherent for $i \leq n$, or equivalently, there exists an open neighborhood V of Y such that for every closed point x of $U \cap V$, one has $\operatorname{prof} F_x > n+1$.

Proposition XII.5.7. *Let* S *be a locally Noetherian prescheme, let* $g : U \to X$ *be a morphism of preschemes of finite type over* $S^{(*)}$*, with structural morphisms* h *and* f*, let* F *be a quasi-coherent module on* U*, and let* n *be an integer. The following conditions are equivalent:*

(i) For every base change $S' \to S$*, with* S' *Noetherian, the module* $\mathrm{R}^n g'_*(F')$ *on* X' *is coherent.*

(ii) For every base change as above and every coherent ideal $\mathcal{J}$ *on* S'*, denoting by* $\mathcal{I}$ *the ideal* $\mathcal{J}\mathcal{O}_{X'}$ *on* X'*, the graded module*

$$\bigoplus_{m \geq 0} \mathrm{R}^n g'_*(\mathcal{I}^m F')$$

over $\bigoplus_{m \geq 0} \mathcal{I}^m$ *is of finite type.*

(iii) For every base change $S' \to S$, and $\mathcal{J}$ as above, the graded module 48

$$\bigoplus_{m \geq 0} \mathrm{R}^n g'_*(\mathcal{I}^m F'/\mathcal{I}^{m+1}F')$$

over $\mathrm{gr}_I(\mathcal{O}_{X'}) = \bigoplus_{m \geq 0} \mathcal{I}^m/\mathcal{I}^{m+1}$ *is of finite type.*

Obviously, (ii) $\Longrightarrow$ (i) and (iii) $\Longrightarrow$ (i), as one sees by taking $\mathcal{I} = 0$ in conditions (ii) and (iii). The reverse implications are obtained by applying (i) to the composite base change $S'' \to S' \to S$, where S'' is equal to $\operatorname{Spec} \bigoplus_{m \geq 0} \mathcal{J}^m$ resp. $\operatorname{Spec} \bigoplus_{m \geq 0} \mathcal{J}^m/\mathcal{J}^{m+1}$.

The point of this proposition is that conditions of the form (ii) are those that occur in the "algebraic-formal comparison theorems", whereas conditions of the form (iii) occur in the "existence theorems"that complement them; cf. Exposé IX. A first interesting case is that in which $f : X \to S$ is the identity, so that one is dealing with conditions on a morphism $h : U \to S$ locally of finite type and a quasi-coherent module F on U, flat over S. To obtain sufficient conditions, we shall assume that U embeds by $g : U \to X$ as an open subprescheme of an X that is *proper* over S. Applying XII.5.1, we thus see:

Corollary XII.5.8. *Let $f : X \to S$ be a proper morphism, with S locally Noetherian, let U be an open subset of X, let $g : U \to X$ be the canonical immersion, let $h = fg : U \to S$, and let F be a quasi-coherent module on X, flat over S. Assume that, for every base change $S' \to S$, with S' locally Noetherian, $\mathrm{R}^i g'_*(F')$ is coherent on X' for $i \leq n$. Then the following hold:*

(i) For every base change $S' \to S$, with S' locally Noetherian, $\mathrm{R}^i h'_*(F')$ is coherent on S' for $i \leq n$.

(ii) For every $S' \to S$ as above, and every coherent ideal $\mathcal{J}$ on S', the graded modules 49

$$\bigoplus_{m \geq 0} \mathrm{R}^i h'_*(\mathcal{J}^m F')$$

over $\bigoplus_{m \geq 0} \mathcal{J}^m$ are of finite type for $i \leq n$.

(iii) For every $S' \to S$ and $\mathcal{J}$ as above, the graded modules

$$\bigoplus_{m \geq 0} \mathrm{R}^i h'_*(\mathcal{J}^m F'/\mathcal{J}^{m+1}F')$$

over $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_{S'}) = \bigoplus_{m \geq 0} \mathcal{J}^m/\mathcal{J}^{m+1}$ are of finite type for $i \leq n$.

Moreover, under the conditions of (ii), and by the comparison theorem 1.1, denoting by $\widehat{S'}$ the formal completion of S' with respect to $\mathcal{J}$, and by $\widehat{U'}$ that of U' with respect to $\mathcal{J}\mathcal{O}_{U'}$, the canonical homomorphisms

$$\widehat{\mathrm{R}^i h'_*(F')} \to \mathrm{R}^i \widehat{h'}_*(\widehat{F'}) \to \varprojlim_k \mathrm{R}^i h'_*(F'_k)$$

are isomorphisms for $i \leq n - 1$.

Remark XII.5.9. Assume in addition that the conditions of XII.5.8 hold with F *coherent*, and consider a base change $S' \to S$ as in XII.5.9 (i). Assume further that S' is locally embeddable in a regular scheme, or, more generally, that there exists a faithfully flat and quasi-compact morphism $S'' \to S$ such that S'' is locally embeddable in a regular scheme; this condition holds in particular if S' is local. Then the conclusion of XII.5.8 (i) and (ii) remains valid when F' is replaced there by a module G' on U' such that every point of U' has an open neighborhood on which G' is isomorphic to a module of the form F'^m. Indeed, one reduces to the case where S' itself is locally embeddable in a regular scheme, and hence so are $S'' = \operatorname{Spec}\left(\bigoplus_{m \geq 0} \mathcal{J}^m\right)$ and

$X'' = X' \times_{S'} S'' = X \times_S S''$, which are of finite type over it. One then applies the finiteness criterion VIII VIII.2.3 to the direct images for $i \leq n$ of G'' under the immersion $U'' \to X''$, noting that its conditions hold by hypothesis for F'' , and therefore also for G'', since they 50
are expressed in terms of depth and G'' is locally isomorphic to an F''^n. The same argument shows that, if $\mathcal{G}'$ is a coherent module on $\widehat{U'}$ (the completion of U' with respect to the ideal $\mathcal{J}\mathcal{O}_{U'}$), such that $\mathcal{G}'_0 = \mathcal{G}'/\mathcal{J}\mathcal{G}'$ is locally of the form $F_0'^n$, then the conclusion of (iii) remains valid when F' is replaced there by $\mathcal{G}'$. Using the results of Exposé IX, one thus obtains the following result:

Corollary XII.5.10. *Let $f : X \to S$ be a proper morphism, with S locally Noetherian, and let U be an open subset of X; assume that U is flat over S, and that, for every base change $S' \to S$, with S' locally Noetherian, $R^i g'_*(\mathcal{O}_{U'})$ is coherent on X' for $i = 0, 1$. Assume now that S' is of the form* Spec(A), *where A is a Noetherian ring equipped with an ideal J such that A is separated and complete for the J-adic topology. Under these conditions:*

(i) The functor $F \longmapsto \widehat{F}$ from the category of locally free modules on U' to the category of locally free modules on $\widehat{U'}$ is fully faithful.

(ii) For every locally free module $\mathfrak{F}$ on $\widehat{U'}$, there exists a coherent module F on U' (not necessarily locally free, alas), and an isomorphism $\widehat{F} \simeq \mathfrak{F}$.

It remains only to prove (ii), by XII.5.9. But it follows from this remark and 2.1 that $\mathfrak{F}$ is induced by a coherent module $\mathfrak{G}$ on $\widehat{X'}$. By the existence theorem EGA III 5.1.4, $\mathfrak{G}$ is of the form $\widehat{F}$, where F is coherent on X, which proves the conclusion.

Remarks XII.5.11. 1°) Using XII.5.10, XII.4.7, and a suitable hypothesis saying that certain local rings of the geometric fibers of $X' \to S'$ are "pure"resp. parafactorial, one should be able to obtain statements saying that the functor $Z' \longmapsto \widehat{Z'}$ from the category of finite ètale coverings of X' to the category of finite ètale coverings of $\widehat{X'}$ (or, equivalently, of X'_0) is 51
an equivalence of categories, resp. that the functor $L \longmapsto \widehat{L}$ from the category of invertible modules on X' to the category of invertible modules on $\widehat{X'}$ is an equivalence. Using recent results of Murre, it is likely that one should be able to deduce existence theorems for Picard schemes of certain *nonproper* algebraic schemes[(*)]. In general, the removal of purity hypotheses

[(*)]It is in fact enough that g be quasi-compact and quasi-separated (EGA IV1.2.1), with no condition on U, X.

[(*)]*Editor's note.* Of course, in the projective case one should refer to Grothendieck's existence theorems in FGA; cf. Grothendieck A., "Technique de descente et théorèmes d'existence en géométrie algébrique. VI. Les schémas de Picard: propriétés générales", in *Séminaire Bourbaki*, vol. 7, Société mathématique de France, Paris, 1995, Exp. 236, p. 221–243 and "Technique de descente et théorèmes d'existence en géométrie algébrique. V. Les schémas de Picard: théorèmes d'existence", in *Séminaire Bourbaki*, vol. 7, Société mathématique de France, Paris, 1995, Exp. 232, 143–161. The nine finiteness conjectures found there are proved in Exposés XII and XIII by Mme Raynaud and Kleiman in SGA 6. For an excellent elementary account of the subject, see Kleiman's expository article (Kleiman S., "The Picard scheme", forthcoming in Contemp. Math.). For an application of these techniques to the global generalized Jacobians of a relative smooth curve, see (Contou-Carrère C., "La jacobienne généralisée d'une courbe relative; construction et propriété universelle de factorisation", *C. R. Acad. Sci. Paris Sér. A-B* **289** (1979), no. 3, A203–A206 and "Jacobiennes généralisées globales relatives", in *The Grothendieck Festschrift, Vol. II*, Progr. Math., vol. 87, Birkhäuser, Boston, 1990, p. 69–109). See also, by the same author and in the purely local setting, the construction and study of the "local generalized Jacobian"functor ("Jacobienne locale, groupe de bivecteurs de Witt universel, et symbole modéré", *C. R. Acad. Sci. Paris Sér. I Math.* **318** (1994), no. 8, p. 743–746). Moreover, although in the case of a projective smooth morphism the connected components of the Picard scheme are proper, this is no longer so in the singular case. The problem of compactifying Picard schemes then arises naturally; this problem has been studied in detail, notably in (Altman A.B. & Kleiman S., "Compactifying the Picard scheme", *Adv. in Math.* **35** (1980), no. 1, p. 50–112, and "Compactifying the Picard scheme. II", *Amer. J. Math.* **101** (1979), no. 1, p. 10–41). The case of curves had been studied earlier (D'Souza C., "Compactification of generalised Jacobians", *Proc. Indian Acad. Sci. Sect. A Math. Sci.* **88** (1979), no. 5, p. 419–457). One even knows exactly when the compactified Jacobian of a curve is irreducible (Rego C.J., "The compactified Jacobian", *Ann. Sci. Éc. Norm. Sup. (4)* **13** (1980), no. 2, p. 211–223), the latter being the closure of the ordinary Jacobian when the curve is geometrically integral and locally planar; for a construction in families of compactified Jacobians, see (Esteves E., "Compactifying the in various existence theorems, especially representability theorems for functors such as the Hilbert or Picard functors, by means of the techniques developed in this seminar, deserves systematic study.

2°) One may seek manageable necessary and sufficient conditions, in terms of depth, for the universal finiteness condition considered in XII.5.10 to hold. When S is the spectrum of a field, it follows easily from EGA III 1.4.15 that it is necessary and sufficient for the $\mathrm{R}^i g_*(F)$ $(i \leq n)$ to be coherent, which is indeed expressed in terms of depth by VIII VIII.2.3. In the general case, however, note that it is not enough to require the preceding condition to hold for all the fibers $U_s \subset X_s$ $(s \in S)$, even when $n = 0$. For example, take $X = S$, let S be the spectrum of a discrete valuation ring, let U be the open subset consisting only of the generic point, and let $F = \mathcal{O}_U$.

3°) Here, however, is a *sufficient* condition ensuring that the hypotheses of XII.5.10 hold: it is enough that f be flat and that, for every $s \in S$ and every $x \in Y_s = X_s - U_s$, one have

$$\operatorname{prof} \mathcal{O}_{X_s,x} \geq n + 2.$$

Indeed, taking Lemma XII.2.5 into account (cf. relation (16) following XII.2.5), it follows that $g_*(\mathcal{O}_U) \simeq \mathcal{O}_X$ and $\mathrm{R}^i g_*(\mathcal{O}_U) = 0$ for $0 < i \leq n$, and the same relations will clearly hold after every base change $S' \to S$.

relative Jacobian over families of reduced curves", *Trans. Amer. Math. Soc.* **353** (2001), no. 8, p. 3045–3095). Since then, existence results for the Picard scheme in the proper case have advanced beyond the original edition of SGA 2; cf. (Murre J.P., "On contravariant functors from the category of pre-schemes over a field into the category of abelian groups (with an application to the Picard functor)", *Publ. Math. Inst. Hautes Études Sci.* **23** (1964), p. 5–43), and especially (Artin M., "Algebraization of formal moduli. I", in *Global Analysis (Papers in Honor of K. Kodaira)*, Univ. Tokyo Press, Tokyo, 1969, p. 21–71). See also (Raynaud M., "Spécialisation du foncteur de Picard", *Publ. Math. Inst. Hautes Études Sci.* **38** (1970), p. 27–76) for a scheme proper over a discrete valuation ring but not necessarily flat. For a discussion of more recent results, in particular Artin's, for the Picard functor of proper flat schemes, especially in the cohomologically flat case in dimension 0, see Chapter VIII of (Bosch S., Lütkebohmert W. & Raynaud M., *Néron models*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 21, Springer-Verlag, Berlin, 1990) and the references cited there. Much more recently, very precise results have been obtained for relative curves $f : X \to S$ over the spectrum S of a discrete valuation ring with perfect residue field. More precisely, one assumes that f is proper and flat, that X is regular, and that $f_*\mathcal{O}_X = \mathcal{O}_S$. On the other hand, one does not assume that f is cohomologically flat in dimension 0, i.e. one does not assume that $H^1(X, \mathcal{O})$ is torsion-free. The Picard functor is then not representable, either by a scheme or by an algebraic space, as soon as the torsion in question is nonzero. Let J be the Néron model of the generic fiber of f; it is a quotient of the Picard functor P. Raynaud then showed that the kernel of the tangent map $H^1(X, \mathcal{O}) = \mathrm{Lie}(P) \to \mathrm{Lie}(J)$ coincides with the torsion subgroup of H^1, and that the cokernel has the same length (see Theorem 3.1 of (Liu Q., Lorenzini D. & Raynaud M., "Néron models, Lie algebras, and reduction of curves of genus one", *Invent. Math.* **157** (2004), p. 455–518)). This result enables those authors to study the relation between the Birch–Swinnerton-Dyer and Artin–Tate conjectures (see Th.. 6.6 of *loc. cit.*). As for the local Picard scheme, see Boutot's thesis, cited in editor's note ((*)) on page 131.

Exposé XIII

Problems and conjectures

XIII.1 Relations between global and local results. Affine problems connected with duality

It is well known that many statements concerning a projective scheme X can be formulated **52**
in terms of statements concerning a certain graded ring, or better, a complete local ring, namely the homogeneous coordinate ring of X (i.e. the affine ring of the cone $\tilde{X}$ over X), or its completion (i.e. the completion of the local ring at the vertex of $\tilde{X}$). The value of this reformulation is that, starting from known global results, it often makes it possible to conjecture, or even prove, analogous results for complete Noetherian local rings more general than those which actually occur in the global statement, for example for local rings that are not necessarily of equal characteristic. Thus, Serre's duality theorem for projective space XII XII.1.1 suggested the useful local duality theorem V 2.1. Serre's fundamental theorem on the cohomology of coherent Modules on projective space (finiteness, asymptotic behavior for large n of $\mathrm{H}^i(X, F(n))$, cf. EGA III 2.2.1) generalizes to a structure theorem for the local invariants $\mathrm{H}^i_{\mathfrak{m}}(M)$; see V 3. Likewise, the Lefschetz theorems for the fundamental group and the Picard group ("equivalence criteria"), well known in the classical case and subsequently extended to an arbitrary base field, suggested the "local"Lefschetz theorems of Exposés X and XI. Of course, the local theorems are in turn valuable tools for obtaining global statements. For example, local duality allows a global asymptotic property XII XII.1.3 (i) to be formulated by the vanishing of certain local invariants $\mathrm{H}^i(F_x)$. More substantially, the local Lefschetz theorems, implying for example the "purity"or parafactoriality of certain complete-intersection local rings (X 3.4 and XI XI.3.14), make it possible in the global Lefschetz theorems to dispense with certain nonsingularity hypotheses, as in X 3.5, 3.6, 3.7.

Another useful generalization of theorems concerning projective schemes over a field k **53**
consists in replacing k by a general base scheme. Thus, the sequel to EGA III will give(*) a generalization in this direction of Serre duality(*); the finiteness and asymptotic-behavior theorems for $\mathrm{H}^i(X, F(n))$ were stated in EGA III 2.2.1 over a general base scheme; finally, the Lefschetz theorems can likewise be developed for a projective morphism, as we saw in XII XII.4.9, thanks to the local theorem XII XII.4.7. Of course, working over a general base scheme also leads to essentially new statements, such as the "comparison theorem"EGA III 4.15 and the existence theorem for sheaves EGA III 5.1.4 (which, as we saw elsewhere in IX, arise from the same key theorems of a cohomological nature as the Lefschetz theorems for π_1 and Pic).

It is therefore necessary to isolate theorems that simultaneously encompass the two

(*) *Editor's note.* in fact, this generalization is not found there; see the note below and the editor's note (5) on page 2

(*) cf. the Hartshorne Seminar, cited at the end of Exp. IV.

generalizations just mentioned of statements concerning projective schemes over a field. The natural objects for such a common generalization are the *Noetherian rings separated and complete for an I-adic topology.* From this point of view, their study has not yet been seriously undertaken and seems to me at present the most interesting subject in the local theory of coherent sheaves. Here is a typical problem in this direction:

Conjecture 1.1 ("Second affine finiteness theorem"[**] [(*)]). *Let M be a module of finite type over a Noetherian ring A (which may, if necessary, be assumed to be a quotient of a regular ring), and let J be an ideal of A. Prove that the modules $\mathrm{H}^i_J(M)$ are "J-cofinite", i.e. that the modules*

$$\mathrm{Hom}_A(A/J, \mathrm{H}^i_J(M))$$

are of finite type.

Recall that $\mathrm{H}^i_J(M)$ denotes the module $\mathrm{H}^i_Y(X, \tilde{M})$ (where $X = \mathrm{Spec}(A)$, $Y = V(J)$) of Exp. I, interpreted in II in terms of an inductive limit of the cohomology of Koszul complexes, **54**
or, for $i \geq 2$, the module $\mathrm{H}^{i-1}(X - Y, \tilde{M})$. To tell the truth, XIII.1.1 should be a consequence of a more precise statement, implying that the $\mathrm{H}^i_J(M)$ belong to a suitable abelian subcategory $\underline{D}_J$ of the category $\underline{C}_J$ of A-modules with support $\subset Y = V(J)$, such that $H \in \mathrm{Ob}\,\underline{D}_J$ implies that H is J-cofinite. (N.B. The category of modules H with support contained in $V(J)$ and which are J-cofinite is unfortunately not stable under passage to quotients!). The essential problem would then consist in defining $\underline{D}_J$. More precisely, the solution of problem XIII.1.1 should emerge (at least if A is a quotient of a regular ring) from a duality theory generalizing both local duality and the duality theory for projective morphisms alluded to above, and which would be of the following kind:

Conjecture 1.2 ("Affine duality"[(*)]). *Suppose that A is regular, separated, and complete for the J-adic topology. Let $C^\bullet(A)$ be an injective resolution of A.*

(i) Prove that the functor

$$D_J : L_\bullet \longmapsto \mathrm{Hom}_J(L_\bullet, C^\bullet(A))$$

from the category of complexes of A-modules that are free of finite type in every dimension and bounded above (where the morphisms are homomorphisms of complexes up to homotopy) to the category of complexes of A-modules $K^\bullet$ that are injective in every dimension and bounded above (where the homomorphisms are defined in the same way), is fully faithful.

[**]This conjecture, and Conjecture 1.2 below, are false, as was shown by R. Hartshorne, "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164, Section 3.

[(*)]*Editor's note.* However, if A is complete local (respectively regular of positive characteristic) and J is the maximal ideal, the statement is true for M of finite type (respectively $M = A$), cf. (Hartshorne R., "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164, Corollary 1.4) (respectively (Huneke C. & Sharp R., "Bass numbers of local cohomology modules", *Trans. Amer. Math. Soc.* **339** (1993), no. 2, p. 765–779), which moreover contains much stronger results). For completely different methods (D-modules) for treating characteristic zero, see (Lyubeznik G., "Finiteness properties of local cohomology modules (an application of D-modules to commutative algebra)", *Invent. Math.* **113** (1993), no. 1, p. 41–55); see also, by the same author, "Finiteness properties of local cohomology modules for regular local rings of mixed characteristic: the unramified case", *Comm. Algebra* **28** (2000), no. 12, p. 5867–5882, Special issue in honor of Robin Hartshorne, and "Finiteness properties of local cohomology modules: a characteristic-free approach", *J. Pure Appl. Algebra* **151** (2000), no. 1, p. 43–50. The notion of a cofinite module has since evolved under Hartshorne's leadership. One says that M is J-cofinite if its support is contained in $V(J)$ and if all the $\mathrm{Ext}^i_A(A/J, \mathrm{H}^i_J(M))$ are of finite type. On this subject, see for example (Delfino D. & Marley Th., "Cofinite modules and local cohomology", *J. Pure Appl. Algebra* **121** (1997), no. 1, p. 45–52).

[(*)]This conjecture, false as stated, was nevertheless established in a fairly similar form by R. Hartshorne, "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164.

(ii) *Prove that, for every $K^\bullet$ of the form $D_J(L_\bullet)$, the* $\mathrm{H}^i(K^\bullet)(= \mathrm{Ext}^i_Y(X; L_\bullet, \mathcal{O}_X))$ *are J-cofinite.*

(iii) *More precisely, prove that the $K^\bullet$ which are homotopic to a complex of the form $D_J(L_\bullet)$ can be characterized by finiteness properties of the $\mathrm{H}^i(K^\bullet)$ stronger than the one considered in (ii), for example by the property $\mathrm{H}^i(K^\bullet) \in \mathrm{Ob}\,\underline{D}_J$, where $\underline{D}_J$ is a suitable abelian category, as contemplated above.* **55**

Note that the problem has an affirmative solution when A is local and J is an ideal of definition (cf. Exp. IV), and also when J is the zero ideal. In these two exceptional cases, one may take for $\underline{D}_J$ simply the category of Modules with support $V(J)$ which are J-cofinite (which in the second case means simply that one takes the category of Modules of finite type over A). An affirmative solution of Conjecture XIII.1.2 in general would give one for XIII.1.1, by taking for $L_\bullet$ the dual of a finite free resolution of M. On the other hand, an affirmative solution of XIII.1.1 would give an affirmative answer to the first part of the following conjecture, which we formulate in "global"form:

Conjecture XIII.1.3. *Let $X \subset \mathbf{P}^r_k$ be a closed subscheme of the standard projective scheme which is locally a complete intersection and each of whose irreducible components has codimension $\geq s$. Let $U = \mathbf{P}^r_k - X$.*

(i) *Prove that, for every coherent Module F on U, one has*

$$\dim \mathrm{H}^i(U, F) < +\infty \text{ for } i \geq s.^{(*)}$$

(ii) *Give an example, with X connected and regular, for which one has*

$$\mathrm{H}^s(U, F) \neq 0.$$

To see that (i) is a special case of XIII.1.1, consider

$$\mathrm{H}^i(U, F(\cdot)) = \bigoplus_n \mathrm{H}^i(U, F(n)) = \mathrm{H}^i(E^{r+1} - \tilde{X}, \tilde{F})$$

as a module over the affine ring $k[t_0, \ldots, t_r]$ of the cone E^{r+1} over $\mathbf{P}^r$. This module is none other than $\mathrm{H}^{i+1}_J(M)$, where J is the ideal of the cone $\tilde{X}$ over X in E^{r+1}. On the other hand, the hypothesis on X, which implies that $\tilde{X}$ is also a complete intersection of codimension $\geq s$ at every point of E^{r+1} distinct from the origin, implies that $\mathrm{H}^{i+1}_J(M)$ is zero away from the origin for $i \geq s$. Thus, if it is J-cofinite as required by XIII.1.1, it is *a fortiori* $\mathfrak{m}$-cofinite, which readily implies that it is finite-dimensional in every degree (*) Note moreover that Conjecture **56**

(*) Partie (i) of this conjecture was proved by R. Hartshorne when X is smooth cf. *Ample subvarieties of algebraic varieties*, Notes written in collaboration with C. Musili, Lect. Notes in Math, vol. 156, Springer-Verlag, Berlin-New York, 1970, Theorem III.5.2. This author also found an example for (ii), cf. R. Hartshorne, "Cohomological dimension of algebraic varieties", *Ann. of Math. (2)* **88** (1968), p. 403–450, example on page 449.

(*) *Editor's note.* Hartshorne proved (Hartshorne R., "Cohomological dimension of algebraic varieties", *Ann. of Math. (2)* **88** (1968), p. 403–450) that the cohomology $H^{n-1}(\mathbf{P}^n_k - X, F)$ is zero for F coherent and X of positive dimension (k algebraically closed). In fact, thanks essentially to Serre duality and the Lichtenbaum theorem — the vanishing of the cohomology of coherent sheaves in the maximal dimension of irreducible noncomplete quasi-projective varieties —, one is reduced to proving that the formal completion $\hat{\mathbf{P}}^n_k$ and X have the same field of rational functions. This is the difficult point (Theorem 7.2 of *loc. cit.*); in other words, $\mathbf{P}^n_k$ is $G3$ in the terminology of (Hironaka H. & Matsumura H., "Formal functions and formal embeddings", *J. Math. Soc. Japan* **20** (1968), p. 52–82). These authors independently proved the preceding results, and in fact much more. They proved that X is universally $G3$ and calculated the field of rational functions of the formal completion of an abelian variety along a positive-dimensional subvariety. It is in this article that the now-classical conditions $G1, G2, G3$ appear for the first time.

XIII.1.3 already arises for a nonsingular irreducible curve X in $\mathbf{P}^3$: it is unknown whether in this case the $\mathrm{H}^2(\mathbf{P}^3 - X, \mathcal{O}_X(n))$ are finite-dimensional, or whether they are necessarily zero(*). It is not even known whether there exists an irreducible curve in $\mathbf{P}^3$ which is not set-theoretically the intersection of two hypersurfaces(*).

Problem XIII.1.4. *Give an affine variant of the "comparison theorem"* EGA III 4.1.5 *as a theorem on the commutation of the functors* H^i_J *with certain inverse limits.*

Finally, in the present line of thought, I had posed the following problem: let A be a *complete* regular Noetherian local ring, and K its field of fractions; prove that $\mathrm{Ext}^i_A(K, A) = 0$ for every i. An affirmative answer was given on the spot by M. Auslander; the regularity hypothesis may be replaced by the hypothesis that A is integral, and in fact $\mathrm{Ext}^i_A(K, M) = 0$ for every i whenever M is of finite type over A. This follows immediately from the following statement, due to Auslander: if A is a complete Noetherian local ring, then for every module of finite type M over A, the functors $\mathrm{Ext}^i_A(., M)$ transform inductive limits into inverse limits(*).

XIII.2 Problems related to π_0: local Bertini theorems

Let A be a complete noetherian local ring, f an element of its maximal ideal, $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A/fA)$. The use of the local "Lefschetz" technique makes it possible to give criteria for $Y' = X' \cap Y$ (where $X' = X - \{\mathfrak{m}\}$) to be connected, in terms of hypotheses on X'. Thus, it suffices that: a) X' is connected, b) $\mathrm{prof}\, \mathcal{O}_{X',x} \geq 2$ for every closed point x of X', c) f is **57**
A-regular. Note, however, that hypotheses b) and c) are not purely topological in nature; for example, they are not invariant under replacement of X by X_{red}. In the analogous situation for a projective scheme X' over a field and a hyperplane section Y' of X', the use of the "Bertini theorem" and Zariski's "connectedness theorem" in fact gives results of a distinctly more satisfactory form, which had led me, in the oral seminar, to state a conjecture that I have since solved affirmatively. Let us therefore state here:

Theorem XIII.2.1. *Let A be a complete noetherian local ring, X its spectrum, $\mathfrak{a}$ the closed point of X, $X' = X - \{\mathfrak{a}\}$. Suppose that X satisfies the following conditions (where k denotes an integer ≥ 1):*

a_k) The irreducible components of X' have dimension $\geq k + 1$.

b_k) X' is connected in dimension $\geq k$, i.e. it cannot be disconnected by a closed subset of dimension $< k$ (cf. III 3.8).

Let m be an integer, $0 \leq m \leq k$, and let $f_1, \dots, f_m \in \mathfrak{r}(A)$; set $B = A/\sum_i f_i A$, $Y = \mathrm{Spec}(B) = V(f_1) \cap \cdots \cap V(f_m)$, $Y' = X' \cap Y = Y - \{\mathfrak{a}\}$. Then Y satisfies conditions a_{k-m}), b_{k-m}). In particular, for every sequence of $m \leq k$ elements $f_1, \dots, f_m$ of $\mathfrak{r}(A)$, $Y' = X' \cap V(f_1) \cap \cdots \cap V(f_m)$ is connected.

It is moreover easy to see that if the last conclusion holds (it plainly suffices to take $m = k$ there), and if the case where X is irreducible of dimension 0 or 1 is excluded, it follows that the irreducible components of X' have dimension $\geq k + 1$, and X' is connected in dimension $\geq k$, so that, in a sense, XIII.2.1 is a "best possible" result.

Let us give the principle of the proof of XIII.2.1. Only condition b_{k-m}) for Y presents a problem. For a given k, one is easily reduced to the case where X is integral, and even (by passing to the normalization, which is finite over X) to the case where X is *normal.* If $k = 1$, **58**

(*)The question has just been answered affirmatively by R. Hartshorne (Hartshorne R., "Ample vector bundles", *Publ. Math. Inst. Hautes Études Sci.* **29** (1966), p. 63–94, Theorem 8.1) and H. Hironaka.

(*)*Editor's note.* see Conjecture XII.3.5 and the corresponding note.

(*)*Editor's note.* write $K = \varinjlim_{a \neq 0} A[1/a]$ and observe that $a \neq 0$ is A-regular.

hence $\dim X' \geq 2$, then X' has depth ≥ 2 at its closed points and one may apply the result recalled at the beginning of n°, which shows that $Y' = X' \cap V(f)$ is connected. In the case $k \geq 1$, suppose the theorem proved for $k' < k$. By induction on m, one is reduced to the case $m = 1$, i.e. to verifying that, for $f_1 \in \mathfrak{r}(A)$, $X' \cap V(f_1)$ is connected in dimension $\geq k-1$. If it were not, i.e. if it were disconnected by a Z' of dimension $< k-1$, there would exist a sequence $f_2, \ldots, f_k$ such that $X' \cap V(f_1) \cap \cdots \cap V(f_k)$ is disconnected, and in this sequence one may choose $f_2 \in \mathfrak{r}(A)$ arbitrarily, subject only to the condition that it not vanish at any point of a certain finite subset F of X' (namely the set of maximal points of Z'). Moreover, one easily verifies, using the fact that X' is normal and therefore satisfies Serre's condition (S_2)(*), that there exists a finite subset F' of X' such that $f \in \mathfrak{r}(A)$, $V(f) \cap F' = \emptyset$ implies that $V(f) \cap X'$ also satisfies condition (S_2). One may then choose f_2 so that f_2 vanishes neither on F nor on F', and hence so that $X' \cap V(f_2)$ satisfies (S_2). But then, by the theorem of Hartshorne III 3.6, $X' \cap V(f_2)$ is connected in codimension 1 and thus (since every component of $X' \cap V(f_2)$ has dimension $\geq k$) it is connected in dimension $\geq k-1$. Applying the induction hypothesis to $V(f_2) = \operatorname{Spec}(A/f_2A)$, it follows that $X' \cap V(f_2) \cap V(f_1) \cap V(f_3) \cap \cdots \cap V(f_k)$ is connected, whereas it had been constructed disconnected, a contradiction.

Let us mention some interesting corollaries:

Corollary XIII.2.2. *Let $f\colon X \to Y$ be a proper morphism of locally noetherian preschemes, with Y integral, $y_0 \in Y$, and y_1 the generic point of Y. Assume that*

a) Y is unibranch at y_0, and every irreducible component of X dominates Y.

b) The irreducible components of X_1 have dimension $\geq k+1$, and X_1 is connected in dimension $\geq k$.

Then the irreducible components of X_0 have dimension $\geq k+1$, and X_0 is connected in dimension $\geq k$. **59**

Indeed, Zariski's connectedness theorem (cf. EGA III 4.3.1) implies that X_0 is connected; to show that it is not disconnected by a closed subset of dimension $< k$, one is reduced to showing that the local rings at points $x \in X_0$ such that $\dim \overline{x} < k$ have a spectrum that is not disconnected by x. But this is true without assuming either that f is proper or that Y is unibranch at y_0. To see this, one is reduced to the case where X is integral and dominates Y, and, if desired, where Y is affine of finite type over $\mathbf{Z}$, so that one is under the hypotheses of the dimension formula for $\mathcal{O}_{X,x}$ over $\mathcal{O}_{Y,y_0}$. Using the finiteness of the normalization in this case, one may even suppose X *normal*; hence, by a theorem of Nagata(*), the completion of a local ring $\mathcal{O}_{X,x}$ of X' is still normal, and therefore (if $\mathcal{O}_{X,x}$ has dimension N) $\operatorname{Spec}(\hat{\mathcal{O}}_{X,x})$ is connected in dimension $\geq N-1$. Let $n = \dim \mathcal{O}_{Y,y_0}$; then $\operatorname{deg\,tr}\ k(x)/k(y) < k$ implies $\dim \mathcal{O}_{X,x} > n + (k+1) - k = n+1$, in view of $\dim X_1 \geq k+1$, and, taking a system $f_1, \ldots, f_n$ of parameters of $\mathcal{O}_{Y,y_0}$ and lifting them to elements of $\mathcal{O}_{X,x}$, one sees from XIII.2.1 that $\operatorname{Spec}(\hat{\mathcal{O}}_{X,x}/\sum f_i \hat{\mathcal{O}}_{X,x})$ is connected in dimension ≥ 1, i.e. is not disconnected by its closed point, or, equivalently, $\operatorname{Spec}(\hat{\mathcal{O}}_{X_0,x})$ is not disconnected by its closed point; *a fortiori* the same holds for $\operatorname{Spec}(\mathcal{O}_{X_0,x})$.

As in the case of the ordinary connectedness theorem, one may vary XIII.2.2 by taking geometric fibers (over algebraic closures of the residue fields), provided that one assumes Y to be *geometrically* unibranch at y_0, or (with no hypothesis other than that Y is noetherian) that f is universally open. Applying this to the case where Y is the dual scheme of a projective scheme $\mathbf{P}^r_k$ over a field, one recovers a strengthened form of the global result that had inspired XIII.2.1, namely:

(*)Cf. EGA IV 5.7.2.

(*)Cf. EGA IV 7.8.3 (i) (ii) (v).

Corollary 2.3(*). *Let X be a closed subscheme of $\mathbf{P}^r_k$ (k a field); suppose that the irreducible components of X have dimension $\geq l+1$, and that X is geometrically connected in dimension $\geq$* 60
l. Then for every sequence $H_1, \dots, H_m$ of m hyperplanes of $\mathbf{P}^r_k$ $(0 \leq m \leq l-1)$, $X \cap H_1 \cap \cdots \cap H_m$ satisfies the same condition with $l - m$; in particular, it is geometrically connected in dimension $\geq l-1$.

One may moreover modify this statement in the obvious way for the case where one is given a proper morphism $X \to \mathbf{P}^r_k$, which is not necessarily an immersion, and an analogous extension is possible for XIII.2.1 (by considering a scheme proper over X'). These statements moreover follow formally from the statements given here, in view of the ordinary connectedness theorem, which reduces us to the case of a finite morphism.

Corollary XIII.2.4. *Let A be a complete normal noetherian local ring of dimension $\geq k+2$. Let $X = \operatorname{Spec}(A)$, $X' = X - \{\mathfrak{a}\}$, and let $f_1, \dots, f_k$ be elements of $\mathfrak{r}(A)$; then $Y' = X' \cap V(f_1) \cap \cdots \cap V(f_k)$ is connected, and $\pi_1(Y') \to \pi_1(X')$ is surjective.*

One proceeds as in SGA 1 X 2.11.

Throughout the preceding discussion, only questions of *connectedness* were involved. In the global case, however, well-known theorems assert that for an irreducible projective variety $X \subset \mathbf{P}^r_k$, with k algebraically closed, its intersection with a sufficiently "general"hyperplane H is irreducible (and not merely connected): this is the *Bertini theorem*, proved by Zariski, which in turn implies, by Zariski's connectedness theorem, that for *every* H, $H \cap X$ is connected (although not necessarily irreducible). One may moreover proceed in the opposite direction, by proving the latter result using a Lefschetz-type technique and deducing the Bertini theorem by reducing to the case where X is normal and using the following result: for a "sufficiently general"H, $X \cap H$ is also normal. This suggests the

Conjecture 2.5(*). *Let A be a complete normal noetherian local ring. Show that there exists a nonzero $f \in \mathfrak{r}(A)$ such that $Y' = X' \cap V(f) = Y - \{\mathfrak{a}\}$ (where $Y = \operatorname{Spec}(A/fA)$) is normal* 61
(and hence irreducible by XIII.2.1 if $\dim A \geq 3$).

Ideally, one should show that, in a suitable sense, there are even "many"elements f having the property in question; for example, that f may be chosen in an arbitrary power of the maximal ideal. Using Serre's normality criterion and the remark made above concerning Serre's property (S_2), one sees that XIII.2.5 would have an affirmative answer if the same were true for the

Conjecture 2.6(*). *Let A be a complete noetherian local ring, U an open subset of its spectrum X, and F a finite subset of $X' = X - \{\mathfrak{a}\}$. Suppose that U is regular. Prove that there exists an $f \in \mathfrak{r}(A)$ such that $V(f) \cap U$ is regular, and $V(f) \cap F = \emptyset$.*

For a result of "local Bertini"type, see Chow [2]

(*) *Editor's note.* for a very beautiful direct proof, see (Fulton W. & Lazarsfeld. R., "Connectivity and its applications in algebraic geometry", in *Algebraic geometry (Chicago, Ill., 1980)*, Lect. Notes in Math., vol. 862, Springer, Berlin-New York, 1981, p. 26–92, Theorem 2.1). Cf. also [HL], cited in the editor's note ((*)) on page 135.

(*) *Editor's note.* see the following editor's note.

(*) *Editor's note.* a proof of this conjecture can now be found in the literature, and the preceding conjecture must therefore also be regarded as proved, as indicated above. One can also find two earlier published but unfortunately unsuccessful proof attempts, by Flenner and Trivedi. See Trivedi V., "Erratum: "A local Bertini theorem in mixed characteristic"", *Comm. Algebra* **25** (1997), no. 5, p. 1685-1686. However, the editor has not verified that the proof is now complete.

XIII.3 Problems related to π_1

Here again there are many questions, suggested by the global results or by the transcendental results.

Conjecture 3.1$^{(*)}$**.** *Let A be a complete noetherian local ring with algebraically closed residue field, let $X = \mathrm{Spec}(A)$, let $X' = X - \{\mathfrak{a}\}$, where $\mathfrak{a}$ is the closed point. Suppose that the irreducible components of X have dimension ≥ 2 and that X' is connected.*

(i) Prove that $\pi_1(X')$ is topologically finitely generated.

(ii) If p is the characteristic exponent of the residue field k of A, prove that the largest topological quotient group of $\pi_1(X')$ that is "of order prime to p"is finitely presented.

For part (i), using the descent theory of SGA 1 IX 5.2 and Theorem 2.4, one is reduced to **62**
the case in which A is normal of dimension 2. In this case, a systematic method for studying the fundamental group of X', introduced by Mumford [5] in the transcendental setting, consists in *desingularizing* X, i.e. considering a projective birational morphism $Z \to X$, with Z integral and regular, which induces an isomorphism $Z' = Z|X' \to X'$. It is plausible that such a Z always exists; in any case this is what Abhyankar's method [1] proves in the case of "equal characteristic"$^{(*)}$. Let C be the fiber over the closed point of X under $Z \to X$; it is an algebraic curve over the residue field k, connected by the connectedness theorem. The solution of XIII.3.1 then seems to be related to the following problem.

Problem XIII.3.2. *With the preceding notation, relate $\pi_1(X')$ to the topological invariants of C, in particular to its fundamental group, so as to exhibit the topological finite generation of $\pi_1(X')$, using for example* SGA I*, Theorem* X 2.6*.*

Another method would be to regard A as a finite algebra over a complete *regular* local ring B of dimension 2, ramified along a curve C contained in $\mathrm{Spec}(B) = Y$. This leads to the following problem.

Problem XIII.3.3. *Let A be a complete regular local ring of dimension* 2*, with algebraically closed residue field k; let X be its spectrum, and let C be a closed subset of X of dimension* 1*. Define local invariants of the embedded curve C that have a meaning independent of the residue characteristic and whose knowledge makes it possible to calculate the fundamental group of $X - C$ by generators and relations when k has characteristic zero. Prove that when k has characteristic $p > 0$, the "tame"fundamental group of $X - C$ is a quotient of the preceding one, and that the two fundamental groups (in characteristic* 0 *and in characteristic $p > 0$) have the same maximal quotient of order prime to p.*

Of course, XIII.3.3 shows that in XIII.3.1 one should also replace X' by a scheme of the form $X - Y$, where Y is a closed subset of X that has codimension ≥ 2 in every component of **63**
X that contains it. When this restriction on Y is dropped, there should still be an analogous finiteness result, provided that "tame"-type restrictions are imposed on the ramification at the maximal points of the irreducible components of Y that have codimension 1.

Problem XIII.3.4. *Let A be a complete noetherian local ring of dimension* 2*, with algebraically closed residue field. Again let $X = \mathrm{Spec}(A)$ and $X' = X - \{\mathfrak{a}\}$. Find special structural properties of $\pi_1(X')$ in the case where A is a complete intersection.*

$^{(*)}$*Editor's note.* The analogous statement is true for (connected) schemes X of finite type over a separably closed field k, under the hypothesis of strong desingularization for all $\bar{k}$-schemes of finite type; in particular, this holds if k has characteristic zero or if X has dimension ≤ 2. One reduces to the case of quasi-projective surfaces by Lefschetz-type techniques developed by Mme Raynaud; cf. the notes *supra*; see SGA 7.1, Theorem II.2.3.1.

$^{(*)}$The possibility of "resolving"A has now been proved in full generality by Abhyankar [8].

A satisfactory solution of this problem might make it possible to solve the following old problem:

Conjecture 3.5$^{(*)}$**.** *Find an irreducible curve in* $\mathbf{P}^3_k$ *(where* k *is an algebraically closed field), preferably nonsingular, which is not set-theoretically the intersection of two hypersurfaces.*

(Kneser [4] shows that it can always be obtained as the intersection of three hypersurfaces).

XIII.4 Problems related to higher π_i: local and global Lefschetz theorems for complex analytic spaces$^{(*)}$

Let X be a scheme locally of finite type over the field of complex numbers $\mathbf{C}$. One can associate with it a complex analytic space X^h, and hence homotopy and homology invariants $\pi_i(X^h)$, $\mathrm{H}_i(X^h)$, $\mathrm{H}^i(X^h)$, etc. Moreover, X is known to be connected if and only if X^h is connected, so there is a bijection

$$\pi_0(X^h) \to \pi_0(X).$$

Similarly, since every étale cover X' of X defines an étale cover X'^h of X^h , there is a canonical homomorphism 64

$$\pi_1(X^h) \to \pi_1(X),$$

which, by a theorem of Grauert–Remmert, is known to identify the second group with the completion of the first for the topology defined by subgroups of finite index. This simply expresses the fact that $X' \longmapsto X'^h$ is an equivalence from the category of étale covers of X to the category of finite étale covers of X^h. It follows that the results of this seminar, obtained by purely algebraic methods, concerning $\pi_0(X)$ and $\pi_1(X)$ imply results for $\pi_0(X^h)$ and $\pi_1(X^h)$, which are transcendental in nature. Moreover, if X is proper, the well-known exact sequence $0 \to \mathbf{Z} \to \mathbf{C} \to \mathbf{C}^* \to 0$ shows that the Néron–Severi group of X (the quotient of its Picard group by the connected component of the identity) is isomorphic to a subgroup of $\mathrm{H}^2(X^h, \mathbf{Z})$; in the nonsingular Kähler case, it is the subgroup denoted $\mathrm{H}^{(1,1)}(X^h, \mathbf{Z})$ (classes of type $(1,1)$):

$$\mathrm{Pic}(X)/\operatorname{Pic}^0(X) \subset \mathrm{H}^2(X, \mathbf{Z}).$$

Consequently, the information that we have obtained about Picard groups implies information, admittedly very partial, about the groups $\mathrm{H}^2(X^h, \mathbf{Z})$. It is tempting to complete all these fragmentary results with conjectures.

Very precise indications pointing in the same direction as those just mentioned are supplied by a classical theorem of Lefschetz [7]. It asserts that if X is a nonsingular irreducible projective analytic space of dimension n, and if Y is a nonsingular hyperplane section, then the inclusion

$$Y^{n-1} \to X^n$$

induces a homomorphism 65

$$\pi_i(Y^{n-1}) \to \pi_i(X^n)$$

that is an *isomorphism for* $i \leq n-2$ and an *epimorphism for* $i = n-1$. The analogous statement follows for the homomorphisms

$$\mathrm{H}_i(Y^{n-1}) \to \mathrm{H}_i(X^n)$$

$^{(*)}$*Editor's note.* As of autumn 2004, this problem is still open.

$^{(*)}$*Editor's note.* The statements are made more precise in the Comments (Section XIII.6). The conjectures appearing there have become theorems; cf. the footnotes to Section XIII.6.

on homology (integral homology, to be definite), while in cohomology

$$\mathrm{H}^i(X^n) \to \mathrm{H}^i(Y^{n-1})$$

is an isomorphism in degree $i \leq n-2$ and a monomorphism in degree $i = n-1$. We have obtained variants of these results in the setting of schemes, for π_0, π_1, and Pic, valid moreover, to a large extent, without nonsingularity hypotheses; cf. Exp. XII. Furthermore, in removing the nonsingularity hypotheses, we made essential use of "local"variants of these global Lefschetz theorems. All this suggests the following problems, which will doubtless have to be attacked simultaneously(*).

Problem XIII.4.1. *Let X be an analytic space and Y a closed analytic subset of X (or merely a closed subset?)(*) such that for every $x \in Y$, the local ring $\mathcal{O}_{X,x}$ is a complete intersection. Let n be the complex codimension of Y in X. Is the canonical homomorphism*

$$\pi_i(X - Y) \to \pi_i(X)$$

an isomorphism for $i \leq n-2$ and an epimorphism for $i = n-1$?

In this problem and the following ones, a base point is of course implicitly understood to have been chosen in order to define the homotopy groups. To state the following problem, one must define, for an analytic space X (more generally, for a locally path-connected space) and a point $x \in X$, local invariants $\pi_i^x(X)$(*). To do this, choose a nonconstant map f from the 66
interval $[0,1]$ to X such that $f(0) = x$ and $f(t) \neq x$ for $t \neq 0$; such a map exists if x is not an isolated point. Then, for every neighborhood U of x, there is an $\varepsilon > 0$ such that $0 < t < \varepsilon$ implies $f(t) \in U$, and the homotopy groups $\pi_i(U - x, f(t))$ are essentially independent of t: as t varies, they are connected by a transitive system of isomorphisms. They may be denoted by $\pi_i(U - x, f)$. Set

$$\pi_i^x(X) = \varprojlim_U \pi_{i-1}(U - x, f),$$

where the inverse limit is taken over the system of open neighborhoods U of x. Strictly speaking, this limit depends on f and should be denoted $\pi_i^x(X, f)$, but one verifies that as f varies these groups are isomorphic to one another(*); more precisely, they form a local system on the space of paths of the kind under consideration issuing from x. These invariants are the homotopical counterpart of the local cohomology invariants $\mathrm{H}_x^i(F)$ for a sheaf F on X, introduced in I, and should play the role of *relative local homotopy groups* of X modulo $X - x$. Their vanishing for $i \leq n$ and every $x \in Y$, where Y is a closed subset of X of topological dimension $\leq d$, should imply that the homomorphisms

$$\pi_i(X - Y) \to \pi_i(X)$$

are bijective for $i < n - d$ and surjective for $i = n - d$(*). From this point of view, XIII.4.1 would imply, when Y is reduced to a point, a conjecture of purely local nature, expressed by

$$\pi_i^x(X) = 0 \text{ for i} \leq \text{n} - 1$$

when X is a complete intersection of dimension n at x.

As an example of the local invariants $\pi_i^x(X)$, note that if x is a nonsingular point of X of complex dimension n, then 67

(*)The formulations XIII.4.1 to XIII.4.3 below are provisional. See Conjectures A to D below, in the "Comments on Exp. XIII", for more satisfactory formulations, and also Exp. XIV.

(*)*Editor's note.* The meaning of this question is not clear; indeed, the statement of the problem itself does not seem to make sense in this case, since the codimension of Y in X is not defined when Y is no longer assumed analytic.

(*)If $i \geq 2$. For the case $i \leq 1$, cf. the *Comments* in no. 6 below, page 134.

(*)At least if x does not locally disconnect X; cf. the *Comments* below, page 134.

(*)For a corrected formulation, cf. the *Comments* below, page 134.

$$\pi_i^x(X) = \pi_{i-1}(S^{2n-1}),$$

where S^{2n-1} denotes the sphere of dimension $2n-1$. In particular, in this case $\pi_i^x(X) = 0$ for $i \leq 2n-1$, which corresponds to the fact that if a closed subset Y of codimension $\geq m$ is removed from a topological *manifold* X, then $\pi_i(X - Y) \to \pi_i(X)$ is an isomorphism for $i \leq m-2$ and an epimorphism for $i = m-1$.

This being said:

Problem XIII.4.2. *Let X be an analytic space, let $x \in X$, let t be a section of $\mathcal{O}_X$ vanishing at x, and let Y be the zero set of t. Suppose that the following conditions are satisfied:*

a) t is regular at x (i.e. it is not a zero divisor at x; this hypothesis may moreover be superfluous).

b) At the points x' of $X - Y$ near x, $\mathcal{O}_{X,x'}$ is a complete intersection. If XIII.4.1 is true, this hypothesis should be replaceable by the following more general one: for x' as above, $\pi_i^{x'}(X) = 0$ for $i \leq n-1$.

c) At the points y of $Y - \{x\}$ near x, one has

$$\operatorname{prof} \mathcal{O}_{X,y} \geq n$$

(it suffices, for example, that $\operatorname{prof} \mathcal{O}_{X,x} \geq n$*).*

Under these conditions, is the canonical homomorphism

$$\pi_i^x(Y) \to \pi_i^x(X)$$

an isomorphism for $i \leq n-2$ and an epimorphism for $i = n-1$?

Here, finally, is a global variant of XIII.4.2, which should follow from it by considering the affine cone at its vertex and which would generalize the classical Lefschetz theorems:

Problem XIII.4.3. *Let X be a projective analytic space, equipped with an ample invertible Module L; let t be a section of L, and let Y be the zero set of t. Suppose that:* **68**

a) t is a regular section (a hypothesis that may be superfluous).

b) For every $x \in X - Y$, $\mathcal{O}_{X,x}$ is a complete intersection. This should be replaceable by $\pi_i^x(X) = 0$ for $i \leq n-1$.

c) For every $x \in Y$, $\operatorname{prof} \mathcal{O}_{X,x} \geq n$*.*

Under these conditions, is the homomorphism

$$\pi_i(Y) \to \pi_i(X)$$

an isomorphism for $i \leq n-2$ and an epimorphism for $i = n-1$?

We leave it to the reader to state analogous conjectures of a *cohomological* nature(*), with hypotheses and conclusions concerning local cohomological invariants (with coefficients in a given group). In any event, the key result seems to be XIII.4.2, with hypothesis b) taken in the more general form just indicated, whether one adopts the point of view of homology or that of homotopy.

We have stated these conjectures in the transcendental setting in the hope of interesting topologists in them and convincing them that "Lefschetz"-type questions are far from closed. Of course, now that we are about to possess a good theory of the cohomology of schemes (with finite coefficients), thanks to the recent work of M. Artin, the same questions arise in the setting of schemes, and it is difficult to doubt that they will receive an affirmative answer in the near future(*).

(*) Cf. Exp. XIV for the corresponding results in the theory of schemes.

(*) See the preceding note.

XIII.5 Problems related to local Picard groups

A first fundamental problem, pointed out for the first time by Mumford [5] in a particular case, is the following. Let A be a complete local ring with residue field k, let $X = \mathrm{Spec}(A)$, and let $U = \mathrm{Spec}(A) - \{a\}$, where a is the maximal ideal of A, i.e. the closed point of $\mathrm{Spec}(A)$. The problem is to construct a strict projective system G of locally algebraic groups G_i over k, and a natural isomorphism **69**

$$\mathrm{Pic}(U) \simeq G(k) \tag{$+$}$$

where, of course, we set $G(k) = \varprojlim G_i(k)$. Heuristically, the problem is to "put an algebraic-group structure"(or at least a pro-algebraic one, in a suitable sense) on the group $\mathrm{Pic}(U)$.

It is clear that, as stated, the problem is not sufficiently precise, since the datum of an isomorphism $(+)$ is far from characterizing the pro-object G. If A contains a subfield, again denoted by k, that is a coefficient field, the problem can be made more precise by requiring that, for a varying extension k' of k, there be an isomorphism, functorial in k':

$$\mathrm{Pic}(U') \simeq G(k') \tag{$+'$}$$

where U' is the open subset analogous to U in $\mathrm{Spec}(A')$, with $A' = A\hat{\otimes}_k k'$. One can proceed analogously even if A has no coefficient field, provided that k is perfect, since one can then construct an A' functorially "by residue-field extension k'/k". Moreover, when A admits a coefficient field, the algebraic structure obtained on $\mathrm{Pic}(U)$ will depend essentially on the choice of that coefficient field (as one sees already for the projective cone over an elliptic curve); it therefore seems necessary to start from a "pro-algebraic ring over k"in the sense of Greenberg [3], in order to define the pro-object G. It is moreover conceivable that, when no coefficient field is given, one obtains only a projective system of *quasi*-algebraic groups in **70**
the sense of Serre, or rather quasi-locally algebraic groups (the groups G_i obtained will not in general be of finite type over k, but only locally of finite type over k). It is even possible that in general one obtains only a still weaker structure on $\mathrm{Pic}(U)$, of the kind encountered by Néron [6] in his theory of degeneration of abelian varieties defined over local fields.

One method of attacking the problem, likewise introduced by Mumford, consists in resolving the singularities of X, i.e. in considering a projective birational morphism $Y \to X$ with Y regular. When U is regular (i.e. a is an isolated singular point), one can often find Y such that $Y_{|}U = V \to U$ is an isomorphism. In that case one therefore has

$$\mathrm{Pic}(U) \simeq \mathrm{Pic}(V) \simeq \mathrm{Pic}(Y)/\Im\mathbf{Z}^I,$$

where I is the set of irreducible components of the fiber Y_a (each of which defines an element of $\mathrm{Pic}(Y)$, since it is a locally principal divisor, by the regularity of Y). On the other hand, using the techniques of Formal Geometry in EGA III 4 and 5, in particular the existence theorem, one finds

$$\mathrm{Pic}(Y) \simeq \varprojlim \mathrm{Pic}(Y_n),$$

where $Y_n = Y \otimes_A A_n$ and $A_n = A/\mathfrak{m}^{n+1}$. When A admits a coefficient field k, the theory of Picard schemes of the projective k-schemes Y_n is available, and hence

$$\mathrm{Pic}(Y_n) \simeq \mathbf{Pic}_{Y_n/k}(k).$$

This therefore gives a construction of a projective system of locally algebraic groups $\mathbf{Pic}_{Y_n/k}/\Im\mathbf{Z}^I$, which is the desired system. (*) In the case considered here, one can moreover see (using the fact that a is an isolated singular point) that the connected components of the universal-image subgroups in this projective system form an *essentially constant* projective **71** system; thus, in this case, one obtains a locally algebraic group G as a solution of the problem. If one further assumes that A is normal of dimension 2, then a remark of Mumford (stating that the intersection matrix of the components of Y_a in X is negative definite(*)) implies that G is even an algebraic group, i.e. of finite type over k (the number of its connected components being equal, moreover, to the determinant of the intersection matrix just considered).

If, on the contrary, a is not an isolated singularity, examples (with A of dimension 2) show that one obtains a projective system of algebraic groups which does not reduce to a single algebraic group.

Once a good notion of "*local Picard scheme*"were available, one should strengthen the notion of parafactoriality by saying that A is "*geometrically parafactorial*"when not only A, and even $\widehat{A}$, is parafactorial, but the local Picard scheme $G(\widehat{A})$ is the trivial group (which, when the residue field is not algebraically closed, is stronger than saying that G has no rational point over k other than the identity). The necessity of a strengthened notion of parafactoriality becomes apparent on recalling that there exist complete normal local rings of dimension 2 which are factorial but admit finite étale algebras which are not(*). A "*geometrically factorial*"local ring would then be a normal ring A such that all the localizations of dimension ≥ 2 are geometrically parafactorial, or better, such that the localizations of $\widehat{A}$ are parafactorial(*). Of course, it would be interesting to find a "good"definition of these notions, independent of the still undeveloped theory of local Picard schemes.(*)

It is in any event plausible that these notions will be needed if one wishes to obtain statements of the following type: Let A be a "good ring"(for example an algebra of finite type over $\mathbf{Z}$, or over a complete local ring, for example over a field). Let U be the set of $x \in X = \operatorname{Spec}(A)$ such that $\mathcal{O}_{X,x}$ is "geometrically factorial"; is U open? Or again: Let **72** $f\colon X \to Y$ be a flat morphism of finite type with Y locally noetherian, and let U be the set of $x \in X$ such that $\mathcal{O}_{X_{f(x)},x}$ is "geometrically factorial"; is U open, at least under suitable additional hypotheses on f? I doubt that true statements of this type exist with the usual notion of factorial ring.

We have raised here, in a particular case, the question of studying the geometric properties of "varying"local rings, for example the $\mathcal{O}_{X,x}$ as x ranges over a prescheme X. When X is, for example, a scheme of finite type over a field, it is known(*) that there exists on X a projective system of finite algebras $P^n_{X/k}$ (obtained by completing $X \times_k X$ along the diagonal),

(*)*Editor's note.* The question was greatly clarified by the results of Boutot (Boutot J.-F., *Schéma de Picard local*, Lect. Notes in Math., vol. 632, Springer, Berlin, 1978). In particular, if A is a complete (noetherian) local k-algebra of depth ≥ 2 such that $H^2_{\mathfrak{m}}(A)$ has finite dimension over k, the local Picard group is a group scheme locally of finite type over k, whose tangent space at the origin is $H^2_{\mathfrak{m}}(A)$. If A is moreover normal of dimension ≥ 3, Serre's normality criterion XI XI.3.12, together with Corollary V 3.6, ensures the required finiteness and hence the existence of the local Picard scheme. See also (Lipman J., "The Picard group of a scheme over an Artin ring", *Publ. Math. Inst. Hautes Études Sci.* **46** (1976), p. 15–86) for an approach closer to Grothendieck's outline above.

(*)*Editor's note.* Mumford D., "The topology of normal singularities of an algebraic surface and a criterion for simplicity", *Publ. Math. Inst. Hautes Études Sci.* **9** (1961), p. 5–22.

(*)*Editor's note.* Factorial rings with nonfactorial henselization arise naturally in the study of moduli spaces of vector bundles. See, for example, (Drézet J.-M., "Groupe de Picard des variétés de modules de faisceaux semi-stables sur $\mathbf{P}_2$", in *Singularities, representation of algebras, and vector bundles (Lambrecht, 1985)*, Lect. Notes in Math., vol. 1273, Springer, Berlin, 1987, p. 337–362). *Strictly speaking*, Drézet shows that the completion is not factorial, but in fact the proof gives the result for the henselization: the point is that Luna's étale slice theorem (Luna D., "Slices étales", in *Sur les groupes algébriques*, Mém. Soc. math. France, vol. 33, Société mathématique de France, Paris, 1973, p. 81–105) describes the local ring of a quotient in the sense of invariant theory near a semistable point, locally for the étale topology.

(*)For a more flexible notion of "geometrically factorial"local ring, cf. *Comments*, page 133.

(*)*Editor's note.* See page 133: a local ring is geometrically factorial (resp. parafactorial) if its strict henselization is factorial (resp. parafactorial).

(*)*Editor's note.* See EGA IV.16.

whose fiber at every point $x \in X$ rational over k is isomorphic to the projective system of the $\mathcal{O}_{X,x}/\mathfrak{m}_X^{n+1}$. It is then natural to relate the study of the completions of the local rings $\mathcal{O}_{X,x}$, for varying x, to that of the "algebraic family of complete local rings"given by the P^n, noting that for every $x \in X$ (rational over k or not), one obtains a complete local ring

$$P^\infty(x) = \varprojlim P^n(x)$$

(where $P^n(x) =$ reduced fiber $P^n \otimes_{\mathcal{O}_{X,x}} k(x)$). Of particular interest, for example, will be the complete ring thus associated with the generic point, and one expects its algebro-geometric properties (expressed, for example, by its local Picard, homotopy, or homology groups) to be essentially those of the completions $\widehat{\mathcal{O}}_{X,x}$ for x in a suitable dense open subset U.

More generally, one may undertake the simultaneous study of the complete local rings obtained in this way from an adic projective system (P_n) of finite algebras over a given scheme X. It is plausible that, under certain regularity conditions (such as flatness of the P_n), the **73**
local homotopy groups arise from a projective system of finite group schemes over X, and that analogous results hold for the local Picard groups. As regards the latter, a first interesting case deserving investigation is that in which one starts with an algebraic surface X having singular curves, and seeks to study the local Picard schemes at varying points of these curves in terms of a suitable pro-group scheme defined over the singular locus.

XIII.6 Comments(*)

The viewpoint of the "étale cohomology"of schemes, and recent progress in this theory, leads **74**
us both to make some of the problems posed more precise and to broaden them. For the notion of "topology"and of the "étale topology of a scheme", I refer to M. Artin, *Grothendieck Topologies*, Harvard University 1962 (mimeographed notes)(*).

By providing a finer notion of localization than that furnished by the traditional "Zariski topology", this theory gives particular importance to *strictly local rings* , i.e. henselian local rings with separably closed residue field. For every local ring A with residue field k, and every separable closure k' of k, one can find a local homomorphism from A to a strictly local ring A', the *strict local closure* of A, with residue field k', having an evident universal property. The ring A' is henselian and flat over A, and $A' \otimes_A k \simeq k'$; it is noetherian if and only if A is. (Cf. *loc. cit.* Chap. III, section 4)(*). If X is a prescheme, x a point of X, and x' a point above x which is the spectrum of a separable closure k' of $k = k(x)$, one is led to define the *strictly local ring* of X at x', $\mathcal{O}'_{X,x'}$, as the strict local closure of the usual local ring $\mathcal{O}_{X,x}$, relative to the residue-field extension k'/k. It is the *strictly* local rings at the "geometric"points of X which, from the viewpoint of the étale topology, are meant to reflect the local properties of the prescheme X. They also play, in many respects, the role formerly assigned to the *completions* of the local rings of X (say, at points with algebraically closed residue field), while remaining "closer"to X and allowing an easier passage to "neighboring points".

One should therefore reconsider for noetherian henselian local rings (resp. noetherian strictly local rings) many questions generally posed for complete local rings (possibly restricted to those having an algebraically closed residue field). Thus the topological problems raised **75**
in n^{os} XIII.2 and XIII.3 arise more generally for strictly local rings. One can moreover state conjecturally, for "good"strictly local rings, certain simple-connectedness and acyclicity properties for the geometric fibers of the canonical morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$; these would show that, for many properties of a "topological"nature, it amounts to the same thing to

(*)Written in March 1963.
(*)Or preferably to SGA 4.
(*)Or EGA IV 18.8.

prove them for the ring A or for its completion $\widehat{A}$. Certain results already obtained in this direction(*) give reason to hope that complete results in this direction will soon be available.

The notion of étale localization provides a seemingly reasonable definition of a "*geometrically parafactorial*"or "*geometrically factorial*"local ring (the need for which was indicated in no. XIII.5, p. 131): this will mean a local ring whose strict local closure is parafactorial, resp. factorial. Hypotheses of this nature arise naturally in the study of the étale cohomology of preschemes(*). Thus, if X is a locally noetherian prescheme whose strictly local rings are factorial (i.e. whose ordinary local rings are "geometrically factorial"), one proves that the $\mathrm{H}^i(X_{\text{ét}}, \mathbf{G}_m)$ are torsion groups for $i \geq 2$ (which sometimes makes it possible to express these groups in terms of cohomology groups with coefficients in the groups $\boldsymbol{\mu}_n$ of n-th roots of unity), and if X is integral with fraction field K, the natural homomorphism $\mathrm{H}^2(X_{\text{ét}}, \mathbf{G}_m) \to \mathrm{H}^2(K, \mathbf{G}_m) = \mathrm{Br}(K)$ is injective(*); examples show that these conclusions may fail, even for local X, if one assumes only that X is factorial rather than geometrically factorial(*).

As regards the problems of local and global Lefschetz type raised in XIII.3.4, and their analogues in scheme theory, the homological version of these questions has been considerably clarified: everything follows formally from three general theorems. The first concerns the cohomological dimension of certain affine schemes (resp. Stein spaces), such as affine schemes **76**
X of finite type over an algebraically closed field: their cohomological dimension is $\leq \dim X$ ("affine Lefschetz theorem")(*); the second is a duality theorem for the cohomology (with discrete coefficients) of a projective morphism(*); and the last is a *local duality* theorem of an analogous nature(*). In Algebraic Geometry, only the last of these has not been proved at the time of writing (it is, however, proved in characteristic 0, using resolution of singularities by Hironaka). Moreover, in the transcendental setting, global and local duality are already available, having recently been proved by Verdier(*). Let us merely indicate that, in the statements of the homological versions of Problems XIII.4.2 and XIII.4.3 (which henceforth deserve the name of conjectures), the "at infinity"conditions a) and c) are certainly superfluous; only the *local cohomological structure* of $X - Y$ is important, and one may assume, for example, that it is locally a complete intersection of dimension $\geq n$. Moreover, in XIII.4.3, say, the fact that Y is a hyperplane section should play no role and should be replaceable by the sole

(*)Cf. M. Artin in SGA 4 XIX

(*)*Editor's note.* See, for example, (Strano R., "The Brauer group of a scheme", *Ann. Mat. Pura Appl. (4)* **121** (1979), p. 157–169), where the hypothesis that the local rings of a scheme X are geometrically parafactorial sometimes makes it possible to prove that the Brauer groups of X (computed in terms of Azumaya algebras) coincide with the cohomological Brauer group of X.

(*)*Editor's note.* The connection between the Brauer group and the Picard group is a close one. In this connection, let us cite the following results of Saito (Saito S., "Arithmetic on two-dimensional local rings", *Invent. Math.* **85** (1986), no. 2, p. 379–414) in the case of surfaces, the first local and the other global. Let A be an excellent normal henselian local ring of dimension 2 with *finite* residue field, and let X be the complement of the closed point in $\mathrm{Spec}(A)$. Then there is a perfect duality of *torsion* groups $\mathrm{Pic}(X) \times \mathrm{Br}(X) \to \mathbf{Q}/\mathbf{Z}$ — by the Brauer group of X we mean the *cohomological* Brauer group $\mathrm{Br}(X) = H^2_{\text{ét}}(X, G_m)$. In the global case one has the following generalization of a result of Lichtenbaum (Lichtenbaum S., "Duality theorems for curves over p-adic fields", *Invent. Math.* **7** (1969), p. 120–136): let k be the fraction field of a complete discrete valuation ring $\mathcal{O}$ with finite residue field, and let X be a projective, smooth, and geometrically complete curve over k. The group $\mathrm{Pic}^0(X)$ is endowed with the topology induced by the adic topology of k, and $\mathrm{Pic}(X)$ is the topological group for which $\mathrm{Pic}^0(X)$ is an open subgroup. Then there is a perfect duality of topological groups $\mathrm{Pic}(X) \times \mathrm{Br}(X) \to \mathbf{Q}/\mathbf{Z}$. Note that this statement, which concerns curves, is of course proved by considering a regular (proper and flat) model of X over $\mathcal{O}$: it is a result about surfaces.

(*)Cf. A. Grothendieck, le groupe de Brauer II (Séminaire Bourbaki no. 297, Nov. 1965), notably 1.8 and 1.11 b.

(*)Cf. SGA 4 XIV.

(*)Cf. SGA 4 XVIII.

(*)Cf. SGA 5 I.

(*)*Editor's note.* See Verdier J.-L., "Dualité dans la cohomologie des espaces localement compacts", in *Séminaire Bourbaki*, vol. 9, Société mathématique de France, Paris, 1995, Exp. 300, p. 337–349.

hypothesis that X is compact and $X - Y$ is Stein (i.e. in Algebraic Geometry, X is proper over k and $X - Y$ is affine; as we said, the homological version of this conjecture is proved for algebraic spaces over the field $\mathbf{C}$)(*).

In the definition (p. 128) of the $\pi_i^x(X)$, one must assume $i \geq 2$. For $i = 0, 1$, there is no reasonable definition of the $\pi_i^x(X)$; they should be replaced by $\mathrm{H}_0^x(X)$ and $\mathrm{H}_1^x(X)$, defined respectively as the cokernel and kernel in the natural homomorphism

$$\varprojlim \mathrm{H}_0(U - (x), \mathbf{Z}) \to \varprojlim \mathrm{H}_0(U, \mathbf{Z}).$$

At a pinch, and for convenience of formulation, one may set $\pi_i^x(X) = \mathrm{H}_i^x(X)$ for $i \leq 1$; otherwise, subsequent assertions concerning the π_i^x must be supplemented by the corresponding assertions for $\mathrm{H}_0^x, \mathrm{H}_1^x$. If x is an isolated point of X, one should set $\pi_i^x(X) = 0$ for $i \neq 0$ and $\pi_0^x(X) = \mathrm{H}_0^x(X) = \mathbf{Z}$.

The assertion that the $\pi_i^x(U, f)$ are mutually isomorphic is true only when X is not **77**
disconnected by x in a neighborhood of x, i.e. if $\pi_i^x(X) = 0$ for $i = 0, 1$. In the general case $\pi_i^x(X)$ can designate only a *family* of groups, not necessarily mutually isomorphic; nevertheless the notation $\pi_i^x(X) = 0$ retains an evident meaning.

Page 128, where I predict that the vanishing of the local homotopy invariants $\pi_i^x(X)$ for $x \in Y$, $i \leq n$, should imply that $\pi_i(X - Y) \to \pi_i(X)$ is bijective for $i < n - d$ and surjective for $i = n - d$, one must be cautious, since in the present context (as in Algebraic Geometry) one does not have at one's disposal "general"points at which the local conditions would also have to apply. For this reason it will doubtless be necessary to use relative local homotopy invariants

$$\pi_i^Y(X, f) = \pi_i^Y(X, x) = \varprojlim_U \pi_{i-1}(U - U \cap Y, f(t)) \qquad \text{for } i \geq 2,$$

(with an *ad hoc* definition as above for $i = 0, 1$), where Y is a closed subset of X; or else to compensate for the absence of general points by expressing the hypotheses on X in terms of properties, topological in nature (for the étale topology), of the spectra of the local rings of X, which makes it possible to recover general points. The same reservation applies to the generalization of Conjectures XIII.4.2 and XIII.4.3 to the case in which $X - Y$ is not assumed to be locally a complete intersection, a generalization suggested in the statements of conditions b) of those conjectures.

To formulate the expurgated versions of Conjectures XIII.4.2 and XIII.4.3 suggested by the results mentioned above, we make the following

Definition XIII.6.1. Let X be a topological space, let Y be a locally closed subset of X, and let n be an integer. One says that X has *homotopical depth* $\geq n$ along Y, and writes $\operatorname{prof\,hpt}_Y(X) \geq n$, if for every $x \in Y$ one has $\pi_i^Y(X, x) = 0$ for $i < n$.

It should be equivalent to say that, for every open subset X' of X and every $x \in X' \cap U = U'$ **78**
(where $U = X - Y$), the canonical homomorphism

$$\pi_i(U', x) \to \pi_i(X', x)$$

is an isomorphism for $i < n - 1$ and a monomorphism for $i = n - 1$(*).

Definition XIII.6.2. Let X be a complex analytic space and let n be an integer. One says that the *rectified homotopical depth* of X is n(*) if, for every locally closed analytic subset Y of X, one has

$$\operatorname{prof\,htp}_Y(X) \geq n - \dim Y \tag{x}$$

(*)Cf. Exp. XIV.

(*)*Editor's note.* When the pair (X, Y) is moreover polyhedral, this equivalence holds; cf. (Eyral C., "Profondeur homotopique et conjecture de Grothendieck", *Ann. Sci. Éc. Norm. Sup. (4)* **33** (2000), no. 6, p. 823–836).

(*)In the first edition of these notes, we used the term "true homotopical depth". In the present version we follow EGA IV 10.8.1.

(where, of course, $\dim Y$ denotes the *complex* dimension of Y).

It should be equivalent to say that, for every irreducible locally closed analytic subset Y of X, there exists a closed analytic subset Z of Y, of dimension $< \dim Y$, such that relation (x) holds with $Y - Z$ in place of Y. This would make it possible, for example, to restrict in Definition XIII.6.2 to the case where Y is nonsingular. (*)

The following conjecture, purely topological in nature, is in the spirit of a "*local Hurewicz theorem*".

Conjecture A ("local Hurewicz theorem"(*)). *Let X be a topological space and Y a locally closed subset, subject if necessary to "smoothness"conditions such as local triangulability of the pair (X, Y), and let n be an integer ≥ 3. In order that* prof $\mathrm{htp}_Y(X) \geq n$, *it is necessary and sufficient that* 79

$$\mathrm{H}^i_Y(\mathbf{Z}_X) = 0 \quad \text{for } i < n$$

(one then says that X has cohomological depth $\geq n$ *along Y), and that the local fundamental groups*

$$\pi_2^Y(X, x) = \varprojlim_{U \ni x} \pi_1(U - U \cap Y)$$

vanish (one then says that X is "pure"along Y).

Note that if X is an analytic space, Y an analytic subspace, and X is pure along Y, then for every $x \in Y$ the local ring $\mathcal{O}_{X,x}$, as well as its localizations with respect to prime ideals containing the ideal defining the germ Y at x (i.e. in the inverse image Y_x of Y under $\mathrm{Spec}(\mathcal{O}_{X,x}) = X_x \to X$), are pure in the sense of Exp. X; it seems plausible that the converse is also true. Analogous remarks apply to cohomological depth, it being understood that one works with the étale topology on the $\mathrm{Spec}(\mathcal{O}_{X,x})$.

Conjecture XIII.4.1 then generalizes to

Conjecture B ("Purity"(*)). *Let E be an analytic space and X an analytic subset of E. Suppose that E is nonsingular of dimension N at $x \in X$, and that X can be described by p analytic equations in a neighborhood of every point. Then the rectified homotopical depth of X is $\geq N - p$.*

In particular, a local complete intersection of dimension n at every point would have rectified homotopical depth $\geq n$, which is precisely Conjecture XIII.4.1.

Conjectures XIII.4.2 and XIII.4.3 generalize respectively to:

Conjecture C ("Local Lefschetz"(*)). *Let X be an analytic space, let Y be a closed analytic subset, and let x be a point of Y. Suppose that $X - Y$ is Stein in a neighborhood of x (for example, that Y is defined by one equation at x), and that $X - Y$ has rectified homotopical depth $\geq n$ in a neighborhood of x (for example, that at every point of $X - Y$ near x it is a complete intersection of dimension $\geq n$; cf. Conjecture B). Then the canonical homomorphism*

$$\pi_i^x(Y) \to \pi_i^x(X)$$

(*) *Editor's note.* All the conjectures that follow, suitably rectified, if I may say so, became theorems through the work of Hamm and Lê Dũng Tráng (Hamm H.A. & Lê Dũng Tráng, "Rectified homotopical depth and Grothendieck conjectures", in *The Grothendieck Festschrift, Vol. II*, Progr. Math., vol. 87, Birkhäuser, Boston, 1990, p. 311–351), cited as [HL] below. As for the two conjecturally equivalent definitions of rectified depth, they are in fact equivalent to a third, expressed in terms of Whitney stratification (cf. *loc. cit.*, Theorem 1.4).

(*) *Editor's note.* As observed in [HL], Example 3.1.3, this conjecture is already false for $X = \{\boldsymbol{z} \in \mathbf{C}^n \mid z_1^2 + z_2^3 + \cdots + z_n^3 = 0\}, n \geq 4$, and Y reduced to the origin. Suitably modified, however, it is true (Theorem 3.1.4 of *loc. cit.*).

(*) *Editor's note.* This conjecture is proved, even when E is singular, in [HL]: it is Theorem 3.2.1.

(*) *Editor's note.* This conjecture is proved in [HL], indeed even in the strong form of the remark that follows; cf. Theorem 3.3.1 of *loc. cit.*.

is an isomorphism for $i < n - 1$ and an epimorphism for $i = n - 1$.

Conjecture D ("Global Lefschetz"(*)). *Let X be a* compact *analytic space and let Y be an analytic subspace of X such that $U = X - Y$ is Stein and has rectified homotopical depth $\geq n$ (for example, is a complete intersection of dimension $\geq n$ at every point). Then the canonical homomorphism* **80**

$$\pi_i(Y) \to \pi_i(X)$$

is an isomorphism for $i < n - 1$ and an epimorphism for $i = n - 1$.

Remark. When, in the statements C and D, the hypothesis that $X - Y$ is Stein is replaced by the hypothesis that $X - Y$ is a union of $c + 1$ Stein open subsets (which plays the role of a topological "concavity"hypothesis), the conclusions should simply be modified by replacing n there by $n - c$.(*)

Finally, let us make explicit, in the "global case"D, the conjecture concerning the fundamental group (obtained by taking $n = 3$):

Conjecture D′ (Global Lefschetz for the fundamental group(*)). *Let X be a compact analytic space over the field of complex numbers and let Y be a closed analytic subset such that $U = X - Y$ is Stein. Suppose moreover that the following conditions hold:*

(i) For every $x \in U$, the local fundamental group $\pi_2^x(X, x)$ vanishes (i.e. X is "pure at x"*), or merely the local ring $\mathcal{O}_{X,x}$ is pure.*

(ii) The local rings at the points of U are "connected in dimension ≥ 2".

(iii) The local rings at the points of U have dimension ≥ 3.

Under these conditions, for every $x \in Y$, the homomorphism

$$\pi_1(Y, x) \to \pi_1(X, x)$$

is an isomorphism (and $\pi_2(Y, x) \to \pi_2(X, x)$ is an epimorphism). **81**

Note that the local conditions (i), (ii), and (iii) on U hold if U is locally a complete intersection of dimension ≥ 3. From the viewpoint of Algebraic Geometry (when U arises from a scheme, again denoted by U), conditions (i) through (iii) correspond to hypotheses on the local invariants $\pi_i^x(U)$, namely $\pi_i^x(U) = 0$ for $i < 3 - \deg\operatorname{tr} k(x)/k$, at points x for which one has respectively $\deg\operatorname{tr} k(x)/k = 0, 1, 2$. The global condition on U (U Stein) is satisfied if X is projective and Y is a hyperplane section.

(*) *Editor's note.* This conjecture is again proved in [HL], indeed even in the strong form of the remark that follows; cf. Theorem 3.4.1 of *loc. cit.*.

(*) *Editor's note.* Finally, let us mention the following result of Fulton, to be compared with the Fulton–Hansen result cited in the editor's note ((*)) on page 113: let X and H be closed subschemes of $\mathbf{P}^m_{\mathbf{C}}$, let n be the dimension of X, and let d be the codimension of H. Then the map

$$\pi_i(X, X \cap H) \to \pi_i(\mathbf{P}^n_{\mathbf{C}}, H)$$

is an isomorphism if $i \leq n - d$ and is surjective if $i = n - d - 1$; see (Fulton W., "Connectivity and its applications in algebraic geometry", in *Algebraic geometry (Chicago, Ill., 1980)*, Lect. Notes in Math., vol. 862, Springer, Berlin–New York, 1981, p. 26–92).

(*) *Editor's note.* This conjecture is proved in [HL]; cf. Theorem 3.5.1 of *loc. cit.*.

Bibliography

[1] S. Abhyankar – "Local uniformisation on algebraic surfaces over ground fields of characteristics $p \neq 0$", *Annals of Math.* **63** (1956), p. 491–526. 82

[2] W.L. Chow – "On the theorem of Bertini for local domains", *Proc. Nat. Acad. Sci. U.S.A.* **44** (1958), p. 580–584.

[3] M. Greenberg – "Schemata over local rings", *Annals of Math.* **73** (1961), p. 624–648.

[4] M. Kneser – "Über die Darstellung algebraischer Raumkurven als Durchschnitte von Flächen", *Archiv der Math.* **XI** (1960), p. 157–158.

[5] D. Mumford – "The topology of normal singularities of an algebraic surface, and a criterion for simplicity", *Publ. Math. Inst. Hautes Études Sci.* **9** (1961), p. 5–22.

[6] A. Néron – "Modèles minimaux des variétés abéliennes sur les corps locaux et globaux", *Publ. Math. Inst. Hautes Études Sci.* **21** (1964), p. 5-128.

[7] A.H. Wallace – *Homology Theory of algebraic Varieties*, Pergamon Press, 1958.

[8] S. Abhyankar – "Resolution of singularities of arithmetical surfaces", in *Arithmetical Algebraic Geometry*, Harper and Row, New-York, 1965, p. 111–152.

[HL] H.A. Hamm & Lê Dũng Tráng – "Rectified homotopical depth and Grothendieck conjectures", in *The Grothendieck Festschrift, Vol. II*, Progr. Math., vol. 87, Birkhaüser Boston, 1990, p. 311–351.

Exposé XIV

Depth and Lefschetz theorems in étale cohomology

by Mrs M. Raynaud[(*)]

In no. XIV.1, we define a notion of "*étale depth*"that is the analogue in étale cohomology of the notion of depth studied in III, in the cohomology of coherent sheaves. After a technical part, we prove "Lefschetz theorems"in no. XIV.4, the central theorem being XIV.4.2. Let X be a scheme, Y a closed subset of X, U the complementary open subset $X - Y$, and F an abelian sheaf on X for the étale topology; in general, the purpose of Lefschetz theorems is to show that, if F satisfies certain local conditions on U, expressible in terms of étale depth at the points of U, then, under certain additional conditions of a global nature on U (for example, U *affine*), the natural map of étale cohomology groups **83**

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, F_{|Y})$$

is an isomorphism for values $i < n$, where n is a certain specified integer. By taking F to be a constant sheaf, one thereby obtains conditions for $\pi_0(X)$ to equal $\pi_0(Y)$ and conditions for the abelianized fundamental groups of X and Y to be the same. In no. XIV.5, the introduction of a notion of "geometric depth"makes it possible to give useful special cases of the Lefschetz theorems (XIV.5.7). Finally, in no. XIV.6, we mention some conjectures, concerning in particular "noncommutative"variants of the theorems obtained.

XIV.1 Cohomological and homotopical depth

XIV.1.0

Fix the following notation. Let X be a scheme[(*)], Y a closed subset of X, U the complementary open subset, and $i : Y = X - U \to X$ the canonical immersion. Let $\underline{\Gamma}_Y$ be the functor that associates to an abelian sheaf on X the "sheaf of sections with support in Y", that is, $\underline{\Gamma}_Y = i_* i^!$ (cf. SGA 4 IV 3.8 and VIII 6.6), and let Γ_Y be the functor $\Gamma \cdot \underline{\Gamma}_Y$ (where Γ is the "global sections"functor). Consider the derived category $D^+(X)$ and the derived functor $\mathrm{R}\underline{\Gamma}_Y$ (resp. $\mathrm{R}\Gamma_Y$) of $\underline{\Gamma}_Y$ (resp. of Γ_Y) (cf. [3]). Given a complex of abelian sheaves F on X, bounded below, it may be regarded as an element of $D^+(X)$; we then denote by $\underline{H}^p_Y(F)$ the p-th cohomology sheaf of $\mathrm{R}\underline{\Gamma}_Y(F)$ and by $\mathrm{H}^p_Y(X, F)$ the p-th cohomology group of $\mathrm{R}\Gamma_Y(F)$. The results of (SGA 4 V 4.3 and 4.4) extend trivially to $\underline{H}^p_Y(F)$ and $\mathrm{H}^p_Y(X, F)$. **84**

[(*)]Based on unpublished notes by A. Grothendieck.

[(*)]In accordance with the new terminology (cf. the reissue of EGA I), here we shall call "scheme"what was formerly called "prescheme", and "separated scheme"what was called "scheme".

Proposition XIV.1.1. *Let X be a scheme, Y a closed subset of X, U the complementary open subset, and $i : U \to X$ the canonical immersion. Let F denote either a sheaf of sets on X, a sheaf of groups on X, or a complex of abelian sheaves on X, bounded below. Fix the following notation: if $X' \to X$ is a morphism, U' and F' denote the inverse images of U and F on X'; moreover, if $\bar{y}$ is a geometric point of X, $\bar{X}$ denotes the strict localization of X at $\bar{y}$, and $\bar{U}$ and $\bar{F}$ the inverse images of U and F in $\bar{X}$.*

1°) *Let F be a sheaf of sets on X and n an integer ≤ 2; then the following conditions are equivalent:* **85**

(i) *The canonical morphism*

$$F \to i_* i^* F$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

(ii) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^0(X', F') \to \mathrm{H}^0(U', F')$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

Suppose in addition that the open subset U is retrocompact in X; then the preceding conditions are equivalent to the following one:

(iii) *For every geometric point $\bar{y}$ of Y, the canonical morphism*

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{U}, \bar{F})$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

2°) *Let F be a sheaf of groups on X and n an integer ≤ 3; then the following conditions are equivalent:*

(i) *The canonical morphism*

$$F \to i_* i^* F$$

is injective if $n \geq 1$, bijective if $n \geq 2$, and if $n \geq 3$, in addition to the preceding conditions, the sheaf of pointed sets $\mathrm{R}^1 i_*(i^* F)$ is zero.

(ii) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^0(X', F') \to \mathrm{H}^0(U', F')$$

is injective if $n \geq 1$, bijective if $n \geq 2$, and moreover the canonical morphism **86**

$$\mathrm{H}^1(X', F') \to \mathrm{H}^1(U', F')$$

is injective if $n \geq 2$, and bijective if $n \geq 3$.

(ii bis) *The same as* (ii)*, except in the case $n = 2$, where it is only assumed that* $\mathrm{H}^0(X', F') \to \mathrm{H}^0(U', F')$ *is bijective.*

Suppose in addition that U is retrocompact in X; then the preceding conditions are also equivalent to the following one:

(iii) *For every geometric point $\bar{y}$ of Y, the canonical morphism*

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{U}, \bar{F})$$

is injective if $n \geq 1$, and bijective if $n \geq 2$; finally, if $n \geq 3$, in addition to the preceding conditions, $\mathrm{H}^1(\bar{U}, \bar{F})$ *is zero.*

3°) *Let F be a complex of abelian sheaves, bounded below, and let n be an integer; then the following conditions are equivalent:*

(i) *We have $\underline{H}^p_Y(F) = 0$ for $p < n$ (cf. 1.0).*

(ii) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^p(X', F') \to \mathrm{H}^p(U', F')$$

is bijective for $p < n-1$, and injective for $p = n-1$.

Suppose that U is retrocompact in X; then the preceding conditions are also equivalent to the following one:

(iii) *For every geometric point $\bar{y}$ of Y, the canonical morphism*

$$\mathrm{H}^p(\bar{X}, \bar{F}) \to \mathrm{H}^p(\bar{U}, \bar{F})$$

is bijective for $p < n-1$, and injective for $p = n-1$.

In the case where F is an abelian sheaf and $n \geq 2$, conditions (i) *and* (ii) *are also equivalent to the following one:* **87**

(ii bis) *For every scheme X' étale over X, the canonical morphism*

$$\mathrm{H}^p(X', F') \to \mathrm{H}^p(U', F')$$

is bijective for $p < n-1$.

Proof. 1°) It is clear that (i) $\Longleftrightarrow$ (ii). Let us show that, if U is quasi-compact in X, (i) $\Longleftrightarrow$ (iii). Indeed, (i) is equivalent to saying that, for every geometric point $\bar{y}$ of X, the morphism $F_{\bar{y}} \to (i_* i^* F)_{\bar{y}}$ is injective if $n \leq 1$ and bijective if $n \leq 2$ (SGA 4 VIII 3.6). Since this morphism is in any case bijective when $\bar{y}$ is a geometric point of U, it is enough to consider the geometric points $\bar{y}$ of Y. Now it follows from the fact that i is quasi-compact and from (SGA 4 VIII 5.3) that the morphism

$$F_{\bar{y}} \to (i_* i^* F)_{\bar{y}}$$

is canonically identified with the morphism

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{U}, \bar{F}),$$

which proves the equivalence of (i) and (iii).

2°) (i) $\Longrightarrow$ (ii). The assertions concerning H^0 follow from 1°). Let i' be the canonical immersion of U' into X'; the assertions concerning H^1 follow from the exact sequence (SGA 4 XII 3.2)

$$0 \to \mathrm{H}^1(X', i'_* i'^* F') \to \mathrm{H}^1(U', F') \to \mathrm{H}^0(X', \mathrm{R}^1 i'_*(i'^* F')).$$

(ii bis) $\Longrightarrow$ (i) . By 1°), it is enough to show that, for $n \geq 3$, one has $\mathrm{R}^1 i_*(i^* F) = 0$. Now $\mathrm{R}^1 i_*(i^* F)$ is the sheaf associated with the presheaf $X' \longmapsto \mathrm{H}^1(U', F')$, that is, by hypothesis, the sheaf associated with the presheaf $X' \longmapsto \mathrm{H}^1(X', F')$, and the latter is zero. **88**

(i) $\Longleftrightarrow$ (iii). Taking 1°) into account, the only thing left to verify is that the relation $\mathrm{R}^1 i_*(i^* F) = 0$ is equivalent to the condition that $\mathrm{H}^1(\bar{U}, \bar{F}) = 0$ for every geometric point $\bar{y}$ of Y. Since $\mathrm{R}^1 i_*(i^* F))$ is zero outside Y, saying that $\mathrm{R}^1 i_*(i^* F) = 0$ is equivalent to saying that $(\mathrm{R}^1 i_*(i^* F))_{\bar{y}} = 0$ for every geometric point $\bar{y}$ of Y. It is then enough to note that, since i is quasi-compact, one has $(\mathrm{R}^1 i_*(i^* F))_{\bar{y}} = \mathrm{H}^1(\bar{U}, \bar{F})$ (SGA 4 VIII 5.3).

3°) (i) $\Longrightarrow$ (ii). Let X' be a scheme ètale over X; one has the exact sequence (SGA 4 V 4.5)

$$\to \mathrm{H}^p_{Y'}(X', F') \to H^p(X', F') \to H^p(U', F') \to; \qquad (*)$$

hence (ii) is equivalent to $H^p_{Y'}(X', F') = 0$ for $p < n$ and for every scheme X' ètale over X. Consider now the spectral sequence

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(X', \underline{H}^q_Y(F)) \Longrightarrow \mathrm{H}^*_{Y'}(X', F');$$

by hypothesis, $\underline{H}^q_Y(F) = 0$ for $q < n$, whence $\mathrm{E}_2^{pq} = 0$ for $p + q < n$ and consequently $\mathrm{H}^p_{Y'}(X', F') = 0$ for $p < n$.

(ii) $\Longrightarrow$ (i). The sheaf $\underline{H}^p_Y(F)$ is associated with the presheaf $X' \longmapsto \mathrm{H}^p_{Y'}(X', F')$; since we have already observed that (ii) is equivalent to the relation $\mathrm{H}^p_{Y'}(X', F') = 0$ for $p < n$ and for every scheme X' ètale over X, it follows that $\underline{H}^p_Y(F) = 0$ for $p < n$.

(i) $\Longleftrightarrow$ (iii). The sheaves $\underline{H}^p_Y(F)$ are concentrated on Y; consequently, saying that $\underline{H}^p_Y(F) = 0$ is equivalent to saying that, for every geometric point $\bar{y}$ of Y, the fiber $(\underline{H}^p_Y(F))_{\bar{y}}$ 89
is zero. Since i is quasi-compact, (SGA 4 VIII 5.2) gives $(\underline{H}^p_Y(F))_{\bar{y}} = \underline{H}^p_{\bar{Y}}(\bar{X}, \bar{F})$. The equivalence of (i) and (iii) follows, in view of the analogue on $\bar{X}$ of the exact sequence $(*)$.

(ii bis) $\Longrightarrow$ (ii) when F is an abelian sheaf. The only thing left to show is that $\underline{H}^{n-1}_Y(F) = 0$. Now, for $n > 2$, the sheaf $\underline{H}^{n-1}_Y(F)$ is associated with the presheaf $X' \longmapsto \mathrm{H}^{n-2}(U', F') = \mathrm{H}^{n-2}(X', F')$ and is therefore zero. The case $n = 2$ follows from the fact that $\underline{H}^1_Y(F)$ is the cokernel of the morphism $F \to i_* i^* F$.

Definition XIV.1.2. The notation is that of XIV.1.1. One says that F has ètale Y-depth $\geq n$ and writes

$$\operatorname{prof}_Y(F) \geq n$$

if F satisfies the equivalent conditions (i) and (ii) of XIV.1.1. If F is a complex of abelian sheaves, the ètale Y-depth of F is the supremum of the n for which $\operatorname{prof}_Y(F) \geq n$; the same notation will be used if F is a sheaf of sets resp. of not necessarily commutative groups (so that one then has $0 \leq \operatorname{prof}_Y(F) \leq 2$, resp. $0 \leq \operatorname{prof}_Y(F) \leq 3$, when the context does not make clear which of the three variants considered here is being used).

If $\mathbf{L}$ is a set of prime numbers, one says that the ètale Y-depth of X for $\mathbf{L}$ is $\geq n$ and writes

$$\operatorname{prof}^{\mathbf{L}}_Y(X) \geq n$$

if, for every constant sheaf of the form $\mathbf{Z}/\ell\mathbf{Z}$ with $\ell \in \mathbf{L}$, one has $\operatorname{prof}_Y(\mathbf{Z}/\ell\mathbf{Z}) \geq n$. The ètale 90
Y-depth of X for $\mathbf{L}$ is defined in the evident way. If $\mathbf{L} = \mathbf{P}$, the set of all prime numbers, and there is no risk of confusion with the notation of (EGA IV 5.7.1) (which concerns the case $F = \mathcal{O}_X$), $\mathbf{L}$ is omitted from the notation; otherwise one writes $\operatorname{prof\,ét}(X)$.

Finally, one says that X has homotopical Y-depth ≥ 3 for $\mathbf{L}$ and writes

$$\operatorname{prof\,hop}^{\mathbf{L}}_Y(X) \geq 3$$

if, for every constant finite sheaf F of $\mathbf{L}$-groups on X, one has $\operatorname{prof}_Y(F) \geq 3$. If $\mathbf{L} = \mathbf{P}$, $\mathbf{L}$ is omitted from the notation.

Corollary XIV.1.3. *Under the conditions of XIV.1.1, if* $\operatorname{prof}_Y(F) \geq n$*, then, for every closed subset Z of Y, one has*

$$\operatorname{prof}_Z(F) \geq n.$$

For example, let us give the argument when F is a complex of abelian sheaves bounded below. We use XIV.1.1 3°) (ii). Let $V = X - Z$ and, for every integer p, consider the sequence of morphisms

$$\mathrm{H}^p(X, F) \xrightarrow{f} \mathrm{H}^p(V, F) \xrightarrow{g} \mathrm{H}^p(U, F).$$

By hypothesis, g and $f \circ g$ are bijective for $p < n - 1$ and injective for $p = n - 1$; hence the same is true of f. Since the argument remains valid when X is replaced by a scheme X' ètale over X, this proves XIV.1.3.

Corollary XIV.1.4. *The notation is that of XIV.1.1 2°). If X' is a scheme over X, let Φ'* 91
denote the functor that assigns to an ètale covering of X' its restriction to U', and let $\Phi'_{F'}$

denote the functor that assigns to a torsor[(*)] *under F' its restriction to U'. Then the following conditions are equivalent:*

(i) *One has* $\operatorname{prof}_Y(F) \geq 1$ *(resp.* $\operatorname{prof}_Y(F) \geq 2$*, resp.* $\operatorname{prof}_Y(F) \geq 3$*).*

(ii) *For every scheme X' ètale over X, the functor $\Phi'_{F'}$ is faithful (resp. fully faithful, resp. an equivalence of categories).*

In particular, one has $\operatorname{prof}_Y(X) \geq 1$ *(resp.* $\operatorname{prof}_Y(X) \geq 2$*, resp.* $\operatorname{prof\,hop}_Y(X) \geq 3$*) if and only if the functor Φ' is faithful (resp. fully faithful, resp. an equivalence of categories).*

This follows from XIV.1.1 2°) (ii), in view of the interpretation of $\mathrm{H}^1(X', F')$ as the set of classes (up to isomorphism) of torsors under F' (SGA 4 VII 2), and of ètale coverings Z of degree n of a scheme as being associated with principal Galois coverings whose group is the symmetric group $\mathfrak{S}_n$: to Z one associates the covering $\underline{\mathrm{Isom}}_X(Z_0, Z)$, where Z_0 is the trivial covering of X of degree n.

Corollary XIV.1.5. *Under the conditions of XIV.1.1* 3°)*, suppose that* $\operatorname{prof}_Y(F) \geq n$*; then*

$$\mathrm{H}^n_Y(X, F) \simeq \mathrm{H}^0(X, \underline{H}^n_Y(F)).$$

The corollary follows from the spectral sequence

$$\mathrm{E}_2^{pq} = H^p(X, \underline{H}^q_Y(F)) \Longrightarrow H^*_Y(X, F).$$

Indeed, by hypothesis $\mathrm{E}_2^{pq} = 0$ for $q < n$, and hence

$$\mathrm{H}^n_Y(X, F) = \mathrm{E}_2^{0n} = \mathrm{H}^0(X, \underline{H}^n_Y(F)).$$

Remarks XIV.1.6. a) The notion of Y-depth, in the form of the equivalent conditions (i) and (ii) of XIV.1.1, makes sense on any site. In the particular case in which X is a locally Noetherian scheme equipped with the Zariski topology and F is a sheaf of coherent $\mathcal{O}_X$-modules, one recovers the usual notion of Y-depth as the infimum of the depths at the points of Y (III). **92**

b) For $n \leq 2$, the notion of ètale Y-depth of X is independent of $\mathbf{L}$. For $n = 1$, it simply means that U is dense in X. Indeed, this condition is necessary for $\operatorname{prof}_Y(F) \geq 1$, and it is also sufficient because one may assume that X is reduced, in which case the condition that U be dense in X is preserved by ètale base change (EGA IV 11.10.5) (ii) b)). If U is quasi-compact in X, the relation $\operatorname{prof}_Y(X) \geq 1$ is also equivalent to saying that Y contains no generic point of X (EGA I 6.6.5). For $n = 2$ and U quasi-compact in X, the condition $\operatorname{prof}_Y(X) \geq 2$ is equivalent to the condition that, for every geometric point $\bar{y}$ of Y, $\bar{U}$ be nonempty and connected, that is, that "Y does not disconnect X locally for the ètale topology".

c) If X has Y-depth $\geq n$ for $\mathbf{L}$ and U is quasi-compact in X, then for every locally constant abelian sheaf of $\mathbf{L}$-torsion F on X, one has $\operatorname{prof}_Y(X) \geq n$. Indeed, since the property $\operatorname{prof}_Y(F) \geq n$ is local for the ètale topology, one may assume that F is constant; then F is a filtered direct limit of sheaves that are finite direct sums of sheaves of the form $\mathbf{Z}/p^m\mathbf{Z}$, where m is an integer > 0 and $p \in \mathbf{L}$. Using XIV.1.1 (iii) and (SGA 4 VII 3.3), one sees that one may reduce to the case $F = \mathbf{Z}/p^m\mathbf{Z}$, and then , by induction on m, to the case $F = \mathbf{Z}/p\mathbf{Z}$, for **93**
which the assertion follows from the definition.

d) By XIV.1.4, if $\operatorname{prof}_Y(X) \geq 3$, the pair (X, Y) is *pure* in the sense of X 3.1. In fact, the pure pairs encountered in practice (cf. X 3.4) satisfy the stronger condition of homotopical depth ≥ 3, and this notion can therefore advantageously replace that of a pure pair.

e) Let F be a complex of abelian sheaves and $T(F)$ the complex obtained by applying the translation functor to F ([3]); then one plainly has:

$$\operatorname{prof}_Y(T(F)) = \operatorname{prof}_Y(F) - 1.$$

[(*)]That is, a "homogeneous principal bundle"in older terminology.

f) Let us point out that the recent work of Artin–Mazur ([1]) makes it possible to define the notion of homotopical depth $\geq n$ for every integer n (not only for $n \geq 3$).

g) Under the conditions of XIV.1.1 3), one has $\mathrm{prof}_Y(F) = \infty$ if and only if $F \xrightarrow{\sim} Ri_*(i^*F)$ in $D^+(X)$. Indeed, the $\underline{H}^p_Y(F)$ are the cohomology sheaves of the cone (= mapping cylinder) of the canonical morphism $F \to Ri_*i^*(F)$.

Definition XIV.1.7. Let X be a scheme, x a point of X, $\bar{x}$ a geometric point above x, and $\bar{X}$ the strict localization of X at $\bar{x}$. As before, F denotes either a sheaf of sets on X, a sheaf of groups on X, or a bounded-below complex of abelian sheaves on X; $\bar{F}$ denotes its inverse image on $\bar{X}$, and $\mathbf{L}$ a set of prime numbers. We say that F has *étale depth* $\geq n$ *at the point* x (respectively that the *étale depth for* $\mathbf{L}$ *of* X *at* x *is* $\geq n$, respectively that the **94**
homotopical depth for $\mathbf{L}$ *of* X *at* x *is* ≥ 3) and write $\mathrm{prof}_x(F) \geq n$ (respectively $\mathrm{prof}^{\mathbf{L}}_x(X) \geq n$, respectively $\mathrm{prof\,hop}^{\mathbf{L}}_x(X) \geq 3$) if $\mathrm{prof}_{\bar{x}}(\bar{F}) > n$ (respectively $\mathrm{prof}^{\mathbf{L}}_{\bar{x}}(\bar{X}) > n$, respectively $\mathrm{prof\,hop}^{\mathbf{L}}_{\bar{x}}(\bar{X}) \geq 3$). The integer $\mathrm{prof}^{\mathbf{L}}_x(X)$ and, if F is a complex of abelian sheaves, the integer $\mathrm{prof}_x(F)$ are defined in the evident way. If $\mathbf{L}$ is the set of all prime numbers, we omit $\mathbf{L}$ from the notation $\mathrm{prof}^{\mathbf{L}}_x(X)$, unless there is a risk of confusion with the notation of (EGA IV 5.7.1), in which case we write $\mathrm{prof\,\acute{e}t}_x(X)$.

We then have the following *pointwise characterization* of depth:

Theorem XIV.1.8. *Let X be a scheme and Y a closed subset of X such that the open subset $U = X - Y$ is retrocompact in X. If F is either a sheaf of sets on X, a sheaf of groups on X, or a bounded-below complex of abelian sheaves on X, then*

$$\mathrm{prof}_Y(F) = \inf_{y \in Y} \mathrm{prof}_y(F).$$

Let us first show that, for every point y of Y, one has the inequality $\mathrm{prof}_y(F) \geq \mathrm{prof}_Y(F)$. Indeed, let $\bar{y}$ be a geometric point above y, $\bar{X}$ the strict localization of X at $\bar{y}$, and $\bar{F}$ and $\bar{Y}$ the inverse images of F and Y on $\bar{X}$. By XIV.1.7 and XIV.1.3, one has

$$\mathrm{prof}_y(F) = \mathrm{prof}_{\bar{y}}(\bar{F}) \geq \mathrm{prof}_{\bar{Y}}(\bar{F}) \geq \mathrm{prof}_Y(F),$$

the last inequality using the hypothesis "U retrocompact", through conditions (iii) in XIV.1.1 and transitivity in the formation of strict localizations.

Conversely, suppose that, for every point y of Y, one has $\mathrm{prof}_y(F) \geq n$ (n an integer), and let us show that $\mathrm{prof}_Y(F) \geq n$.

First recall the following well-known results (SGA 4 VIII): **95**

Lemma XIV.1.8.1. *Let X be a scheme and F a sheaf of sets on X (respectively let $G \to F$ be a monomorphism of sheaves of sets on X). Then two sections s and s' of F coincide (respectively a section s of F comes from a section of G) if and only if this is so locally. In particular, if s and s' are two sections of F, there is a largest open subset V of X on which they coincide (respectively, if s is a section of F on X, there is a largest open subset V of X such that $s|V$ comes from a section of G on V). This open subset is also the set of points x of X such that, denoting by $\bar{x}$ a geometric point above x, the sections s and s' have the same image in the fiber $F_{\bar{x}}$ (respectively the image of s in $F_{\bar{x}}$ comes from an element of $G_{\bar{x}}$).*

Let us return to the proof of XIV.1.8.

1°) Case where F is a sheaf of sets. If $n = 1$, it is enough to show that the canonical morphism

$$\mathrm{H}^0(X, F) \to \mathrm{H}^0(U, F)$$

is injective, since the result also applies when X is replaced by a scheme étale over X. Let s and s' be two sections of F over X which have the same image in $\mathrm{H}^0(U, F)$, and let V be the largest open subset over which they are equal; clearly $V \supset U$. Suppose $V \neq X$, and let y be a maximal point of $X - V$, $\bar{y}$ a geometric point above y, $\bar{X}$ the strict localization of X at $\bar{y}$, **96**
and $\bar{V}$ and $\bar{F}$ the inverse images of V and F on $\bar{X}$. By the choice of y, one has $\bar{X} - \bar{y} = \bar{V}$, and by hypothesis the morphism

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{X} - \bar{y}, \bar{F}) = \mathrm{H}^0(\bar{V}, \bar{F})$$

is injective. It follows that s and s' coincide at the point y, which is absurd. If $n = 2$, it is enough, in view of the preceding, to show that the morphism

$$\mathrm{H}^0(X, F) \to \mathrm{H}^0(U, F) = \mathrm{H}^0(X, i_* i^* F)$$

is surjective (where i is the canonical immersion of U into X). Let s be a section of $i_* i^* F$ over X, and let V be the largest open subset over which it comes from a section of F. Suppose $V \neq X$, and let y be a maximal point of $X - V$; with the preceding notation, it follows from the hypothesis that the canonical morphism

$$\mathrm{H}^0(\bar{X}, \bar{F}) \to \mathrm{H}^0(\bar{X} - \bar{y}, \bar{F}) = \mathrm{H}^0(\bar{V}, \bar{F})$$

is bijective; consequently $s|\bar{V}$ extends to the point $\bar{y}$, which is absurd and completes the proof in case 1°).

2°) Case where F is a sheaf of groups. In view of 1°), the only thing left to show is that, in the case $n = 3$, the morphism

$$\mathrm{H}^1(X, F) = \mathrm{H}^1(X, i_* i^* F) \to \mathrm{H}^1(U, F)$$

is bijective. We already know that it is injective, by 1°) and XIV.1.1 2°) (ii bis). For surjectivity, we use the exact sequence (SGA 4 XII 3.2)

$$0 \to \mathrm{H}^1(X, i_* i^* F) \to \mathrm{H}^1(U, F) \xrightarrow{d} \mathrm{H}^0(X, \mathrm{R}^1 i_*(i^* F)).$$

Let $s \in \mathrm{H}^1(U, F)$, and let $V \supset U$ be the largest open subset over which $d(s) = 0$; this is also **97**
the largest open subset such that s comes from an element of $\mathrm{H}^1(V, F)$. Suppose $V \neq X$, and let y be a maximal point of $X - V$; if $\bar{X}$ is the strict localization of X at a geometric point $\bar{y}$ above y, then, with the evident notation, we have the exact sequence

$$0 \to \mathrm{H}^1(\bar{X}, \bar{i}_*(\bar{i}^* \bar{F})) \to \mathrm{H}^1(\bar{U}, \bar{F}) \xrightarrow{\bar{d}} \mathrm{H}^0(\bar{X}, \mathrm{R}^1 \bar{i}_*(\bar{i}^* \bar{F})).$$

Since $i : U \to X$ is quasi-compact, $\mathrm{R}^1 \bar{i}_*(\bar{i}^* \bar{F})$ is the inverse image of $\mathrm{R}^1 i_*(i^* F)$ under the morphism $\bar{X} \to X$, whence $\mathrm{H}^0(\bar{X}, \mathrm{R}^1 \bar{i}_*(\bar{i}^* \bar{F})) = (\mathrm{R}^1 i_*(i^* F))_{\bar{y}}$. By hypothesis, and since $y \in Y$, the morphism

$$\mathrm{H}^1(\bar{X}, \bar{F}) \to \mathrm{H}^1(\bar{V}, \bar{F})$$

is bijective. The image $\bar{s}$ of s in $\mathrm{H}^1(\bar{U}, \bar{F})$, which extends to $\bar{V}$ by the definition of V, therefore also extends to $\bar{X}$; it follows that $\bar{d}(\bar{s}) = 0$, and hence the image of $d(s)$ in the geometric fiber $(\mathrm{R}^1 i_*(i^* F))_{\bar{y}}$ is zero; but this contradicts the definition of V, proving case 2°).

3°) Case where F is a bounded-below complex of abelian sheaves. We argue by induction on n. The conclusion holds for sufficiently small n, since F is bounded below. Suppose, therefore, that $\mathrm{prof}_Y(F) \geq n-1$, and let us show that $\mathrm{prof}_Y(F) \geq n$, knowing that for every point y of Y one has $\mathrm{prof}_y(F) \geq n$. It is enough to see that the canonical morphism

$$\mathrm{H}^{n-2}(X,F) \to \mathrm{H}^{n-2}(U,F) \tag{$*$}$$

is surjective and that

$$\mathrm{H}^{n-1}(X,F) \to \mathrm{H}^{n-1}(U,F) \tag{$**$}$$

is injective (the result also applying when X is replaced by a scheme étale over X). **98**

a) Surjectivity of $(*)$. The proof is analogous to that of 2°). In view of XIV.1.5 and (SGA 4 V 4.5), one has the exact sequence

$$\mathrm{H}^{n-2}(X,F) \to \mathrm{H}^{n-2}(U,F) \xrightarrow{d} \mathrm{H}^{n-1}_Y(X,F) = \mathrm{H}^0(X, \underline{H}^{n-1}_Y(F)).$$

Let $s \in \mathrm{H}^{n-2}(U,F)$, and let $V \supset U$ be the largest open subset over which $d(s)=0$, which is also the largest open subset such that s extends to $\mathrm{H}^{n-2}(V,F)$. Suppose $V \neq X$, and let y be a maximal point of $X-V$ and $\bar{X}$ the strict localization of X at a geometric point $\bar{y}$ above y. Since $i : U \to X$ is quasi-compact, the formation of $\underline{H}^{n-1}_Y(F)$ commutes with the base change $\bar{X} \to X$, and hence one has (with the evident notation) the exact sequence

$$\mathrm{H}^{n-2}(\bar{X},\bar{F}) \to \mathrm{H}^{n-2}(\bar{U},\bar{F}) \xrightarrow{\bar{d}} \mathrm{H}^{n-1}_{\bar{Y}}(\bar{X},\bar{F}) = (\underline{H}^{n-1}_Y(F))_{\bar{y}},$$

the last equality following from the retrocompactness hypothesis on U.

But by hypothesis one has the isomorphism

$$\mathrm{H}^{n-2}(\bar{X},\bar{F}) \xrightarrow{\sim} \mathrm{H}^{n-2}(\bar{X}-\bar{y},\bar{F}) = \mathrm{H}^{n-2}(\bar{V},\bar{F});$$

consequently the image $\bar{s}$ of s in $\mathrm{H}^{n-2}(\bar{U},\bar{F})$, which extends (by definition of V) to $\mathrm{H}^{n-2}(\bar{V},\bar{F})$, also extends to $\mathrm{H}^{n-2}(\bar{X},\bar{F})$; but this shows that $\bar{d}(\bar{s})=0$, that is, that $d(s)$ is zero at $\bar{y}$, which is absurd.

b) Injectivity of $(**)$. Using the surjectivity of $(*)$, one obtains the exact sequence

$$0 \to \mathrm{H}^0(X, \underline{H}^{n-1}_Y(F)) \to \mathrm{H}^{n-1}(X,F) \to \mathrm{H}^{n-1}(U,F)$$

and we must show that every element $s \in \mathrm{H}^0(X, \underline{H}^{n-1}_Y(F))$ is zero. Let V be the largest open **99**
subset over which $s=0$. Suppose that $V \neq X$, and let y be a maximal point of $X-V$ and $\bar{X}$ a strict localization of X at a geometric point $\bar{y}$ above y. By the induction hypothesis and 84, one has $\mathrm{prof}_{\bar{Y}}(\bar{F}) \geq \mathrm{prof}_Y(F) \geq n-1$, and hence the map e in the following diagram is injective:

$$\mathrm{H}^0(\bar{X}, \underline{H}^{n-1}_Y(\bar{F})) = (\underline{H}^{n-1}_Y(F))_{\bar{y}} \xrightarrow{e} \mathrm{H}^{n-1}(\bar{X},\bar{F}) \xrightarrow{f} \mathrm{H}^{n-1}(\bar{V},\bar{F}).$$

The same is true of f by the hypothesis; the equality on the left follows from the retrocompactness hypothesis on U. Let $\bar{s}$ be the image of s in $(\underline{H}^{n-1}_Y(F))_{\bar{y}}$; since s vanishes over V, one has $f \cdot e(\bar{s}) = 0$, whence $\bar{s}=0$, which contradicts the choice of y and completes the proof.

Remark XIV.1.9. A result analogous to XIV.1.8 is probably valid when the étale topos of a scheme X is replaced by a "locally finite-type topos", that is, one definable by a locally finite-type site (SGA 4 VI 1.1). To see this, one must use a result of P. Deligne (SGA 4 VI.9) asserting that there are "enough fiber functors"in such a topos.

We shall deduce from XIV.1.8 important cases in which the étale depth can be determined.

Theorem 1.10 (Cohomological semipurity theorem[(*)]). *Let X denote either a scheme smooth over a field k, or an excellent regular scheme* (EGA IV 7.8.2) *of characteristic zero (N.B. if resolution of singularities in the sense of* (SGA 4 XIX) *is assumed, it is enough, more*

generally, to suppose that X is an excellent regular scheme of equal characteristic). Let Y **100**
be a closed subset of X and **L** *the set of prime numbers distinct from the characteristic of X. Then*

$$\operatorname{prof}_Y^{\mathbf{L}}(X) = 2\operatorname{codim}(Y, X).$$

Proof. It follows from XIV.1.8 that

$$\operatorname{prof}_Y^{\mathbf{L}}(X) = \inf_{y\in Y} \operatorname{prof}_y(X).$$

On the other hand, since $\operatorname{codim}(Y, X) = \inf_{y\in Y} \dim \mathcal{O}_{X,y}$, we are reduced to showing that

$$\operatorname{prof}_y^{\mathbf{L}}(X) = 2\dim \mathcal{O}_{X,y},$$

which follows from (SGA 4 XVI 3.7 and XIX 3.2).

Theorem XIV.1.11 (Homotopical purity theorem(*)). *If X is a locally noetherian scheme that is regular (resp. whose local rings are complete intersections), and Y is a closed subset of X such that* $\operatorname{codim}(Y, X) \geq 2$ *(resp.* $\operatorname{codim}(Y, X) \geq 3$*), then one has*

$$\operatorname{prof}\operatorname{hop}_Y(X) \geq 3.$$

Indeed, it follows from XIV.1.8 that $\operatorname{prof}\operatorname{hop}_Y(X) = \inf_{y\in Y} \operatorname{prof}\operatorname{hop}_y(X)$. But the strictly local rings of X at the various points of Y are regular rings of dimension ≥ 2 (resp. complete intersections of dimension ≥ 3). It then follows from the purity theorem X 3.4 that $\operatorname{prof}\operatorname{hop}_y(X) \geq 3$, which proves the theorem.

Example XIV.1.12. Let X be a locally noetherian scheme, Y a closed subset of X, and $n = 1$ or 2. Then, if $\operatorname{prof}_Y(\mathcal{O}_X) \geq n$ ($\operatorname{prof}_Y(\mathcal{O}_X)$ denoting the Y-depth in the sense of coherent sheaves (cf. 1.6 a)), one also has $\operatorname{prof}_Y(X) \geq n$; this is obvious for $n = 1$, and for $n = 2$ it is precisely Hartshorne's theorem (III 1). On the other hand, the analogous assertion is false for **101**
$n \geq 3$. For example, take an affine space of dimension ≥ 3 over a field of characteristic $\neq 2$ and let the group $\mathbf{Z}/2\mathbf{Z}$ act by reflection through the origin. Let X be the quotient and $Y = \{x\}$ the image of the origin in X. Then $\mathcal{O}_{X,x}$ is a Cohen–Macaulay ring, so $\operatorname{prof}_x(\mathcal{O}_X) \geq 3$; but the affine space with the origin removed is an étale covering of $X - \{x\}$ that does not extend to an étale covering of X; hence, by XIV.1.4, $\operatorname{prof}_Y(X) = 2$.

The following theorem is the analogue of (EGA IV 6.3.1):

(*) *Editor's note.* Gabber subsequently proved — in 1994 — Grothendieck's absolute cohomological purity conjecture: if Y is a closed subscheme of absolute Noetherian schemes of pure codimension c and n is an integer invertible on X, then $\mathcal{H}_Y^q(\Lambda)$ is zero if $q \neq 2c$ and is $\Lambda_Y(-c)$ (Tate twist) otherwise, where $\Lambda = \mathbf{Z}/n\mathbf{Z}$. See (Fujiwara K., "A Proof of the Absolute Purity Conjecture (after Gabber)", in *Algebraic geometry 2000, Azumino (Hotaka)*, Adv. Stud. in Pure Math., vol. 36, 2002, p. 153-183). For applications to the existence of the dualizing complex, see (SGA 5, Lect. Notes in Math., vol. 589, Springer-Verlag, 1977, p. 1672), Exposé 1 and *loc. cit.*, §8. This conjecture had been proved in the case $n = \ell^\nu$, with ℓ a sufficiently large prime invertible on X, by making crucial use of K-theory (Thomason R.W., "Absolute cohomological purity", *Bull. Soc. Math. France* **112** (1984), no. 3, p. 397–406). K-theory reappears in Gabber's proof through the Atiyah-Hirzebruch-Thomason spectral sequence relating étale cohomology and K-theory, a method already used in Thomason's approach. Besides this result, the other fundamental argument is the generalization of the Lefschetz theorem cited in note ((*)), page 156.

(*) *Editor's note.* recently, de Jong and Oort obtained the following purity statement: let $\tilde{S} \to S$ be a resolution of the singularities of the spectrum S of a normal noetherian local ring of dimension 2, and let U be the complement of the closed point s in S. Suppose in addition that $k(s)$ is algebraically closed. Then, for every prime number p, in particular if S has characteristic p, the restriction morphism $H^1_{\text{ét}}(\tilde{S}, \mathbf{Q}_p) \to H^1_{\text{ét}}(U, \mathbf{Q}_p)$ is bijective (de Jong A.J. & Oort F, "Purity of the stratification by Newton polygons", *J. Amer. Math. Soc.* **13** (2000), no. 1, p. 209–241, Theorem 3.2). If $k = \mathbf{C}$ and A is the completion of a surface singularity, this result is due to Mumford (see page 137, [5]).

Theorem XIV.1.13. *Let $f : X \to S$ be a morphism of schemes, Y a closed subset of X, and Z a closed subset of S, such that $f(Y) \subset Z$. Suppose that the local rings of X at the various points of Y are noetherian and that the open subsets $X - Y$ and $S - Z$ are retrocompact in X and S, respectively. Let p, q, r be integers such that $p \geq -r$, $q \geq 0$, let $\mathbf{L}$ be a set of prime numbers, and let F be a complex of abelian sheaves of $\mathbf{L}$-torsion on S, such that the cohomology sheaves $\mathrm{H}^i(F)$ vanish for $i < -r$. Suppose that*

a) The morphism f is locally $(p+q+r-2)$-acyclic for $\mathbf{L}$ (SGA 4 XV 1.11).

b) One has

$$\operatorname{prof}_Z(F) \geq p.$$

c) For every point s of Z, one has

$$\operatorname{prof}^{\mathbf{L}}_{Y_s}(X_s) \geq q.$$

Then one has **102**

$$\operatorname{prof}_Y(f^*F) \geq p+q.$$

We shall need the following lemma:

Lemma XIV.1.13.1. *Let $\mathbf{L}$ be a set of prime numbers, let n and r be integers, and let $f : X \to S$ be a morphism locally n-acyclic for $\mathbf{L}$. Let F be a complex of abelian sheaves whose cohomology sheaves are of $\mathbf{L}$-torsion, such that $\mathrm{H}^i(F) = 0$ for $i < -r$, let Z be a closed subset of S such that $S - Z$ is retrocompact in S, and let $T = f^{-1}(Z)$. Then the canonical morphism*

$$f^*(\underline{H}^i_Z(F)) \to \underline{H}^i_T(f^*F)$$

is bijective for $i < n - r + 2$ and injective for $i = n - r + 2$.

Set $U = S - Z$ and $V = X - T$, so that one has the cartesian square

$$\begin{tikzcd} V \arrow[r, "g"] \arrow[d, "k"'] & U \arrow[d, "j"] \\ X \arrow[r, "f"] & S \end{tikzcd}$$

Consider the following commutative diagram whose rows are exact

$$\begin{tikzcd} {} \arrow[r] & f^*(\underline{H}^i_Z(F)) \arrow[r] \arrow[d] & f^*(\underline{H}^i(F)) \arrow[r] \arrow[d, "\wr"] & f^*(\underline{H}^i(\mathrm{R}j_*(j^*F))) \arrow[r] \arrow[d] & {} \\ {} \arrow[r] & \underline{H}^i_T(f^*F) \arrow[r] & \underline{H}^i(f^*F) \arrow[r] & \underline{H}^i(\mathrm{R}k_*(k^*f^*F)) \arrow[r] & {} ; \end{tikzcd}$$

it follows that one is reduced to showing that the morphism

$$f^*(\underline{H}^i(\mathrm{R}j_*(j^*F))) \to \underline{H}^i(\mathrm{R}k_*(k^*f^*F))$$

is bijective for $i < n - r + 1$ and injective for $i = n - r + 1$. But such a morphism arises from **103**
the following morphism between hypercohomology spectral sequences

$$\begin{tikzcd} f^*E_2^{p,q} = f^*(\mathrm{R}^p j_*(\underline{H}^q(j^*F))) \arrow[r, Rightarrow] \arrow[d] & f^*(\underline{H}^*(\mathrm{R}j_*(j^*F))) \arrow[d] \\ E_2'^{p,q} = \mathrm{R}^p k_*(\underline{H}^q(k^*f^*F)) \arrow[r, Rightarrow] & \underline{H}^*(\mathrm{R}k_*(k^*f^*F)). \end{tikzcd}$$

Since j is quasi-compact, it follows from (SGA 4 XV 1.10) that the morphism $f^*(E_2^{p,q}) \to E_2'^{p,q}$ is bijective for $p \leq n$ and injective for $p = n+1$; in particular, it is bijective for $p+q \leq n-r$ and injective for $p+q = n-r+1$. The conclusion follows at once.

Let us return to the proof of XIV.1.13. Let $T = f^{-1}(Z)$. By XIV.1.13.1 and condition a), the canonical morphism $f^*(\underline{H}^i_Z(F)) \to \underline{H}^i_T(f^*F)$ is an isomorphism for $i \leq p+q$. It therefore follows from b) that $\underline{H}^i_T(f^*F) = 0$ for $i < p$ and, for $i < p+q$, the restriction of $\underline{H}^i_T(f^*F)$ to T is the inverse image of a sheaf G^i on Z. Let

$$f_T : T \to Z$$

be the restriction of f to T. It then follows from c) and the corollary below that $\underline{H}^j_Y(f_T^*(G^i)) = 0$ for $j < q$. We conclude that

$$\underline{H}^j_Y(\underline{H}^i_T(f^*F)) = 0 \text{ for } i+j < p+q,$$

because the inequality $i+j < p+q$ implies either $i < p$, in which case $\underline{H}^i_T(f^*F) = 0$, or $j < q$, in which case $\underline{H}^j_Y(\underline{H}^i_T(f^*F)) = 0$. Given that, with the notation of XIV.1.0, $\underline{\Gamma}_Y = \underline{\Gamma}_Y.\underline{\Gamma}_T$, one has the spectral sequence

$$E_2^{i,j} = \underline{H}^j_Y(\underline{H}^i_T(f^*F)) \Longrightarrow \underline{H}^*_Y(f^*F); \tag{1.13.2}$$

since $E_2^{i,j} = 0$ for $i+j < p+q$, one sees that $\underline{H}^k_Y(f^*F) = 0$ for $k < p+q$. The theorem will therefore be proved if we prove the following corollary (which is the special case of 1.13 obtained by taking $Z = S$, $r = p = 0$ and F concentrated in degree 0). **104**

Corollary XIV.1.14. *Let $f : X \to S$ be a morphism, let Y be a closed subset such that the complementary open subset $X - Y$ is quasi-compact in X and the local rings of X at the various points of Y are Noetherian. Let $\mathbf{L}$ be a set of prime numbers, let q be an integer, and let F be an abelian sheaf of $\mathbf{L}$-torsion on S. Suppose that f is locally $(q-2)$-acyclic for $\mathbf{L}$ and that, for every point s of S, one has $\operatorname{prof}^{\mathbf{L}}_{Y_s}(X_s) \geq q$. Then $\operatorname{prof}_Y(f^*F) \geq q$.*

1°) Reduction to the case in which X and S are strictly local schemes, f is a local morphism, and Y consists of a closed point of X.

By XIV.1.8, to prove XIV.1.14, one must show that, for every point y of Y, one has:

$$\operatorname{prof}_y(f^*F) \geq q.$$

Let $s = f(y)$, let $\bar{s}$ be a geometric point above s, let $\bar{y}$ be a geometric point above y and $\bar{s}$, let $\bar{X}$ and $\bar{S}$ be the strict localizations of X and S at $\bar{y}$ and $\bar{s}$, respectively, let $\bar{f} : \bar{X} \to \bar{S}$ be the canonical morphism, and let $\bar{F}$ be the inverse image of F on $\bar{S}$. Since $\operatorname{prof}_y(f^*F) = \operatorname{prof}_{\bar{y}}(\bar{f}^*\bar{F})$, it is enough to show that the hypotheses of XIV.1.14 are preserved when f (resp. Y, resp. F) is replaced by $\bar{f}$ (resp. $\{\bar{y}\}$, resp. $\bar{F}$). The quasi-compactness condition follows from the Noetherian hypothesis on $\mathcal{O}_{X,x}$, which implies that $\bar{X}$ is Noetherian . By (SGA 4 XV 1.10 (i)), $\bar{f}$ is still locally $(q-2)$-acyclic for $\mathbf{L}$. Moreover, the fiber $(\bar{X})_{\bar{s}}$ of $\bar{X}$ above $\bar{s}$ identifies with the strict localization of X_s at $\bar{y}$, and therefore satisfies $\operatorname{prof}_{\bar{y}}((\bar{X})_{\bar{s}}) \geq q$. Since an analogous relation is trivially satisfied for the fibers of the $\bar{S}$-scheme $\bar{X}$ other than the closed fiber, this completes the reduction. **105**

2°) The case in which X and S are strictly local, f is a local homomorphism, and Y consists of the closed point of X. Let

$$g : Y = X - \{y\} \to S$$

be the structural morphism of U. We must show that the canonical morphism

$$u_i : H^i(X, f^*F) \to H^i(U, f^*F)$$

is bijective for $i \leq q-2$ and injective for $i = q-1$. Consider the commutative diagram

$$\begin{tikzcd}
H^i(X, f^*F) \arrow[rr, "u_i"] & & H^i(U, f^*F) \\
& H^i(S, F) \arrow[ul, "v_i"] \arrow[ur, "w_i"'] &
\end{tikzcd}$$

The morphism v_i is plainly bijective for every i. Moreover, g is locally $(q-2)$-acyclic for $\mathbf{L}$; its fibers are also $(q-2)$-acyclic for $\mathbf{L}$, as follows from the fact that $\operatorname{prof}_y(X_s) \geq q$ and that the fibers of f are $(q-2)$-acyclic for $\mathbf{L}$. Since g is quasi-compact because X is Noetherian, (SGA 4 XV 1.16) shows that g is $(q-2)$-acyclic for $\mathbf{L}$. Consequently w_i, and hence also u_i, is bijective for $i \leq q-2$ and injective for $i = q-1$, which completes the proof of XIV.1.14. **106**

Corollary XIV.1.15. *Let $f : X \to S$ be a morphism of schemes, let $\mathbf{L}$ be a set of prime numbers, let m and r be integers, and let F be a complex of abelian sheaves of $\mathbf{L}$-torsion on S such that $\underline{H}^i(F) = 0$ for $i < -r$. Let x be a point of X, let $s = f(x)$, and suppose that the local ring $\mathcal{O}_{X,x}$ is Noetherian. Then, if f is locally m-acyclic for $\mathbf{L}$, one has*

$$\operatorname{prof}_x(f^*F) \geq \inf(\operatorname{prof}_s(F) + \operatorname{prof}^{\mathbf{L}}_x X_s), n) \text{ where } n = m - r + 2. \tag{$*$}$$

In particular, if $n \geq \operatorname{prof}_s(F) + \operatorname{prof}^{\mathbf{L}}_x(X_s)$, for example if f is locally acyclic for $\mathbf{L}$, then

$$\operatorname{prof}_x(f^*F) \geq \operatorname{prof}_s(F) + \operatorname{prof}^{\mathbf{L}}_x(X_s). \tag{$**$}$$

If $\mathbf{L}$ consists of a single element ℓ and $n \geq \operatorname{prof}_s(F) + \operatorname{prof}^{\mathbf{L}}_x(X_s)$, the preceding inequality is an equality.

One reduces to the case in which s and x are closed points by taking the strict localizations of S and x at geometric points $\bar{s}$ above s and $\bar{x}$ above x and $\bar{s}$. If $n \geq \operatorname{prof}_s(F) + \operatorname{prof}^{\mathbf{L}}_x(X_s)$, then $(*)$ follows from XIV.1.13 by taking $p = \operatorname{prof}_s(F)$ and $q = \operatorname{prof}^{\mathbf{L}}_x(X_s)$ there (the hypothesis that $S - \{s\}$ is quasi-compact in S follows from the fact that $X - X_s$ is quasi-compact in X and that f is surjective, since it is (-1)-acyclic, except possibly when the conclusion of 1.15 is empty). If $n < \operatorname{prof}_s(F) + \operatorname{prof}^{\mathbf{L}}_x(X_s)$, inequality $(*)$ again follows from XIV.1.13, for example by taking $p = \operatorname{prof}_s(F)$ and $q = n - p$. It remains to prove the last assertion. Let **107**
$p = \operatorname{prof}_s(F)$ and $q = \operatorname{prof}_x(X_s)$; it follows from (1.13.2) that

$$\underline{H}^{p+q}_x(f^*(F)) \simeq \underline{H}^q_x(f^*(\underline{H}^p_s(F))).$$

Since $\operatorname{prof}_s(F) = p$, the sheaf $\underline{H}^p_s(F)$ is a nonzero sheaf of ℓ-torsion, constant on s. Consequently the sheaf $G = f^*(\underline{H}^p_s(F))$ is a nonzero sheaf of ℓ-torsion, constant on X_s, and therefore contains a subsheaf isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$; since $\underline{H}^q_x(\mathbf{Z}/\ell\mathbf{Z})$ is nonzero, one indeed has $\underline{H}^q_x(G) \neq 0$.

Corollary XIV.1.16. *Let $f : X \to S$ be a regular morphism of excellent schemes (*EGA IV 7.8.2*) of characteristic zero, let ℓ be a prime number, and let F be a complex of sheaves of ℓ-torsion on S. Let $x \in X$ and $s = f(x)$; then*

$$\operatorname{prof}_x(f^*F) = \operatorname{prof}_s(F) + 2\dim(\mathcal{O}_{X,x}).$$

Indeed, f is locally acyclic (SGA 4 XIX 4.1). It then follows from XIV.1.15 that

$$\operatorname{prof}_x(f^*F) = \operatorname{prof}_s(F) + \operatorname{prof}_x(X_s).$$

But by XIV.1.10,

$$\operatorname{prof}_x(X_s) = 2\dim \mathcal{O}_{X_s,x},$$

which gives the result.

Remark XIV.1.17. It follows from XIV.1.15 that XIV.1.13 remains valid when b) and c) are replaced by the conditions: **108**

b′) For every point $s \in f(Y)$, one has $\operatorname{prof}_s(F) \geq p$.

c′) For every point $x \in Y$, if $s = f(x)$, one has $\operatorname{prof}_x^{\mathbf{L}}(X_s) \geq q$.

For a sheaf of sets or groups, one has the following theorem, analogous to XIV.1.13.

Theorem XIV.1.18. *Let $f : X \to S$ be a morphism of schemes and let Y be a closed subset of X such that $X - Y$ is quasi-compact in X and, for every point x of Y, the local ring $\mathcal{O}_{X,x}$ is Noetherian.*

1°) *Let F be a sheaf of sets on S and let n be an integer equal to* 1 *or* 2*. Suppose that f is locally $(n-2)$-acyclic and that, for every point s of $f(Y)$,*

$$\operatorname{prof}_{Y_s}(X_s) + \operatorname{prof}_s(F) \geq n.$$

Then:

$$\operatorname{prof}_Y(f^*F) \geq n.$$

2°) *Let $\mathbf{L}$ be a set of prime numbers and let F be a sheaf of ind-$\mathbf{L}$-groups. Suppose that f is locally* 1*-aspherical for $\mathbf{L}$ (*SGA 4 XV 1.11*) and that, for every point s of $f(Y)$,*

$$\operatorname{prof\,hop}_{Y_s}^{\mathbf{L}}(X_s) + \operatorname{prof}_s(F) \geq 3.$$

Then:

$$\operatorname{prof}_Y(f^*F) \geq 3.$$

Proof. We reduce, as in XIV.1.14 and XIV.1.15, to the case where X and S are strictly local schemes, f is a local morphism, and Y is the closed point x of X. Let $s = f(x)$ be the closed point of S; we have the commutative diagram: **109**

$$\begin{tikzcd}
X - X_s \arrow[r, "i"] \arrow[d, "h"'] & X - \{x\} \arrow[r, "j"] \arrow[dr, "g"] & X \arrow[d, "f"] \\
S - \{s\} \arrow[rr, "k"'] & & S.
\end{tikzcd}$$

1°) a) *Case* $n = 1$.

If $\operatorname{prof}_s(F) \geq 1$, then the morphism $F \to k_*k^*F$ is injective, and hence the morphism $f^*F \to f^*(k_*k^*F)$ is also injective. On the other hand, it follows from the fact that f is locally (-1)-acyclic that the morphism $f^*(k_*k^*F) \to (j.i) * (f^*F_{|X-X_s}))$ is injective. Finally, the composite morphism $f^*F \to (j.i)*(f^*F_{|X-X_s}))$ is injective, which shows that $\operatorname{prof}_{X_s}(f^*F) \geq 1$, and consequently also $\operatorname{prof}_x(f^*F) \geq 1$.

If $\operatorname{prof}_x(X_s) \geq 1$, consider the commutative diagram

$$\begin{tikzcd}
H^0(X, f^*F) \arrow[r, "v"] \arrow[d, "\wr"'] & H^0(X - \{x\}, f^*F) \arrow[d] \\
H^0(X_s, f^*F) \arrow[r, "v'"] & H^0(X - \{x\}, f^*F);
\end{tikzcd} \tag{$*$}$$

By hypothesis, v' is injective, and therefore so is v.

b) *Case* $n = 2$. Consider the commutative diagram

$$\begin{tikzcd}
H^0(S, F) \arrow[rr, "u"] \arrow[d, "m"', "\wr"] & & H^0(S - \{s\}, F) \arrow[d, "n"', "\wr"] \\
H^0(X, f^*F) \arrow[r, "v"] & H^0(X - \{x\}, f^*F) \arrow[r, "w"] & H^0(X - X_s, f^*F);
\end{tikzcd} \tag{$**$}$$

we must show that v is bijective. The morphism m is obviously bijective, and, since f is **110**
0-acyclic, n is also bijective.

If $\operatorname{prof}_s(F) \geq 2$, u is bijective. As we saw in a), the sole hypothesis $\operatorname{prof}_s(F) \geq 1$ implies the relation $\operatorname{prof}_{X_s}(f^*F) \geq 1$; consequently v and w are injective, and it then follows from $(**)$ that v is bijective.

If $\operatorname{prof}_x(X_s) \geq 2$, then g is 0-acyclic (because it is locally 0-acyclic and its fibers are 0-acyclic). It follows that $v \cdot m$ is bijective, and therefore v is bijective.

If $\operatorname{prof}_s(F) \geq 1$ and $\operatorname{prof}_x(X_s) \geq 1$, then we already know that v and w are injective. Let z be a maximal point of $X_s - \{x\}$ (such a point exists by the hypothesis $\operatorname{prof}_x(X_s) \geq 1$), let $\bar{Z}$ be the strict localization of X at a geometric point above z, and let $\overline{f^*F}$ be the inverse image of f^*F on $\bar{Z}$. Consider the commutative diagram

$$\begin{tikzcd}
 & H^0(S,F) \arrow[rr] \arrow[dl, "m"', "\sim"] & & H^0(S-\{s\},F) \arrow[dr, "n", "\sim"'] & \\
H^0(X,f^*F) \arrow[rr, "v"] \arrow[dr, "m'"', "\sim"] & & H^0(X-\{x\},f^*F) \arrow[rr, "w"] \arrow[dl, "r"] & & H^0(X-X_s,f^*F) \arrow[dl, "n'", "\sim"'] \\
 & H^0(\bar{Z},f^*F) \arrow[rr] & & H^0(\bar{Z}-\{\bar{z}\},\overline{f^*F}) &
\end{tikzcd}$$

the morphism $m' \cdot m$ is obviously bijective, and it follows from the fact that f is locally 0-acyclic that $n' \cdot n$ is bijective; consequently m' and n' are also bijective. Since w is injective, r is also injective, and therefore v is bijective.

2°) In view of b), we already know that $\operatorname{prof}_x(f^*F) \geq 2$.

If $\operatorname{prof}_s(F) \geq 3$, then $\mathrm{R}^1k_*(k^*F) = 1^{(*)}$. Since f is locally 1-aspherical, one has $\mathrm{R}^1(j \cdot$ **111**
$i)_*(f^*F_{|X-X_s}) = f^*(\mathrm{R}^1k_*(k^*F)) = 1$. Thus $\operatorname{prof}_{X_s}(f^*F) \geq 3$ and consequently $\operatorname{prof}_x(f^*F) \geq 3$.

If $\operatorname{prof}_x(X_s) \geq 3$, then g is 1-aspherical (because g is locally 1-aspherical and its fibers are 1-aspherical). Thus $H^1(X - \{x\}, f^*F) = H^1(S,F) = 1$, and consequently $\operatorname{prof}_x(f^*F) \geq 3$.

If $\operatorname{prof}_s(F) \geq 2$ and $\operatorname{prof}_x(X_s) \geq 1$, we use the exact sequence (SGA 4 XII 3.2):

$$1 \to \mathrm{R}^1j_*(i_*(f^*F_{|X-X_s})) \to \mathrm{R}^1(j.i)_*(f^*F_{|X-X_s}) \to j_*(\mathrm{R}^1i_*(f * F_{|X-X_s})).$$

Since f and g are locally 1-aspherical, one has

$$\begin{aligned} \mathrm{R}^1(j \cdot i)_*(f^*F_{|X-X_s}) &\simeq f^*(\mathrm{R}^1k_*(k^*F)) \\ \mathrm{R}^1i_*(f^*F_{|X-X_s}) &\simeq g^*(\mathrm{R}^1k_*(k^*F)); \end{aligned}$$

the preceding exact sequence can then be written in the form

$$1 \to \mathrm{R}^1j_*(i_*(f^*F_{|X-X_s})) \to f^*(\mathrm{R}^1k_*(k^*F)) \xrightarrow{a} j_*(j^*(f^*(\mathrm{R}^1k_*(k^*F)))). \qquad (***)$$

The hypothesis $\operatorname{prof}_s(F) \geq 2$ shows that the morphism $F \to k_*k^*F$ is bijective; applying g^*, and taking into account the fact that g is locally 0-acyclic, we find $f^*F_{|X-\{x\}} = i_*(f^*F_{|X-X_s})$. The hypothesis $\operatorname{prof}_x(X_s) \geq 1$ shows that the morphism a is injective (note that $f^*(\mathrm{R}^1k_*(k^*F))$ is a sheaf equal to 1 outside X_s and constant on X_s). It then follows from $(***)$ that $\mathrm{R}^1j_*(f^*F_{|X-\{x\}}) = 1$, and hence $\operatorname{prof}_x(f^*F) \geq 3$. **112**

If $\operatorname{prof}_s(F) \geq 1$ and $\operatorname{prof}_x(f^*F) \geq 2$, consider the sheaf of homogeneous spaces G defined by the exact sequence

$$1 \to F \to k_*k * F \to G \to 1.$$

(*) *Editor's note.* the trivial torsor is subsequently denoted successively by 0 or 1; we have retained this double notation, which, on reflection, causes no ambiguity.

Applying the exact functor g^* to this exact sequence and using (SGA 4 XII 3.1), we obtain the following commutative diagram with exact rows:

$$\begin{tikzcd} f^*(k_*k^*F) \arrow[r] \arrow[d] & f^*G \arrow[r] \arrow[d, "b"] & 1 & \\ j_*(g^*(k_*k^*F)) \arrow[r, "u"] & j_*(g^*G) \arrow[r] & \mathrm{R}^1 j_*(g^*F) \arrow[r] & \mathrm{R}^1 j_*(g^*(k_*k^*F)). \end{tikzcd}$$

Since $\mathrm{prof}_x(X_s) \geq 2$, the morphism b is bijective, so u is surjective, and we thus have a map whose kernel is reduced to the neutral element:

$$1 \to \mathrm{R}^1 j_*(g^*F) \to \mathrm{R}^1 j_*(g^*(k_*k^*F)) = R.$$

Since $g^*(k_*k^*F) \simeq i_*(f^*F_{|X-X_s})$ (because g is locally 0-acyclic), R is identified with the first term of the exact sequence $(*{*}*)$; but in the preceding case we saw that $R = 1$ as soon as $\mathrm{prof}_x(X_s) \geq 1$, which proves that $\mathrm{prof}_x(f^*F) \geq 3$ and completes the proof of XIV.1.18.

The following corollaries are generalizations of (SGA 4 XVI 3.2 and 3.3).

Corollary XIV.1.19. *Let $f : X \to S$ be a flat morphism with separable fibers between locally Noetherian schemes, and let Y be a closed subset of X. Suppose that, for every point $s \in f(Y)$,* **113**
the fiber Y_s is rare[(*)] in X_s and that one of the following two conditions holds:

a) the closure of $f(Y)$ is rare in S.

b) X_s is geometrically unibranch at the points of Y_s.

Then

$$\mathrm{prof}_Y(X) \geq 2.$$

Indeed, it follows from the hypothesis on f that f is locally 0-acyclic (SGA 4 XV 4.1). We then apply XIV.1.13. The hypothesis that Y_s is rare in X_s (respectively that $\overline{f(Y)}$ is rare in S) is equivalent, by XIV.1.6 b), to the relation $\mathrm{prof}_{Y_s}(X_s) \geq 1$ (respectively $\mathrm{prof}_{\overline{f(Y)}}(S) \geq 1$). The hypothesis that X_s is geometrically unibranch at every point of Y_s is equivalent to saying that the strict localization of X_s at a geometric point of Y_s is irreducible; knowing that Y_s is rare in X_s, this clearly implies $\mathrm{prof}_{Y_s}(X_s) \geq 2$, by XIV.1.8. In either case XIV.1.13 indeed gives $\mathrm{prof}_Y(X) \geq 2$.

Corollary XIV.1.20. *Let $f : X \to S$ be a regular morphism (*EGA IV 6.8.1*) between locally Noetherian schemes, and let Y be a closed subset of X. Suppose that, for every point $s \in f(Y)$, one of the following conditions holds:*

a) One has $\mathrm{codim}(Y_s, X_s) \geq 2$.

b) One has $\mathrm{codim}(Y_s, X_s) \geq 1$ *and* $\mathrm{prof}_s(S) \geq 1$.

c) One has $\mathrm{prof\,hop}_s(S) \geq 3$.

Then **114**

$$\mathrm{prof\,hop}_Y(X) \geq 3.$$

Indeed, this follows from XIV.1.18, since hypothesis a) implies $\mathrm{prof\,hop}_{Y_s}(X_s) \geq 3$ (cf. XIV.1.11), and the condition $\mathrm{codim}(Y_s, X_s) \geq 1$ clearly implies $\mathrm{prof}_Y(X) \geq 2$.

(*) *Editor's note.* "rare"= "having empty interior", cf. Bourbaki TG IX.52.

XIV.2 Technical lemmas

XIV.2.1

Let S be a locally noetherian scheme, $f : X \to S$ a morphism locally of finite type, and t a point of S. If $x \in X$ is such that $s = f(x) \in \operatorname{Spec} \mathcal{O}_{S,t}$, set

$$\delta_t(x) = \deg\operatorname{tr} k(x)/k(s) + \dim(\overline{\{s\}}),$$

where $\overline{\{s\}}$ denotes the closure of s in $\operatorname{Spec} \mathcal{O}_{S,t}$, and $k(x)$ and $k(s)$ the residue fields of x and s, respectively. If S is a local ring with closed point t, we also write $\delta(x)$ instead of $\delta_t(x)$ (cf. SGA 4 XIV 2.2).

Lemma XIV.2.1.1. *Let there be a cartesian square*

$$\begin{tikzcd} X' \arrow[r, "h"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S \end{tikzcd} ,$$

where S and S' are noetherian local rings, with closed points t and t', respectively, g is a faithfully flat morphism such that $g^{-1}(t) = t'$, and f is a morphism locally of finite type. Let **115**
$x' \in X',\ x = h(x'),\ s = f(x),\ s' = f'(x')$; then one has

$$\delta(x') \leq \delta(x).$$

Moreover, the preceding inequality is an inequality if and only if one has:

$$\deg\operatorname{tr} k(x)/k(s) = \deg\operatorname{tr} k(x')/k(s') \text{ and } \dim(\overline{\{s\}}) = \dim(\overline{\{s'\}}).$$

In particular, given $x \in X$, one can find x' such that $\delta(x) = \delta(x')$.

Indeed, one has (EGA IV 6.11)

$$\dim(\overline{\{s\}}) = \dim g^{-1}(\overline{\{s\}}).$$

It follows that, for every point s' of $g^{-1}(s)$, one has the relation $\dim(\overline{\{s'\}}) \leq \dim(\overline{\{s\}})$, and that, given s, one can find $s' \in g^{-1}(s)$ for which equality holds. Let Z then denote the schematic closure of x in the fiber X_s of X at s, and let $Z' = Z \times_{\operatorname{Spec} k(s)} \operatorname{Spec} k(s')$. Then Z' is equidimensional of dimension $\deg\operatorname{tr} k(x)/k(s)$; therefore, for every point $x' \in Z'_x$, one has

$$\deg\operatorname{tr} k(x')/k(s') \leq \deg\operatorname{tr} k(x)/k(s), \text{ with equality}$$

when x' is a maximal point of Z'_x. The asserted conclusion follows at once.

XIV.2.2

Let $f : X \to S$ be a morphism locally of finite type and T a closed subset of S. Let $x \in X, s = f(x)$; we set

$$\delta_T(x) = \deg\operatorname{tr} k(x)/k(s) + \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) = \inf_{t \in T \cap \overline{\{s\}}} \delta_t(x).$$

Lemma XIV.2.2.1. *Let there be a cartesian square* **116**

$$\begin{tikzcd} X' \arrow[r, "h"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S \end{tikzcd} ,$$

where the schemes S and S' are locally noetherian and catenary, the morphism f is locally of finite type, and g is faithfully flat. Let T be a closed subset of S and T' a closed subset of S', such that $g(T') \subset T$, let x' be an element of X', let $x = h(x')$, and let

$$h_{x'} : \operatorname{Spec} \mathcal{O}_{X',x'} \to \operatorname{Spec} \mathcal{O}_{X,x}$$

be the morphism induced by h. Then one has:

$$\delta_T(x) - \delta_{T'}(x') \leq \dim h_{x'}^{-1}(x).$$

Let $s' = f'(x'), s = f(x)$. By definition, one has:

$$\begin{aligned}\delta_T(x) - \delta_{T'}(x') = \deg\operatorname{tr} k(x)/k(s) - \deg\operatorname{tr} k(x')/k(s')&\\ + \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) - \operatorname{codim}(\overline{\{s'\}} \cap T', \overline{\{s'\}}).&\end{aligned}$$

Since g is faithfully flat, it follows from (EGA IV 6.1.4) that one has

$$\begin{aligned}\operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) &= \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap g^{-1}(T), g^{-1}(\overline{\{s\}}))\\ &\leq \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap T', g^{-1}(\overline{\{s\}}));\end{aligned} \tag{$*$}$$

since S' is catenary, one has, by (EGA 0_{IV} 14.3.2 b)):

$$\begin{aligned}\operatorname{codim}(\overline{\{s'\}} \cap T', g^{-1}(\overline{\{s\}})) &= \operatorname{codim}(\overline{\{s'\}} \cap T', \overline{\{s'\}}) + \operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}}))\\ &= \operatorname{codim}(\overline{\{s'\}} \cap T', g^{-1}(\overline{\{s\}}) \cap T') + \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap T', g^{-1}(\overline{\{s\}})).\end{aligned}$$

From this relation and $(*)$, one deduces **117**

$$\delta_T(x) - \delta_{T'}(x') \leq \deg\operatorname{tr} k(x)/k(s) - \deg\operatorname{tr} k(x')/k(s') + \operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})).$$

Let us calculate $\operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})) = \dim \mathcal{O}_{S'_s,s'}$ (where S'_s is the fiber of S' at s). Let Z be the closed image of x in X_s and let $Z' \subset X'_s$ be the scheme defined by the cartesian square

$$\begin{tikzcd} Z' \arrow[r] \arrow[d] & Z \arrow[d] \\ S'_s \arrow[r] & \operatorname{Spec} k(s) \end{tikzcd} .$$

The morphism $Z \to \operatorname{Spec} k(s)$ is flat, locally of finite type, and $\dim Z = \deg\operatorname{tr} k(x)/k(s)$. It then follows from (EGA IV 6.1.2) that

$$\dim(\mathcal{O}_{Z',x'}) = \dim(\mathcal{O}_{S'_s,s'}) + \deg\operatorname{tr} k(x)/k(s) - \deg\operatorname{tr} k(x')/k(s');$$

taking into account the fact that $Z'_{s'} \simeq Z \otimes_{k(s)} k(s')$, one then obtains:

$$\delta_T(x) - \delta_{T'}(x') \leq \dim(\mathcal{O}_{Z',x'}).$$

But $\operatorname{Spec}(\mathcal{O}_{Z',x'})$ identifies with the fiber at x of the morphism

$$(\operatorname{Spec}(\mathcal{O}_{X',x'}))_s \to (\operatorname{Spec}(\mathcal{O}_{X,x}))_s,$$

and hence also with the fiber at x of $h_{x'}$, which proves the theorem.

XIV.2.3

The proofs of the theorems in no. XIV.4 are based on duality theory; they use the following lemmas. Let m be an integer that is a power of a prime number ℓ; if X is a scheme, all the **118** sheaves considered on X are sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules; there is then a notion of a dualizing complex on X (SGA 5 I 1.7). Suppose that such a complex K exists on X; then, for every geometric point $\bar{x}$ over a point x of X, one deduces from K (cf. SGA 5 I 4.5) a dualizing complex $K_{\bar{x}}$ on $\operatorname{Spec} k(\bar{x})$, so that $K_{\bar{x}} \simeq \mathbf{Z}/m\mathbf{Z}[n]$ (the brackets denoting the translation functor) for a certain integer n depending only on x. We set

$$\delta_K(x) = n.$$

If K is normalized at the point x (SGA 5 I 4.5), then $n = 0$.

Lemma XIV.2.3.1. *Let X be a locally noetherian scheme endowed with a dualizing complex K. If x and x' are two points of X, such that x is a specialization of x' and* $\operatorname{codim}(\overline{\{x\}}, \overline{\{x'\}}) = 1$, *then one has*

$$\delta^K(x) = \delta^K(x') - 2.$$

One may first reduce to the case where X is a strictly local scheme. Indeed, let $\bar{X}$ be the strict localization of X at a geometric point $\bar{x}$ over x, let $i : \bar{X} \to X$ be the canonical morphism, and let $\bar{x}'$ be a geometric point of $\bar{X}$ over x'. Then i^*K is a dualizing complex on $\bar{X}$, and one has (SGA 5 I 4.5)

$$(i^*K)_{\bar{x}} \simeq K_{\bar{x}} \text{ and } (i^*K)_{\bar{x}'} \simeq K_{\bar{x}'},$$

which completes the reduction to the strictly local case. **119**

If $j : \overline{\{x'\}} \to X$ denotes the immersion of the reduced closed subscheme of X whose underlying space is $\overline{\{x'\}}$, then $\mathrm{R}^! j(K)$ is a dualizing complex on $\overline{\{x'\}}$, and one sees at once, using (SGA 5 I 4.5), that it suffices to prove the lemma for $\overline{\{x'\}}$. One is thus reduced to the case where X is an integral strictly local scheme of dimension 1.

Let X' then be the normalization of X and let $f : X' \to X$ be the canonical morphism; f is integral, surjective, and radicial, and it follows that f^*K is a dualizing complex on X' and that it suffices to prove the lemma for X' and for the points over x and x'. One is thus reduced to the case where X is a local, integral, regular scheme of dimension 1, but it is known (cf. SGA 5 I 4.6.2 and 5.1) that then $\mu_m[2]$ and $\mathbf{Z}/m\mathbf{Z}$ are dualizing complexes, normalized at the points x and x', respectively; the lemma follows at once.

Lemma XIV.2.3.2. *Let S be a noetherian local scheme and $f : X \to S$ a morphism of finite type. If K is a dualizing complex on S, normalized at the closed point t of S, and if* $\mathrm{R}^! f(K) = K'$ *is a dualizing complex on X (cf.* SGA *5 I 3.4.3), one has, for every point x of X:*

$$\delta^{K'}(x) = 2\delta(x).$$

Indeed, let $s = f(x)$ and let x' be a closed point of the fiber X_s; then $\delta^{K'}(x') = \delta^K(s)$ and, by XIV.2.3.1,

$$\delta^K(s) = 2 \operatorname{codim}(\overline{\{t\}}, \overline{\{s\}}) = 2 \dim(\overline{\{s\}}).$$

Since x' may be chosen to be a specialization of x, one has, by XIV.2.3.1, **120**

$$\delta^{K'}(x) = \delta^{K'}(x') + 2 \operatorname{codim}(\overline{\{x\}}, \overline{\{x'\}}) = \delta^{K'}(x') + 2 \operatorname{deg\,tr} k(x)/k(s);$$

the lemma follows at once.

The following lemma will be used only for the converse of the Lefschetz theorem, in no. XIV.4:

Lemma XIV.2.3.3. *Let there be a cartesian square*

$$\begin{tikzcd} X' \arrow[r, "h"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S \end{tikzcd} ,$$

where S is an excellent strictly local scheme of characteristic zero, S' is the completion of S, and f is a morphism of finite type. Let ℓ be a prime number, $x \in X$, let Z be the schematic closure of X'_x in X', and let $i : X'_x \to Z$, $j : Z \to X$ be the canonical morphisms. Then, if $k : X' \to R$ is a closed immersion of X' into an excellent regular scheme R of characteristic zero, the complex

$$K' = i^*(R^!(k.j)(\mathbf{Z}/\ell\mathbf{Z}))$$

is a constant dualizing complex on X'_x (that is, it has only one nonzero cohomology sheaf, isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$).

In view of (SGA 5 I 3.4.3), the only thing to prove is that K' is constant. But, since Z is excellent, the set of points of Z whose local rings are regular is an open subset U (EGA IV **121**
7.8.3 (iv)), and U plainly contains X'_x, which is regular. Let

$$u : U \to R$$

then be the canonical immersion of U into R; it follows from the purity theorem (SGA 4 XIX 3.2 and 3.4) and the isomorphism

$$(\mu_l)_S \simeq (\mathbf{Z}/\ell\mathbf{Z})_S$$

(S strictly local) that one has

$$\mathrm{R}^!u(\mathbf{Z}/\ell\mathbf{Z}) \simeq \mathbf{Z}/\ell\mathbf{Z}[2c],$$

where c is a locally constant function on U, necessarily constant in a neighborhood of X'_x, because the fibers of g are geometrically integral by (EGA IV 18.9.1), and hence X'_x is integral. The lemma follows at once.

XIV.3 Converse to the affine Lefschetz theorem

The present section will be used in no. XIV.4 to prove a converse to the "Lefschetz theorem"; a reader interested only in the direct part of that theorem may therefore omit the present section.

XIV.3.1

Recall the statement of the affine Lefschetz theorem[(*)] (SGA 4 XIX 6.1 bis):

Let S be a strictly local excellent scheme of characteristic zero, $f : X \to S$ an affine morphism of finite type, and F a torsion sheaf on X. Then, setting **122**

[(*)] *Editor's note.* Gabber proved the following generalization. Let Y be a strictly local scheme of arithmetic type over a regular Noetherian scheme S of dimension ≤ 1. Let $f : X \to Y$ be an affine morphism of finite type, $\Lambda = \mathbf{Z}/n\mathbf{Z}$ with n invertible on X, and F a Λ-sheaf. Then $H^q(X, F) = 0$ if $q > \delta(F)$. The following local Lefschetz theorem follows. Let $\mathcal{O}$ be strictly local of arithmetic type over S. For every non-zero-divisor $f \in \mathcal{O}$ and every Λ-sheaf F on $\mathrm{Spec}(\mathcal{O}[f^{-1}])$, one has $H^q(\mathrm{Spec}(\mathcal{O}[f^{-1}]), F) = 0$ for $q > \dim(\mathcal{O})$. Cf. (Fujiwara K., "A Proof of the Absolute Purity Conjecture (after Gabber)", in *Algebraic geometry 2000, Azumino (Hotaka)*, Adv. Stud. in Pure Math., vol. 36, 2002, p. 153-183, §5), and especially Illusie's article (Illusie L., "Perversité et variation", *Manuscripta Math.* **112** (2003), p. 271-295). This result is one of the crucial points used by Gabber in his proof of Grothendieck's purity theorem (cf. note ((*)), page 146).

$$\delta(F) = \sup\{\delta(x) | x \in X \text{ and } F_{\bar{x}} \neq 0\},$$

one has

$$H^q(X, F) = 0 \text{ for } q > \delta(F).$$

Before stating the converse, let us prove a few lemmas.

Lemma XIV.3.2. *Let K be a field, ℓ a prime number distinct from the characteristic of K, and F a nonzero constructible sheaf of ℓ-torsion on K. Suppose that the ℓ-cohomological dimension of K (*SGA 4 X 1*) is equal to n (this holds, for example, if K is the field of fractions of an integral strictly local excellent ring of characteristic zero and dimension n (*SGA 4 XIX 6.3*), or if K is an extension of finite type and transcendence degree n over a separably closed field (*SGA 4 X 2.1*)). Then one can find a finite separable extension L of K such that:*

$$H^n(L, F_{|L}) \neq 0.$$

One can find a finite extension K' of K such that the restrictions of F and μ_ℓ to $\operatorname{Spec} K'$ are constant sheaves. Then $\mathrm{cd}_\ell(K') = \mathrm{cd}_\ell(K) = n$ (SGA 4 X 2.1), and it follows from ([2] II §3 Prop. 4 (iii)) that one can find a finite extension L of K' such that **123**

$$H^n(L, \mu_\ell) \neq 0, \quad \text{i.e. } H^n(L, \mathbf{Z}/\ell\mathbf{Z}) \neq 0.$$

Now the functor $H^n(L, \cdot)$ is right exact on the category of sheaves of ℓ-torsion, since $\mathrm{cd}_\ell(L) = n$; as F has a quotient isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$, one also has $H^n(L, F_{|L}) \neq 0$.

Corollary XIV.3.3. *Let k be a field, K an extension of finite type and transcendence degree n over k, and F a nonzero constructible sheaf of ℓ-torsion on K, with ℓ prime to the characteristic of k. Then one can find a finite separable extension L of K such that, if $u : \operatorname{Spec} L \to \operatorname{Spec} k$ denotes the canonical morphism, one has*

$$R^n u_*(F_{|\operatorname{Spec} L}) \neq 0.$$

When the field k is separably closed, the corollary is a special case of XIV.3.2. In the general case, one can find a finite separable extension k_1 of k such that the irreducible components of $K \otimes_k k_1$ are geometrically irreducible (EGA IV 4.5.11); let K_1 be one of them. If k' is a separable closure of k_1, then $K' = K_1 \otimes_{k_1} k'$ is a field, and by (EGA IV 4.2) one has

$$\deg\operatorname{tr} K'/k' = \deg\operatorname{tr} K/k = n.$$

It then follows from XIV.3.2 that one can find a finite separable extension L' of K' such that $H^n(L', F_{|L'}) \neq 0$. But $k' = \varinjlim_i k_i$, where k_i runs through the finite extensions of k_1 contained in k', and consequently $K' = \varinjlim_i (k_i \otimes_{k_1} K_1)$. It follows that one can find an index i and a **124**
finite separable extension L of $k_i \otimes_{k_1} K_1 = K_i$ such that $L' \simeq L \otimes_{K_i} K'$. The extension L of K has the required property; indeed, it follows from the commutative diagram

$$\begin{tikzcd}
 & \operatorname{Spec} L \arrow[dl, "v"'] \arrow[dr, "u"] & \\
\operatorname{Spec} k_i \arrow[rr, "w"] & & \operatorname{Spec} k\,,
\end{tikzcd}$$

with w finite and hence $R^q w_* = 0$ if $q > 0$, that

$$R^n u_*(F_{|\operatorname{Spec} L}) \simeq w_*(R^n v_*(F_{|\operatorname{Spec} L})).$$

Now $R^n v_*(F_{|\operatorname{Spec} L}) \neq 0$, since $H^n(L', F_{|L'}) \neq 0$; therefore one also has $R^n u_*(F_{|\operatorname{Spec} L}) \neq 0$.

Recall the following familiar lemma (cf. EGA 0_{III} 10.3.1.2 and EGA IV 18.2.3):

Lemma XIV.3.4. *Let X be a scheme, x a point of X, and K a finite separable extension of $k(x)$. Then there exist a scheme X_1 étale over X and affine, and a point $x_1 \in X_1$ above x, such that $k(x_1)$ is $k(x)$-isomorphic to K.*

In no. XIV.4 we shall use the following technical form of the converse to XIV.3.1.

Proposition XIV.3.5. *Let there be a Cartesian square* **125**

$$\begin{tikzcd} X' \arrow[r, "h"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd}$$

*where the schemes S and S' are strictly local excellent schemes of characteristic zero, the morphism f is locally of finite type, g is regular (*EGA IV 6.8.1*) and surjective, and the closed fiber of g is reduced to the closed point of S'. Given an S-scheme X_1 (respectively an S-morphism f_1, etc.), we denote by X_1' (respectively f_1', etc.) the scheme $X_1 \times_S S'$ (respectively the morphism $(f_1)_{(S')}$, etc.). Let F be a sheaf of $\mathbf{Z}/m\mathbf{Z}$-modules on X' (m a power of a prime number ℓ), constructible and satisfying the following conditions:*

(i) For every point $x \in X$, one can find a finite separable extension K of $k(x)$ such that the restriction of F to the fiber $(X')_{(\operatorname{Spec} K)}$ comes by inverse image from a constructible sheaf on $\operatorname{Spec} K$.

(ii) For every morphism $f_1 : X_1 \to S$, with X_1 affine and étale over X, for every point $s \in S$, and for every integer $q > 0$, one can find a finite separable extension K of $k(s)$ such that the restriction of $R^q f'_{1*}(F|X_1')$ to the fiber $S'_{(\operatorname{Spec} K)}$ comes by inverse image from a constructible sheaf on $\operatorname{Spec} K$.

Let n be an integer, and suppose that for every affine scheme X_1 étale over X one has **126**

$$H^i(X_1', F) = 0 \text{ for } i > n.$$

Then, if $\bar{x}'$ is a geometric point above the point $x' \in X'$ such that $F_{\bar{x}'} \neq 0$, one has

$$\delta(x') \leq n.$$

Let Z' be the set of points x' of X' such that $F_{\bar{x}'} = 0$. Then, if $Z = h(Z')$, one has, by (i), $Z' = h^{-1}(Z)$; let $x' \in X'$, $x = h(x')$, $s' = f'(x')$, $s = f(x)$. It follows from XIV.2.1.1 and the fact that the function δ decreases under specialization that it suffices to prove the inequality $\delta(x') \leq n$ when x is a maximal point of Z and x' is such that

$$r = \operatorname{deg\,tr} k(x)/k(s) = \operatorname{deg\,tr} k(x')/k(s') \text{ and } d = \dim \overline{\{s\}} = \dim \overline{\{s'\}}$$

Let x' be such a point; it suffices to show that one can find an affine scheme X_1 étale over X such that

$$H^{d+r}(X_1', F) \neq 0.$$

The set Z' is constructible (SGA 4 IX 2.4), and hence so is Z (EGA IV 1.9.12); one may then suppose, after restricting X to a neighborhood of x, that Z is an irreducible closed subset with generic point x. Let $T = f(Z)$; T is a constructible set contained in $\overline{\{s\}}$; one can therefore find an affine open subset U of S such that $s \in U$ and $T \cap U = T_U$ is an irreducible closed subset of U with generic point s.

Let V then be an affine scheme étale over X, whose image in X contains x and whose image in S is contained in U; let Z_V be the inverse image of Z in V and let $u : Z_V \to T_U$ be **127** the canonical morphism. Let W be an affine scheme étale over U; denote by T_W the inverse image of T_U in W and let $X_1 = W \times_U V$. Since F is zero outside Z', one has the spectral sequence

$$E_2^{pq} = H^p((T_W)', R^q u'_*(F_{|(Z_V)'})) \Longrightarrow H^*(X'_1, F).$$

We shall show that V and W may be chosen so that

a) $E_2^{pq} = 0$ for $p > d$ and for $q > r$.

b) $E_2^{dr} \neq 0$.

The relation $H^{d+r}(X'_1, F) \neq 0$ will then follow from the spectral sequence.

1°) Set $G_q = R^q u'_*(F_{|(Z_V)'})$; then one has:

$$(G_q)_{\bar{s}'} = H^q((Z_V)'_{\bar{s}'}, F_{|(Z_V)'_{\bar{s}'}}),$$

because s' is a maximal point of $(T_U)'$. Since the fiber $(Z_V)'_{\bar{s}'}$ is an affine scheme of finite type of dimension r over a separably closed field, it follows from XIV.3.1 that

$$(G_q)_{\bar{s}'} = 0 \text{ for } q > r.$$

For $q > r$, let Y'_q be the set of points of $(T_U)'$ where the geometric fiber of G_q is $\neq 0$, **128**
and let $Y_q = g(Y'_q)$; then $Y'_q = g^{-1}(Y_q)$ by (ii), so Y_q is a constructible subset of T_U (SGA 4 XIX 5.1 and EGA 1.9.12) that does not contain s; after restricting U to an open neighborhood of s, one may suppose that $G_q = 0$ for $q > r$, and hence that $E_2^{pq} = 0$ for $q > r$.

Moreover, since $(T_W)'$ is an affine scheme of finite type over $g^{-1}(\overline{\{s\}})$, one has, for every q (cf. XIV.3.1):

$$H^p((T_W)', G_q) = 0 \text{ for } p > \dim g^{-1}(\overline{\{s\}}) = d,$$

whence condition a).

2°) Let us show that V may be chosen so that $(G_r)_{\bar{s}'} \neq 0$. By (i), there exists a constructible sheaf I, defined over a finite separable extension K of $k(x)$, whose inverse image on $(X')_{(\operatorname{Spec} K)}$ is isomorphic to $F_{|(X')_{(\operatorname{Spec} K)}}$. By XIV.3.3, one finds a finite separable extension L of K such that, if $v : \operatorname{Spec} L \to \operatorname{Spec} k(s)$ denotes the canonical morphism, then $R^r v_*(I) \neq 0$. Since the morphism $S'_s \to \operatorname{Spec} k(s)$ is regular, one has, by (SGA 4 XIX 4.2):

$$R^r v'_*(F_{|(\operatorname{Spec} L)'}) \simeq (R^r v_*(I))' \neq 0.$$

By Lemma XIV.3.4, one can find an affine scheme X_2 étale over X, and a point x_2 of X_2 over x, such that L is $k(x)$-isomorphic to $k(x_2)$, and one may suppose that X_2 lies over U. Since x is a maximal point of Z, one has **129**

$$\operatorname{Spec} L \simeq \varprojlim_V Z_V,$$

where V ranges over the affine open neighborhoods of x_2. By passage to the limit (SGA 4 VII 5.8), after restriction to the geometric fiber at $\bar{s}'$, one deduces:

$$(R^r v'_*(F_{|\operatorname{Spec} I})')_{\bar{s}'} = \varinjlim_V (R^r u'_*(F_{|(Z_V)'})_{\bar{s}'},$$

which indeed shows that one can find V such that $(G_r)_{\bar{s}'} \neq 0$.

3°) With the scheme V having been chosen in 2°), let us show that the scheme W may be chosen so that

$$E_2^{dr} = H^d((T_W)', G_r) \neq 0.$$

By (ii), there exists a constructible sheaf J, defined over a finite separable extension K of $k(s)$, whose inverse image on $(S')_{(\operatorname{Spec} K)}$ is isomorphic to $G_{r|(S')_{(\operatorname{Spec} K)}}$. By Lemma XIV.3.2, one can find a finite separable extension L of K such that $H^d(\operatorname{Spec} L, J) \neq 0$. Since the morphism $(S')_{(\operatorname{Spec} L)} \to \operatorname{Spec} L$ is acyclic (SGA 4 XIX 4.1 and XV 1.10 and 1.16), one has

$$H^d((\operatorname{Spec} L)', G_{r|(S')_{(\operatorname{Spec} L)'}}) = H^d(\operatorname{Spec} L, J) \neq 0.$$

By XIV.3.4, one can find an affine scheme U_1 étale over U, and a point s_1 over s, such that $k(s_1)$ is $k(s)$-isomorphic to L. Since s is a maximal point of T_U, one has

$$\operatorname{Spec} L \simeq \varprojlim_W T_W,$$

where W ranges over the affine open neighborhoods of s_1. One deduces that $(\operatorname{Spec} L)' \simeq \varprojlim_W (T_W)'$, and by passage to the limit (SGA 4 VII 5.8): **130**

$$H^d((\operatorname{Spec} L)', G_{r|(\operatorname{Spec} L)'}) \simeq \varinjlim_W H^d((T_W)', G_{r|(T_W)'});$$

consequently, one can find W such that

$$H^d((T_W)', G_{r|(T_W)'}) \neq 0,$$

which completes the proof of the theorem.

Corollary XIV.3.6. *The hypotheses concerning S, S', f, f', m are those of XIV.3.5. Let F now denote a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X', bounded below and with constructible cohomology, and whose cohomology sheaves satisfy conditions* (i) *and* (ii) *of XIV.3.5. Let n be an integer, and suppose that, for every affine scheme X_1 étale over X, one has*

$$H^i(X_1', F) = 0 \textit{ for } i > n.$$

Then, if $\bar{x}'$ is a geometric point lying over a point x' of X' such that, for an integer j, $(\underline{H}^j(F))_{\bar{x}'} \neq 0$, one has

$$\delta(x') \leq n - j.$$

Let T' be the set of points of X' where the conclusion of XIV.3.7 fails, and suppose $T' \neq \emptyset$; let $T = f(T')$, let x be a maximal point of T, and let x' be a point of X' lying over x. Let j be the largest integer such that $(\underline{H}^j(F))_{\bar{x}'} \neq 0$; thus $r = \delta(x) > n - j$. Let Z_q' be the set **131**
of points where the geometric fiber of $\underline{H}^q(F)$ is $= 0$, and let $Z_q = h(Z_q')$; as in the proof of XIV.3.5, one sees that Z_q is constructible. Clearly $Z_q' = \emptyset$ for $q > n$ and for q sufficiently small. The other values of q fall into three subsets. Let

$$Q_1 = \{q \mid x \in Z_q \text{ and a generization of } x, \text{ distinct from } x, \notin Z_q\}.$$

One has $j \in Q_1$, and one can find an affine open neighborhood U_1 of x such that, for every $q \in Q_1$, $U_1 \cap Z_q$ is an irreducible closed subset with generic point x. If $q \in Q_1$, one has

$$\delta(\underline{H}^q(F)_{|U_1'}) = \delta(x) \quad \text{(for the definition of } \delta(\underline{H}^q(F)) \text{, cf. XIV.3.1).} \tag{$*$}$$

Let

$$Q_2 = \{q \mid \text{no generization of } x \text{ belongs to } Z_q\}.$$

Then, if $j < q \leq n$, one has $q \in Q_2$, and one can find an affine open neighborhood U_2 of x such that, for every $q \in Q_2$, $Z_q \cap U_2 = \emptyset$; thus one has

$$\underline{H}^q(F)_{|U_2'} = 0 \text{ for } q \in Q_2. \tag{$**$}$$

Finally, let

$$Q_3 = \{q \mid Z_q \text{ contains strict generizations of } x\}.$$

Then one can find an affine open neighborhood of x, U_3, such that, for every $q \in Q_3$, all the maximal points of $Z_q \cap U_3$ are generizations of x. If $q \in Q_3$, one has

$$\delta(\underline{H}^q(F)_{|U_3'}) \leq n - q. \tag{$***$}$$

For every affine scheme X_1 étale over $U_1 \cap U_2 \cap U_3$, consider the hypercohomology spectral sequence **132**

$$E_2^{pq} = H^p(X_1', \underline{H}^q(F)) \Longrightarrow H^*(X_1', F).$$

One has $E_2^{pq} = 0$ for $q \in Q_2$ by $(**)$. One has $E_2^{pq} = 0$ for $p + q \geq r + j$ except possibly for $p = r, q = j$. Indeed, this is clear if $q \in Q_2$; if $q \in Q_1$, then $p > r$ except when $p = r, q = j$, and this follows from XIV.3.1 in view of $(*)$; finally, if $q \in Q_3$, since $r > n - j$, one has $p > n - q$, and the assertion follows from XIV.3.1 in view of $(***)$. Since $H^{r+j}(X_1', F) = 0$, it follows from the spectral sequence that

$$H^r(X_1', \underline{H}^j(F)) = 0;$$

but, by XIV.3.5, this implies $\delta(x) < r$, which is absurd.

Corollary XIV.3.7. *Let S be an excellent strictly local scheme of characteristic zero, $f : X \to S$ a morphism locally of finite type, m a power of a prime number, F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below, with constructible cohomology, and n an integer. Then the following conditions are equivalent:*

(i) For every affine scheme X_1 étale over X, one has

$$H^i(X_1, F) = 0 \textit{ for } i > n.$$

(ii) For every geometric point $\bar{x}$ lying over the point x of X, and for every integer j such that $(\underline{H}^j(F))_{\bar{x}} \neq 0$, one has **133**

$$\delta(x) \leq n - j.$$

(i) $\Longrightarrow$ (ii) is the special case of XIV.3.6 obtained by taking $S = S'$.

(ii) $\Longrightarrow$ (i) follows immediately from XIV.3.1, using the hypercohomology spectral sequence

$$H^p(X_1, \underline{H}^q(F)) \Longrightarrow H^*(X_1, F).$$

XIV.4 Main theorem and variants

XIV.4.0

Let $g \colon X \to S$ be a separated morphism of finite type, let T be a closed subset of S, let $Z = g^{-1}(T)$, and let F be a complex of abelian sheaves on X, bounded below. We call the group

$$\mathrm{H}^i_{Z!}(X/S, F) = \mathrm{H}^i_T(S, \mathrm{R}_! g(F))$$

the *i-th cohomology group of F, with proper support and support in Z*, where $\mathrm{R}_! g$ denotes the "direct image with proper support"(SGA 4 XVII). In the particular case in which g is proper, one simply has

$$\mathrm{H}^i_{Z!}(X/S, F) = \mathrm{H}^i_Z(X, F).$$

Proposition XIV.4.1. *Let $f\colon U \to S$ be a morphism of finite type and let F be a complex of abelian sheaves on U, bounded below. Suppose that a factorization of f is given:* **134**

$$\begin{tikzcd} U \arrow[rr, "i"] \arrow[dr, "f"'] & & X \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

where i is an open immersion and g is a separated morphism of finite type, and let G be a complex of abelian sheaves on X, bounded below, that extends F. Let Y be a closed subscheme of X whose underlying space is $X - U$, so that one has a commutative diagram:

$$\begin{tikzcd} Y \arrow[rr, "j"] \arrow[dr, "h"'] & & X \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

Finally, let n be an integer and let T be a closed subset of S. Then the following conditions are equivalent:

(i) One has $\operatorname{prof}_T(\mathrm{R}_! f(F)) \geq n$.

(ii) The canonical morphism

$$\underline{H}^i_T(\mathrm{R}_! g(G)) \to \underline{H}^i_T(\mathrm{R}_! h(j^* G))$$

is bijective for $i < n-1$ and injective for $i = n-1$.

(iii) For every scheme S' ètale over S, if X' (resp. f', resp. etc.) denotes the scheme $X \times_S S'$ (resp. the morphism $f_{(S')}$, resp. etc.), the canonical morphism **135**

$$\mathrm{H}^i_{g'^{-1}(T')!}(X'/S', G') \to \mathrm{H}^i_{h'^{-1}(T')!}(Y'/S', j'^* G')$$

is bijective for $i < n-1$ and injective for $i = n-1$.

In the derived category $\mathrm{D}^+(X)$ (cf. [3]), consider the distinguished triangle

$$\begin{tikzcd} & j_* j^* G \arrow[dl] & \\ i_! F \arrow[rr] & & G. \arrow[ul] \end{tikzcd}$$

Applying the functor $\mathrm{R}_! g$ to this triangle gives the triangle

$$\begin{tikzcd} & \mathrm{R}_! h(j^* G) \arrow[dl] & \\ \mathrm{R}_! f(F) \arrow[rr] & & \mathrm{R}_! g(G). \arrow[ul] \end{tikzcd} \tag{$*$}$$

Let us show that (i) $\Longleftrightarrow$ (ii). Indeed, by Definition XIV.1.2, (i) is equivalent to the relation

$$\underline{H}^i_T(\mathrm{R}_! f(F)) = 0 \quad \text{for } i < n;$$

while $(*)$ gives the exact sequence of sheaves

$$\to \underline{H}^i_T(\mathrm{R}_! f(F)) \to \underline{H}^i_T(\mathrm{R}_! g(G)) \to \underline{H}^i_T(\mathrm{R}_! h(j^* G)) \to,$$

which proves the equivalence of (i) and (ii). **136**

(i) $\Longleftrightarrow$ (iii). Indeed, (i) is equivalent to saying that, for every scheme S' ètale over S, one has

$$\mathrm{H}^i_{T'}(S', \mathrm{R}_! f'(F')) = 0 \quad \text{for } i < n. \tag{$**$}$$

But $(*)$ gives the exact sequence of abelian groups

$$\to \mathrm{H}^i_{T'}(S', \mathrm{R}_! f'(F')) \to \mathrm{H}^i_{T'}(S', \mathrm{R}_! g'(G')) \to \mathrm{H}^i_{T'}(S', \mathrm{R}_! h'(j'^* G')) \to ;$$

in view of XIV.4.0, this exact sequence takes the form

$$\to \mathrm{H}^i_{T'}(S', \mathrm{R}_! f'(F')) \to \mathrm{H}^i_{g'^{-1}(T')!}(X'/S', G') \to \mathrm{H}^i_{h'^{-1}(T')!}(Y'/S', j'^* G') \to .$$

The equivalence of (i) and (iii) follows, taking into account the form $(**)$ of (i).

When $f\colon U \to S$ is affine, we shall give *local* conditions on F under which conditions (i) through (iii) of XIV.4.1 hold. In what follows, all schemes considered are excellent schemes of characteristic zero, and all sheaves are sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules, where m is a power of a prime number. If resolution of singularities in the sense of (SGA 4 XIX) were available, the stated results and their proofs would remain valid for excellent schemes of equal characteristic, with m prime to the characteristic.

Theorem XIV.4.2. *Let S be an excellent scheme of characteristic zero and let $f\colon U \to S$ be a separated morphism of finite type. Let F be a bounded-below complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on U with constructible cohomology, let n be an integer, and let T be a closed subset of S. Then the following conditions are equivalent:* **137**

(i) For every scheme U_1 ètale over U and affine over S, denoting by f_1 the structural morphism of U_1 and by F_1 the restriction of F to U_1, one has

$$\mathrm{prof}_T(\mathrm{R}_! f_1(F_1)) \geq n$$

(cf. Proposition. XIV.4.1 for the meaning of this relation).

(ii) For every point u of U, one has

$$\mathrm{prof}_u(F) \geq n - \delta_T(u),$$

where one sets (cf. XIV.2.2): $\delta_T(u) = \deg\mathrm{tr}(k(x)/k(s)) + \mathrm{codim}\left(\overline{\{s\}} \cap T, \overline{\{s\}}\right)$.

Proof. 1°) Let t be a point of T, let $\overline{S}$ be the strict localization of S at a geometric point above t, and let S' be the completion of $\overline{S}$, with closed point t'; then $\overline{S}$ is excellent by (EGA IV 7.9.5), so S' is an excellent complete strictly local scheme. Given a scheme U over X (respectively an S-morphism f, respectively etc.), we shall denote by U' (respectively f', respectively etc.) the scheme $U \times_S S'$ (respectively the morphism $f_{(S')}$, respectively etc.). We have the Cartesian square

$$\begin{tikzcd} U' \arrow[r, "h"] \arrow[d, "f'"'] & U \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd}$$

in which the morphism g is regular (EGA IV 7.8.2). Let us show that it is enough to prove that (for every point $t \in T$) the following two properties are equivalent: **138**

$(\mathrm{i})_t$ For every scheme U_1 étale over U and affine over S, setting $f_1\colon U_1 \to S$, one has

$$\operatorname{prof}_{t'}(\mathrm{R}_! f_1'(F_1')) \geq n.$$

$(\mathrm{ii})_t$ For every point u' of U', one has

$$\operatorname{prof}_{u'}(F') \geq n - \delta_{t'}(u').$$

It is enough to prove the following lemma:

Lemma XIV.4.2.1. *One has* (i) $\Longleftrightarrow$ $(\mathrm{i})_t$ *for every* $t \in T$ *and* (ii) $\Longleftrightarrow$ $(\mathrm{ii})_t$ *for every* $t \in T$.

(i) $\Longleftrightarrow$ $(\mathrm{i})_t$ for every $t \in T$. Indeed, (i) is equivalent to saying that, for every scheme U_1 étale over U and affine over S, one has

$$\operatorname{prof}_T(\mathrm{R}_! f_1(F_1)) \geq n\ ;$$

now, by XIV.1.8,

$$\operatorname{prof}_T(\mathrm{R}_! f_1(F_1)) = \inf_{t\in T} \operatorname{prof}_t(\mathrm{R}_! f_1(F_1)).$$

Since $g^*(\mathrm{R}_! f_1(F_1)) \simeq \mathrm{R}_! f_1'(F_1')$ (SGA 4 XVII), one has, by XIV.1.16,

$$\operatorname{prof}_t(\mathrm{R}_! f_1(F_1)) = \operatorname{prof}_{t'}(\mathrm{R}_! f_1'(F_1')),$$

and hence (i) is equivalent to saying that, for every $t \in T$, one has $\operatorname{prof}_{t'}(\mathrm{R}_! f_1'(F_1')) \geq n$, which **139**
is precisely $(\mathrm{i})_t$.

$(\mathrm{ii})_t$ for every $t \in T \Longrightarrow$ (ii). Indeed, let $u \in U$; we must show the relation

$$\operatorname{prof}_u(F) \geq n - \delta_T(u),$$

where $\delta_T(u) = \inf_{t\in T\cap\overline{\{s\}}} \delta_t(u)$ (cf. XIV.2.2); we are therefore reduced to showing that, for every $t \in T \cap \overline{\{s\}}$, one has

$$\operatorname{prof}_u(F) \geq n - \delta_t(u).$$

Let u' be a point of U' such that $h(u') = u$ and $\delta_{t'}(u') = \delta_t(u)$ (cf. XIV.2.1.1). Since h is locally acyclic (SGA 4 XIX 4.1), it follows from XIV.1.16 and the fact that u' is a generic point of U'_u that

$$\operatorname{prof}_{u'}(F') = \operatorname{prof}_u(F).$$

But by $(\mathrm{ii})_t$ one has $\operatorname{prof}_u(F) = \operatorname{prof}_{u'}(F') \geq n - \delta_t(u)$, which proves (ii).

(ii) $\Longrightarrow$ $(\mathrm{ii})_t$ for every t. With the notation of XIV.2.2.1, for every point u' of U' one has, by XIV.1.16,

$$\operatorname{prof}_{u'}(F') \geq \operatorname{prof}_u(F) + 2\dim h_{u'}^{-1}(u) \geq \operatorname{prof}_u(F) + \dim h_{u'}^{-1}(u).$$

Taking into account XIV.2.2.1 and (ii), we obtain

$$\operatorname{prof}_{u'}(F') \geq n - \delta_T(u) + \dim h_{u'}^{-1}(u) \geq n - \delta_{t'}(u'),$$

which is precisely $(\mathrm{ii})_t$. **140**

2°) $(\mathrm{ii})_t \Longleftrightarrow (\mathrm{i})_t$. We reduce immediately to the case where F is bounded, by truncating F at a sufficiently high degree. We can realize S' as a closed subscheme of a complete regular local scheme, hence an excellent one; it then follows from (SGA 5 I 3.4.3) that there exists a dualizing complex K on S' and that $\mathrm{R}^! f'(K) = K'$ is a dualizing complex on U'. We choose K so that $\delta^K(t') = 0$ (for the definition of $\delta^K(t')$, cf. XIV.2.3), and denote by $\mathrm{D}\,F'$ the dual of F' with respect to K'. We can reformulate the hypothesis $(\mathrm{ii})_t$ as follows:

Lemma XIV.4.2.2. *Let u' be a point of U'; then the following conditions are equivalent:*

(i) One has $\operatorname{prof}_{u'}(F') \geq n - \delta_{t'}(u')$.

(ii) One has $\big(\underline{H}^q(\mathrm{D}\,F')\big)_{\overline{u}'} = 0$ *for* $q > -n - \delta_{t'}(u')$ *($\overline{u}'$ a geometric point above u').*

Let $\overline{U}'$ be the strict localization of U' at $\overline{u}'$, and let $\overline{F}'$ be the inverse image of F under the morphism $\overline{U}' \to U'$. The relation $\operatorname{prof}_{u'}(F') \geq n - \delta_{t'}(u')$ is, by definition, equivalent to the following:

$$\mathrm{H}^i_{\overline{u}'}(\overline{F}') = 0 \quad \text{for } i > n - \delta_{t'}(u'). \tag{$*$}$$

Let $\mathrm{D}\left(\mathrm{H}^i_{\overline{u}'}(\overline{F}')\right)$ be the dual of the abelian group $\mathrm{H}^i_{\overline{u}'}(\overline{F}')$ with respect to $\mathbf{Z}/m\mathbf{Z}$. By XIV.2.3.2, $K'[-2\delta_{t'}(u')] = K''$ satisfies $\delta^{K''}(u') = 0$; since F' has constructible cohomology, one has $\overline{\mathrm{D}\,F'} = \mathrm{D}(\overline{F}')$, and the local duality theorem (SGA 5 I 4.5.3) then shows that 141

$$\mathrm{D}\left(\mathrm{H}^i_{\overline{u}'}(\overline{F}')\right) \simeq \left(\mathrm{H}^{-i-2\delta_{t'}(u')}(\mathrm{D}\,F')\right)_{\overline{u}'}.$$

Thus $(*)$ is equivalent to the relation

$$\big(\underline{H}^q(\mathrm{D}\,F')\big)_{\overline{u}'} = 0 \quad \text{for } q > -n - \delta_{t'}(u'). \tag{$**$}$$

We are now in a position to prove the theorem. The relation $(\mathrm{ii})_t$ is equivalent to relation $(**)$. Let $G^q = \underline{H}^q(\mathrm{D}\,F')$; the affine Lefschetz theorem (XIV.3.1) implies in particular that, for every scheme U_1 étale over U and affine over S, one has

$$\mathrm{H}^p(U_1', G^q) = 0 \quad \text{for } p > \delta(G^q),$$

where $\delta(G^q)$ is the upper bound of the $\delta_{t'}(u')$ for those u' such that $G^q_{\overline{u}'} \neq 0$; by $(**)$, one has $\delta(G^q) \leq -n - q$, so $(\mathrm{ii})_t$ implies the relation

$$\mathrm{H}^p(U_1', \underline{H}^q(\mathrm{D}\,F')) = 0 \quad \text{for } p > -q - n.$$

Taking into account the hypercohomology spectral sequence for the functor "sections on U_1'"with respect to the complex $\mathrm{D}\,F'$:

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(U_1', \underline{H}^q(\mathrm{D}\,F')) \Longrightarrow \mathrm{H}^*(U_1', \mathrm{D}\,F'),$$

we obtain the relation

$$\mathrm{H}^i(U_1', \mathrm{D}\,F') = 0 \quad \text{for } i > -n. \tag{$***$}$$

Conversely, suppose that the preceding relation holds for every U_1 étale over U and affine over S. Apply Proposition XIV.3.6, replacing S there by $\overline{S}'$; the hypotheses of XIV.3.6 concerning $\overline{S}$ are satisfied, since, for every affine scheme $\overline{U}_1$ étale over $\overline{U}$, one can find a scheme above $\overline{U}_1$ which comes by inverse image from a scheme étale over U and affine over S; as for the hypotheses concerning F, they are satisfied by XIV.2.3.3. Thus, for every point u' of U' such that $\big(\underline{H}^q(\mathrm{D}\,F')\big)_{\overline{u}'} \neq 0$, one has: 142

$$\delta_{t'}(u') \leq -n - q,$$

which is precisely relation $(**)$; we have therefore proved the equivalence

$$(\mathrm{ii})_t \Longleftrightarrow (***).$$

We shall transform relation $(***)$; first, one has

$$\mathrm{H}^i(U_1', \mathrm{D}\,F') = \big(\underline{H}^i(\mathrm{R}f'_{1*}(\mathrm{D}\,F_1'))\big)_{t'}\ ;$$

but by (SGA 5 I 1.12), there is a canonical isomorphism

$$\mathrm{R}f'_{1*}(\mathrm{D}\,F'_1) \simeq \mathrm{D}\left(\mathrm{R}_!f'_1(F'_1)\right),$$

where $\mathrm{D}\left(\mathrm{R}_!f'_1(F'_1)\right)$ denotes the dual of $\mathrm{R}_!f'_1(F'_1)$ with respect to K. We thus see that $(\mathrm{ii})_t$ is equivalent to

$$\left(\underline{H}^i(\mathrm{D}(\mathrm{R}_!f'_1(F'_1)))\right)_{t'} = 0 \quad \text{for } i > -n.$$

Applying the local duality theorem (SGA 5 I 4.5.3) once again, but this time at the point t', **143**
we find that

$$\left(\underline{H}^i(\mathrm{D}(\mathrm{R}_!f'_1(F'_1)))\right)_{t'} \simeq \mathrm{D}\left(\mathrm{H}^{-i}_{t'}(\mathrm{R}_!f'_1(F'_1))\right),$$

and finally $(\mathrm{ii})_t$ is equivalent to the relation

$$\mathrm{H}^i_{t'}(\mathrm{R}_!f'_1(F'_1)) = 0 \quad \text{for } i < n,$$

that is, $\mathrm{prof}_{t'}(\mathrm{R}_!f'_1(F'_1)) \geq n$, which completes the proof of the theorem.

Remark XIV.4.2.3. The argument simplifies quite considerably when one assumes that S admits (at least locally) a dualizing complex (for example, that it is locally immersible in a regular scheme). This avoids recourse to a completion (the passage to the case where S is strictly local being immediate), to XIV.2.3.3, and to the rather unappealing technical statement XIV.3.6, which may then be replaced by the more congenial reference XIV.3.7.

Corollary XIV.4.3. *Let S be an excellent scheme of characteristic zero and $f\colon U \to S$ a separated morphism of finite type such that U is the* union of $c+1$ open subsets affine over S. *Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules, bounded below and with constructible cohomology, let n be an integer, and let T be a closed subset of S. Suppose that, for every point $u \in U$, one has*

$$\mathrm{prof}_u(F) \geq n - \delta_T(u).$$

Then one has **144**

$$\mathrm{prof}_T(\mathrm{R}_!f(F)) \geq n - c.$$

Indeed, let U_j, $0 \leq j \leq c$, be a covering of U by open subsets U_j affine over S. Adopting again the notation of the proof of XIV.4.2, one has, for every j,

$$\mathrm{H}^i(U'_j, \underline{H}^q(\mathrm{D}\,F')) = 0 \quad \text{for } i > -n.$$

Using the spectral sequence that relates the cohomology of U to that of the covering formed by the U_j (SGA 4 V 2.4), the preceding relation shows that

$$\mathrm{H}^i(U', \underline{H}^q(\mathrm{D}\,F')) = 0 \quad \text{for } i > -n + c.$$

The corollary then follows from the end of the proof of XIV.4.2.

Corollary XIV.4.4. *Let S be an excellent scheme of characteristic zero, $g\colon X \to S$ a morphism, U an open subset of X that is the union of $c+1$ open subsets affine over S, Y a closed subscheme with underlying space $X - U$, and $j\colon Y \to X$ the natural morphism. Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below and with constructible cohomology, let T be a closed subset of S, and let n be an integer. Suppose that, for every point u of U, one has*

$$\mathrm{prof}_u(F) \geq n - \delta_T(u).$$

Then the canonical morphism **145**

$$\mathrm{H}^i_{g^{-1}(T)!}(X/S, F) \to \mathrm{H}^i_{(g^{-1}(T)\cap Y)!}(Y/S, j^*F)$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

This follows immediately from XIV.4.1 and XIV.4.3.

Corollary XIV.4.5 (Local Lefschetz theorem). *Let S be an excellent henselian local scheme of characteristic zero, t the closed point of S, X a scheme proper over $S' = S - \{t\}$, and U an open subset of X that is the union of $c+1$ affine open subsets. Let Y be a closed subscheme of X with underlying space $X - U$, let $j\colon Y \to X$ be the canonical morphism, let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below and with constructible cohomology, and let n be an integer. Suppose that, for every point u of U, one has*

$$\operatorname{prof}_u(F) \geq n - \delta'_t(u), \quad \textit{where } \delta'_t(u) = \delta_t(u) - 1.$$

Then the canonical morphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, j^*F)$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

Let $f\colon U \to S$ be the canonical morphism; it follows from XIV.4.2, applied with n replaced by $n+1$, that

$$\operatorname{prof}_t(\mathrm{R}_! f(F_{|U})) \geq n + 1 - c.$$

The preceding relation shows that the canonical morphism **146**

$$\mathrm{H}^i(S, \mathrm{R}_! f(F_{|U})) \to \mathrm{H}^i(S', \mathrm{R}_! f(F_{|U}))$$

is bijective for $i < n - c$, and injective for $i = n - c$. Since $\mathrm{R}_! f(F_{|U})$ is zero outside S', one has $\mathrm{H}^i(S, \mathrm{R}_! f(F_{|U})) \simeq \big(\underline{H}^i(\mathrm{R}_! f(F_{|U}))\big)_t = 0$, and consequently

$$\mathrm{H}^i(S', \mathrm{R}_! f(F_{|U})) = 0 \quad \text{for } i < n - c. \tag{$*$}$$

Let $g\colon X \to S'$, $h\colon Y \to S'$, $f'\colon U \to S'$ be the canonical morphisms. It follows from the distinguished triangle

$$\begin{tikzcd}
& \mathrm{R}h_*(j^*F) \arrow[dl] & \\
\mathrm{R}_! f'(F_{|U}) \arrow[rr] & & \mathrm{R}g_*(F) \arrow[ul]
\end{tikzcd}$$

that condition $(*)$ is equivalent to the assertion that the morphism

$$\mathrm{H}^i(S', \mathrm{R}g_*(F)) \to \mathrm{H}^i(S', \mathrm{R}h_*(j^*F))$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$. Since this morphism identifies canonically with the morphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, j^*F),$$

the conclusion follows at once.

Corollary XIV.4.6 (Global Lefschetz theorem). *Let S be the spectrum of a field, X a* **147** *scheme proper over S, and U an open subset of X that is the union of $c+1$ affine open subsets. Let Y be a closed subscheme of X with underlying space $X - U$, let $j\colon Y \to X$ be the canonical morphism, let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X, bounded below and with constructible cohomology, and let n be an integer. Suppose that, for every point u of U, one has*

$$\operatorname{prof}_u(F) \geq n - \dim\big(\overline{\{u\}}\big).$$

Then the canonical morphism

$$\mathrm{H}^i(X, F) \to \mathrm{H}^i(Y, j^*F)$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

More generally, if $g\colon X \to S$ is a separated morphism of finite type, with the hypotheses on S, U, Y, and F being the same as above, then the canonical morphism

$$\mathrm{H}^i_!(X/S, F) \to \mathrm{H}^i_!(Y/S, j^*F)$$

(where $\mathrm{H}^i_!$ denotes cohomology with proper support, that is, $\mathrm{H}^i_!(X/S, F) = \mathrm{H}^i(S, \mathrm{R}_!g(F))$) is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

The corollary is a special case of XIV.4.4, with $T = S$.

Here is a partial converse to XIV.4.3:

Proposition XIV.4.7. *Let S be a Noetherian scheme and let $f\colon U \to S$ be a morphism of finite type. Suppose that there exists a dualizing complex K on S and that $\mathrm{R}^!f(K)$ is a dualizing complex on U. Let T be a closed subset of S and let c be an integer. Then the following conditions are equivalent:* 148

(i) For every bounded-below complex F of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on U with constructible cohomology, and every integer n such that, for every point u of U,

$$\mathrm{prof}_u(F) \geq n - \delta_T(u),$$

one has

$$\mathrm{prof}_T(\mathrm{R}_!f(F)) \geq n - c.$$

(ii) For every constructible sheaf G of $\mathbf{Z}/m\mathbf{Z}$-modules on U, and every point $t \in T$, one has

$$(\mathrm{R}^p f_*(G))_{\overline{t}} = 0 \qquad \text{for } p > \delta(G, f, t) + c$$

*(recall from (*SGA 4 XIX 6.0*) that $\delta(G, f, t) = \sup\{\delta_t(u) | t \in \overline{\{u\}}$ and $G_{\overline{u}} \neq 0\}$).*

N.B. Condition (ii) is satisfied by XIV.3.1 if f is separated and, locally on S for the ètale topology, U is a union of $c+1$ open subsets affine over S; thus XIV.4.7 contains XIV.4.3(*).

We may plainly assume that S is local and that T is the closed point t of S. The proof of (ii) $\Longrightarrow$ (i) is essentially identical to part 2°) of the proof of XIV.4.2. Let us briefly prove that (i) $\Longrightarrow$ (ii). The local duality theorem (SGA 5 I 4.3.2), applied to $\mathrm{D}\,G$, shows that 149

$$\mathrm{D}\left(\mathrm{H}^i_{\overline{u}}(\mathrm{D}\,G)\right) \simeq \left(\underline{H}^{-i-2\delta_t(u)}(G)\right)_{\overline{u}}.$$

Since G is concentrated in degree 0, one therefore has $\mathrm{H}^i_{\overline{u}}(\mathrm{D}\,G) = 0$ except possibly for $i = -2\delta_t(u)$; more precisely,

$$\mathrm{prof}_u(\mathrm{D}\,G) = \begin{cases} -2\delta_t(u) & \text{if } G_{\overline{u}} \neq 0, \\ \infty & \text{if } G_{\overline{u}} = 0. \end{cases}$$

It follows that, for every $u \in U$,

$$\mathrm{prof}_u(\mathrm{D}\,G) \geq -n - \delta_t(u).$$

Hypothesis (i) then gives $\mathrm{prof}_t(\mathrm{R}_!f(\mathrm{D}\,G)) \geq -n - c$. We transform this relation using the isomorphism $\mathrm{R}_!f(\mathrm{D}\,G) \simeq \mathrm{D}(\mathrm{R}f_*(G))$ (SGA 5 I 1.12) and applying the local duality theorem at t; this gives

$$(\underline{H}^i(\mathrm{R}f_*(G))_{\overline{t}} = 0 \qquad \text{for } i > n + c,$$

which is precisely (ii).

XIV.4.8

Under the hypotheses of XIV.4.4 with g proper (resp. XIV.4.5, resp. XIV.4.6 with g proper), if V is an open neighborhood of Y in X, the morphism **150**

$$\mathrm{H}^i(V, F) \to \mathrm{H}^i(Y, j^*F)$$

is bijective for $i < n-c-1$ and injective for $i = n-c-1$. Indeed, if $\iota\colon V \to X$ is the canonical morphism, it is enough to apply XIV.4.4 (resp. XIV.4.5, resp. XIV.4.6) to the complex $\mathrm{R}\iota_*(F_{|V})$. One may ask whether the preceding morphism is bijective for $i = n-c-1$ and injective for $i = n-c$. It plainly suffices that the hypotheses hold with n replaced by $n+1$; the following proposition shows that slightly less is enough.

Proposition XIV.4.9. *Let S be an excellent local scheme of characteristic zero with closed point t (resp. assume in addition to the preceding conditions that S is Henselian), let $f\colon X \to S$ be a scheme proper over S (resp. proper over $S - \{t\}$), and let U be an open subset of X that is the union of $c+1$ affine open subsets. Let Y be a closed subscheme of X whose underlying space is $X - U$, let $j\colon Y \to X$ be the canonical morphism, let F be a bounded-below complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X with constructible cohomology, and let n be an integer. Suppose that, for every point u of U,*

$$\mathrm{prof}_u(F) \geq \inf(n-1, n-\delta_t(u)) \quad (\textit{resp.}\ \ \mathrm{prof}_u(F) \geq \inf(n-1, n+1-\delta_t(u))).$$

Then, for every open neighborhood V of Y in X, the canonical morphism **151**

$$\mathrm{H}^i_{f^{-1}(t)}(V, F) \to \mathrm{H}^i_{f^{-1}(t)\cap Y}(Y, j^*F) \quad (\textit{resp.}\ \ \mathrm{H}^i(V, F) \to \mathrm{H}^i(Y, j^*F))$$

is bijective for $i < n-c-2$ and injective for $i = n-c-2$. Moreover, there exists an open neighborhood V_0 of Y in X such that, for every such V with $V \subset V_0$, the canonical morphism

$$\mathrm{H}^i_{f^{-1}(t)\cap V}(V, F) \to \mathrm{H}^i_{f^{-1}(t)\cap Y}(Y, j^*F) \quad (\textit{resp.}\ \ \mathrm{H}^i(V, F) \to \mathrm{H}^i(Y, j^*F))$$

is bijective for $i < n-c-1$ and injective for $i = n-c-1$.

Proof. For simplicity, set $\delta'_t(u) = \delta_t(u)$ (resp. $\delta'_t(u) = \delta_t(u) - 1$). The first assertion of XIV.4.9 follows from XIV.4.8, since the hypotheses of XIV.4.4 (resp. XIV.4.5) hold there with n replaced by $n-1$. They also hold for n itself, except at points u such that $\delta'_t(u) = 0$. Now, for $u \in U$, the condition $\delta'_t(u) = 0$ is equivalent to saying that u is a *closed point of* U_t (resp. a *closed point of* X). Let E be the set of points u of U such that $\delta'_t(u) = 0$; let us show that *for all but finitely many points* $u \in E$, one has $\mathrm{prof}_u(F) \geq n$. Let $\overline{S}$ be the strict localization of S at t, let S' be the completion of $\overline{S}$, with closed point t', and consider the Cartesian square

$$\begin{tikzcd} U' \arrow[r, "h"] \arrow[d, "f'"'] & U \arrow[d, "f"] \\ S' \arrow[r, "g"] & S. \end{tikzcd}$$

The depth hypotheses at the points of U are preserved when U is replaced by U' and F by its inverse image F' on U'. Indeed, let $u' \in U'$ and $u = h(u')$. If $u \notin E$, one has $\mathrm{prof}_u(F) \geq n - \delta'_t(u)$, and XIV.4.2.1 shows that this implies $\mathrm{prof}_{u'}(F') \geq n - \delta'_t(u')$. If $u \in E$, then u' is a closed point of U'_t (resp. a point of $X' = X \times_S S'$ closed in X'), and, since the fiber U'_u of h at u has dimension zero and h is regular, XIV.1.16 gives $\mathrm{prof}_{u'}(F') = \mathrm{prof}_u(F) \geq n-1$. **152**

(*)At least in the case where S locally admits a dualizing complex, for example when S is locally embeddable in a regular scheme.

Let K be a dualizing complex on S' normalized at the closed point t', and let $\mathrm{D}\,F'$ be the dual of F' with respect to $\mathrm{R}^!f(K)$. By XIV.4.2.2, the ètale-depth hypotheses at the points of U' translate into the relations

$$(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} = 0 \quad \text{for } q > -n - \delta'_t(u') \quad (\text{resp. } q > -n - 2 - \delta'_{t'}(u')),$$

if u' is not a point of $E' = h^{-1}(E)$, and

$$(\underline{H}^q(\mathrm{D}\,F'))_{\overline{u}'} = 0 \quad \text{for } q > -(n-1) \quad (\text{resp. } q > -n - 1),\ \text{if } u' \in E'.$$

Let $G = \underline{H}^{-(n-1)}(\mathrm{D}\,F')$ (resp. $G = \underline{H}^{-n-1}(\mathrm{D}\,F')$). Since G is a constructible sheaf, the set of points at which its geometric fiber is nonzero is constructible (SGA 4 IX 2.4 (iv)). By hypothesis this set is contained in the set E' of closed points of $U'_{t'}$ (resp. of points of U' closed in X'); it follows from XIV.4.9.1 below that this set consists of finitely many points. Applying XIV.4.2.2, one sees that $\mathrm{prof}_{u'}(F') \geq n$ for all but finitely many points of E'. By XIV.1.16, it follows that, for all but finitely many points of E, **153**

$$\mathrm{prof}_u(F) \geq n.$$

Let V be an open neighborhood of Y in X, contained in the complement in X of the finite set of points u of E for which $\mathrm{prof}_u(F) = n - 1$. If $\iota\colon V \to X$ is the canonical immersion, set

$$F_1 = \mathrm{R}\iota_*(F_{|V})\ ;$$

then F_1 is a bounded-below complex of sheaves on X with constructible cohomology (SGA 4 XIX 5.1). We shall see that, for every point u of U, the complex F_1 satisfies

$$\mathrm{prof}_u(F_1) \geq n - \delta'_t(u). \qquad (*)$$

If $u \in U \cap V$, then $\mathrm{prof}_u(F_1) = \mathrm{prof}_u(F)$, and relation $(*)$ holds by hypothesis at the points of U that do not belong to E; at the latter points, it also holds by the choice of V. Finally, if $u \in U$ and $u \notin V$, then by XIV.1.6 g), $\mathrm{prof}_u(F_1) = \infty$. Apply XIV.4.4 (resp. XIV.4.5) with F replaced by F_1; this gives the asserted result, since, for every i,

$$\mathrm{H}^i_{f^{-1}(t)}(X, \mathrm{R}\iota_*(F_{|V})) \simeq \mathrm{H}^i_{f^{-1}(t)\cap V}(V, F) \quad (\text{resp. } \mathrm{H}^i(X, \mathrm{R}\iota_*(F_{|V})) \simeq \mathrm{H}^i(V, F).$$

Lemma XIV.4.9.1. *A constructible set E contained in the set of closed points of a Noetherian scheme X consists of finitely many points .* **154**

Indeed, E is a finite union of sets of the form $U \cap \complement V$, where U and V are open subsets of X; by hypothesis, all the points of $U \cap \complement V$ are generic points of this set, so there are only finitely many of them.

XIV.5 Geometric depth

To apply XIV.4.2 and its corollaries in practice, one needs a convenient criterion for verifying the étale-depth hypotheses at the points of U. We shall give such a criterion, using the local Lefschetz theorem XIV.4.5.

XIV.5.1

Let A be a Noetherian local ring; when we speak of the étale depth of A, we mean the depth at the closed point. We shall introduce a notion of "geometric depth of A"and use XIV.4.5 to compare it with the étale depth $\operatorname{prof\,\acute{e}t}(A)$.

Proposition XIV.5.2. *Let A be a Noetherian local ring; suppose that A is isomorphic to a quotient of a regular local ring B by an ideal I (this is true, for example, when A is complete, by Cohen's theorem (*EGA 0_{IV} 19.8.8*)). Let q be the minimal number of generators of I; then the number* $\dim(B) - q$ *is independent of the choice of B.*

The minimal number of generators of I is also equal to the dimension of the k-vector space $I \otimes_B k$, where k denotes the residue field of A. We reduce at once to the case where A is complete, since $\widehat{A} \simeq \widehat{B}/\widehat{I}$ with $\dim \widehat{B} = \dim B$ and $\mathrm{rg}_k(I \otimes_B k) = \mathrm{rg}_k(\widehat{I} \otimes_{\widehat{B}} k)$; for the same reason we may suppose that the ring B is complete. Let B and B' be two complete regular local rings, $f\colon B \to A$ and $f'\colon B' \to A$ two surjective homomorphisms, and $I = \operatorname{Ker}(f)$, $I' = \operatorname{Ker}(f')$. We must show that 155

$$\dim B - \mathrm{rg}_k(I \otimes_B k) = \dim B' - \mathrm{rg}_k(I' \otimes_{B'} k).$$

First consider the case in which there is a factorization of the form

$$\begin{tikzcd} B \arrow[rr, "f"] \arrow[dr, "g"'] & & A \\ & B' \arrow[ur, "f'"'] & \end{tikzcd}$$

with g surjective. Let $J = \operatorname{Ker}(g)$; then $J \subset I$ and $I/J = I'$. Since B' is regular, $\dim(B') = \dim(B) - \mathrm{rg}_k(J \otimes_B k)$, and J is generated by elements belonging to a regular system of parameters of B. It follows that there is an exact sequence

$$0 \to J \otimes_B k \to I \otimes_B k \to J/I \otimes_{B'} k \to 0,$$

and consequently

$$\dim B - \mathrm{rg}_k(I \otimes_B k) = \dim B - \mathrm{rg}_k(J \otimes_B k) - \mathrm{rg}_k(J/I \otimes_{B'} k) = \dim B' - \mathrm{rg}_k(I' \otimes_{B'} k).$$

The general case reduces to the preceding one; to see this, it is enough to show that one can find a complete regular local ring B'' and surjective homomorphisms $g\colon B'' \to B$ and $g'\colon B'' \to B'$ making the diagram commutative 156

$$\begin{tikzcd} & B \arrow[dr, "f"] & \\ B'' \arrow[ur, "g"] \arrow[dr, "g'"'] & & A \\ & B'. \arrow[ur, "f'"'] & \end{tikzcd} \tag{$*$}$$

Now, if W is a Cohen ring with residue field k, there is a local morphism $W \to A$ which lifts to B and B' (EGA IV 19.8.6), so that one has the commutative diagram

$$\begin{tikzcd} & B \arrow[dr] & \\ W \arrow[ur] \arrow[dr] & & A \\ & B'. \arrow[ur] & \end{tikzcd}$$

One can find integers n and n' and surjective morphisms $h\colon W[\![T_1,\dots,T_n]\!]\to B$ and $h'\colon W[\![T'_1,\dots,T'_{n'}]\!]\to B'$ which are morphisms of W-algebras (EGA 0_{IV} 19.8.8); if we then set $B''=W[\![T_1,\dots,T_n,T'_1,\dots,T'_{n'}]\!]$ and define g and g' as morphisms of W-algebras such that

$$g(T_i)=h(T_i),\quad g(T'_i)=b_i,\quad g'(T_i)=b'_i,\quad g'(T'_i)=h'(T'_i),$$

where b_i (respectively b'_i) is an element of B (respectively of B') lifting $(f'\circ h')(T'_i)$ (respectively $(f\circ h)(T_i)$), the diagram $(*)$ is indeed commutative.

Proposition XIV.5.2 justifies the following definition:

Definition XIV.5.3. Let A be a Noetherian local ring and $\widehat{A}$ its completion, which is thus isomorphic to the quotient of a complete regular local ring B by an ideal I; if q is the minimal number of generators of I, the *geometric depth* of A is the number **157**

$$\operatorname{prof\,géom}(A)=\dim B-q$$

Proposition XIV.5.4. *Let A be a Noetherian local ring. Then*

$$\operatorname{prof\,géom}(A)\le\dim A,$$

with equality if and only if A is a complete intersection.

We may suppose that A is complete. Let $A=B/I$, where B is a complete regular local ring and I an ideal of B. If $(x_1,\dots,x_q)$ is a minimal system of generators of I, then $\dim A\ge\dim B-q$, and saying that $\dim A=\dim B-q$ is equivalent to saying that $(x_1,\dots,x_q)$ belongs to a system of parameters of B (EGA 0_{IV} 16.3.7); the proposition follows immediately.

Proposition XIV.5.5. *Let A and A' be Noetherian local rings, and $f\colon A\to A'$ a local homomorphism. Suppose that f is flat and that, denoting by k the residue field of A, $A'\otimes_A k$ is a field, separable over k. Then*

$$\operatorname{prof\,géom}(A)=\operatorname{prof\,géom}(A').$$

After replacing A and A' by their completions, we may suppose that A and A' are complete **158**
(it follows from (EGA 0_{III} 10.2.1) that the flatness hypothesis is preserved, and this is evident for the other hypotheses). Let $A=B/I$, where B is a regular local ring and I an ideal of B. Since A' is formally smooth over A (EGA 0_{IV} 19.8.2), it follows from (EGA 0_{IV} 19.7.2) that one can find a complete Noetherian local ring B' and a local homomorphism $B\to B'$ such that B' is a flat B-module and $B'\otimes_B A\simeq A'$. Thus $A'\simeq B'/IB'$; moreover the ring B' is regular: indeed, if $\mathfrak{m}$ is the maximal ideal of B, $\mathfrak{m}B'$ is the maximal ideal of B', and, since $\mathfrak{m}$ is generated by a regular sequence by the definition of "regular", $\mathfrak{m}B'$ is generated by a B'-regular sequence (EGA 0_{IV} 15.1.14). Since one clearly has $\dim B=\dim B'$, and since I and IB' have the same minimal number of generators, the assertion follows.

Theorem 5.6(*)**.** *Let A be an excellent local ring of characteristic zero. Then*

$$\operatorname{prof\,ét}(A)\ge\operatorname{prof\,géom}(A).$$

One may suppose that A is strictly local and complete, since geometric depth and étale depth are preserved under passage to the strict henselization and to the completion by XIV.5.5 and XIV.1.16. Let $A\simeq B/I$, where B is a complete regular local ring, and let $(f_1,\dots,f_q)$ be a minimal system of generators of the ideal I. Thus one has **159**

(*)*Editor's note.* Illusie subsequently proved the inequality $\operatorname{prof}_x(\mathbf{Z}/\ell^\nu\mathbf{Z})\ge\operatorname{prof\,géom}_x(X/S)-\delta(x)+1$ for x a point of X, where X is a scheme of finite type over a trait S whose residual characteristic is prime to ℓ, and $\nu\ge1$. If S has characteristic zero, this is a consequence of Theorem XIV.5.6; see (Illusie L., "Perversité et variation", *Manuscripta Math.* **112** (2003), p. 271-295).

$$\pi = \text{prof géom}(A) = \dim B - q.$$

Consider the closed immersion

$$Y = \operatorname{Spec} A \to X = \operatorname{Spec} B,$$

and let $U = X - Y = \bigcup_{1\le i\le q} X_{f_i}$. If a denotes the closed point of X, one must show that, for every prime number p, one has

$$\mathrm{H}^i_a(Y, \mathbf{Z}/p\mathbf{Z}) = 0 \quad \text{for } i < \pi.$$

Since B is regular and excellent, one has prof ét$(B) = 2\dim X$ (cf. XIV.1.10) and consequently $\mathrm{H}^i_a(X, \mathbf{Z}/p\mathbf{Z}) = 0$ for $i < 2\dim X$. Thus, to prove the theorem, it suffices to prove that the morphism

$$\mathrm{H}^i_a(X, \mathbf{Z}/p\mathbf{Z}) \to \mathrm{H}^i_a(Y, \mathbf{Z}/p\mathbf{Z}) \tag{$*$}$$

is bijective for $i < \pi$. For this, apply the local Lefschetz theorem XIV.4.5 with $n = \pi + q - 1$, $c = q - 1$, and hence $n - c = \pi$. Note that $U = X - Y$ is the union of the q affine open subsets X_{f_i}. Let us show that, for every point u of U, one has:

$$\text{prof ét}_u(X) \ge \pi + q - 1 - \dim\left(\overline{\{u\}}\right) = \dim \mathcal{O}_{X,u}$$

(where $\overline{\{u\}}$ denotes the closure of u *in* $X - \overline{\{a\}}$). Indeed, it follows from XIV.1.10 that

$$\text{prof ét}_u(X) = 2\dim \mathcal{O}_{X,u} \ge \dim \mathcal{O}_{X,u}.$$

Using XIV.4.5, one sees that $(*)$ is bijective for $i < \pi$, which completes the proof of the theorem. **160**

Corollary XIV.5.7. *Let S be the spectrum of a field of characteristic zero (resp. an excellent henselian local scheme of characteristic zero), and let $f\colon X \to S$ exhibit X as a scheme proper over S (resp. over $S - \{s\}$). Let U be a union of $c+1$ open subsets of X that are* affine, *let Y be a closed subscheme with underlying space $X - U$, and let n and m be integers > 0. Suppose that, for every point u of U, one has*

$$\text{prof géom}(\mathcal{O}_{X,u}) \ge n - \dim\left(\overline{\{u\}}\right)$$

($\overline{\{u\}}$ being the closure of u in X). Then the canonical morphism

$$\mathrm{H}^i(X, \mathbf{Z}/m\mathbf{Z}) \to \mathrm{H}^i(Y, \mathbf{Z}/m\mathbf{Z})$$

is bijective for $i < n - c - 1$, and injective for $i = n - c - 1$.

Apply XIV.4.5 and XIV.4.6. The hypotheses on étale depth at the points of U are satisfied because, by XIV.5.6, one has

$$\text{prof ét}_u(X) \ge \text{prof géom}(\mathcal{O}_{X,u}) \ge n - \dim\left(\overline{\{u\}}\right).$$

XIV.6 Open questions

XIV.6.1

One may ask whether the implication (ii) $\Longrightarrow$ (i) of XIV.4.2 holds more generally for torsion sheaves F, not necessarily annihilated by a given integer m and not necessarily constructible . In the case where S is not of characteristic zero, it seems possible that this implication **161** remains valid, even for sheaves of p-torsion (p being the residue characteristic). Finally, it is also unclear whether the hypothesis that S is excellent can be removed.

XIV.6.2

Let X be a scheme proper over a field k, or else the complement of the closed point of a henselian local scheme, and let $j : Y \to X$ be a closed subscheme of X whose complement U is affine. Then, if F is a sheaf of sets on X or a sheaf of not necessarily commutative groups, the statements XIV.4.5 or XIV.4.6 and XIV.4.9 still make sense for such an F, provided that one restricts to small values of n. If u is a point of U, denote by $\bar{u}$ a geometric point lying over u, by $X(\bar{u})$ the strict localization of X at $\bar{u}$, and by $F_{\bar{u}}$ the fiber of F at $\bar{u}$. Then, possibly making certain hypotheses on X and F, for example by assuming X excellent (possibly of characteristic zero, or of equal characteristic using resolution of singularities) and F ind-finite (or, if necessary, even **L**-ind-finite with **L** prime to the characteristic of X), one would like to prove the following statements:

a) Let F be a sheaf of sets (resp. a sheaf of groups), and suppose that, for every point u of U, one has
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) \leq 1$$
(that is, for such a u, one has $\operatorname{prof}_u(F) \geq 1$). Then, as V ranges over the set of open neighborhoods of Y, the canonical morphism **162**
$$\varinjlim_V H^0(V, F) \to H^0(Y, j^*F)$$
is bijective (resp. one has the preceding conclusion and, moreover, the morphism $\varinjlim_V H^1(V, F) \to H^1(Y, j^*F)$ is injective). If F is constructible, one may replace the $\varinjlim$ by the cohomology of V for V "sufficiently small".

b) Let F be a sheaf of sets (resp. a sheaf of groups), and suppose that, for every point u of U, one has $\operatorname{prof}_u(F) \geq 2 - \dim(\overline{\{u\}})$, which is also expressed by the relations
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ is bijective if } \dim(\overline{\{u\}}) = 0$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ is injective if } \dim(\overline{\{u\}}) = 1.$$
Then the canonical morphism
$$H^0(X, F) \to H^0(Y, j^*F)$$
is bijective (resp. one has the preceding conclusion and, moreover, the morphism $H^1(X, F) \to H^1(Y, j^*F)$ is injective).

c) Let F be an ind-finite sheaf of groups. Suppose that, for every point u of U, one has
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective if } \dim(\overline{\{u\}}) = 0 \text{ or } 1,$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) = 2.$$
Then , as V ranges over the set of open neighborhoods of Y, the canonical morphisms **163**
$$\varinjlim_V H^0(V, F) \to H^0(Y, j^*F) \text{ and } \varinjlim_V H^1(V, F) \to H^1(Y, j^*F)$$
are bijective. If F is constructible, one may replace the $\varinjlim$ by the cohomology of V for V sufficiently small.

d) Let F be a sheaf of groups. Suppose that, for every point u of U, one has $\operatorname{prof}_u(F) \geq 3 - \dim(\overline{\{u\}})$, which is also expressed by the conditions

$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective, and } H^1(X(\bar{u}) - \bar{u}, F) = 0 \text{ if } \dim(\overline{\{u\}}) = 0,$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective if } \dim(\overline{\{u\}}) = 1,$$
$$F_{\bar{u}} \to H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) = 2.$$

Then the canonical morphisms

$$H^0(X, F) \to H^0(Y, j^*F) \text{ and } H^1(X, F) \to H^1(Y, j^*F)$$

are bijective.

As evidence in favor of these statements [(*)], let us mention XIII XIII.2.1, X 3.4, and XII XII.3.5. Note that, by the argument of XIV.4.8 and XIV.4.9, statement a) (resp. c)) would follow from b) (resp. d)).

XIV.6.3

The following statement, analogous to XIV.5.6, would follow from d): if A is a noetherian local ring (possibly excellent) and $\operatorname{prof\,géom}(A) \geq 3$, then $\operatorname{prof\,hop}(A) \geq 3$. To see this, realize $Y' = \operatorname{Spec} A$ as a closed subscheme of a regular local scheme $X' = \operatorname{Spec} B$, whose complement is the union of q affine open subsets, with the relation $\dim B - q = \operatorname{prof\,géom}(A)$. For every point x of X', if $n = \dim B$, one has $\operatorname{prof\,hop}_x(X) \geq \inf(3, n - \dim(\overline{\{x\}}))$ (cf. XIV.1.11), and one deduces from d) that this implies $\operatorname{prof\,hop}_y(Y') \geq \inf(3, n - q - \dim(\overline{\{y\}}))$ for every point y of Y'. The result is then obtained by taking y to be the closed point of Y'. **164**

XIV.6.4

A variant of XIV.4.2, at least of the implication (ii) $\Longrightarrow$ (i), should also hold in the complex analytic case, provided one works with "analytically constructible"sheaves (cf. XIII); the proof would be analogous to that of XIV.4.2, using the duality theory of J.-L. Verdier. Note, however, that for the complex analytic analogue of the noncommutative variants mentioned in XIV.6.2, not even a method of approach is available for the statements concerning the fundamental group suggested by the results of Exposés X, XII, and XIII, recalled at the end of XIV.6.2. Indeed, the methods of the Seminar seem inextricably tied to the case of *finite* coverings (which may be studied in terms of coherent sheaves of algebras).

[(*)] *Editor's note.* all the statements of XIV.6.2, apart from the constructible variants, were proved by Mrs Raynaud; see (Raynaud M., "Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental", *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, Corollary III.1.3).

Bibliography

[1] M. Artin & B. Mazur – "Homotopy of Varieties in Etale Topology", in *Proceedings of a Conference on Local Fields*, Springer, 1967.

[2] J.-P. Serre – *Cohomologie Galoisienne*, Springer-Verlag, 1964.

[3] J.-L. Verdier – *Des catégories dérivées des catégories abéliennes*, with a preface by Luc Illusie, edited by Georges Maltsiniotis, Astérisque, vol. 239, Société mathématique de France, Paris, 1996.

Index of notation

Terminological index